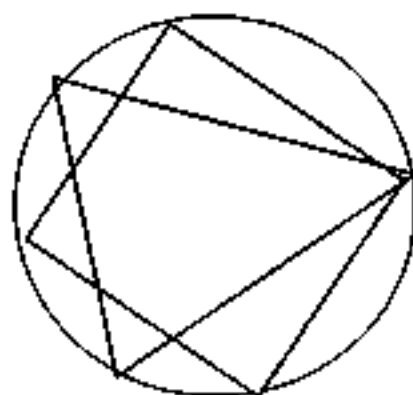


THE MYSTERIES OF GANN

ANALYSIS UNVEILED!



By

DANIEL T. FERRERA

2001

**SACRED SCIENCE INSTITUTE
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FORWARD

Since the death of W. D. Gann in 1955, the legend and mystery surrounding this great market master has continued to grow until it has today reached legendary status. One must ask why the status of the legacy of one so difficult to understand in his written teachings that the majority of people who approach his books and courses either give up right away or spend years, even decades, in an almost obsessive search to understand the theories of this enigmatic man. Certainly Gann himself is responsible for creating much of this legend beginning with his interview in the 1909 Ticker & Investment Digest, the precursor of the Wall Street Journal, where, under observation, he made 286 trades of which 264 were profitable and only 22 losers. He explains his success as being based upon a scientific "Law of Vibration", an integrated empirical science based upon mathematics, and natural philosophy, and at that point the legend was born.

It would be over 10 years before Gann began his writing career producing seven books and many hundreds of pages of advanced stock and commodity market courses, containing varied levels of revelation of his market secrets accordingly as the student paid higher and higher prices, with his Number Three Master Forecasting Method going for \$2,500 in 1939, and other of his "complete" courses priced at \$5,000, with these prices excluding his greatest secrets like the Mathematical Formula For Market Predictions and his astrological system.

What is most perplexing, however, is that even today, with access to most of these rarest and valuable of Gann's teachings publicly available at far more affordable prices than in Gann's time, there are still very, very few individuals who have been able to master, let alone even profitably apply most of Gann's analytical techniques. One wonders whether there were not further courses which explained even more clearly the inner mechanisms of Gann's systems, and there are certainly rumors that such courses did indeed exist. There are subjects like astrology, which are known to have been used by Gann, the clues to which may be found scattered throughout all of his work, yet there is no explicit instruction on the use of any such method anywhere to be found. Even the "Law of Vibration" is almost never referred to again after the 1909 interview, and there is no instruction on the application of natural science in his work. Yet, anyone who has studied these subjects is fully aware that these scientific and metaphysical principles provide the fundamental basis to every element of Gann's theory.

What is most surprising in modern Gann research is that there are so few analysts who follow Gann's clues and look to the underlying principles behind his theories. One finds traders uselessly drawing "Gann Angles" on their charts without having the slightest sense of their meaning or true purpose. The following course was written to rectify this problem, and to "unveil" the mystery of Gann Analysis by revealing the scientific foundations upon which it is based, and their integrated application to trading and forecasting. Outside of the work of Dr. Jerome Baumring, and the classic market masters, this is the first work we have seen of a standard of quality that lives up to that of the Master, W. D. Gann, himself.

W. Bradstreet Stewart
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The Secret of Gann Angles

Much has been published regarding WD Gann's use of geometrical angles. Today, many software programs (incorrectly) have these angles as standard features for technical analysis. However, few (if any) people know what they really are or how to actually use them. I have heard so many stories about these angles that it is clear to me that they are completely misunderstood. Some will say that your charts have to be scaled to a particular price increment to square out the chart. Others will say that it is based purely on geometry and because the Tangent of 45 is 1 then this must be the real 1 x 1 angle. The funny thing is both arguments are correct but not completely correct in a Gann sense! First, lets examine what Gann actually said about this subject:

"After long years of practical experience, I have discovered that Geometrical Angles measure accurately Space, Time, Volume, and Price.

Mathematics is the only exact science, as I have said before. Every nation on the face of the earth agrees that 2 and 2 make 4, no matter what language it speaks. Yet all other sciences are not in accord as mathematical science. We find different men in different professions along scientific lines disagreeing on problems, but there can be no disagreement in mathematical calculation.

There are 360 degrees in a circle, no matter how large or how small the circle may be. Certain numbers of these degrees and angles are of vast importance and indicate when important tops and bottoms occur on stocks; as well as denote important Resistance

Levels. When once you have thoroughly mastered the Geometrical Angles, you will be able to solve any problem and determine the trend of any stock."

"We use geometrical angles to measure Space and Time periods because it is a shorter and quicker method than addition or multiplication, provided you follow the rules and draw the angles or lines accurately from tops and bottoms or extreme highs and lows."

Gann left several clues in this introduction to geometrical angles in his veiled language. Did you spot them? Why was Gann so particular when it came to drawing his charts? What did he mean when he said that these angles are a shorter and quicker method provided that you follow the rules? OK, that is enough; I will translate this into common English.

There are 360 degrees in a cycle (planetary orbit, "space movement"), no matter how large or how small the cycle may be. Certain divisions of these degrees (aspects) are of vast importance. This is why he was so meticulous about his charts. He was scaling them to the planetary orbits or cycles he found useful for each stock and commodity. 1 unit of price / day = the motion of the Earth around the Sun. 1 unit of price / week = to the quarter Moon cycle. 1 unit of price / month = the orbit of Saturn which is also the secret to the hexagon or cube chart but that's another matter. To prove that this interpretation is correct we will study Gann's coffee letter to a private student. I have provided a copy of this letter and would encourage you to read it first before reading my analysis! Gann has some mistakes in this letter and it is my belief that they were

intentional since the math was basic and in his other published works it's hard to find even a single spelling error.

"In making my predictions, I use geometry and mathematics, just as the astronomer does, based on immutable laws". WD Gann, 1927 Tunnel Thru the Air.

Analysis of Gann's Coffee Letter

First off some basic Astronomical or Astrology background is needed. There are 360 degrees in the circle of the Zodiac. The zodiac is divided into 12 equal parts of 30 degrees each. The starting point is Aries where the sun crosses the ecliptic at the March 21st Vernal Equinox. Then come Taurus, Gemini, Cancer (Summer Solstice), Leo, Virgo, Libra (Autumnal Equinox), Scorpio, Sagittarius, Capricorn (Winter Solstice), Aquarius, and Pieces.

0 to 30 = Aries ♈	150 to 180 = Virgo ♍	300 to 330 = Aquarius ♒
30 to 60 = Taurus ♉	180 to 210 = Libra ♎	330 to 360 = Pieces ♏
60 to 90 = Gemini ♊	210 to 240 = Scorpio ♏	360 = 0 Aries ♈ again
90 to 120 = Cancer ♋	240 to 270 = Sagittarius ♐	
120 to 150 = Leo ♌	270 to 300 = Capricorn ♑	

In the first part of the letter, Gann starts breaking down the March 19, 1954 high price into zodiac degrees. Using a scale of one point (minimum tick) to one-degree 8729 equals 29 Gemini. $8729 \text{ ticks} \div 360 = 24.247$. Therefore we reject 24 circles or $360 * 24 = 8640$. This leaves us with $8729 \text{ minus } 8640 = 89$. Because Gemini starts a 60-degrees we get $89 - 60 = 29 \text{ Gemini}$. Using 1 cent to one degree $87.29 = 27\text{-degrees } 17 \text{ minutes of Gemini}$ because $87.29 - 60 = 27.29$ and $0.29 * 60 = 17$. There are 60 minutes to a degree just as there are 60 minutes to an hour or 60 seconds to a minute.

The next line says, "Using a scale of 30 points (Saturn) to one degree, equals 21 Capricorn." $8729/30 = 291$ and because Capricorn starts at 270 degrees of the zodiac $291 - 270 = 21$. Thirty is the number of years it takes Saturn to complete a single orbit around the Sun and this is why Gann used 30 points to a degree.

The dollar Value \$28,171 again is in Capricorn $28,171 - 27,000 \text{ (zero Capricorn)} = 11 \text{ 45 Capricorn}$. One could also argue that this value is 28 Aries depending upon the decimal point.

The average of the 5 options on March 19, 1954 was 8663. The average of the 5 options on March 19, 1954 was actually 8668. Here Gann makes a mistake, and he is also intentionally vague with how he came up with the zodiac degree. But here is the answer 8668 divided by 12 (Jupiter's orbit in years) = 722.333 reject the two circles or 720-degrees and you get $2^{\circ}20'$ Aries. Which is 90 degrees (not 60) or square to helio Jupiter. Actually, if you calculate this angle exactly, you get $87^{\circ}15'$ or 8715! Note: at the bottom of the letter, Gann shows the longitude of Heliocentric-Jupiter as $29^{\circ}35'$ Gemini. This means that the current price of 8729 at one tick per degree, i.e. the minimum fluctuation

is equal to 29° Gemini. This is within a half of a degree to being exactly equal to Jupiter's longitude. Helio-Uranus is at $21^{\circ} 52'$ Cancer and the Saturn price at 21° Capricorn is opposite to this location. In astrology, the squares and the opposition points are strong negative / resistant forces. It is not my intention to teach Astrology, as there are numerous books on the subject but it is an important subject for understanding W.D. Gann and the subject was a dominant feature in Gann's suggested reading list.

In the next two paragraphs, Gann starts throwing out all kinds of points per month without any explanation as to why he is using the numbers or the significance of the numbers. It appears to be random and of no use to the student. Again, I believe this is intentional, as Gann was known for making his student work through his examples to gain understanding. Based on many of today's Gann experts I'm confident Gann's geometric angles are still veiled in perpetual misunderstanding. Now we will remove the mystery.

Definition: The *True Gann Angles* are the minimum fluctuation of the underlying commodity, stock, or index multiplied by the number of years it takes a planet to complete a full 360 degree orbit. This value defines the 45-degree angle. All of the other angles are calculated from this master angle by dividing the value by 45° to get a price to degree value. This unifies a relationship with both price and time and is a Vector that can be projected into the future. Lets look at an example that will show where Gann is coming up with the numbers in his letter and how he is creating planetary orbit vectors that unite price with time.

It takes Saturn 30 years to complete an orbit around the Sun. The minimum fluctuation in coffee is 1 tick so $30 \times 1 = 30$ and therefore the 45 degree angle = 30 points or ticks / one degree of Saturn. The price degree ratio is $30 / 45 = 0.666667$ so as an example, the 75 degree angle would = 75×0.666667 or 50 points / degree of Saturn. (Gann said in the coffee letter: "that at 50 pts. Per month the 45 -degree angle would cross at 8715 on March 19, 1954". The math shows us that the 50 points per month number comes from Gann's 75-degree angle not the 45). However, Gann said that this date was 276 months from April 16, 1931 low of \$435. If you multiply 276×50 , you get 13,800, so this is an error or typo. At 30 points per month you get $276 \times 30 = 8280 +$ the 435 low equals Gann's calculation of 8715! Gann also says in the letter that from May 15, 1940 to March 19, 1954 - Saturn moved $181^{\circ}35'$, which gives a price of 6990. This is another typo! Converting $181^{\circ}35'$ to a decimal, we get 181.583. This amount times 30 is 5447.5. If you add this to the May 15, 1940 low of \$544 you get 5990 not 6990. He then says at 45 points per degree that $181^{\circ}35'$ of Saturn would yield 8715. Why did he use 45 points to a degree? Because it is his $67\frac{1}{2}^{\circ}$ angle based upon Saturn's orbit. $181.583^{\circ} \times 45 = \$8171.235 +$ the \$544 low = 8715! Here is the calculation for the 45 points per degree. The $67\frac{1}{2}$ degree angle is $67.5 \times 0.666667 = 45$ points / degree of Saturn. A 60-degree angle would be 60×0.666667 or 40 points / degree. Can you see why Gann was using these numbers or points per degree in his letter? All of the other Gann angles are calculated in this fashion. For convenience, the other angles that Gann used are:

Gann AnglesTangent Function

11.25	67.5	8X1= 82.87	1X8= 7.12
22.5	71.25	4X1= 75.96	1X4= 14.03
33.75	75	3X1= 71.56	1X3= 18.43
45	78.75	2X1= 63.43	1X2= 26.56
56.25	82.5	1X1= 45	
60	86.25		
63.75	90		

Note: Gann did not use the Tangent function; all his Angles are 3.75° apart.

Side Note: This tangent issue has been confusing for many Gann students. A friend of mine Ken Turkin, recently figured out that Gann took a square that was 12 by 12 and drew an arc on the inside of this square from the top left hand corner to the bottom right hand corner, i.e. the height of the square is the radius of the arc and the lower left hand corner of the square would represent the center of this arc. If you do this on graph paper, you will find that all the infamous 1 X 1, 2 X 1, 1 X 2, etc., will all be 3.75° apart! Why did Gann choose this particular square? The largest planet in the solar system is Jupiter; in fact, Jupiter is larger in mass than all of the remaining 8 planets combined. Jupiter takes 12 years to orbit the sun and the number twelve is mentioned multiple times in the Bible i.e. twelve tribes of Israel, twelve Apostles, etc. In terms of music, there are 12 and only 12 notes. You can have an infinite variation in terms of frequency (by multiplying or dividing by 2) but there are still only 12 tones or vibrations! Getting back to Jupiter, his

price degree ratio is 12/45 or 0.266. The reciprocal of this is 1/0.266 or 45°/12, equals 3.75°. On page 82 of The Tunnel Thru The Air, Gann says, "To those who understand numbers, '266' reveals some remarkable events that are to follow the election of the Pope". This is interesting since 12/45 = 0.266. The Galactic center is 266 or 26-Sagittarius.

<u>Gann Angle</u>	<u>Mars</u>	<u>Jupiter</u>	<u>Saturn</u>	<u>Uranus</u>
11.25	0.47	3.00	7.50	21.00
22.5	0.94	6.00	15.00	42.00
33.75	1.41	9.00	22.50	63.00
45=orbit around the sun in years	1.88	12.00	30.00	84.00
56.25	2.35	15.00	37.50	105.00
60	2.50	16.00	40.00	112.00
67.5	2.82	18.00	45.00	126.00
71.25	2.98	19.00	47.50	133.00
75	3.14	20.00	50.00	140.00
78.75	3.29	21.00	52.50	147.00
82.5	3.45	22.00	55.00	154.00
86.25	3.60	23.00	57.50	161.00
90	3.76	24.00	60.00	168.00
Price Degree Ratio	0.0418	0.266	0.666	1.867

- Move the decimals around on some of these numbers, they are very interesting! For example: The square root of 2, Pi, Phi or Fibonacci ratio, reciprocal of hours in a day, number of the beast, degrees in a circle, days in a month, weeks in a year, hours in a day, synodic cycles, etc....

To make sure that you understand the concept of Gann's Angles, let's say you want to figure the 60-degree angle for a NYSE listed stock for the planet Jupiter.

Answer: Jupiter takes 12 years to complete one orbit.

The minimum fluctuation of most stocks is an eighth:

$12 \times 0.125 = \$1.50$ per one degree of Jupiter for the 45 degree angle. $\$1.50/45 = 0.033333$ price degree ratio. $60^\circ \times 0.03333 = \$2.00 / 1$ degree of Jupiter. So lets say the low price took place one year ago and Jupiter has since moved 28° degrees. $28^\circ \times 2 = \$56.00$ price up from zero. If the low was $\$22.75$, then we would also look at the price $56 + 22.75 = \$78.75$. Also note that the square of the low would be $\$22.75/0.125 = 182^\circ$ degrees of the Sun, lunar quarters, Jupiter, Saturn and Solar years. Note that the entire zodiac is 360° and that $360 \times 1/8^{\text{th}}$ is 45° . Therefore, $\$45.00$ is equal to 360° in $1/8^{\text{th}}$ increments or $1/8^{\text{th}}$ per degree scale. The price low of $\$22.75/45 = 0.5055555$ and $360^\circ \times 0.5055555 = 182^\circ$ or 2° Libra or $\$22.75/0.125 = 182^\circ$. $\$22.75/\1.50 (Jupiter scale) = $\$15.166$ and this in degrees would be $\$15.166/45 = 0.33703703$ and $360^\circ \times 0.33703703$ is 121.33 , which equals $121^\circ 20'$ of the circle or $1^\circ 20'$ Leo. We would watch the stock closely as Jupiter casts aspects (especially 90, 180, 270-degrees) to the longitude date of the low, i.e. $1^\circ 20'$ Leo. The Saturn conversion would work out like this $30 \times .125 = \$3.75$ and $\$22.75/\3.75

= \$6.0666. This amount divided by 45 is $6.0666/45 = 0.13481481$ and this multiplied by 360° is 48.53 or $48^\circ 32'$ of the circle, which is $18^\circ 32'$ Taurus. This is the same procedure illustrated in the coffee letter with the only difference being that stocks have a different minimum fluctuation (vibration) than coffee. Because $\$22.75/0.125$ (an 8°) = 182° degrees we would project the orbits of the planets this amount into the future which is a $\frac{1}{2}$ complete orbit or opposition aspect plus 2 extra degrees. As an example, let's say that Mercury was at 3-degrees of Scorpio on the date of the low. We would simply go through an ephemeris and find Mercury at 5-degrees of Taurus, which is 182° from our first date. Then we would add the 182° degrees again, which would be 5-Scorpio, and this would be another important date for a trend change. We would do this with all the planets knowing that as each one moves 182° in longitude they are squaring the low price by time. I believe this is how Gann converted price to time and to my knowledge this technique has never been published in any written form. I will talk about this subject further when I cover Gann's Squaring of Price with Time or Converting Price to Longitude. It is extremely important that you understand this concept to it's fullest. The Gann Angles along with planetary time/price relationships are the *Key* to converting prices to degrees in the zodiac. From the few examples that exist, and my own experiments with the technique, it appears that it is important that the price conversion has an angular or astrology aspect relationship to the specific planet you used to convert the price. Then a secondary factor would be if other planets had an angular relationship to the same degree. We can see this in Gann's example above where the price of the 5 coffee options converted to the Jupiter scale are making a square aspect to the current heliocentric position of Jupiter. Also on this particular day, heliocentric Jupiter is square

to the Earth (or geocentric Sun) and the North Node of the Moon is at 20° 37' Capricorn conjunct the Saturn converted Price, which is in opposition to Uranus in Cancer. Also 8729 at 1 tick per degree is at 29° of Gemini, which is the current location of Heliocentric Jupiter. I have seen many people use pure zodiac longitude converted to price and there is nothing wrong with this technique if the price scales workout and have some mathematical relationship. However, it is extremely important that you understand the method of converting price based upon planetary orbits, especially the big guys Jupiter & Saturn.

Another important timing feature of this coffee letter, that was not discussed and to my knowledge was never disclosed by W.D. Gann, has to do with Geocentric Jupiter and the eclipse of the Sun that took place on January 5th 1954. I will discuss this again in detail when I get to the analysis of Tunnel Thru The Air, but for now we will illustrate this set up. If we look up the geocentric position of Jupiter during the eclipse on January 5th, 1954 we will find the location to be 18° 33' Gemini & in retrograde motion. On March 18, 1954, Jupiter in direct motion returns to the exact location it had at the eclipse! Providing the entry timing for the trade. In addition, Gann's personal Horoscope for Coffee (copy included) March 7 1882, has Saturn at 8°38' degrees of Taurus. On March 19, 1954, (retrograde) Saturn is at 8°34' of Scorpio in opposition (180-degrees) to natal Saturn. Jupiter's return to the eclipse location is Square to both the natal Sun and natal Uranus in Gann's coffee horoscope.

It is important to note that this letter is one of the only known examples of the true mathematical angles! Gann was about 74 years old when he wrote this letter and was very ill. Also, the recipient of the letter was a private student who signed an agreement with Gann not to reveal these rules and instructions to anyone! Because we are left with so little to go on we have to make some assumptions about using this technique. (1) From the evidence available it seems that we should primarily use this method with the following planets: Jupiter, Saturn, and the Sun as these support the instructions in the letter (I also use Uranus because Uranus's orbit divided by Jupiter's orbit = 7 & this number is mentioned most in the Bible.). Chart examples of Soybeans & Coffee prove that Gann also used Mars for geometric angles. The other planets are used for measuring the time element and also for financial astrology. These three planets (Jupiter, Saturn and the Sun) have a unique mathematical relationship with each other.

Cosmology 17

In one year, Saturn moves about 12 degrees, the Sun moves 360, and Jupiter 30 and there are 12 months in the year of about 30 days each. Therefore we have a nice harmony of numbers $360/12 = 30$ which is the number of days in a month and the number of years in Saturn's orbit. $360/30 = 12$ which is Jupiter's orbit in years and the number of months in a year. Or inversely, $30 * 12 = 360$. $30/12 = 2.5$ which is Jupiter's monthly average movement. In one month, the Sun will move 30 degrees, Saturn will move 1 degree and the Moon will complete an orbit around the Earth. $12/30 = 0.40$ and this amount times 360 = 144 or the square of 12 a very important Gann number! $144 * 2.5$ will take us back to 360. $144/2.5 = 144^{2/3}/360$. If we square 144, i.e. $144 * 144 = 20736$, which Gann

called the great cycle because 20,736 days approximates the 3 conjunctions of Jupiter with Saturn. The geometric mean of Jupiter's orbit & Saturn's orbit = $\text{Sqrt. } (12 \times 30) = 18.97$ years. This is equal to the Metonic Lunar Cycle, which causes Full Moons, New Moons, and eclipses to occur at approximately the same calendar day and zodiac degree every 19 years. The harmonic mean = $2 \times (12 \times 30) / (12 + 30) = 17.143$ years. (Readers of Bradley Cowan's Four Dimensional Stock Market Structures & Cycles will want to study this in connection with his market cube theory as it relates directly to this time period), which is an extremely important cycle in the markets. 17 years is related to the Saturn cycle by the square root of 3 (diagonal of a cube). 17 years is related to Jupiter's orbit by the square root of 2 (diagonal of a square). 17 years is one 5th of Uranus's orbit. 17 years is 9 times larger than the Mars cycle & also 1/9th of the Neptune's cycle. 17 years expressed in days is 6209. This number divided by 225 (Venus's orbit in days) = 27.59 which is close to the orbit of the Moon around the Earth. There are 17 steps to the entrance of the Great Pyramid of Gizeh. The sum of the digits 1 through 17 = 153 (a number that appears in the Pyramid and in the Bible multiple times), i.e. The Grand Gallery is 153 feet long, the number of fish caught by Jesus' disciples was 153, Jesus' public ministry lasted 918 days (153 * 6). 153 equals 9 times 17. Note: The United States was formed with the signing of The Declaration of Independence, July 1776. If we add 153 years or 9 cycles of 17, we get the year 1929 the Peak of a Major period of U.S. economic expansion followed by the "Great Depression". The Florida Bank Failures occurred in July of 1929. Working backwards 153 years from 1776, we get the year 1623. Another important cycle is 37 years. Thirty seven is two complete lunar node cycles of 18.6 years, Two geometric mean cycles of Jupiter and Saturn, and is also 12 (Jupiter's

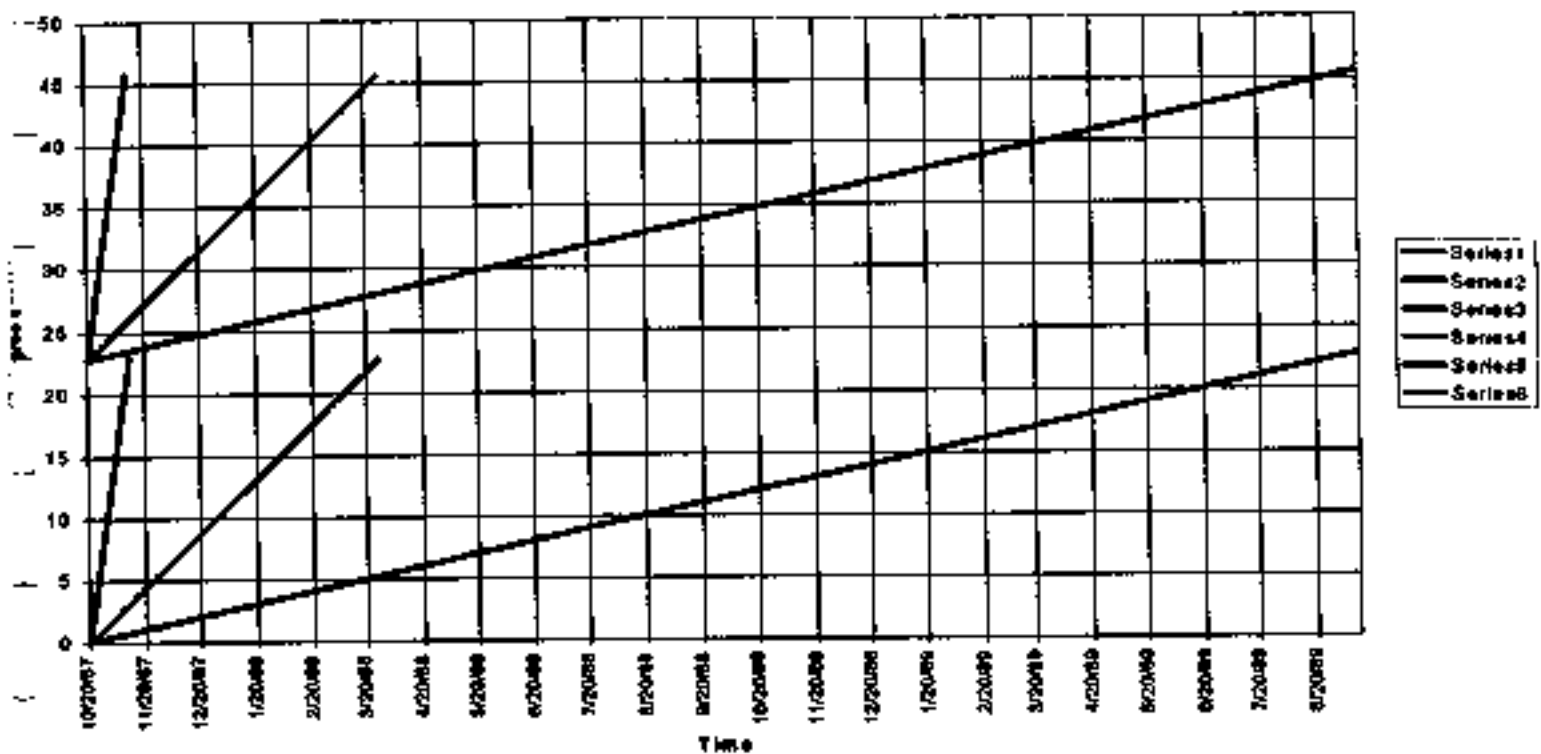
orbit) multiplied by Pi, i.e. $12 * 3.14 = 37$. Adding 37 years to dates of major highs and lows usually define future major turning points! Now that we have laid down this theory, and have shown astronomical/astrological proof direct from Gann's personal letter to a private student let us now look at his non-astronomical approach.

Gann said that: "the first and most important angle to draw was the 45-degree angle that moves up one point per day, one point per week or one point per month". He says that this is a 45 because it divides the space and time periods into 2 equal parts. As long as the market or a stock stays above the 45-degree angle, it is in a strong position and indicates higher prices. Below the 45-degree angle indicates a weak position and lower prices.

This is a very interesting quote because in terms of geometry these angles would all have different slopes and could not be equal (or 45-degrees). On the other side, in terms of the concept of cycles the slopes are equal (1 unit of price / 1 unit of time) as each has a common relationship with price / time which = the slope.

If we take the stock example above, the low price of \$22.75 would take 22.75 days (Sun), Weeks (lunar quarters), Months (Saturn) and years. Let us say that the low occurred on October 20th 1987. Therefore we have the following: 22.75 days = November 11th, 1987, 22.75 weeks = March 27th, 1988 and 22.75 Months is September 11th, 1989 (22.75 years would be out in July 2010)

Time Price



The lower portion illustrates the 45-degree lines up from zero and the upper half illustrates the 45 lines up from the low price of \$22.75. It is easy to see that in terms of traditional geometry that these trend lines are completely different. However, all are achieving the same objective of relating a time cycle to price. These are supposed to be the guidelines for determining the current position (strength or weakness) of a stock or commodity under analysis according to Gann's instructions (Gann Angle Course, copy included). This is another reason why I believe Gann actually used the planets. By using the planets, he could simply look in an ephemeris and locate the dates that were related to the price. However, based on this course, we would be interested in non-planetary motion of 22.75 degrees ($60\text{minutes} \times 0.75 = 45 \text{ min.}$) i.e. 22-degrees, 45 minutes or 22.75 days, weeks, months and years as is drawn out in the graph above. This seems to

contradict his coffee letter especially when we consider that the geometric angles are supposed to be an easier and faster way to measure "space movement", and this method is certainly faster and easier. If we use the instructions in the coffee letter we would need 182°-degrees or 8 times as much space movement to balance price with time (remember the minimum fluctuation on stocks is an 8th) with each 22°45' equaling 1/8th of the actual time cycle. Therefore, you would still get market hits using Gann's course instructions because each turning point is the one-eighth harmonic of the actual price cycle in terms of degrees, longitude or time. However, you would never realize that the second square of price, i.e. the second cycle of 22.75 days, weeks, months or years represents only the first 90° of the price cycle, and the 4th square of price would be the opposition of 180°, etc., with 182 days, weeks, months and years equaling the full 360°. The non-astronomical approach is definitely faster and easier and will quickly put you in the approximate time area to be on the look out for market reversals or changes in trend. The price expressed as a time cycle of solar years is the strongest cycle in terms of Gann's quick non-astronomical approach because it has the strongest correlation to an astronomical relationship, i.e. the Earth's orbit around the Sun. If you take the harmonics of this cycle, or every 1/8th of 22.75 years, you would look for the stock to change trend every 2.843 years. If you take the 1/8th harmonic of 2.843 years, you get 0.35546875 and this multiplied by 365.25 (days in a year) is every 130 days.

If we study the example in his geometric angle course, on page GA-16, we find him using \$1 / month for U.S. Steel Corporation. He starts the angle up from the date of the low (October of 1907) from a price of zero and the stock advanced (+240 or two

triangles) to a price of \$261.75 ($\approx 24^\circ$ Capricorn) in 262 months (Degrees of Saturn or Lunar months?). Saturn would have moved 275 degrees from 25 Pieces to 29 32 Sagittarius. $30 \times .125 = \$3.75/45 = 0.08333 \times 11.25$ (angle) = .9375 per degree of Saturn $\times 275$ degrees = \$257 & 7/8ths (pretty close). Lunar months would = 288 (the second square of 12, or 144×2) and the 11.25 angle would cross at about \$270. Using solar months we get his answer of 262. However, the low in 1907 was late October around the 30° and this would really be \$261. In any event, when the price of U.S. Steel met with resistance at these levels and dropped below the (geometric angles) longitude price lines, it indicated weakness to Gann. Another very important part of this example is that the time period in days is approx. 7980 divided by the low of \$21.875 = 365 or a solar year. Therefore: we take the low price of 21.875 and add it to the year 1907 and we get 1929.875, or the year 1929 + 7/8ths. $365 \text{ days} \times 0.875 = 320 \text{ days}$ (7/8ths of a year) and this amount added too late October 1907 would put us in September of 1929 (Squaring out Price with Time)! The price advance from the low would = $262 - 22 = 240$ or Trine ($120 + 120$) a harmonic point in the circle. The price of \$262 is equal to 22 Sagittarius and geocentric Saturn is located at 23 55 Sagittarius on September 3, 1929. This means that the price failure took place at the geocentric longitude of Saturn and then fell below the Heliocentric-Saturn angle @ \$257 & 7/8ths during a time that was 21 & 7/8ths years from the 1907 low of \$21.875! As you can see from this explanation, there is a lot of detail between the lines of Gann's example. Gann left clues but no answers. He wanted his students to discover the secrets of his methods by working through the techniques he was describing and gaining knowledge from actually doing the work for themselves. Make sure that you understand the above concepts & techniques before reading further. I

should also point out that the 1907 price low of \$22.875 converted to degrees is $22.875/0.125 = 183^\circ$ (3° Libra) and Heliocentric-Venus has since moved $12,783^\circ$. If you reject the 35 circles or complete orbits, you get 183° or 3° Libra, i.e. $12,783 - 12,600 = 183$. The position of Heliocentric Saturn is 30° aspect to the \$261.875 (Φ^2), or 24° Capricorn and Heliocentric Mars at 25° Libra is 90° aspect to the \$261.875 price. Pluto is at $18^\circ 06'$ Cancer, which is 75° aspect to the \$22.875 (3° Libra) low. Jupiter and Venus are at $3^\circ 15'$ and $6^\circ 43'$ Gemini, respectively or 120° (Trine) the 1907 low of 3° Libra. The \$261.875 in a Saturn scale is $18^\circ 40'$ Libra, which is square to Pluto in Cancer and 45° to Neptune at 1° Virgo. Gann's angle course page GA-27 says, "U.S. Steel was incorporated February 25, 1901". If we look up the heliocentric positions of the planets on this date (Natal Chart), and compare them with the positions on September 3, 1929, we get the following astrological aspects: Uranus is near exact square natal Saturn, Mars is square natal Mercury, Venus is square the natal Sun, Mercury is conjunct natal Uranus, Jupiter is 144° to natal Saturn and the Sun is 45° to natal Mercury. Most of these aspects are near exact.

Now that we have explained these concepts lets get back to Gann's Coffee letter:

- April 16, 1931 to March 19, 1954 – Geocentric Saturn moved 285° & 38' minutes (to convert minutes in decimal form, you divide by 60). $38\text{min}/60 = 0.633333$ therefore, Saturn moved 285.633. Because it takes Saturn 30 years to complete 1 orbit the 45-degree angle would equal

- $285.6333 * 30 = 8569$. Gann has 8572, which is off by 3 ticks. Next he has October 1, 1931 to March 19, 1954 – Geocentric Saturn moved 231, which would $= 7230$. Here again, Gann is trying to confuse and teach at the same time. Here is how he calculated this price: $231 * 30 = 6930$. The low price on October 1, 1931 was \$300 and $300 + 6930 = 7230$, which is his price calculation. The next line reads May 15, 1940 to March 19, 1954 – Saturn moved $181^{\circ} \& 35'$ minutes using 45 points to 1 degree $= 8715$. This would be the $67 \& \frac{1}{2}$ -degree geometric angle because $30/45 =$
- 0.666667 for the Price/Degree ratio and $67.5 * 0.666667 = 45$ points per degree. $45 * 181.583 = 8145.58 +$ the May 15, 1940 low price of 544 $= 8715$. The rest of the geocentric calculations should be easy for you to understand at this point.

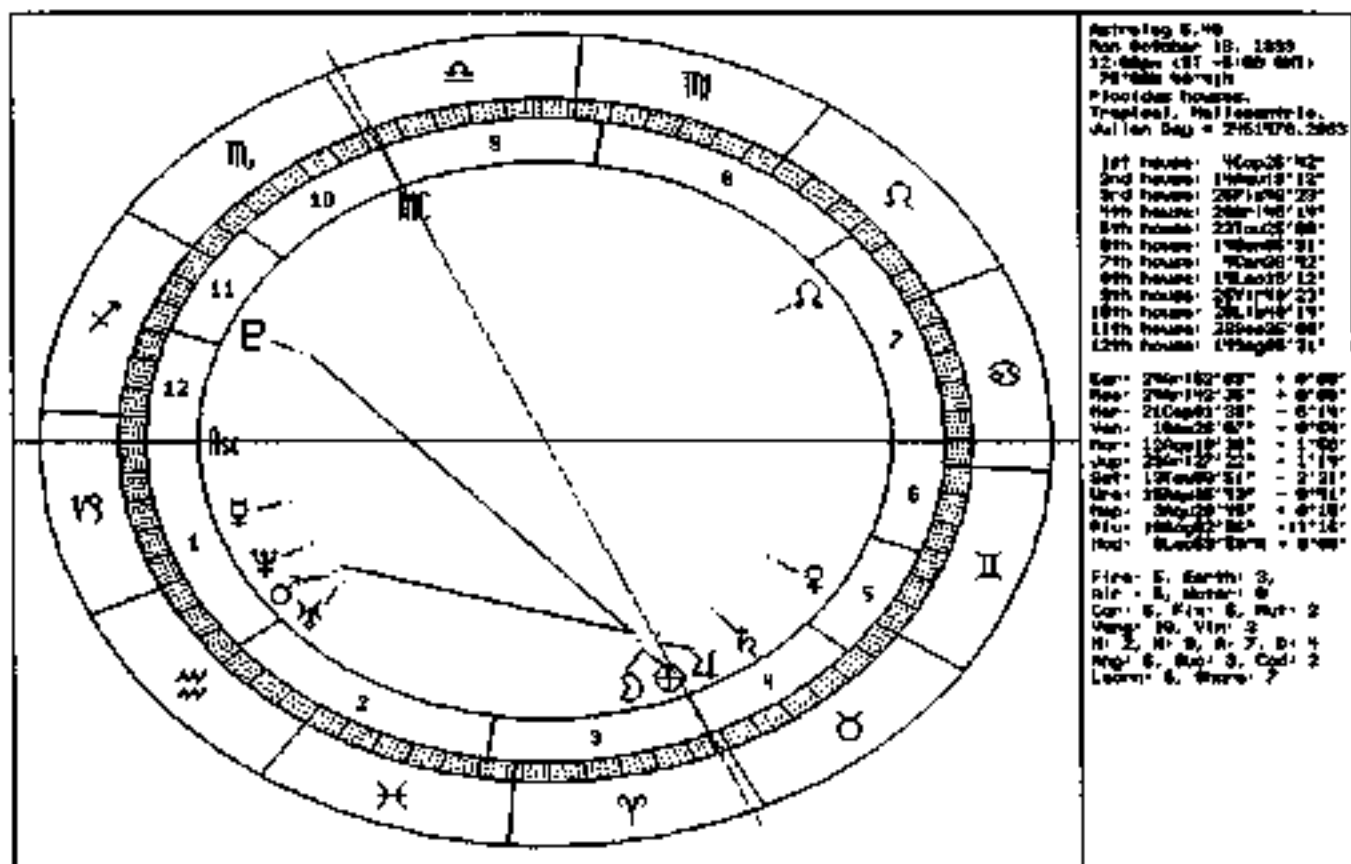
From here, Gann moves on to the heliocentric degrees traveled by Saturn. With the background that I have given you, it should be extremely simple to see where Gann is generating these price levels. One very interesting discovery that I made in the process of reviewing my notes and writing down this information is Gann's first heliocentric example. It reads: April 16, 1931 to March 19, 1954 – Heliocentric Saturn moved 287-degrees & 15 minutes, which equals 17-degrees 13 minutes Capricorn, Price 8632. This seems very innocent and obvious from our mathematical calculation procedure but I think there is one more gem of Gann's wisdom hidden in this tiny sentence! Here is my discovery: If we look up the date April 16, 1931 in a heliocentric ephemeris, we will find Saturn located at 17-degrees 18 minutes of Capricorn, which is equal to 287.283-degrees of the circle on the date of a major low. From this date to the March 19, 1954 date,

heliocentric Saturn moved this exact amount (287-degrees), which in my experience with studying Gann is well beyond coincidence! Although I have not had ample time to perform a study of this specific technique, i.e., of relating a zodiac position to a future time cycle. I have utilized a similar idea (that I think Gann would be proud of) where I try to locate future sensitive degrees in the zodiac based on the Harmonic Mean and Geometric mean of a single planets longitude on the date of a major high or low price. This works very well, which gives me great confidence in the discovery I just presented because the points are all related. To illustrate the concept I will provide an example:

- The Dow Jones Industrial Average made an important low on October 18, 1999.
- The heliocentric position of the planets are illustrated in the following horoscope:

Heliocentric Mercury is at 21 Capricorn = 291.

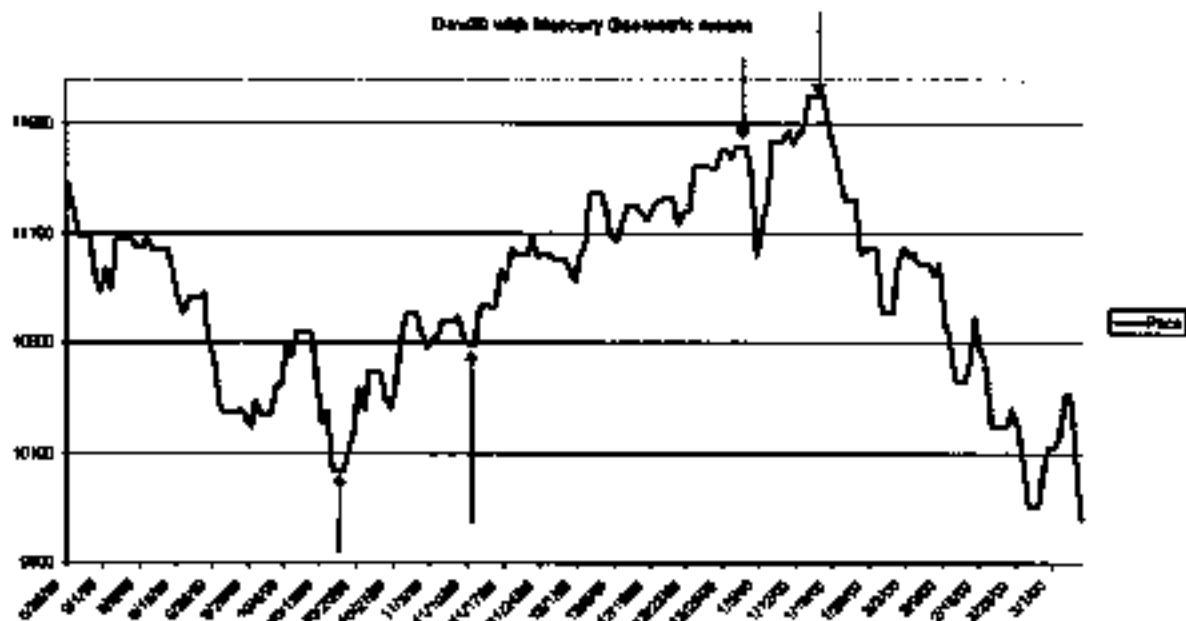
Heliocentric Mars is at 12 Aquarius = 312.



Now lets locate our sensitive degrees using the harmonics of the square & triangle.

To locate Mercury's geometric mean points that are square to 21 Capricorn we raise our aspect by the power of 2 and divide by the October 18, 1999 position. So $90^{\circ} \wedge 2 = 8100$ divided by 291 = 27.83 (Aries). To convert the decimal to minutes multiply by 60. Sixty times 0.83 = 50. So we have 27-degrees 50-minutes of Aries for the first square location. This location was hit November 11, 1999. The next square is $270^{\circ} \wedge 2 = 72,900/291 = 250.51$ or 10-degrees 31-minutes of Sagittarius. This location was hit by Mercury on December 31, 1999. Mars geometric mean square would be $8100/312 = 25.96$ or 25-degrees 57-min. of Aries. This point was hit on Friday, February 11, 2000 and the Market dumped 260 points setting in a major low that was 12% off the January 14th (Mercury) high. The market then reversed its downtrend and advanced nearly 300 points on February 14th & 15th. I will go into this subject again later in the material so make sure you understand the math involved. Suffice it to say I believe Gann was showing this student a very valuable technique via an Astronomical Balancing of Price & Time. I doubt that anyone has ever noticed this before including the recipient of this unique letter. It appears that a balancing of Zodiac Longitude was important to Gann. If for example, the low in coffee occurred when Saturn was at 11 degrees of Libra or 191 of the circle, then a balancing point would occur 191 degrees into the future when Saturn hits 22-degrees Aries.

Note: The highest point on the chart below is January 14, 2000 and Helio-Mercury is again at 21 Capricorn!



The rest of this coffee letter is written clearly and in my opinion needs no further explanation. If you not understand the rest of the information, you should study some basic astrology. Preferably, titles listed on Gann's recommended reading list.

Forecasting With Cycles

Because the coffee letter ends with geocentric movements of Mars, it is appropriate to discuss cycles. The reason for using planets as a measurement for timing potential changes in market trends is that they are natural occurring units of time. This is not astrology! Many people will laugh at this concept and say that it is a ridiculous theory. But these same people have no problem in accepting standard measurements of time such as days, weeks, months, and years. This is why Gann did not talk or write about astrology or planetary cycles even though it was his main forecasting tool. If we examine the typical measurements of time, we will see that they are all based upon planetary movement. It just so happens to be the planet we live on.

A day or 24 hours is simply the amount of time it takes the Earth to rotate or spin 360 degrees on it's axis. A week is based on the Lunar quarters, a month is the amount of time it takes the Earth to move through a zodiac sign or 30-degrees which also completes a 360-degree revolution of the Moon around the Earth. And finally a year is the amount of time it takes the Earth to complete an orbit around the Sun. Our entire concept of time is based upon planetary movement. So why should it seem "weird" to use other naturally occurring time measurements! Is it so far fetched to believe that cycles are actually caused by planetary orbits? Even if the planets do not cause cycles the orbits still represent a natural unit of time that may have some universal harmony that we are

unaware of. Whatever your personal beliefs are, the fact is that Gann used planetary orbits to time market movements. In the appendix, you will find Gann's Forecasting course, which is simply titled Forecasting. Gann produced two different forecasting courses; this one is more rare than the other one titled Forecasting by Time Cycles, which is typically found in Gann's Master Courses for either Stocks or Commodities. In the course provided, Gann gives very specific rules for figuring out when the next top or bottom should be expected. It is important to know that he is not talking about the typical calendar units of time. This should be evident to you by now based on the Coffee letter alone. Since Gann's rules for using these time cycles is clearly laid out in the course, I am not going to discuss how to use these cycles. What I am going to do is define in my own opinion the actual planetary measurements that Gann was really pointing the student towards (to increase accuracy). All of these cycles are heliocentric orbits, not geocentric. Because of the elliptical nature of planetary orbits and changing speeds due to perihelion and aphelion, these important time periods do not typically equal the fixed units of time that Gann wrote in the course. This is why you must learn how to use these planetary clocks!

Great Cycle – 60 Years: This cycle is the Third conjunction of the two largest planets in the solar system, Jupiter & Saturn. A conjunction between the two massive planets takes place every twenty years and is also Gann's 20-year cycle. (The appendix has a chart from Man in Cosmos that illustrates the importance of conjunctions of Jupiter & Saturn on repeating patterns in the Dow Jones Industrial Average. Note: This chart centers the calendars of the price charts with geocentric Jupiter conjunct the Sun. This time period is also two complete Saturn orbits.

50-Year Cycle: This cycle of 49 to 50 years (a period of "jubilee") is related to 210-degrees of Uranus which moves 30 degrees in 7 years. Other potential cycles are 3 & ½ conjunctions of Jupiter & Uranus (every 14 years) or the geometric mean of Saturn & Uranus, which is also equal to this time period. Gann also describes this as being 225° to 240° in the second circle of 360°, the period of being 586 months.

30-Year Cycle: This is the orbit of Saturn

20-Year Cycle: This is the conjunction of Jupiter & Saturn. This period is also 2/3rds of Saturn's orbit.

15-Year Cycle: This is ½ Saturn's orbit also conjunctions of Jupiter & Uranus (14 yrs.) should be watched.

10-Year Cycle: ½ cycle of Jupiter / Saturn conjunctions. Also = 120-degrees of Saturn's orbit.

7-Year Cycle: 30-degrees of Uranus & also ½ cycle of conjunctions with Jupiter & Uranus.

5-Year Cycle: This is the secret of the Hexagon Chart. Each 5-year period is based on 60 degrees of Saturn completing all six sides of a cube or hexagon. It is also $\frac{1}{4}$ of the Jupiter Saturn conjunction cycle.

Minor Cycles 3 & 6 Years: This is 90 & 180-degrees of Jupiter, respectively.

2-Year Cycle: One Mars Saturn conjunction, one Mars orbit 1.88 years.

7-Month Cycle: One Venus orbit is 225 days

3-Month Cycle: One Mercury orbit (such as 10/18/99 – 1/14/00 Dow Jones Mercury moved 360-degrees). This is also 90 degrees of the Earth's orbit.

1-Month Cycle: One lunar orbit around the Earth, 30-degrees of the Earth & 1-degree of Saturn

1-Week Cycle: 90-degrees of the Moon's orbit.

Gann goes on to say: "Stocks move in 10-year cycles, which are worked out in 5-year cycles – a 5-year cycle up and a 5-year cycle down. Begin with extreme tops and extreme bottoms to figure all cycles, either major or minor". From here on he explains his rules for adding specific time periods to tops and bottoms. This is where you need to use the planetary periods above to calculate the future turning point.

Pay close attention to Gann's instructions because he is specific about whether the future date is supposed to be a top or bottom. Also, note that you may get conflicting cycles, i.e. one cycle may be forecasting a top and another a bottom. This is why Gann said: "Time is more important than price". Because when the time element comes in you will be able to visually determine if the market is topping or bottoming from it's price action and trend. Cycles are additive, so you must weigh the total number of topping cycles against

the total number of bottoming cycles to be completely accurate in forecasting a top or bottom. This is really not important as far as the timing is concerned.

The 37-Year Stock Market Cycle

Earlier in the material, I discussed the importance of the 37-year cycle and the mathematical relationship it has to Jupiter & Saturn the largest planets in our solar (radiant energy) system. I also illustrated the cosmic harmony or cosmological key of 17 years as it relates to other planetary orbital periods. In its simplest form, you can add 37 years to any major high or low to estimate when major turns are due. For example, if you add 37 years to the 1907 "Rich mans Panic" low, you get 1944 and adding 37 again, you get 1981. The top in 1929 + 37 = 1966 and next would be 2003. These are just estimations, the actual low in the cycle may vary by + or - a year so it is very important that you understand planetary cycles as fundamental timekeepers. The 37-year cycle, which equals two Lunar Node cycles, is broken up into three distinct phases:

- **Phase 1:** This is the very beginning of the 37-Year cycle. This phase lasts 18 years (Jup/Sat geometric mean) bottom to bottom or one complete cycle of the North Node of the Moon. It typically consists of the classic Elliott Wave up pattern (5 waves to a top) over a 14 year (60-degrees of Uranus, and also 360-degrees of Jup/Urn) time period. Followed by a 4 year (two Mars orbits or two Mars Saturn conjunctions) "Bear Market", completing phase one.

- **Phase 2:** Starting from the end of phase one, this phase lasts about 7-years bottom to bottom and is the shortest phase in the 37-Year cycle. This phase reaches its market peak 5-years (60-degrees of Saturn) from it's starting point. This phase calls for a 5-year "Bull" move, which consists of two up legs (R.N.Elliott's a-b-c pattern) with one intervening "Bear" market or narrow trading range market. From this 5-year (Saturn) high, a 2-year (Mars) Mars/Saturn "Bear" market or sideways market begins completing 7-years total (Uranus 30-degrees).
- **Phase 3:** This is the final and usually most spectacular phase of the cycle. It begins at the end of phase 2 and lasts a total of 12-years (Jupiter) low to low. The average time span from this phase's beginning to its ultimate high is about 8 years (240-degree of Jupiter). From this major "1929 style peak, a 4-year (720-degrees of Mars, 120-degrees of Jupiter, 2 Mars/Saturn conjunctions) "Bear" market begins, typically with a market crash. At the end of the 4-year period the 37-Year cycle starts all over again.

I will give you some highs and lows of each phase in terms of typical calendar years. I strongly urge you to calculate these time periods with the planets as outlined above. Only then will you see and believe the value of planetary movements. At the time of this writing, we are in the 8 year "Bull" market of phase 3 which began after a sideways "Bear" market pattern that lasted from January 29th, 1994 to March 10th, 1995. If this cycle continues to hold, we should reach a 1929 style top near the year 2003. This time

period also correlates with the Uranus Square as described in Four-Dimensional Stock Market Structures & Cycles by Bradley F. Cowan. This point will be reached when Heliocentric Uranus reaches 0-degrees Pisces.

Phase 1	<u>Lows</u>	<u>Highs</u>
	1859-1896	1872-1909
	1896-1933	1909-1946
	1933-1970	1946-1983
	1970-2007	1983-2020
Phase 2	1877-1914	1882-1919
	1914-1951	1919-1956
	1951-1988	1956-1993
	1988-2025	1993-2030
Phase 3	1884-1921	1892-1929
	1921-1958	1929-1966
	1958-1995	1966-2003
	1995-2032	2003-2040

Please study these periods in terms of the actual highs and lows and planetary orbits. For example the phase one low shown above as 1933 actually occurred July 8th, 1932, the phase two low shown as 1988 above actually occurred October 19th, 1987. I am providing both charts in the appendix and historical data in Excel for the Dow Jones Industrial Average from January 1, 1900 to December 31, 1999. This will allow you to study the 37-Year cycle properly. The pure calendar year version of this cycle was originally published by Larry Williams in his book The Secret of Selecting Stocks For Immediate & Substantial Gains (Windsor Books 1995). The astronomical correlation's and refining of this cyclical pattern is the author's own work (Great job Larry!) and is very useful in forecasting market moves. My personal belief is that the year 2003 will probably be a low and not a high. There are many interesting numeric and astrological cycles that support this view. The correlation between the Lunar North Node and American Economic expansion indicates that the US economy will peak out when the Lunar Node is in Leo from September 17th, 1998 to April 4th, 2000. From here the economy should really start to cool down. Saturn enters Gemini on January 27th of the year 2001, which has historically been bad for the United States economically. Numerically, every major low from August of 1982 has followed a Fibonacci sequence in years. $1982 + 3 = 1985$, $1982 + 5 = 1987$, $1982 + 8 = 1990$, $1982 + 13 = 1995$ and $1982 + 21 = 2003$. Cycles also indicate a Republican President that is assassinated in office and strong prospects of a major US military conflict near 2003. My advanced Square of 9 course goes into this subject in detail.

Important Formulas & Techniques for Working with Planetary Cycles

First, it is a necessity that you learn the zodiac signs, zodiac symbols & degree of the circle, and the planetary symbols. Next you should memorize the time required for a planet to complete an orbit around the Sun. You should also memorize the time period required for conjunctions of the outer planets with each other. To illustrate the importance of this lesson, I will provide a few simple examples.

At the Stock Market Low August 12, 1982 Saturn was at 23-Degrees Libra. Adding 180-degrees to this position gives 23-degrees Aries which was hit April 22, 1998 where the vast majority of Stock Market Indexes topped before correcting over 20%. Also, Jupiter was at 13-degrees 45' Scorpio on August 12, 1982. Adding 540-degrees ($360 + 180$) yields 13-degrees 45' of Taurus, which will be hit on March 23, 2000.

The Asian Currency Panic occurred on October 22, 1997 with Mars at 18-degrees Capricorn. Adding 180-degrees to Mars gives 18-degrees Cancer which was the Russian Currency Panic August 31, 1998! This is Gann's 1-year cycle that often shows a change in trend in the 10th or 11th month.

Now we will look at an example with the relationship of two planets. The first thing you must remember is that a pair of planets can never be more than 180-degrees apart. Second, you must observe if the planets are getting closer together (applying) or getting

farther apart (separating). I will give an example of each; both will be updates from

Four-Dimensional Stock Market Structures & Cycles.

At the Market top January 29th, 1994 Saturn is @ 2 06' Pisces and Uranus is @ 22 28' Capricorn. The angle between the two is $332.10 - 292.47 = 39.63$ degrees or 39-degrees 37' minutes of separation. The planets are getting farther apart moving towards 180 or opposition. Adding 30 degrees to this amount of separation gives 69 37' which was the top of the S&P 500 index October 7th, 1997 just prior to the Asian Currency Panic.

Adding another 30-degrees gives 99 37', which will hit on January 28, 2001. The next significant turn will come at $7\frac{1}{2}$ ($\frac{1}{2}$ of 15) + 99 37' = 104 37' separation or 7/25/2001.

You can keep adding harmonics of 30-degrees (15, $7\frac{1}{2}$, $3\frac{3}{4}$) or harmonics of 45-degrees ($22\frac{1}{2}$, $11\frac{1}{4}$, $5\frac{5}{8}$ ths) to calculate future cycle turning dates. Once the number gets to 180, you have to subtract because the planets are no longer separating; they are applying. For example $90\text{-degrees} + 89\ 37' = 179\ 37'$. If we want to calculate the next 30-degree cycle date we would do the following: $180 - 179\ 37' = 23'$ minutes. This is the amount of movement left before the planets start applying. 30-degrees minus 23' minutes = 29 37'. This is the amount we have to subtract from 180 (maximum separation) to calculate the next angle relationship.

$180 - 29\ 37' = 150\ 23'$. This example shows the math required for both separating and applying planet scenarios. Remember that there are only 60' minutes in a degree. This process is very similar to calculating time on a non-digital clock i.e. the old fashion second, minute and hour hands. If you try to visualize the fast moving planets as the

second hand and the slower planets as the minute and hour hands, you will have a pretty good understanding of what is physically happening between the planets.

Three Term Proportion

Arithmetic, Geometric & Harmonic Means of Zodiac Longitude

According to Plato, the laws that govern the creation of all things are the very same laws, which allow for our comprehension. Therefore if the *mean* term is understood, then so is the entire series. For example: the geometric mean between any two terms (A and C) is the square root of ($A \times C$) the two terms multiplied together. The Harmonic mean is ($A \times C$) divided by ($\frac{1}{2} (A+C)$) or the arithmetic mean, which is simply the average of any two terms $(A+B)/2$. Earlier, I showed an illustration of geometric mean aspects of zodiac longitude for the planets Mars & Mercury from the October 18th 1999 low. Now I will illustrate what aspects are important and how to calculate them for future timing. The most important aspects to calculate are the Squares & the Triangles (90,180,270) & (120, 240). Solving the above formulas for an unknown we get the following: (1) Let X = the desired aspect or angular relationship.

Solving for geometric means we get QC squared divided by the previous zodiac position where X is equal to the aspect you want to calculate (See the Mars & Mercury examples presented earlier).

Solving *harmonic means* we get $\frac{1}{2} (X)$ times previous zodiac position divided by the previous zodiac position minus $\frac{1}{2} (X)$. Example: lets say we want to find the Harmonic opposition (180) to Mars on the Dow Jones low from October 18th, 1999. Mars was located at 12 Aquarius so the solution is $\frac{1}{2} (180) * (312) / 312 - \frac{1}{2} (180) = 28080 / 222 = 126.49$ or 6-degree Leo 29' minutes. Looking this up in an ephemeris we get: Monday, August 28, 2000 for our market date.

Three Term Proportion of Planetary Pairs

Moving on with this analysis technique we will look at the geometric & harmonic proportion of two planets. The formulas are based on the total distance traveled by the planets you are using. I typically only use Jupiter, Saturn, & Uranus because of their size and mass. Also look at the Pythagorean formula for the hypotenuse of a right triangle.

Examples are:

Major Tops: September 3, 1929 to March 10, 1937. Uranus moved 30-degrees (important all by itself) & Saturn moved 85.51. The Hypotenuse = $\text{Sqrt. } (85.51^2 + 30^2)$ which equals 90.6 degrees or 90-degrees 36' minutes. Making this 1937 top square to the September 1929 market top.

Major Tops: September 3, 1929 to August 13, 1937(2nd top in 1937 before a major decline). Saturn moved 90-degrees 43' (Square) Jupiter moved 231.63, Uranus moved 31.673. The geometric mean of Jupiter & Saturn is $\text{Sqrt } (90.73 * 231.63) = 144$ (square of 12). Remember that Jupiter's orbit of 12 years divided by Saturn's 30 year orbit = 0.40 and that $360\text{-degrees} * 0.40 = 144$ degrees. In the early 1920's, Professor Weston discovered that 18-degree harmonics of the Jupiter Saturn synodic cycle correlated with

turning points in the New York Stock Exchange. 144 is part of the 18-degree cycle that Professor Weston describes. The 18-degree aspects are the 20th harmonic of the circle (360/20), which is also the number of years required for the two planets to conjoin. The cycle starts at zero and grows by 18 until the opposition at 180-degrees. The end of a 10-year cycle. 0, 18, 36, 54, 72, 90, 108, 126, 144, 162, 180, 162, 144, 126, 108, 90, 72, 54, 36, 18 and 0. The Market tends to peak at 18, 90, and 162-degree aspects. The Market tends to trough or bottom on 54, 126, 180, and 0-degree aspects according to Professor Weston's research. Another market analyst and financial astrologer, Luther Jensen also used Weston's Jupiter/Saturn cycle to forecast market moves.

A more recent example is: From the major low August 7, 1982 to the low October 18th, 1999, Saturn moved 72.033 ($\frac{1}{2}$ the square of 12) and Uranus moved 200. The geometric mean is the Sqrt. $(72.03 \times 200) = 120$ degrees or trine. This is a period of 17 years (this important time period has been already discussed) where three-term geometric proportion is forming a trine (or triangle) with a previous major market low.

I could go on with this technique but it is better for you to do (and discover) the work yourself. Use dates of Major Bull Market Tops & Major Bear Market Bottoms. Next calculate how many degrees Jupiter, Saturn, and Uranus have traveled. Then calculate the geometric, harmonic and hypotenuse of the 3 combinations (Jup/Sat, Jup/Urn, Sat/Urn). If your answer is larger than 360, simply subtract this amount (reject the circle). Then look for harmonics of squares & triangles. Also include, the square of twelve (144) for potential geometric harmony. I have provided you with a few more examples of this technique from my hand written notes in the back of this document.

Gann's MOF, CE Formula and Master Pattern Charts

Gann ends one of his grain letters to a private student with the following statement:
"Remember that you have signed an agreement not to reveal these rules and instructions to any one, and by keeping these secret discoveries confidential for you own use, you will later receive the very important CE Average, and the MOF Formula which is only taught to students who have taken the same course as you have and we do not reveal it to students who take the minor courses and pay less money."

These formulas are actually revealed in Gann's 1954 Coffee letter and his Soybean Letter of the same year. The CE Average is the "Cycle of Eight" Average. Gann simply takes the longitude average of the 8 planets that revolve around the sun both geocentric and heliocentric and reduces each to a single longitude.

The MOF formula is the "Mean of Five". In this formula, Gann only uses the average longitude of Jupiter, Saturn, Uranus, Neptune and Pluto or the 5 outer planets with Mars left out. He computes both the geocentric and heliocentric averages and uses the mean or average longitude. Including Mars gives us the mean of six. Gann said: "The averages of the six major planets Heliocentric and Geocentric are the most powerful points for Time and Price resistance. Also the Geocentric and Heliocentric average of the five planets with Mars left out, is of great importance and should be watched".

Gann used these averages to find support and resistance levels for stocks and commodities. He used these averages of planetary longitude to determine if price was

making an astrological aspect (square, trine, conjunction, opposition, etc.) to the longitude averages. He also looked at the position of each individual planet's geocentric and heliocentric longitude to price, especially Jupiter and Saturn (also the midpoint between these two giant planets). Gann would also place the planets and averages on his "Master Charts" to get different numerical patterns from geometric shapes such as the hexagon, square of 9, square of 4, and the circle chart.

As an example, Gann wrote in his Personal Soybean Letter, Dec. 2, 1953, May Soybeans high $31\frac{1}{4}$. This equaled $18\ 45'$ in Pisces, close square of 90 degrees of Jupiter, 135 degrees to Saturn and 180 degrees of the averages, and 120-degrees of Uranus. 300 price equals 30-degrees Virgo. 302 equals 30-degrees Libra. 304 equals 30-degrees Scorpio". These are obviously not straight longitude conversions of price because $31\frac{1}{4}$ price in pure longitude would be 11-degrees 15' minutes Aquarius. However, on Gann's circle chart which is based on the Earth's 24-hour rotation on it's axis $\$31\frac{1}{4}$ is at $18\ 45'$ Pisces. Just as an exercise, on the Square of 4, $\$311$ would be about 20-degrees Capricorn, on the Square of 9, $\$311$ would be 0-degrees Gemini and on the Hexagon, $\$311$ would be 5-degrees Sagittarius. Gann used these shapes to relate price to zodiac longitude. I have provided copies of all the above mentioned "Master Charts" for your use and study.

Another purpose these charts served was for mathematical sequence. Gann had always said that he based his calculations on cycle theory and mathematical sequences (see pg. 75 of Tunnel Thru the Air). By creating these charts, Gann could look at periodic numerical sequences or number cycles on these "Master Charts" to see if market turning points had any mathematical relationship in Time or Price. Unlike ordinary cycles that

are fixed in terms of size or length, periodic number cycles have a geometric growth that takes successively larger units of calendar time or price. Carl A. Faria wrote an interesting report in June of 1982 on just this subject. The report was titled: Predicting Market Trends with Periodic Number Cycles. The main reason for creating a periodic number table i.e. "Master Chart" is to see where the numbers come out in terms of natural angles. For Example: If you look at the Square of Nine, you can easily see that the center of the chart starts at 1 and is surrounded by the eight numbers 2 through 9. If we divide the 360-degrees of a circle by 8, we get 45-degrees. Therefore, the number 2 is on the 0-degree angle (Aries), the number 3 is on the 45-degree angle (15 Taurus), the number 4 is on the 90-degree angle (Cancer), 5 is on the 135-degree angle (15 Gemini), 6 is on the 180-degree angle (Libra), 7 is on the 225-degree angle (15 Scorpio), 8 is on the 270-degree angle (Capricorn) and 9 is on the 315-degree angle (15 Aquarius). 10 is back at the 0-degree angle, and so on as you keep going around the square. The Square of 4 is based on 90-degrees ($360/4 = 90$) per cycle. The Hexagon is based on 60-degree turns ($360/6 = 60$). Finally, the circle chart is based upon 15-degree turns ($360/24 = 15$). Gann said to use these charts for "Price, Time, Space or volume". The main reason for using these charts is to see if time (market turning points) or price is consistently falling on the same geometric progression, i.e. they are all on the same angle To give you some idea of how you would use this information, I'll give you a few examples. First, lets set up some simple tables based on the Square of 9.

(Cycle #)

<u>Angle</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>
<u>0</u>	<u>2</u>	<u>11</u>	<u>28</u>	<u>53</u>	<u>86</u>	<u>127</u>	<u>176</u>	<u>233</u>	<u>298</u>
<u>45</u>	<u>3</u>	<u>13</u>	<u>31</u>	<u>57</u>	<u>91</u>	<u>133</u>	<u>183</u>	<u>241</u>	<u>307</u>
<u>90</u>	<u>4</u>	<u>15</u>	<u>34</u>	<u>61</u>	<u>96</u>	<u>139</u>	<u>190</u>	<u>249</u>	<u>316</u>
<u>135</u>	<u>5</u>	<u>17</u>	<u>37</u>	<u>65</u>	<u>101</u>	<u>145</u>	<u>197</u>	<u>257</u>	<u>325</u>
<u>180</u>	<u>6</u>	<u>19</u>	<u>40</u>	<u>69</u>	<u>106</u>	<u>151</u>	<u>204</u>	<u>265</u>	<u>334</u>
<u>225</u>	<u>7</u>	<u>21</u>	<u>43</u>	<u>73</u>	<u>111</u>	<u>157</u>	<u>211</u>	<u>273</u>	<u>343</u>
<u>270</u>	<u>8</u>	<u>23</u>	<u>46</u>	<u>77</u>	<u>116</u>	<u>163</u>	<u>218</u>	<u>281</u>	<u>352</u>
<u>315</u>	<u>9</u>	<u>25</u>	<u>49</u>	<u>81</u>	<u>121</u>	<u>169</u>	<u>225</u>	<u>289</u>	<u>361</u>

Read the above chart horizontally, i.e. the numbers on the zero degree angle are 2, 11, 28, 53, etc. the 90-degree angle numbers are 4, 15, 34, 61, 96, etc. Now, there are several ways to use these charts. One technique is to use the positions of the planets and the CE Average, MOF Formula, and the Mean of Six as discussed earlier to determine if there is price resistance or support on the "Master Chart" due to Astrological aspects (Squares, Trines, Conjunctions, Oppositions) Gann's "outer circle". I will illustrate this technique when we analyze Gann's Personal Soybean Letter. Another technique is to find the price of a Major Top or Major Bottom on the "Master Chart" and determine the angle that the price fell upon and convert it to time. For example the major low for the Dow Jones Industrial Average, July 8th, 1932 was \$41.70. If you look at the table above (or on the square of 9 itself) you will see that the number 40 is on the 180-degree angle and 43 is on the 225-degree angle. Since \$41.70 is about half way between these two angles, and we

know that they are separated by 45-degrees then we can determine the actual angle that
 this price hit. One half of 45 is $22\frac{1}{2}$ and this amount added to 180-degrees = 202.5-
 degrees (22 30' Libra). Looking at the Square of 9 we can see the other numbers that are
 on this exact angle. 7, 21, 43, 73, 111, 157, etc. These numbers can represent hours, days,
 weeks, months, years, or even planetary degrees traveled. If we look at the Macro scale
 of Solar years we would get 1939 (7+1932), 1953 (21+1932), 1975 (43+1932), 2005
 (73+ 1932), etc. However, it is important to be exact. If you study the Square of 9, you
 will learn that it is a square root calculator. To determine the next number on an angle
 you simple take the square root of the number (the Sqr. Root of 41.7 = 6.457) add two
 (6.457 +2 =8.457) then square the number (8.457 squared = 71.53. To determine the
 previous number you would have subtracted 2 from the root and Then Square the result.
 Therefore, 41.7 years from July of 1932 would really be in the year 1974, which also had
 a Major market low similar to 1932. 71.53 + July 1932 would yield the year 2003 again
 as a potential low (Please reread the 37-year cycle regarding this year & also Brad
 Cowan's Uranus node at 0-degrees Pisces hitting this same year). When converting price
 to time, it is an absolute necessity to have at least two confirmations on the same "Master
 Chart" of time (your target date) from different Major Tops or Major Bottoms. It is not
 my intention, nor was it Gann's to do all the work for you! To truly learn and master
 these techniques you must do the work yourself. I am only trying to remove much of the
 confusion that surrounds Gann's writing style and techniques. Another example is the top
 price of 386 September 3rd, 1929. The square root of 386 is 19.646. This root number
 minus 12, which is 6 cycles or numbers in towards the center on the same geometric
 angle = 7.646. If we square this number we get 58.47 years. 1929 + 58.47 = the year

1987, which also had a 1929 style crash. Remember, you should examine other market highs & lows in the same manor to find more than a single hit for your target period. To narrow down your target date, you would use this same technique with smaller time periods such as months, weeks and days. As you use smaller increments of time, the significance of the turning point can also be decreased. In other words, the turning points are defined as "major" based upon the time scale you are using. Major turning points on a weekly chart might not even show up on a monthly or yearly chart, but they are important when using weeks to narrow down your target date. The same logic is applicable to days or even hours, i.e. the turning point or swing should be a "Major" swing high or low relative to the time period your chart covers. From this information, it should be obvious to you that the square root of a major high or low price creates or is in sync with a rhythm in time. For example, on August 25th, 1987 the Dow topped at 2746 before the October 8th crash. Converting this to a three digit number 274.6 and calculating the square root we get 16.57. If we add this period as years we get $1987 + 16.57 = 2003$ again using the same technique and calculator. You should also pay attention to the number of days (degrees of longitude) from the natal or first trade date (birth), or the number of days from the most recent major low. For example: The birth of the U.S. Stock Market is May 17th, 1792. The stock market top on September 3rd, 1929 is 50,146 days from 5/17/1792. If you look at where this number falls on the square of nine chart you will see that it is on the 123° angle. The Market low on May 19th, 1947 is 56,613 days from the market birth and is also on the 123° angle on the square of nine. The High on December 2nd, 1968 is 64,481 days from the market birth and is again on the 123° angle of the square of nine. The silver panic and stock market low on March 27th, 1980 is 68,614 days from the birth

of the stock exchange and is again on the 123° of the square of nine! November 5th, 2000 is also on the 123° angle.

Another technique using periodic numbers is to simply take the time (or price) difference between dates (or highs to lows, lows to highs) in terms of days, weeks, months or years to determine if there is a sequence between successive turning dates. Microsoft Excel is extremely useful for this task and also for the previous technique mentioned. I typically only use Weeks, Months, and Years. What you do is keep a list of dates of previous market turns and simply add the periodic time units that fall on the cardinal cross of the periodic chart (0, 90, 180, 270). For example: lets take two recent lows in the stock market, October 23, 1997 and September 1, 1998. On the square of 9, we would add the number series of 2, 11, 28, 53, 86, 127, 176, 233, 298 weeks for the zero angle from each of these two dates to see if they had a common time period. Then we would do the same thing for the 90-degree, 180-degree and 270-degree angles (cardinals only). So we would run the series of 4, 15, 34, 61, 96, 139, 190, 249, 316 weeks, 6, 19, 40, 69, 106, 151, 204, 265, 334 weeks, and finally 8, 23, 46, 77, 116, 163, 218, 281, 352 weeks from both lows and circle all the time periods they have in common. Then, we would do this same procedure with months and circle all the months they have in common. These would be time periods that would require us to be on the lookout for reversals of trend. Using weeks, the only common time periods are the week of February 20th, 1999 and May 25th 2012. Using months, the only common time periods are September 1999, March 2000 (important planetary cycle date), March 2001 and August 2013. Remember from above, that there is a Jupiter cycle from the August 1982 low that hit in March of 2000! NOTE: If you use the Hexagon chart, you will get a completely

different series of cardinal numbers. This also applies to the Square of 4 and the Circle chart. Therefore, do not mix or combine forecasts from the Hexagon Chart with the Square of 9 or the Circle with the Hexagon or the Square of 4 with the Square of 9 or any other combination. Rule: Don't mix apples and oranges! Your common turning date must always come off the same "Master Calculator". I have found the Square of 9 and the Hexagon Chart the most useful for this particular technique.



Another technique is to calculate price levels for support and resistance off of these periodic number tables. The strongest support or resistance is always on the same or opposite angle. The next in significance are the squares or 90-degree angles. For example: November Bears recently made a low of \$411, which is on the 180-degree angle of the Square of 9. Therefore, all the numbers on this angle and the zero angle (which is $180 + 180 = 360$ or back to zero) are important levels for support and resistance. These prices are:

334, 411, 496, 589 for the 180-angle and also 298, 371, 452, 541 for the zero angle. Just simply locate your price on the "Master Chart" and find what angle it is on and look at all the numbers that are on the same or opposite angle. For large numbers like the Dow Jones, simply move the decimal point and convert the price to a four-digit number. Another technique that I'm still researching is to divide the price by the minimum fluctuation to convert the price to ticks. For example, the 411 price we just looked at for beans could also be $\$411 / 0.25$ (or multiplied by 4) = 1644 and this number is on a completely different angle. The first opposition to this number is 1726 divided by 4 = \$431. The first conjunction above 1644 is 1810 divided by 4 = \$452.50. You may want to experiment with this idea.

Earlier in the document, I mentioned that the Square of 9 was a square root calculator. I gave an example based on the July 1932 low price of 41.70. To make sure you fully understand this concept, we will go over some of the mathematical relationships of the Square of 9. The Square of 9 is probably W.D. Gann's most famous Master Chart. Many traders use this chart regularly as a primary trading tool. However, it is my personal belief that most traders are only using a few simple techniques and are not fully utilizing the true potential of this multidimensional Master Calculator.

What is the Square of 9

The Square of 9 is basically a spiral of numbers starting with the number one in the center (or apex of the Great Pyramid) with the number 2 immediately to the left. The rest of the numbers spiral around the center in a clockwise fashion to the number 9, which completes the first cycle of numbers around the center. 10 through 25 completes the 2nd cycle, 26 through 49 completes the 3rd, etc. The square is divided into eight 45-degree

angles. On the next page I have provided a copy of the Square of 9. The numbers that run through the center in the shape of a "+" sign are the cardinal numbers. The numbers that run through the center in the shape of a "X" are the corner numbers. In the first cycle around the center, there is one digit separating each 45-degree angle. In cycle number two (10 to 25) there are 2 digits separating each 45-degree angle. In cycle three (26 to 49), there are 3 digits separating each 45-degree angle. In cycle 1000, there would be 1000 digits or cells separating each 45-degree angle. This particular arrangement of numbers creates a very unique square root relationship with other numbers on the chart. My friend, Michael S. Jenkins illustrates some interesting square root trading techniques utilizing the Fibonacci ratios and the square of 9 in his Gann Course and his book Chart Reading for Professional Traders. Basically, if you want to move around the coordinates on the Gann Wheel you take the number you are interested in (such as the all time High or Low price) take the square root of the number, then add or subtract 2 from the root and re-square the result. Example: Lets say that we are interested in the price 664 (which is in the vertical column straight up from the center). The square root is $25.768 + 2 = 27.768^2 = 771$ which is the number directly above 664 or one full 360 degree cycle out from center. If we subtracted 2 from the root and re-squared the number ($25.768 - 2 = 23.768^2 = 565$) we would get 565 which is directly below 664 or one full 360 degree cycle in towards center. Incorporating the Gann Emblem with this technique, allows us to calculate coordinates that are conjunct δ (360 = +/- 2 from the root), opposition ρ (180 = +/- 1 from root #), trine (120 = +/- .666) (240 = +/- 1.333), square $\square$ (90 = +/- .5) (270 = +/- 1.5) and sextile $*$ (60 = +/- .333) (300 = +/- 1.666). This technique is extremely fast and useful for finding coordinate squares on the Gann Wheel that are making hard

aspects to a previous position on the wheel. Also, Gann believed that the numbers that connected the square base of the pyramid (the 4 corners of the square, i.e. corner #'s.) to the "gravity center" and also the numbers that ran straight vertical and horizontal from the "gravity center" in the form of a cross (cardinal numbers) were very important in balancing "Price & Time" on the wheel. He was basically looking for astronomical longitudes to balance with price on these key angles. Pythagoras said "Units in a circle or in a square are related to each other in terms of Price & Time at specific points." Gann often quoted the Bible, Emerson, Pythagoras and Faraday to name a few. Basically, he was pointing the reader of his works to clues that would allow his student to unlock the code of his writing style. To quickly determine what cycle or ring a price is in, you simply take the square root of the number add and subtract $\frac{1}{4}$, then re-square them both and subtract them from the original number. Example: What cycle or ring is the Dow Jones trading in at the price high of 11,750? Answer: the square root of 11,750 = 108.3974 +.25 = 108.6474. Re-squared = 11,804.26. Now take the original root 108.3974 and subtract $\frac{1}{4}$ and we get 108.1474 and this re-squared is 11,695.86. Now let's take the difference between these numbers. 11,804.26 - 11,750 = 54.26. 11,750 - 11,695.86 = 54.14. Therefore, 11,750 is in cycle number 54 on the Square of 9. This is an important technique when you use the square of 9 as a time cycle, which was discussed earlier when I showed that the 1929 top was on the 123-degree angle on the square of nine.

Time

Around the outer perimeter of the square of 9 is a circle with months and days. The circular calendar starts on the right hand side of the wheel on the same horizontal line as the center block. The date is March 21st and refers to the start of the "natural" year in the

season of spring with the Sun in Aries ♈. "Lamb of God you take away the sins of the world" (spring takes away the sins of winter). Starting at the 9 O'clock position, the dates move around the circle clockwise completing the calendar. This simple relationship allows the user to quickly identify dates that are Conjunct (365 days), Opposite (182 days), Square (90 days), Trine (120 days) or Sextile (60 days) to a past turning date relative to the Sun / Earth relationship.

6-May	13510	13352	13394	13435	13477	13519	13560	13602	13644	13686	13727	13769	13811	13852	13894	13936	13977	14019	14061
	13269	13211	10600	10641	10683	10725	10767	10808	10850	10892	10933	10975	11017	11059	11100	11142	11184	11225	14103
	13227	10516	8123	8181	8223	8264	8306	8348	8390	8431	8473	8515	8556	8598	8640	8681	8723	11267	14144
	13185	10475	6099	6054	6096	6138	6179	6221	6263	6305	6348	6388	6430	6471	6513	6555	6597	11309	14186
	13143	10433	6056	6013	6053	6094	6135	6176	6217	6258	6300	6341	6382	6423	6464	6505	6546	11351	14228
	13102	10391	6014	5971	6011	6052	6093	6134	6175	6216	6257	6298	6339	6380	6421	6462	6503	11392	14269
	13060	10350	5973	5929	5970	6011	6052	6093	6134	6175	6216	6257	6298	6339	6380	6421	6462	11434	14311
	13018	10308	5931	5888	5929	5970	6011	6052	6093	6134	6175	6216	6257	6298	6339	6380	6421	11475	14353
	12977	10266	5889	5846	5887	5928	5969	6010	6051	6092	6133	6174	6215	6256	6297	6338	6379	11517	14394
21-May	12935	10224	5847	5804	5845	5886	5927	5968	6009	6050	6091	6132	6173	6214	6255	6296	6337	11559	14436
	12893	10183	5805	5762	5803	5844	5885	5926	5967	6008	6049	6090	6131	6172	6213	6254	6295	11601	14478
	12852	10141	5764	5721	5762	5803	5844	5885	5926	5967	6008	6049	6090	6131	6172	6213	6254	11642	14520
	12810	10099	5722	5679	5720	5761	5802	5843	5884	5925	5966	6007	6048	6089	6130	6171	6212	11684	14561
	12768	10058	5681	5637	5678	5719	5760	5801	5842	5883	5924	5965	6006	6047	6088	6129	6170	11726	14603
	12726	10016	5639	5596	5637	5678	5719	5760	5801	5842	5883	5924	5965	6006	6047	6088	6129	11767	14645
	12685	9974	5597	5554	5595	5636	5677	5718	5759	5800	5841	5882	5923	5964	6005	6046	6087	11809	14686
	12643	9933	5555	5512	5553	5594	5635	5676	5717	5758	5799	5840	5881	5922	5963	6004	6045	11851	14728
	12601	12560	12518	12476	12435	12393	12351	12309	12268	12226	12184	12143	12101	12059	12018	11976	11934	11892	14770
4-Jul	12560	15520	15479	15437	15395	15354	15312	15270	15228	15187	15145	15103	15062	15020	14978	14937	14895	14853	14811
21-Jun																			
24-Jun																			
2-Aug																			
23-Sep																			

Another price technique is to take all the cardinal and corner numbers and multiply them by the minimum fluctuation of the market you are trading. By doing this, you can see what numbers would appear on important angles of the square of nine if you started with the minimum fluctuation and also grew the square by the same number. You can also use the all time low price as the center and multiple it by the cardinal and corner numbers. For example: the July 1932 Dow Jones Low was 41.70. If we multiply this by 281, a cardinal number, we get 11,717.70, which is very close to the extreme price high of 11,750 reached on January 14th 2000. We can also put the extreme high in the center and subtract the cardinal and corner multiples of extreme low of 41.70. For example: If we take the cardinal number 53 and multiply it by the extreme low of 41.7, we get 2,210.10. If we subtract this amount from 11,750, we get 9,539.9 as a possible support number if the Dow breaks 10,000, which I believe will happen in March 2001, but we will have to wait and see. Obviously, you can also add and subtract the cardinal and corner numbers just as they are without multiplying them by an extreme low. By the way, in all the quotes where Gann discusses a forecast he always says that it was based upon a Master Time Factor. He always describes it a singular thing, which is probably a process. The master time factor is not a particular cycle as many Gann students have been lead to believe. For example, many have thought that the Master Time Factor was the 60-year cycle. However, Gann said that one of his forecasts was based upon his discovery of the Great Cycle, which is the 60-year cycle, and a Master Time Factor. This shows that they are not one in the same. If you go through all the Gann material (Books, Courses and Advertisements) you will find that the only other time he uses the words time factor is in

his Master Charts Course when he describes the Square of 9. In the Soybean letter, Gann says: "The square of '1' is 1 and '1' is the sun". The only other place where Gann puts the number 1 in "1" is his description of the Square of 9. In this same letter, Gann tells the recipient to calculate the longitude average of the 8 planets that revolve around the sun. If "1" is the sun and its surrounded by 8 planets (excluding Earth) then isn't this a very similar description to the way Gann describes the Square of 9? I will leave the rest of this investigation alone.

Converting Longitude to Price.

The secret to converting zodiacal longitude into price levels is always based upon the minimum fluctuation of the security, commodity, currency, etc. Many of today's software programs use techniques that convert the current planetary longitudes into price levels. For example if the planet Saturn is a 23° 21 Taurus, the straight longitude conversion is \$53.35 because Taurus starts at 30° of the zodiacal circle. W.D. Gann illustrates this technique in the Coffee Letter. Other techniques are based upon Gann's "Master Charts". Specifically, the Square of 9, Square of 4, The Hexagon Chart, and the Circle Chart. W.D. Gann gives an example of converting current planetary longitudes to price via a "Master Chart" in the Soybean letter. As an example, Gann wrote in his Personal Soybean Letter: Dec. 2, 1953, May Soybeans high 31 1/4. This equaled 18 45' in Pisces, close square of 90 degrees of Jupiter, 135 degrees to Saturn and 180 degrees of the averages, and 120-degrees of Uranus. 300 price equals 30-degrees Virgo. 302 equals 30-degrees Libra. 304 equals 30-degrees Scorpio". These are obviously not straight longitude conversions of price because 31 1/4 price in pure longitude would be 11-degrees

15' minutes Aquarius. However, on Gann's circle chart which is based on the Earth's 24-hour rotation on it's axis \$311¼ is at 18 45' Pisces. As shown in the earlier exercise, on the Square of 4, \$311 would be about 20-degrees Capricorn, on the Square of 9, \$311 would be 0-degrees Gemini and on the Hexagon, \$311 would be 5-degrees Sagittarius. Gann used these shapes to relate price to zodiac longitude.

Gann's Master Egg Course has a hand written sheet of paper that he called the "Table for Eggs". The table starts at the lower left-hand corner at 0°00' Aries and moves up through all 12 signs ending of course in Pisces. Gann takes the minimum fluctuation of Eggs, which was 5 points and sets this amount equal to one zodiac degree (5pts. = 1°00'). Gann gives a quick relationship showing that 30° or one zodiac sign is equal to 150 points in Eggs. The whole circle is 1800 points because 5 times 360° = 1800. This longitude relationship to the minimum fluctuation is (as I have written about previously) also the secret of Gann's Angles. Knowing the minimum fluctuation allows you to create astrological geometry (aspects) relationships to price. As a quick example, from the above we know that 1800 points/cents is equal to 360°

Therefore $1800 / 4 = 450$ pts and is equal to 90° for the same reason that $360°/4 = 90°$.

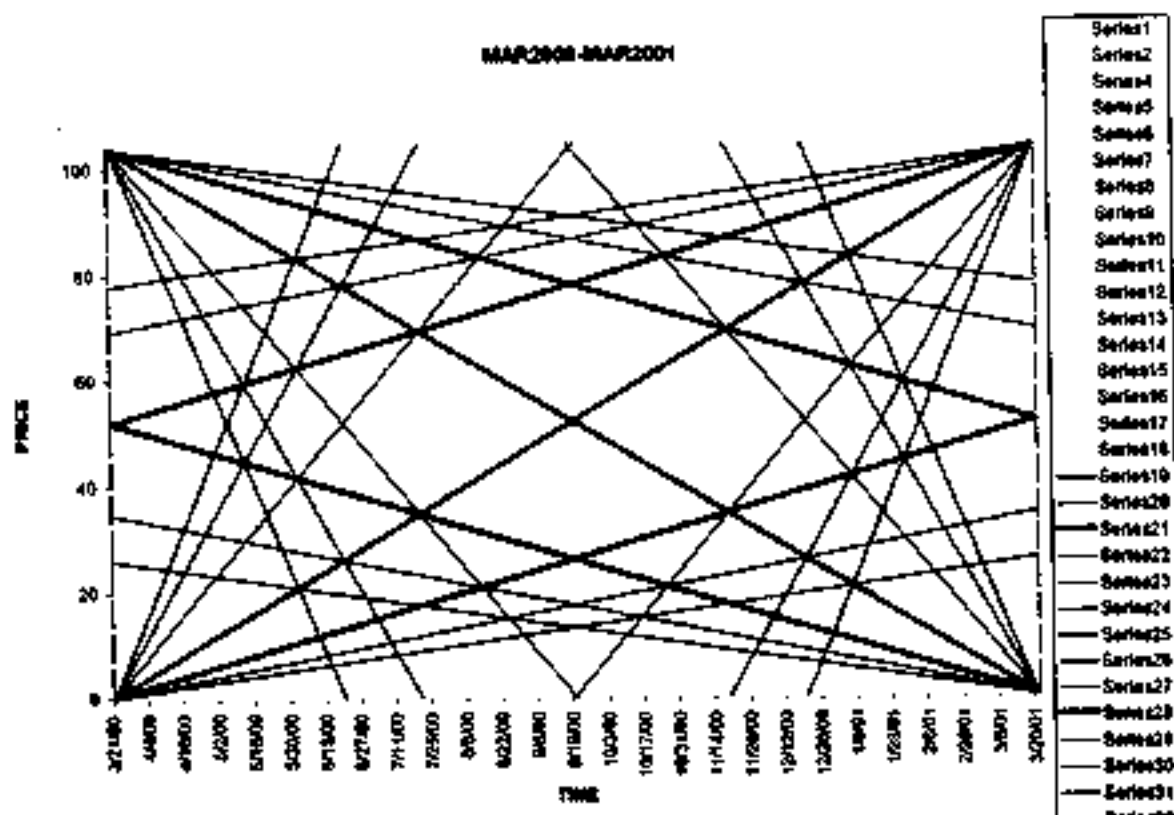
Therefore adding or subtracting 450 points from extreme highs or lows in Eggs will give us our 1st square aspect in price. The next 450 points (total 900) will give us the opposition, the next 450 points or 1350 total will give the next square (270°) and finally the next 450 points will give us the conjunction or 360°. Dividing 1800 by 3 = 600 points and this will calculate the trines or triangles. "Within the circle forms the square and the triangle, which prove the 4th dimension in working out price" W.D. Gann.

Another important technique is to use this same idea based upon natal astrology! The four most important points of a natal horoscope are the Ascendant, Midheaven, Sun longitude, Moon longitude. These are often referred to as the Significators by Separiel, an astrologer on Gann's reading list. For example, let's say that a stock has an Ascendant of $13^{\circ}23'$ Gemini. This longitude is equal to $73^{\circ}23'$ of the circle, because Gemini starts at 60° of the zodiac. The minimum fluctuation of most stocks is $1/8^{\text{th}}$, therefore 73.383° multiplied by $1/8^{\text{th}}$ or $0.125 = \$9.17$ in price. Taking 360° times $1/8^{\text{th}} = \$45$. Therefore $\$9.17$ is conjunct or equal to $13^{\circ}23'$ Gemini and every $\$45.00$ dollars above this price, also is equal to $13^{\circ}23'$ Gemini, i.e. $\$54.17$, $\$99.17$, $\$144.17$, etc. The other Significators are calculated in this same fashion. The squares and trines to this Ascendant are calculated as follows. $\$45.00 = 360^{\circ}$, $\$45/4 = \11.25 or 90° so every addition of $\$11.25$ is square, opposite, square, then finally conjunct the Ascendant respectively. $\$45/3 = \15 , so every addition of $\$15$ is trine, trine, then conjunct the Asc. Just to make sure that you understand the conversions, if we were converting Soybeans, Corn or Wheat, which has a minimum fluctuation of a $1/4$ cent or 0.25 we would find that the Ascendant of $13^{\circ}23'$ Gemini is equal to $73.383 \times 0.025 = \$18.345$. The squares are based upon $360^{\circ}/4$, therefore $360^{\circ} \times .25 = 90$. This means that 90 cents is equal to 360° . 90° is therefore $1/4^{\text{th}}$ of 90 cents or 22.5 cents. Adding this value to the Ascendant value of $\$18.345$ yields 40.845 as square, 63.345 as opposite, 85.845 as the next square, and 108.345 as conjunct or once again equal to $13^{\circ}23'$ Gemini. The trines or triangles are based on $1/3^{\text{rd}}$ of 90 or 30-cent increments, etc. The student should also read and study Gann's definition of "Active Angles" in his Soybean Letter to fully utilize this astrological technique.

Gann's Master Mathematical Formula For Market Predictions

Master Mathematical Price Time and Trend Calculator

In 1954, at the age of nearly 76, W.D. Gann released a course based upon mathematical rules for making market predictions. In the advertisement for this method the author of the add writes: "We will not sell the course of instructions or teach the mathematical formula for market predictions just to get money, which we do not need or have to have; we will only sell the courses to people who meet our requirements, and if we feel after a thorough investigation that a man or woman cannot make a success investing or trading we will refuse to teach them the courses". "The Master Calculator can only be used with the Master Courses. It is not sold separate from the other courses". The price shown for Gann's Master Course in this ad is \$5,000.00, it does not list the additional cost of the Master Mathematical Calculator. The ad describes the calculator as a plastic overlay that accurately measures Time, Price and Trend. Many Gann experts and students have tried to use these overlays, which require the chart to be hand drawn to a proper scale on graph paper that has 8 squares to one inch. This creates a lot of extra tedious work. This has caused one of Gann's greatest discoveries to have been overlooked for many years. I know that some software programs have the ability to make these overlays on price charts but that is still not going to help most people because it turns out that there are also very precise mathematical rules regarding the horizontal axis or Time! In fact, these same mathematical rules apply to price as well but we will leave this topic for latter. The advertisement itself contains clues about the calculator, such as "W.D. Gann's Record for 52 years", "After fifty-two years of experience and research



going back hundreds of years, I have proved to my entire satisfaction that history repeats and that when we know the past, we can determine the future of prices". The calculator that Gann is talking about in this ad is his square of 52 overlay, which is based upon the 52 weeks in a year. The basic overlay looks like that show above.

This is also the same geometric form that Gann used for the Square of 90 and Square of 144 plastic overlays. This basic structure or overlay appears in many books on Sacred Geometry and is also the secret to George Bayer's "Trend Determination" course. Getting back to Gann and the square of 52 course we find that this is the course where Gann made the confusing statement: " Within the circle forms the square, there is an inner circle and an inner square, as well as an outer square and an outer circle which prove the Fourth Dimension in working out price movements". This quote is describing a

fractal nature or structural pattern existing in the markets that is the same on smaller levels as it is at larger levels.

Jumping over to Gann's Angle course, on the last page, GA-32, Gann says:
"Figuring \$100, or per, as a basis for stock prices and changing these prices to degrees, $12\frac{1}{2} = 45$ -degrees, $25 = 90$ -degrees, $37\frac{1}{2} = 135$ -degrees, $50 = 180$ -degrees, $62\frac{1}{2} = 225$ -degrees, $75 = 270$ -degrees, $87\frac{1}{2} = 315$ -degrees, and $100 = 360$ ". This tiny little paragraph is the basis of the entire Murrey Math Trading Course, which readers of Traders World should be very familiar with since, T.H. Murrey has written numerous articles for the magazine. These numbers are all natural 8th numbers off of the base of 10, 100, 1000, 10000, 100000, etc. The part that is most confusing is balancing this with the time element, but based on the information given by Gann himself it should be obvious that he has already shown us part of the relationship. For example: If shares of XYZ stock moved up $12\frac{1}{2}$ dollars in 45 days, weeks or 45 months then the stock is on the price degree of its time angle, that is to say that price and time are equal or balanced. Gann's example: "When a stock sells at 50 on the 180th day, week or month, it is on the degree of its time angle". "On February 1, 1915, U.S. Steel made a low at \$38, which is closest to a price of $37\frac{1}{2}$, which is $\frac{3}{8}$ th of 100 and equals 135-degree angle. Steel was 14 years or 168 months old on February 25, 1915, and hit the angle of 135-degrees, which showed that the price of Steel was behind time, but was in a strong position, holding at \$38 above the 135-degree angle or the price $37\frac{1}{2}$ ". Just to make sure that you understand this information, U.S. Steel would have been balanced or "on the degree of its time angle" at a price of $46\frac{5}{8}$ th because $\$100 / 360\text{-degrees} = 0.27777\text{cents per degree}$ and 168

months multiplied by $0.27777 = \$46.67$ which is closest to $\$46$ or $5/8^{\text{th}}$ in price.

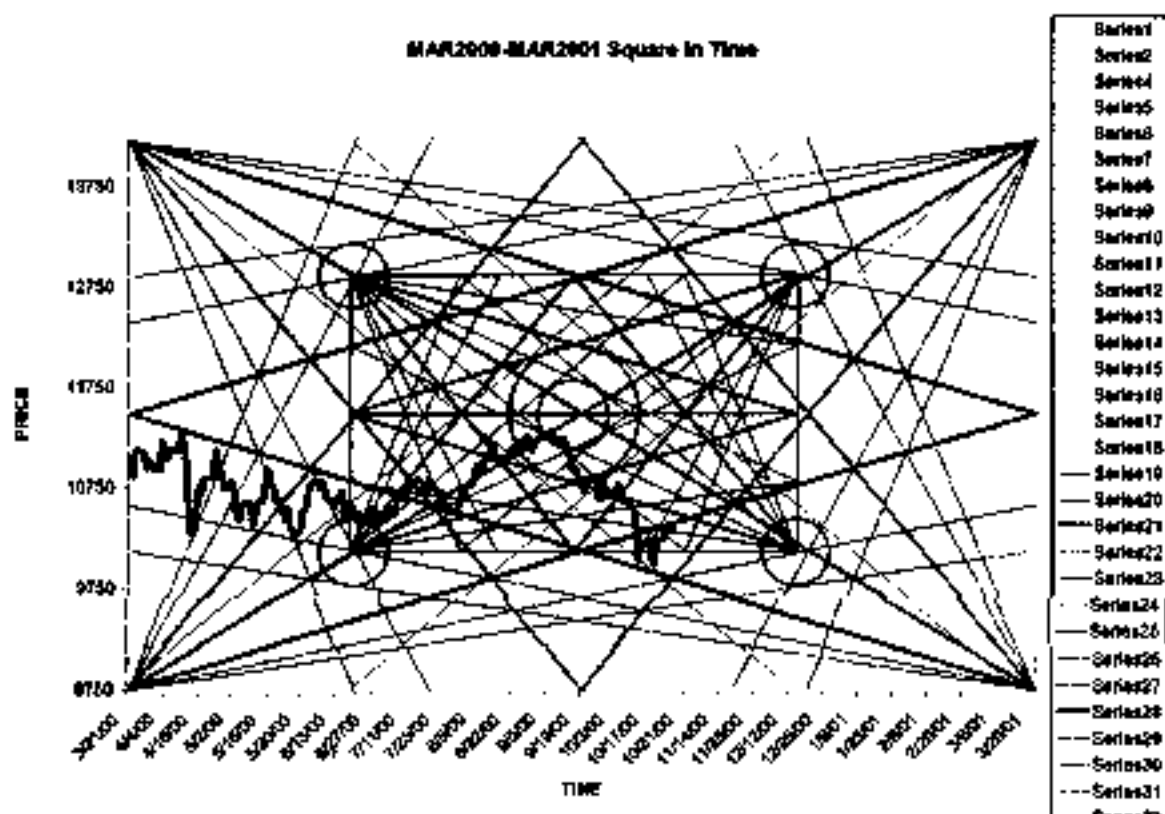
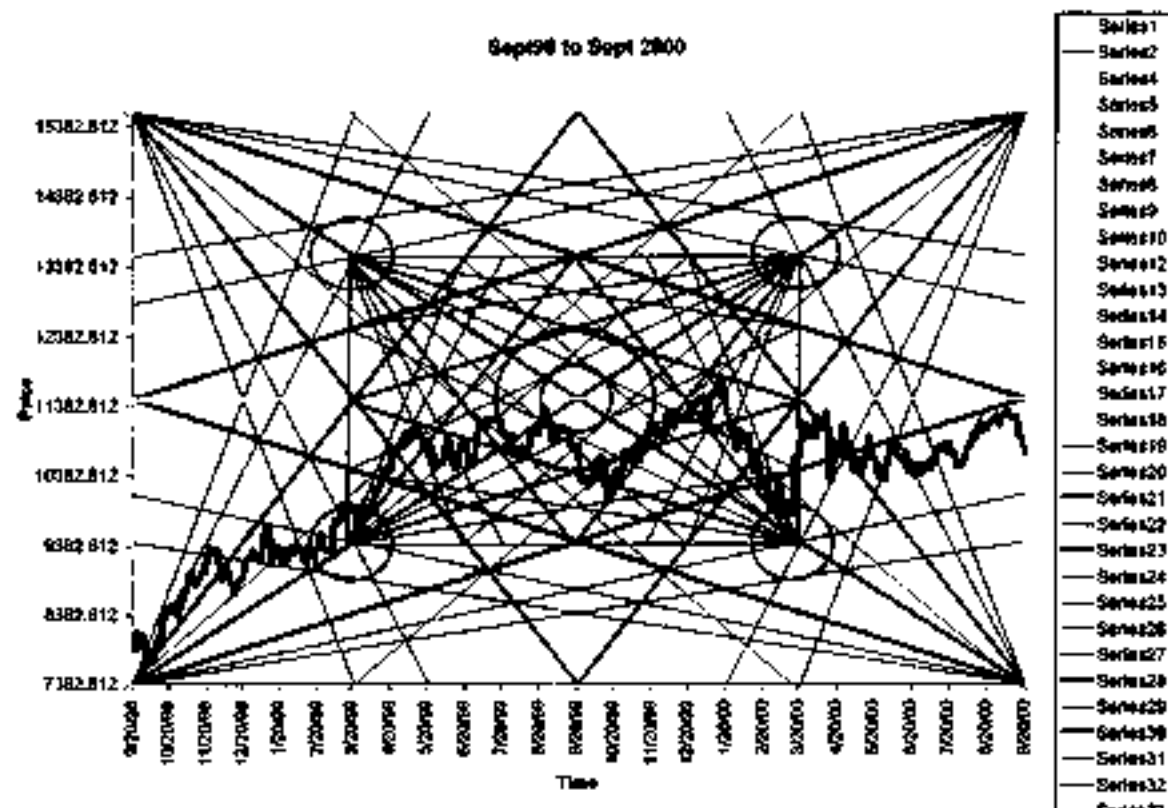
Therefore, Steel is $\$8 \frac{5}{8}^{\text{th}}$ behind time. T.H. Murrey has his time rules based off the "Harvest Moon" but I would stick to W.D. Gann's time definitions when using this natural 8^{th} system and his plastic overlays as his method has a clear degree relationship to both price and time and also to the fractal structure, which he described in the quote mentioned above. I will discuss how to set up these charts properly in terms of time later in the article with specific examples. Gann goes on to say: "When Steel reached $\$200$, it equaled 2 circles of 360-degrees (two complete cycles of $\$100$). When it advanced to $\$261\frac{1}{4}$, it was closest to $\$62\frac{1}{2}$ ($5/8^{\text{th}}$) in the third cycle of $\$100$ or nearest the 225-degree angle or $5/8^{\text{th}}$ point, which is the strongest angle after it crossed the half-way point at $\$250$ or the 180-degree angle". Note that Gann rejects the two circles or cycles of $\$100$ price to convert the time angle. These instructions are also very different than what Murrey Math instructs students to do but I do not want to get into that subject.

Moving on to the subject of time we know that Gann placed great emphasis on the cardinal points of the solar year. These are the Vernal Equinox, Summer Solstice, Autumnal Equinox and Winter Solstice or March 21st, June 21st, September 21st, and December 21st respectively. Gann always said that the year begins at March 21st (not January 1st) and that this was a very important seasonal time. Now this is where Gann's "within the circle forms the square" quote gets real interesting! If we treat each natural year (March 21st to March 21st) as a complete circle, which it is because the earth has gone 360-degrees around the Sun, and take W.D. Gann at his word that there is a square within this orbit then we come up with some very unique geometrical information about

time and squares, i.e. Gann's plastic overlays. Lets build a hypothetical square running from Vernal Equinox to Vernal Equinox as our horizontal axis and we will use 0 to 100 as our vertical. If we draw in the two 45-degree Angles corner to corner we have a shape that looks like a box with an x in it. The two 45's will meet at the exact center of both price and time at \$50 on the Autumnal Equinox or September 21st. If we bisect this square again with a horizontal line going left to right at \$50, and a vertical line going straight up from the Autumn Equinox we can create four smaller squares with half of the x already completed. If we complete these smaller boxes with the missing 45-degree line we will find that these points come out at \$25 and \$75 in price and also 25% and 75% in time or Summer Solstice to Winter Solstice (June 21st to December 21st). If we take these 4 points, which are \$25 on June 21st, \$75 on June 21st, \$25 on December 21st and \$75 on Decembers 21st and make a new square, we find that our new square is exactly half the size of our former square set within the ¼ points in terms of both price and time. Our new square also maintains the exact center of our old square at \$50. This new square is now an exact musical octave of the old square because it's based upon powers of 2 or is exactly ½ our old square. This tells us that if we want to find a square that is within our old square that we will only find it between the two Solstice points in the year! If we wanted to enlarge the original square (to find an outer square) we would again follow the natural law of the musical octave, i.e. powers of 2. The first part of the problem is easy because we know that the next square is going to be twice as large as our old square. So this means that we are moving up from a one-year square to a two-year square in terms of time but this does not answer where in time this new square begins and ends. To calculate where this larger square begins and ends we have to use the rule "as above so below". In

other words, when we made the smaller square, we found that it existed within the larger square at the 25% and 75% points in time, therefore March 21st to March 21st is within the $\frac{1}{4}$ points of the larger square above it. In our one-year square, a $\frac{1}{4}$ of time is 90-degrees or 91.3125 solar days. This means that a $\frac{1}{4}$ of time in the next larger square is 180-degrees or 182.625 solar days, i.e. twice as big. If we add and subtract 180-degrees/days from our smaller square running March to March, we will find that our two-year square begins and ends at the Autumnal Equinox. The same is true for a 4-year, 8-year, 16-year square, etc. Getting back to the inner squares, a 90-degree square (1/2 of our 180-degree Summer Solstice to Winter Solstice and winter back to summer circle) would begin and end 45-degrees after the Summer Solstice and 45-degrees before the Winter Solstice because 45 is $\frac{1}{4}$ of 180. This would be 45.65 days or $\frac{1}{4}$ of 182.625. So this means we would have another (90-degree) square running inside the (180-degree) Solstice squares beginning on August 5th and ending November 5th. Then just as we have two 180-degree Solstice squares running from Summer to Winter, then Winter back to Summer, we also have four 90-degree squares running from August 5th to November 5th, November 5th to February 5th, February 5th to May 5th and May 5th back to August 5th. These are all exact mathematical points to look for squares based upon Gann's plastic overlay pattern. I am providing two charts of the Dow Jones Industrial Average (closing price only) so that you can visually see what Gann was describing in his Square of 52 course. Anyone interested in studying Gann, should purchase their materials from Brad Stewart at Sacred Science Institute (www.Sacredscience.com), as I have found them to contain many unique sections of information that is not provided by others selling Gann related material. I have made many of my best discoveries, including the information

presented in this article, from the course materials provided by the Sacred Science Institute!



“ Within the circle forms the square, there is an inner circle and an inner square, as well as an outer square and an outer circle which prove the Fourth Dimension in working out price movements”. Please note that the inner square is an exact musical octave of the larger outer square.

This next paragraph was not included in the published article:

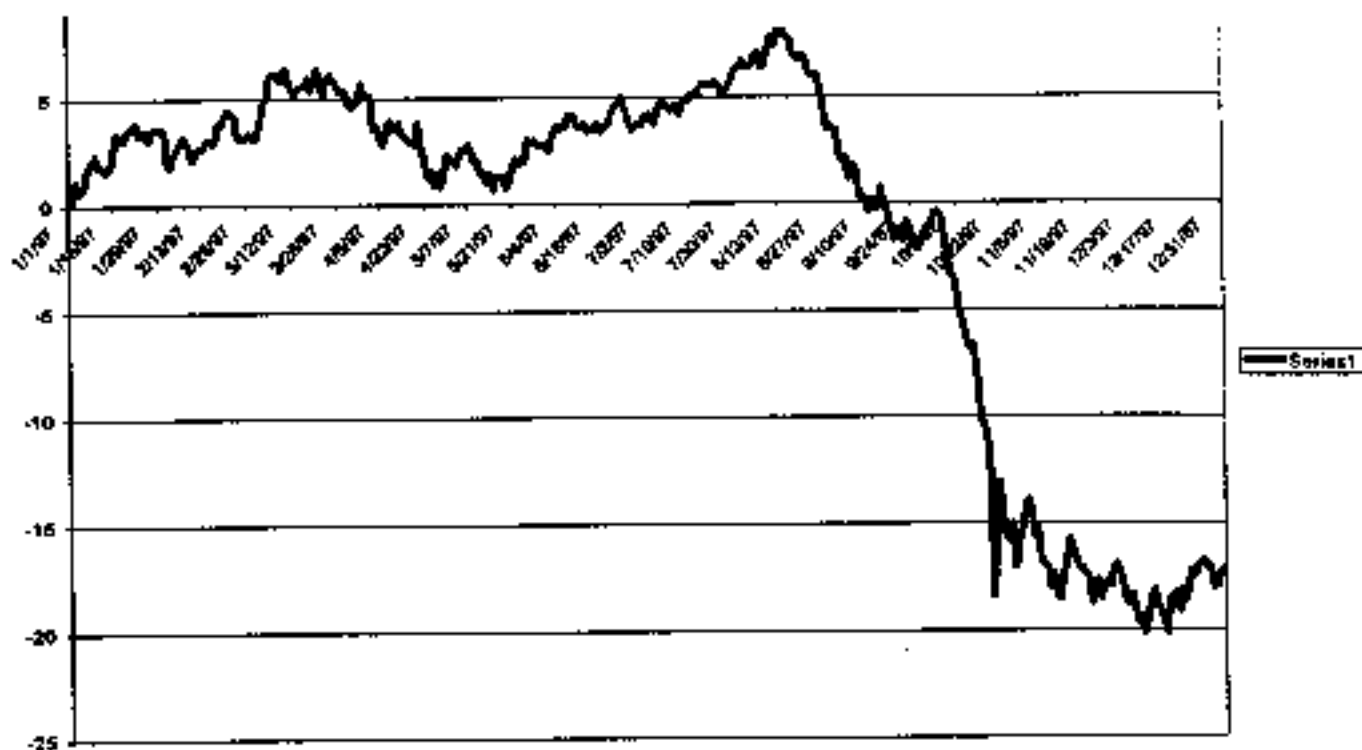
The secret to these charts is to (1) use Gann's base ten system (or Murrey Math) for the vertical price axis, (2) use all the angles presented on his clear plastic geometric overlay chart, i.e. the square of 52 or the square of 144, (3) construct the squares horizontal axis within the natural time periods as mentioned in this article and also within the base 10 system. Note: you will not find these squares unless you use the natural mathematical time periods! Also note that the rules for increasing or decreasing the size of the square on the price axis are the same as those needed for the time axis. In other words, the top and bottom price of each square are the 75%(6/8ths) and 25%(2/8ths respectively) lines of the larger square above itself. Each larger square is exactly 2 times as large as the previous square but are always within each successive larger square on the quarter points of both time and price! This is similar to those Russian dolls that you keep opening and finding smaller versions of the same doll inside each of the preceding larger dolls.

Gann's Mass Pressure Chart

(I've never published this!)

Each year, W.D. Gann would draw up a stick figure forecast of the stock market averages and various commodity futures in his Supply & Demand newsletter service. He would provide commentary that would provide an outline of what he expected the market to do. Gann said that the future was just a repetition of the past and one needed only the right beginning to predict the future. In latter years, Gann provided something he called the Mass Pressure Chart. In his advertisement for his Master Mathematical Time Price and Trend Calculator, Gann said that he would provide a Mass Pressure Chart for the upcoming year. Based upon its vague description, the Mass Pressure Chart is supposed to indicate bullish and bearish trends according to Gann's Master Time Factor. I have talked to many Gann experts about the Mass Pressure Chart and most have no clue as to what it may be. I have never seen an actual original Mass Pressure Chart from W.D. Gann and I do not know of anyone else who has come across one. I have heard rumors and opinions that the Mass Pressure Chart is simply a composite of the Market 20 years ago and 60 years ago. Others have told me that it is actually the Market 80 years ago. In terms of cycles, 80 years would have Mercury and the Sun at approximately the same longitude on the same calendar day. This is also one of George Bayer's rules for forecasting. The 60-year and 20-year cycles deal with the two largest planets in our solar system, Jupiter and Saturn, which conjoin every 20 years. The 3rd conjunction every 60-years typically takes place in the same zodiac sign. In Gann's Forecasting course, he says that the "previous 10-year cycle and 20-year cycle have the most effect in the future". Gann also calls the 60-year cycle the "Master Time Period" and "the Great Cycle" in this

particular course. Gann latter says that he could not have forecast the great advance in 1929 and determine when it would culminate without the 60 and 20 year cycles. In the next paragraph, Gann talks about the forecast for 1935 and he is looking at every 20-year period starting 80 years back, i.e. 1855, 1875, 1895 and 1915. As you go through his forecasts and comparisons of past years, you will find that he consistently leaves out the most recent 10-year cycle. I found this strange since he said that the previous 10-year and 20-year cycles had the most effect on the future. Another interesting point is that he consistently looks back 80-years but does not talk about an 80-year cycle in the beginning of the course where he is outlining the various Major cycles that you should be aware of. I do not know for certain exactly what the Mass Pressure chart is. Although I have my own version, and have been able to duplicate many of Gann's forecast charts, I have never seen an actual Mass Pressure Chart. My personal belief is that it is a composite of the last 10, 20, 60 and 80-year cycles. As an example, when I made a forecast for the year 1997, I looked at the years 1987, 1977, 1937 and 1917. I made notes of when the major highs and lows came out in these years and then went on to make a composite of all these years as a single chart. This chart below is what you get. By the way, do not expect these forecasts to be accurate in terms of amplitude. In other words, they forecast trends and extremes but not the magnitude of the next trend.



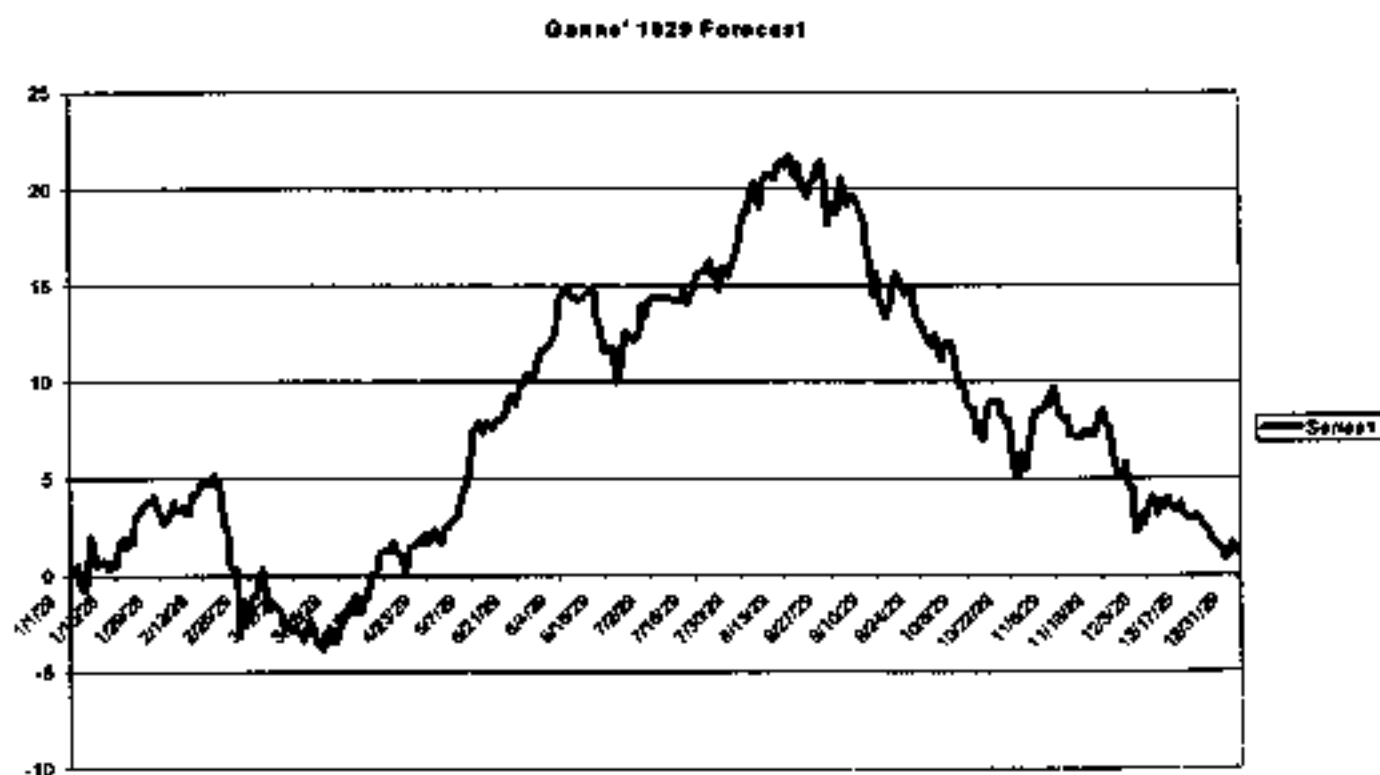
Looking at the above composite chart we get the following forecast: The market is bullish and strong up until March 12, 1997 from here the Dow Jones Industrial Stock Average drops until at least early May with the potential for a double bottom on May 31st presenting secondary buying opportunities. The market then has a strong advance into the first or second week in August where the Majority of stocks should top out for the year. This is where stocks should be sold out to keep the profits accumulated during the preceding advance. The market is bearish and weak into October with a potential short-term top around October 7th to 10th. From October 10th to October 25th bearish cycles accelerate and the potential for a panic type break is indicated between October 21st and October 25th. From here a strong bounce back is indicated for the 26th followed by another drop to test lows before November 1st. The market remains pretty volatile until at least December 3rd where the potential to purchase stocks again at more attractive prices for a Christmas rally is likely. The safest buying point looks like December 21st, 22nd when a seasonal change is indicated, i.e. the winter solstice.

This is what actually happened:



The market first topped on March 12th as indicated in the composite chart. The bearish forecast into early May only lasted until mid April before stocks regained a bullish up-trend. This tells us that the market is stronger or more bullish than the composite. The High for the year came on August 6th before bearish cycles started to weigh on the market, which lines up with the forecast. There was also a second high on October 10th and a panic type break on October 22nd. The market had the strong bounce back as forecast but never tested the panic lows again. Volatility lasted until November 22nd around Thanksgiving when the up-trend accelerated. The market did sell off again providing the buying opportunity indicated for the winter solstice on December 21st, 22nd.

Overall, the forecast predicted the general mood of the market with uncanny accuracy. Even though the forecast chart does not match the actual chart swing for swing the main trends are definitely there. This is my version of the Mass Pressure Chart based upon the information Gann provided in his Master Forecasting Course. I think you will agree that it is a very useful technique. If you combine this type of chart with the actual planetary cycles, you get some fabulous results and remarkable predictions! By the way, these charts can easily be made 10 years in advance. This same type of chart appears to be the technique Gann used to forecast the 1929 Bull Market and Crash. Creating a composite chart of the years 1849+1869+1909 yields a chart that matches up with both Gann's forecast and the instructions he discussed in his forecasting course. The chart looks as follows:



Note that the lows indicated for the end of March and Final top for the end of August followed by a panicky break in the fall of the year are all shown on this chart! This compares very well with what actually happened. You can get an actual copy of Gann's Supply & Demand Letter for 1929 online. The site is www.webtrading.com/gannforecast.htm. My forecast chart lacks some of the details that Gann's contained because I only had monthly data for the year 1869 as opposed to the daily data.

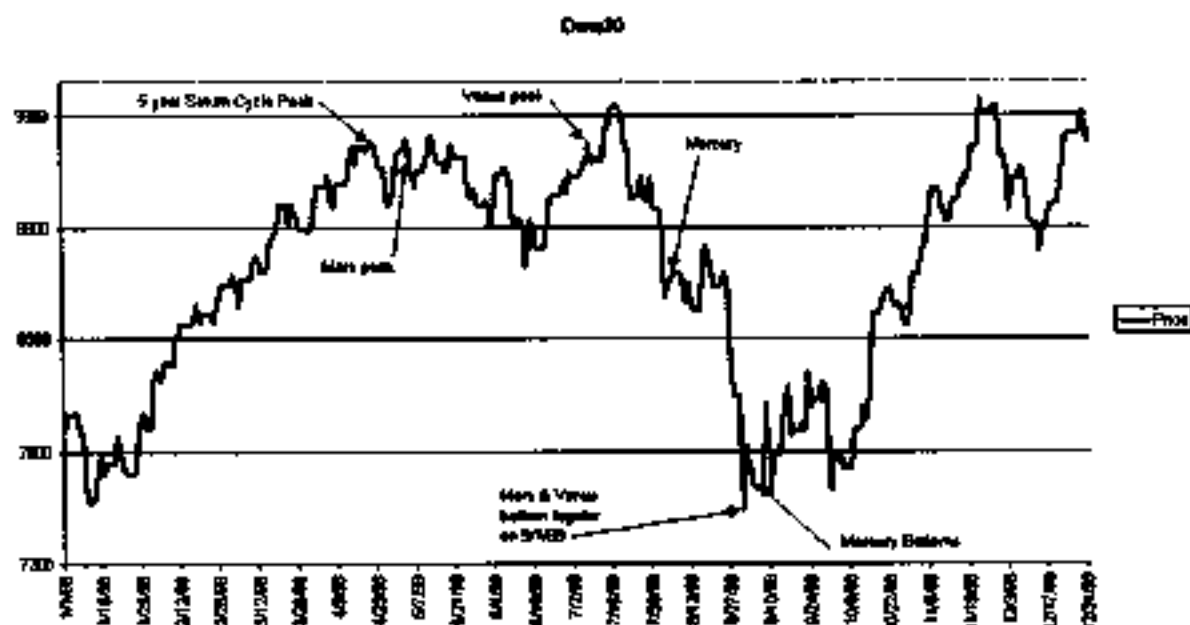
Actual vs forecast



1997 was a fairly easy year to forecast. We were already in a Bull Market and it was a pretty safe bet that we would repeat primarily like the Bull markets of 1987 and 1937. The fact that the market recovered so well after the Asian currency panic is another indication that the bull market will continue in 1998. Gann's basic rules for ascertaining

the type of year the stock market will have on page F-6 of his "Forecasting" course say's that years ending in 8 are typically bull market years. This is another supporting factor for a bullish 1998 forecast. Gann also says in this same course, rule #1 - "A bull campaign generally runs 5 years - 2 years up, 1 year down, and 2 years up, completing a 5-year cycle". This cycle is related to 60°-longitude movement of the planet Saturn. It is also 1/4 of the conjunction cycle of Jupiter with Saturn. Following this basic 2-1-2 rhythm from the August 1982 low, we would expect 1982 to 84 up 2yrs, 1984 to 85 down 1yr, 1985 to 87 up 2yrs, 1987 to 1988 down 1yr, 1988 to 1990 up 2yrs, 1990 to 91 down 1yr, 1991 to 1993 up 2yrs, 1993 to 1994 down 1yr, 1994 to 1996 up 2yrs, 1996 to 1997 down 1yr, 1997 to 1999 up 2yrs, and 1999 to 2000 down 1yr. There are other cycles in play, but the market has followed this general outline very well. The reason I'm going through all of this is because you need to know where you are currently located according to time cycles. For example: Saturn will complete its 3rd 5-year cycle of 60° on 4/22/98 at 23°40' Aries. This is 180° from August of 1982 or one half of Saturn's orbit indicating that we should have a high around this date. The Asian panic occurred on the 22nd of October 1997 when heliocentric Mars was at 18° Capricorn. Gann say's that "the smallest cycle is one year, which often shows a change in the 10th or 11th month". This is 180° of Mars's heliocentric movement, which means that our next major turn will be August 31st or September 1st 1998 when Mars hits 18° Cancer. I know this should be a low because both Saturn and Mars have hexagonal cycles that create the appearance of polarity shifts in their 60° movements. This is because they both actually have two overlapping cycles of 120° that cause market turns every 60°, creating a hexagon or "Star of David". Because this cycle is 1/3rd of the circle (an odd number), you get cycles that

can generate two or three consecutive highs or lows before reversing trend instead of the anticipated saw tooth alternation from high to low that most cycle analysts look for. This is also why the 60° Saturn cycle went low (-) to low (-) from 1982 to 1987 and then went to a high (+) in 1993 and will likely go to another high (+) in 1998, which has been a big help for the bull market. From this information alone, I would expect a major high for stocks on 4/22/98 and a major low on 9/1/98. This will be one of the larger corrections in this bull market because of a harmony with other cycles. If you create a decennial a.k.a. 10-year pattern chart or a Mass Pressure Chart as illustrated earlier in the article, you will see that they will not indicate this strong down move. What is actually happening is that Mars will move a 120° from the low on 10/22/97 to a high on 5/6/98. This will complete a Mars' low to high move. Mars will be negative for the next 60° into 9/1/98 and Saturn will also be negative from 4/22/98 because this date is 120° from the Saturn low in 1988, completing a low to high swing. Mercury should add to the down pressure after 8/5/98 when it passes 21 Capricorn. Venus will add to the negative pressure after 7/7/98 and will bottom on 9/1/98 with Mars, after moving 90° from 0° Taurus to 0° Leo. Mercury will also be 120° from 8/5/98 on 9/3/98 supporting the upward energy for a strong bounce. Both Venus and Mars start an up-move here but there are also other larger cycles in play that support an up-move from 9/1/98 but I will leave these for you to discover. There is a chart on the following page with the cycles illustrated. The following is another article that I never published pertaining to forecasting with planetary cycles. By the way, Saturn will be down for 5-years, which indicates the year 2003 again.



W.D. Gann's Secrets to Forecasting

Today there are so many "Gann Experts" that are cashing in on the mysterious methods of W.D. Gann. There are tons of books, several courses, computer programs, trading systems or methods, etc. that lure in the trading public, especially commodities traders to spend their hard earned money in the hopes of discovering the true "Holy Grail" techniques of Mr. Gann. I have personally spent countless sums of money over the 15 years or more of personal study, in the pursuit of Gann's methods. I really have no idea what the total final bill amounts to but I can easily recall how much of it was a value and how much of it was complete junk. I won't slam anyone or point the finger at authors of material that is of little value. What I am willing to do is share my interpretation of Gann's work with the public in the form of a book. I never planned on writing a book. But after all of the letters I received from the two previous articles that I wrote for Traders World requesting that I offer more information, I decided to put some

information together that would be truly valuable to the "Holy Grail" seeker. To prove my sincerity, and also the value of this work, I will provide you with extremely valuable information in this article on Forecasting.

The main attraction with Gann is forecasting! The idea that the future can be predetermined is a fascinating concept. To truly understand Gann's methods, you must understand his concept of time. Gann said that "Time Proves All Things" and that "Every movement in the market is the result of a natural law and a cause which exists long before the effect takes place and can be determined years in advance. The future is but a repetition of the past, as the Bible plainly states."

"The thing that hath been, it is that which shall be; and that which is done is that which shall be done, and there is no new things under the sun." Eccl 1:9.

Gann's Forecasting Course deals with the use of "Major Time Cycles". Gann believed that everything moves in cycles (circles) as a result of the natural law of action and reaction. He has said " By a study of the past, I have discovered what cycles repeat in the future." The problem that most students of Gann run into is that they get caught up in trying to predict price. Gann always said that between the two, Time was much more important than price. Knowing when to trade is the key to Gann. I will now provide you with a short term and a long-term example based upon the instructions: "Rules For Future Cycles" by W.D. Gann:

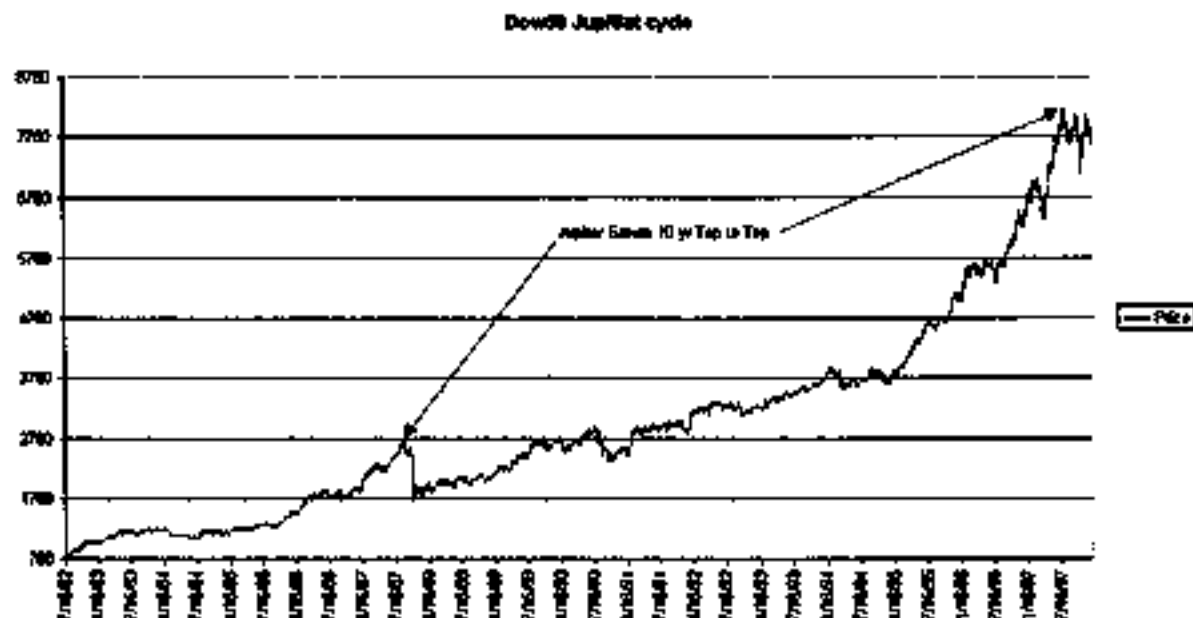
Rule #4 "Adding 10 years to any top, will give you the top of the next 10-year cycle, repeating the same kind of year and about the same average fluctuations"

To really understand what Gann meant by this rule, you have to understand his concept of time or what he means by cause and effect. Gann was a master mathematician, Mason, and also an Astrologer! In fact, most of Gann's recommended reading list deals with Astrology. Now, lets look at what 10 years means to Gann.

Primarily, 10 years is equal to $\frac{1}{2}$ of a Jupiter & Saturn cycle. Secondly, 10 years is equal to 120-degrees of Saturn or Two sides of the cube or his Hexagon chart.

At the market top of August 25th, 1987, you will find that heliocentric Saturn has a longitude of 20-degrees 12' minutes Sagittarius and that heliocentric Jupiter's longitude is 19-degrees 44' minutes of Aries. The angle between these two planets is 119-degrees 32' minutes and they are separating. If we add 120-degrees to Saturn (two five-year sides of the Hexagon), we get 20-degrees 12' minutes Aries, which occurred on January 13th, 1998. If we use the angle between Jupiter & Saturn, then we must find out when these two planets are 60-degrees apart before the conjunction i.e. 180-degrees minus 119 32' = 60-degrees 28' minutes. This occurred around July 16th 1997. If you followed this cycle as it was presented in Bradley Cowan's Book Four-Dimensional Stock Market Structures and Cycles you would have been looking for an angle of 58-degrees, which happened on August 6th, 1997. Brad's cycle work is an excellent resource for understanding Gann's Forecasting course. This example illustrates the Top-to-Top

relationship that Gann presented as rule #4 in his forecasting course. Both dates were exact hits!



Forecasting Monthly Moves

A short-term example or forecasting rule is: "Add three months to an important bottom, then add 4, making 7, to get minor bottoms and reaction points."

Again, we have to decode Gann's definition of time. Three months is 360-degrees of Mercury. Adding 4, making 7 months is 360-degrees of Venus. If you examine the recent low of October 18th, 1999, you will find heliocentric Mercury at 21-degrees Capricorn and Venus is at 1-degree 26' minutes of Gemini. Looking in an ephemeris, we will find Mercury again at 21-Capricorn on January 14th, 2000 (the exact top of the Dow Jones before a 15% + correction). And Venus returns to 1-26' Gemini on Tuesday, May 30th 2000 for another potential turning point. At the time of this writing, March 15th 2000, the Venus forecast is yet to be fulfilled so watch the market around this day! I hope that this information proves to be valuable in your quest for knowledge, as this will be my last

article. One final parting piece of wisdom is that when you think of cycles, think of throwing a stone into a pond of water. Each turning point in the market starts a ripple effect that moves forward in time and effects the future. The more significant the turning point, the larger the ripples. Just like throwing in a small pebble and a large rock. It is obvious that the large rock will overpower the ripples of the small pebble, making its effects non-distinguishable at times. In these instances, record the position of the minor cycles at the time of the "Major" Turn and project these values into the future, along with the old values to see if the minor cycle has remained unchanged or has synchronized itself with the larger cycles "Wheels Within Wheels". This is why Gann said to work all cycles from Major tops and Bottoms in the market.

Also pay attention to April 12, 2000 and July 7, 2000, when Mercury hits 21-capricorn for the 3rd and 4th times, etc. The chart above has been updated so you can now see what happened.

Gann's Permanent Charts

In Gann's Master Course for Stocks, in the Master Charts section, he has a Name Chart for United States Steel. Gann says: "The name United States Steel, contains 17 letters. Therefore, to make up a Permanent Square for United States Steel according to its name, will require 17×17 , or a square of 289. Therefore, 17, which is really $2 \frac{1}{2}$ times the lowest price at which Steel ever sold is an important point.

The prices 34, 51, 68, 85, 102, 119, 136, 153, 170, 187, 204, 221, 238, 255, 272, are all important vibrations on the name and angles of U.S. Steel.

The basic number or low point, for example 9, the lowest digit on U.S. Steel, and its vibrations according to its name, all cause slight variations at times from other stocks, because each stock works according to its own base, beginning point, numbers and name.

"U.S. Steel works good to the natural angles and Master Chart (square of 9), because its digit is 9 and comes out on exact 9 vibrations all the way through, finishing at 261, which was 28 nines from its beginning point, or lowest level." The lowest price that U.S. Steel traded was $8 \frac{3}{8}^{\text{th}}$. $28 \times 9 = 252$ and $252 + 8.375 = 260.375$ or 28 nines from the low. Therefore, the permanent charts are geometric squares based on the total number of letters contained in the name. For example, Gann's permanent chart for the NYSE is a square of 20×20 containing 400 cells. This is because the Name: New York Stock

Exchange contains a total of 20 letters. Gann says that this chart can be used to measure days, weeks, months, or years to determine when tops and bottoms will be made against strong angles, as indicated on this Permanent chart."

"For example: - The New York Stock Exchange was incorporated May 17th, 1792.

Therefore, we begin at "0" on May 17th, 1792. 1793 ends on a 1, when the NYSE was 1 year old. 1812 (War) will come out on 20 which is at the top of the square indicating resistance or change;

1832 on 40	1892 on 100
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1852 on 60	1912 on 120
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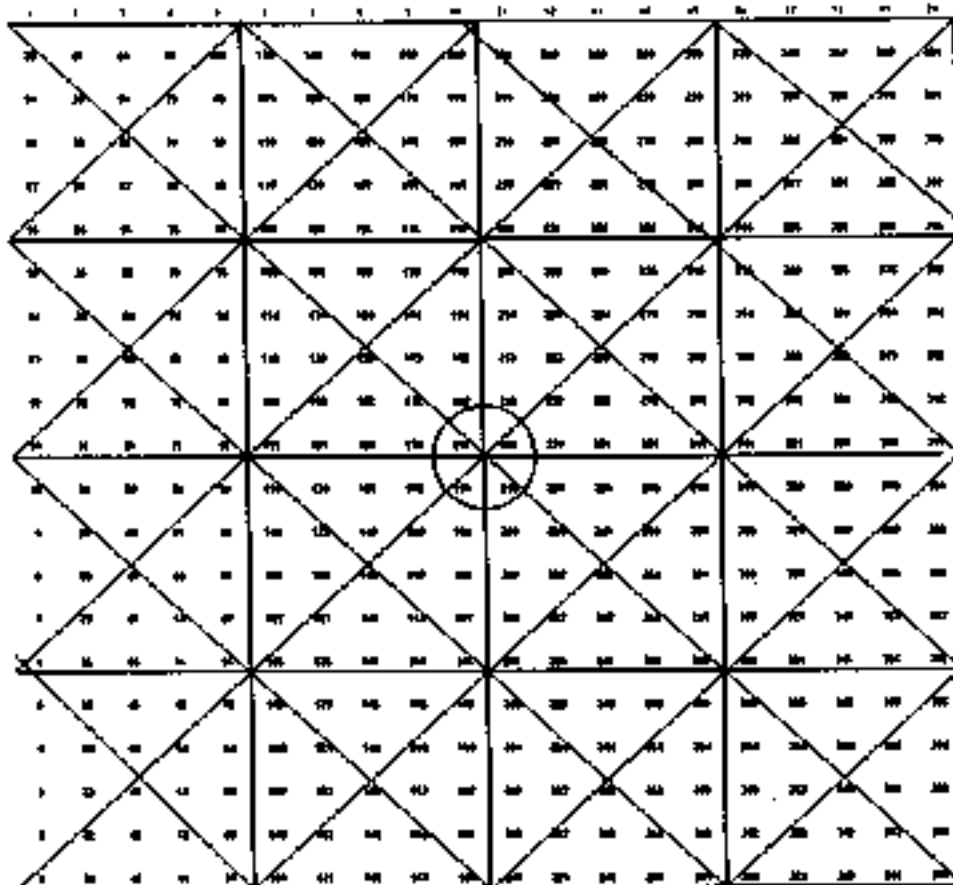
1872 on 80	1932 on 140
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Note that 139, or 1931, strikes the 45-degree angle (on the Permanent Chart), running from 20 down, and that this is the 7th zone; or the 7th space over, which indicates that 1931 is the ending of a bear campaign, and the starting of a bull market. But we must watch for a break around May (Birthday) and June, 1931, when this angle is hit at the end of the 139th year." In the last page of this Permanent Chart section, Gann says that "The more you study these Permanent Charts, the more you will appreciate their great value, and will see that numbers do determine everything in the future, and that geometrical angles and mathematical points measure every Resistance Level, Time, Price, Space or Volume."

My interpretation of this unique esoteric information is as follows: The number of letters in the name defines the permanent chart, which is used to primarily measure time. This coming May 2000 is 208 years from "0" On the NYSE permanent chart. This number shows nothing significant in time resistance if you look at the angles on the chart. The Years 2002 and 2003 (210 & 211 on the chart) show strong resistance as these years are on 45-degree angles and also the midpoint or "gravity center" of the NYSE Permanent Chart! It seems that the year 2003 just keeps raising its head doesn't it?

$208 \times 12 = 2496$ months from May 17th, 1792. If we reject the squares of 20, or 400 we get the number 96 which is just above the 45-degree angle in section 5 of the permanent chart. Note that the number 95 (or one-month prior) is right on a resistance angle indicating that there is resistance in time for April 2000. Also, the months September and October come out on 100 (top of square section 5) & 101 (bottom of square section 6) which indicate changes as both are on 45-degree lines.

NYSE Permanent Chart



$208 \times 52 \text{ (weeks)} = 10816$ Reject the square equals 16 which shows resistance for the week of the May Anniversary, and the week proceeding on the master chart. $208 \times 365.25 = 75972$ - the squares = 372 which again is afflicted by a 45-degree angle on the anniversary.

Also note that Gann sets the price vibration based on the lowest price traded. He also uses this number in time when you read squaring price by time in his angle course. The lowest Price on U.S. Steel was 8 3/8ths on May 14, 1904, which has nothing to do with the number 9. So why does Gann say that 9: "The basic number or low point, for example 9, the lowest digit on U.S. Steel, and its vibrations according to its name, all cause slight variations at times from other stocks, because each stock works according to its own base, beginning point, numbers and name?" I think that I have found the answer!

If we break down the name according to the rules of Numerology we get the following:

1	2	3	4	5	6	7	8	9
A	B	C	D	E	F	G	H	I
J	K	L	M	N	O	P	Q	R
S	T	U	V	W	X	Y	Z	
UNITED			STATES			STEEL		
359254			121251			12553		

Looking at this number breakdown of the name we find that all the numbers in the name repeat at least once except for the numbers 4 and 9. The number 5 repeats the most then #2. I believe the reason that Gann says that 9 is the lowest digit according to its

numbers and name is that the #9 is symbolically the beginning and the end because any number added by 9 reduces numerically back to the original number.

$$1+9=10 \ \& \ 1+0=1 \quad 10+9=19=1+9=10 \ \& \ 1+0=1$$

$$2+9=11 \ \& \ 1+1=2 \quad 11+9=20=2+0=2$$

$$3+9=12 \ \& \ 1+2=3 \quad 12+9=21=2+1=3$$

$$4+9=13 \ \& \ 1+3=4 \quad 13+9=22=2+2=4$$

$$5+9=14 \ \& \ 1+4=5 \quad 14+9=23=2+3=5$$

$$6+9=15 \ \& \ 1+5=6 \quad 15+9=24=2+4=6$$

$$7+9=16 \ \& \ 1+6=7 \quad 16+9=25=2+5=7$$

$$8+9=17 \ \& \ 1+7=8 \quad 17+9=26=2+6=8$$

$$9+9=18 \ \& \ 1+8=9 \quad 18+9=27=2+7=9$$

This can be carried on to infinity and the results will never change. If we apply this same logic to the concept of zero as the beginning point, then we find that $9+0=9$. Therefore nine is both the largest digit and smallest digit or the Alpha and Omega simultaneously because adding 9 to any number must reduce back into the original number! If you read the letters in column 9 you get I R or the statement: "I ARE" which is similar to the Biblical statement I AM that I AM as GOD's response to what his name is. Therefore, in terms of a cycle yields the statement "I AM that I AM, I ARE". This is very interesting in terms of the above concept of adding 9 to a number and reducing it back into the original number through addition.

I believe that because the numbers 4 & 9 have no repeats in the name United States Steel, and that the #9 is a similar concept to the Ace in a deck of cards (i.e. largest and smallest) that this is why it is the "vibration" number for U.S. Steel i.e. it has the lowest frequency in terms of both repeats and also numeric value in an esoteric sense.

If we do this same analysis with the New York Stock Exchange, we get the following:

<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>
A	B	C	D	E	F	G	H	I
J	K	L	M	N	O	P	Q	R
S	T	U	V	W	X	Y	Z	

New York Stock Exchange

555 7692 12632 56381575

Here again we find that all numbers in the name repeat at least once except the number 9 just like U.S Steel. Therefore, the number 9 must also be the fundamental vibration for the NYSE as well.

Gann also uses the name America (square of 7) as a permanent chart beginning October 1492 when Columbus discovered America (1459931), which has the number 3, as the fundamental vibration because it only occurs once in the name and is the lowest value that does not repeat. The Name United States of America begins July 4th, 1776 with

the signing of the Declaration of Independence. This is a square of 21-letters which is 3 times 7 or the letter value of America. It is an interesting coincidence that the new name is exactly the fundamental vibration of America (3) multiplied by its letter value (7) = United States of America! (359254 121251 66 1459931) If you count the repeats and examine the values, you will find that the number three still has the fewest repeats and is also the lowest numeric value of the numbers that repeat twice. Which means that 3 is still the fundamental vibration of the USA. By the way, the year 2003 also comes out on this name square in time as well. Try it yourself!

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
U	N	I	T	E	D	S	T	A	T	E	S	O	F	A	M	E	R	I	C	A
21	42	63	84	105	126	147	168	189	210	231	252	273	294	315	336	357	378	399	420	441
20	41	62	83	104	125	146	167	188	209	230	251	272	293	314	335	356	377	398	419	440
19	40	61	82	103	124	145	166	187	208	229	250	271	292	313	334	355	376	397	418	439
18	39	60	81	102	123	144	165	186	207	228	249	270	291	312	333	354	375	396	417	438
17	38	59	80	101	122	143	164	185	206	227	248	269	290	311	332	353	374	395	416	437
16	37	58	79	100	121	142	163	184	205	226	247	268	289	310	331	352	373	394	415	436
15	36	57	78	99	120	141	162	183	204	225	246	267	288	309	330	351	372	393	414	435
14	35	56	77	98	119	140	161	182	203	224	245	266	287	308	329	350	371	392	413	434
13	34	55	76	97	118	139	160	181	202	223	244	265	286	307	328	349	370	391	412	433
12	33	54	75	96	117	138	159	180	201	222	243	264	285	306	327	348	369	390	411	432
11	32	53	74	95	116	137	158	179	200	221	242	263	284	305	326	347	368	389	410	431
10	31	52	73	94	115	136	157	178	199	220	241	262	283	304	325	346	367	388	409	430
9	30	51	72	93	114	135	156	177	198	219	240	261	282	303	324	345	366	387	408	429
8	29	50	71	92	113	134	155	176	197	218	239	260	281	302	323	344	365	386	407	428
7	28	49	70	91	112	133	154	175	196	217	238	259	280	301	322	343	364	385	406	427
6	27	48	69	90	111	132	153	174	195	216	237	258	279	290	311	342	363	384	405	426
5	26	47	68	89	110	131	152	173	194	215	236	257	278	299	320	341	362	383	404	425
4	25	46	67	88	109	130	151	172	193	214	235	256	277	298	319	340	361	382	403	424
3	24	45	66	87	108	129	150	171	192	213	234	255	276	297	318	339	360	381	402	423
2	23	44	65	86	107	128	149	170	191	212	233	254	275	296	317	338	359	380	401	422
1	22	43	64	85	106	127	148	169	190	211	232	253	274	295	316	337	358	379	400	421
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21

I have included copies of Gann's Master Charts course and also Gann's Mathematical Price Time & Trend Calculators course (The Square of 12 & The Square of 52). The square of 12 works best on the monthly chart because there are 12 months in a year. The square of 52 is used to measure weekly charts, as there are 52 weeks in a year. Again these calculators are primarily designed for time but Gann says that you can use them for price, volume and pitch. The square of 90 is based on the number of days between equinox and solstice points on a calendar, i.e. the cardinal cross-locations in the zodiac. Gann also used a chart called the Square of the Circle, which is $19 \times 19 = 361$ or 1 over a perfect 360. The gravity center of this number chart is 180, which proves the square of the circle. I believe Gann used this chart from Vernal Equinox to Vernal Equinox, i.e. March 21st to March 21st = 360-degrees of longitude. On the last page of Gann's Geometric Angle course, page. GA-32, Gann says: "Figuring \$100, or par, as a basis for stock prices and changing these prices to degrees, $12\frac{1}{2} = 45$ -degrees, $25 = 90$ -degrees, $37\frac{1}{2} = 135$ -degrees, $50 = 180$ -degrees, $62\frac{1}{2} = 225$ -degrees, $75 = 270$ -degrees, $82\frac{1}{2} = 315$ -degrees, and $100 = 360$ ". As previously mentioned, this little paragraph is the basis of the entire Murrey Math Trading Course. These numbers are all natural 8th off of the base of 10, 100, 1000, 10000, 100000, etc. The part that is most confusing is balancing this with the time element, but based on the information given by Gann himself it should be obvious that he has already shown us the relationship. For example: If shares of XYZ stock moved up $12\frac{1}{2}$ dollars in 45 days which are degrees of the sun or 45 months (degrees of Saturn) then the stock is on the price degree of it's time angle, that is to say that price and time are equal or balanced. Gann's example: "When a stock sells at

50 on the 180th day, week or month, it is on the degree of its time angle". "On February 1, 1915, U.S. Steel made a low at \$38, which is closest to a price of \$37½, which is 3/8th of 100 and equals 135-degree angle. Steel was 14 years or 168 months old on February 25, 1915, and hit the angle of 135-degrees, which showed that the price of Steel was behind time, but was in a strong position, holding at \$38 above the 135-degree angle or the price \$37½". Just to make sure that you understand this information, U.S. Steel would have been balanced or "on the degree of its time angle" at a price of \$46 5/8th because \$100/360-degrees = 0.27777cents per degree and 168 months (or degrees of Saturn) multiplied by 0.27777 = \$46.67 which is closest to \$46 or 5/8th in price. Therefore, Steel is 8 5/8th behind time. T.H. Murrey has his time rules based off the "Harvest Moon" but I would stick to Gann's definitions when using this natural 8th system as it has a clear degree relationship to price and time. I have already discussed these charts previously, when I went over the Master Mathematical Price & Trend Calculator, i.e. The Square of 52. "When Steel reached \$200, it equaled 2 circles of 360-degrees. When it advanced to \$261½, it was closest to \$62½ (5/8th) in the third cycle of \$100 or nearest the 225-degree angle or 5/8th point, which is the strongest angle after it crossed the half-way point at \$250 or the 180-degree angle". Note that Gann rejects the two circles or cycles of \$100 price to convert the time angle.

Currently, the Dow Jones Industrial Average is trading between the 1st and 2nd cycle of 10,000. This means that 1/8th or 45-degrees = 1250 points or (1250/45) 27.77 points per day, week, month which as you should have learned earlier, all mean degrees of planetary movement or time. As an example, when the Dow made a low of 9976 on October 18th,

1999 it advanced to a price of 11750 on January 14th, 2000. Heliocentric Mars moved 55.48 degrees @ 27.77 points per degree this would be 1541 points added to the 9976 low is 11517 for the price degree angle. This shows that price is ahead of time by 233 points and a correction is needed to balance price with time. Also remember that heliocentric-Mercury moved exactly 360-degrees from low to high. If you divide the low price of 9976 by 27.77 points, you get 359.23 degrees, which also means that heliocentric mercury has squared out the low price in time or longitude! Because Mercury's orbit is $\frac{1}{4}$ th of a solar year, you can look at this as a 4 X 1 angle. In other words, the Earth moved 90-degrees from 10/18/99 to 01/14/00 and Mercury moved 4 times as much. The next mercury square out of this low would be April 11th, 2000 followed by July 8th, 2000. If you divide the January 14th 2000 high of 11750 by 27.77, you get 423.11 degrees. This would put mercury at 24-degrees of Pisces on July 25th, 2000 and would square out the high price in time or longitude. The next longitude point for mercury would be at 27 Taurus on November 2nd 2000, which is 423.11 degrees added to 24 Pisces. This technique of utilizing the base of 10, along with the Gann Angles technique described earlier allow the analyst the ability to project price and time out into the future. At this point, you should have a clear understanding of what Gann meant when he said that the squaring of price by time was one of his greatest discoveries. You can also quickly calculate aspects of price with this idea. For example: At 27.77 points per degree we can multiply by 30 for the semisextile, 45 for the semisquare, 60 for the sextile, 72 for the biquintile, etc. This would yield 833.10 points (30°), 1250 points (45°), 1666.2 (60°), 1999.4 (72°). Adding these amounts to the low (9976) gives: 10809.10, 11,266, 11,642.20 and 11,975.40.

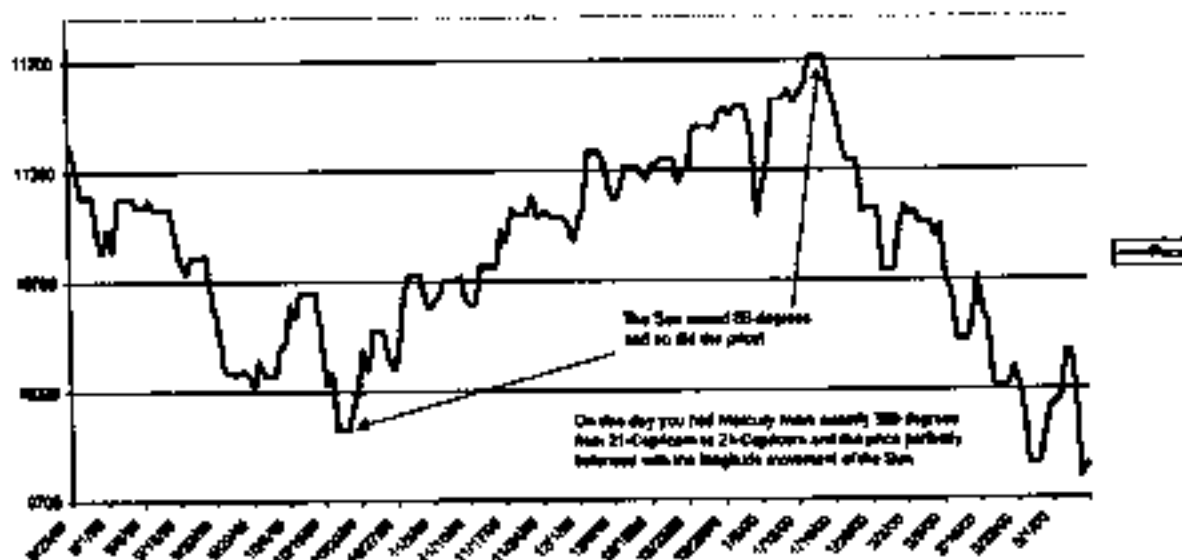
Earlier, when we were going over the Square of 9, I discussed the square root relationships of this Master Chart. If you take the square root of the (January 14, 2000) 11,750 high you get 108.39, which equals $18^{\circ}23'$ Cancer in exact opposition to heliocentric Mercury. If you take the square root of the 9976 low you get 99.87 ($9^{\circ}52'$ Cancer). The difference between these square roots equals 8.52. From 10/18/99 to 1/13/00, Helio-Jupiter moved 7.94 and Helio-Saturn moved 3.14 (Pi). Applying the Pythagorean theorem we get Square root of $(7.94^2 + 3.14^2)$. This is equal to 8.53, very close to our root difference of 8.52. On 1/14/00, this number will be 8.62. The difference between 8.62 and 8.52 equals 0.10, which also would equal 18° on the square of 9 as a root increment. If you add 8.62 to the square root of the low, then re-square, you get 11,772.23 as a price. Remember, increments of 2 added or subtracted to a square root and then re-squared is equal to 360-degrees on the square of 9. So if we look at our number 8.52, we can see that price jumped 4 complete rings out on the square of 9 because 8 divided by 2 is 4. The fractional remainder of 0.52 tells us that price went an additional 94-degrees on the square of 9's rings because 0.50 added or subtracted from a root equals 90-degrees on the square of 9. The remaining balance of 0.02 would equal approximately 4-degrees ($180^{\circ} * 0.02 = 3.6^{\circ}$). As we advance day (1°) by day (1°) from our low date, doesn't it make sense that the relationship for price resistance might have some mathematical relationship to time on the square of 9? In other words, if we were 45° (time) away from 10/18/99 wouldn't you want to know what prices have a 45° relationship on the Square of 9 to the 9976 low price? How would you do it? Simple, you would do it with square root relationships! 45° is calculated by adding or subtracting 0.25 to the square root of the price and then re-square the result. If we do this with the price

low of 9976, we would take the root of 9976, which is 99.87 and add 0.25. This is equal to 100.12. Now we re-square this and get 10,026. The other numbers on this same angle can be calculated by adding or subtracting 2 from the square root and then re-square.

$100.12 + 2 = 102.12$ re-squared = 10,430.52. Get the idea? The next number on the same angle would be 104.12 squared = 10,843.04 etc. These numbers would be carefully observed on December 2nd, 1999 because this date is 45 solar degrees from 10/18/99. If you were looking for all the 88° price relationships on the square of 9, when you were 88° (solar time) away from the 10/18/99 low date in price, you would have been looking for the market to hit the following numbers on January 14th, 2000: 10,479.37, 10,892.85, 11,314.32, 11,743.80, 12,181.27, etc! This is almost a perfect price-time balance. The numbers were calculated as follows: Because adding 2 to a root is equal to 360° on the square of 9, then adding 0.0055555 is equal to 1° because $2/360$ equals .0055555 and this multiplied by 88° = 0.48. Another way this can be calculated is $88°/180 = 0.48888$.

Adding increments of 2 to this number will give us all the numbers on the square of 9 that are on the same angle. So we would add 2.48888 to the square root of our 9976 low and then re-square. This gives us $99.8799 + 2.48888 = 102.3688$. Re-squared equals 10,479.37, as shown above. The other numbers came from adding 4.4888, 6.4888, 8.4888 and 10.4888 to the root of the low price, then re-square the sum. You can also calculate the numbers that would be on the opposite side of the wheel on this date by adding 1.4888, 3.4888, 5.4888, 7.4888, etc. because adding or subtracting 1 to the root number (then re-square) is equal to 180-degrees on the Square of 9.

Condor Bar of 9 price technique



Tunnel Thru The Air

This is probably one of Gann's most important and revealing works. Ironically, it is probably the least studied book that W.D. Gann wrote. It's a fictional novel that was written in 1927, and contains "a valuable secret clothed in a veiled language". This is the first book that Gann wrote, where he "Robert Gordon" gives valuable clues about Bible interpretation and Astrology. The book provides examples of Gann's financial astrology, mundane astrology, natal astrology, progression techniques and more. Gann's instructions are to read this book 3 times and to read the Bible 3 times to gain wisdom & knowledge. Gann also made some very interesting forecasts in this book and covers much of the thought process that went into the predictions.

Robert Gordon's Great Campaign In Major Motors

On page 197 of **Tunnel Thru The Air**, the story reads as follows: On Sunday June 19, 1927, Robert Gordon spent the day studying his charts and working out his cycles for stocks, cotton and grain. He was short of Major Motors and was watching it very closely. On this day, he made a new and great discovery of a time factor from which he figured that Major Motors would decline until about June 30th and then start an advance, which would last until about September 16th, 1927, when the company would be 19 years old and at that time the stock would reach a final high and would then go down to February to April, 1929. He figured that the stock would advance to \$270 by September 16th and made up his mind to watch it closely and cover his shorts if it went down around June 30th, and then start buying the stock. On June 30th, it declined and he bought in his short contracts and bought for long account 500 shares of Major Motors at 192 ½. He decided to pyramid it all the way up. The rest of this section gives specific dates and prices where Gann (Robert Gordon) bought, sold or moved stops. I am not going to reprint the entire book so I recommend that you study it yourself along with my analysis of this and other sections of the book.

First, notice the use of the words "about" and "around" used to describe the date June 30th, 1927. Looking in an ephemeris, you will find a Total Solar Eclipse on June 29th, 1927. On page 56, you will find Robert Gordon's interpretation of Ezekiel 32:7: *And when I shall put thee out, I will cover the heaven, and make the stars thereof dark: I will cover the sun with a cloud, and the moon shall not give her light.* Robert understood this to mean the two great eclipses (Total Lunar & Solar eclipses) that would occur in June

1927. If we erect a natal horoscope for Major Motors based on the above information (and instructions from Sepharial's Arcana) we get the following natal aspects:

The Total Lunar eclipse on June 15th, 1927 created some negative aspects to Major Motors natal chart. This lunar eclipse had the Moon in exact opposition to natal Pluto and the Sun Square to the natal Sun. Gann (Robert Gordon) figured that Major Motors would decline right into the Total Solar eclipse (June 29th, 1927) "around" June 30th. This total eclipse fell with the Sun Square to natal Saturn. The reason Gann expected a Rally from this seemingly very negative aspect is because ^{coinciding} Natal Saturn was Retrograde (moving Backward). When a planet is retrograde, aspects are reversed, i.e. Good aspects such as Trines and Sextiles (120 & 60-degrees) become negative and Bad aspects such as Squares & Oppositions (90, 270 and 180) become positive influences. Gann bought long 500 shares of Major Motors at \$192½ June 30th 1927.

*John Gann
expected
a rally
in 1927.*

The price of \$192½ is in opposition to Jupiter & Uranus (which are moving towards conjunction with each other) on Gann's Hexagon Chart. Gann said that the stock would start an advance, which would last until September 16th, 1927 and reach a price of \$270. On the day of the Total Solar Eclipse, Saturn is retrograde at 2-degrees of Sagittarius. On August 5th, Saturn stops moving retrograde at about 1-degree of Sagittarius and starts moving forward again. On September 16th, 1927 (the 19 year anniversary of Major Motors), Saturn will return to it's exact solar eclipse longitude at 2-degrees Sagittarius, providing the timing of this forecast. The price of \$270 is conjunct Jupiter and Uranus on the Hexagon Chart. The Sun's "Birthday" return on September 16th, 1927 occurs with some other negative astrology: Sun conjunct Natal Sun square to the North Node of the Moon. . If you study the rest of the examples of this "Major Motors Campaign", you will

find that all of the prices (Buys, Sells, and Stops) are right off the Hexagon Chart. The Majority of the prices have an angular relationship with Jupiter & Uranus's longitude. Saturn is second and then Mars. These are the only planets that are used for price targets in this example. I have provided the astrology program Astrolog, which is available for free on the internet so that you may study the natal chart and the angular relationships to price on Gann's Hexagon Chart.

Because I have talked about this eclipse method in the coffee letter and also in the Major Motors Campaign, I will show two more examples with beans. One is directly from Gann, the other is my own.

Gann Example: WD Gann was already short May Soybeans during September 1941. He writes in (How to Make Profits in Commodities, pg. 133) "October 28th, May Soybeans declined to \$155½, making a higher bottom by 1 cent and making this a buying level. The writer covered shorts on this decline". Analysis: on September 21, 1941 there was a total solar eclipse. Jupiter's longitude at the eclipse was 20-degrees 54' minutes of Gemini. On October 28th, 1941 Jupiter returned to it's eclipse location at 20 54' Gemini and held above it's previous bottom by 1 cent, making this a "buying level". Gann covered his short position.

My Example: On August 11th, 1999 there was a total eclipse of the Sun. Soybeans declined immediately! Jupiter's location at the eclipse was 4-degrees 42' minutes of Taurus. On September 7th, 1999 retrograde Jupiter returns to the exact longitude of 4-42' Taurus, timing the exact top of this soybean rally. In other words, soybeans declined immediately (again) when Jupiter returned to the longitude it had during the eclipse!

In 1994, Bonnie Lee Hill, did a seminar on Gann's novel: The Tunnel Thru The Air. In this seminar, she illustrates some of the Mundane Astrology that Gann may have used in his forecasts. Bonnie shows that when there are geocentric aspects between Uranus, which symbolizes change and Neptune, which symbolizes water, the Mississippi River Floods. The aspect must occur in the winter, which means that the Sun is between 0° Capricorn and 0° Aries. This specific period of time is also symbolic of the 3 days Jonah was in the whale's belly and the bible statement that Gann said had a secret meaning: "For as Jonah was three days and three nights in the whale's belly; so shall the Son of man be three days and three nights in the heart of the earth". This is from Matthew 12:38, 39 and 40. Notice that the Saviour does not say the Son of God. He says the "Son of man", which I believe refers to the Sun.

Tunnel Thru the Air page 69 (25), Gann said that this was the key to the interpretation of the future. This allegory regarding three days and nights has been told in many different ways by several religions. I believe that it is a story about the Sun ("the son or Sun of man") as it symbolically dies on the cardinal (cross) sign Capricorn during the Winter solstice on December 21st, 22nd each year. Three days later on December 25th, the birthday of Christ, Apollo, Horus etc., the days start to become noticeably longer. Three months ("days") later, the Sun reaches 0-degrees Aries at the Vernal Equinox on March 21st each year. "Lamb of God, you take away the sins of the world". This is the point where the sun starts to move north of the equator. This is why Easter is always based on the Full moon following the Vernal Equinox in April. From this point on, the amount of daylight increases and the Sun increases in strength as it rises from the grave of winter until the Summer Solstice in the cardinal sign of Cancer June 21st. As far as the

significance for interpretation of the future, I believe that Gann may have been referring to the ingress of the sun into Aries and also the nodes of the other planets. The planetary nodes and conjunctions of the major planets are used for the practice of Mundane Astrology, along with the solar ingress into each of the four cardinal signs with primary emphasis on the vernal equinox. "O, ye hypocrites, ye can discern the face of the sky; but can ye not discern the signs of the times?" Another possibility is that the planets have more influence when the Sun is in the grave of winter.

Bonnie, also shows that droughts are caused by aspects of Saturn, which symbolizes restriction or depression, and Neptune (water). Again, the aspect between Saturn & Neptune must occur while the Sun is "in the grave". This would relate to high prices because droughts hurt supply. Ideal growing conditions (Bumper Crops) are caused by aspects between Jupiter, which symbolizes abundance, and Neptune (water). The aspect between Jupiter and Neptune must occur while the Sun is between Capricorn and Aries. This causes low prices because supply exceeds demand when there are bumper crops. Bonnie made some long term cotton forecasts based upon this information and they have been 100% accurate! The only year that did not come out was the year 1999, but Bonnie said in 1994 that this would probably be a non-event. The reason is that there were aspects between the two indicators, Jupiter-Neptune and Saturn-Neptune. One is calling for a drought, the other ideal conditions. She expected these circumstances to cancel each other out and that is basically what happened. Looking over Bonnie's seminar handout I have found that she allows these aspects to occur as late as the end of April. This would correlate to the mouth of the constellation Cetus, the whale, which would have the Sun in the sign of Taurus. Remember Cetus- The Whale, which symbolically swallowed Jonah?

This constellation runs from Pisces to Taurus, the whales belly is in Pisces when the sun would be south of the equator, corresponding to the month of February and the month where Jonah is spit out is in Taurus (sun is now north of the equator) corresponding to April the month that Jesus rises from the dead on Easter. 40 days later Jesus said: "the son of man" ascends into heaven to be seated at the right hand side of his father.

Another Mundane cycle is the conjunction of Jupiter and Saturn. Every time these planets have conjoined during a U.S. presidential election year, the president has died in office. The only exception to this was Ronald Regan, who was shot but did not die. This was the first time that the conjunction did not take place in an earth sign so this may have some significance. The conjunction of Jupiter-Saturn in June of 2000, takes place in Taurus, an earth sign so the forecast would be that the elected president dies in office.

Luther Jensen's work shows that major wars involving the United States have occurred when Uranus is in the signs of Gemini, Virgo, Sagittarius, and Pisces. Uranus enters Pisces, August of 2003 so we would expect the possibility of a major military event around this time period.

One other Market related mundane cycle is the superior and inferior conjunctions of Mercury with the Sun. In Jeanne Long's book Universal Clock, she says that she found an ancient clay tablet in the British Museum, while researching Gann. She said that the tablet correlated future wheat prices based upon the heliacal rising of Mercury. In other words, it predicted the future price of wheat in terms of "what was in the past will be in the future". She shows several Soybean examples where the price range of Soybeans at the superior conjunction returns at the inferior conjunction. In other words, if the price range was 420 to 425 during the superior conjunction, then expect soybeans to

return to this basic price area at the inferior conjunction. She says that approximately 70% of the time this will be accurate. The times where the price does not return to the old range are usually related by increments of 12 cents. For example, if or second date did not return to the original price area of 420 to 425 it will probably be 12, 24, 36, 48, etc. cents higher or lower. This technique is supposed to work for both Wheat and Soybeans but I would recommend that you do your own research in this area. Gann says "Prove all things to yourself and hold on to what is good".

Gann's Personal Soybean Letter

I have provided a copy of this letter in the back of the material. Please read it over before reading this section. First, I will show you where Gann is getting some of the prices he is using in this letter that are not listed as Major highs or lows. In the first paragraph, Gann says "The one-half point of the highest selling price and one-half between the high and low are very great because these equal the 45-degree angle or the gravity center. For May Beans these prices are $218 \frac{3}{8}$, $240 \frac{3}{8}$, $251 \frac{7}{8}$, and $319 \frac{1}{8}$ ". On this page you will find four prices that are hard to explain. These are $\$218 \frac{3}{8}$, $\$240 \frac{3}{8}$, $\$251 \frac{7}{8}$ and $\$319 \frac{1}{8}$. The first price is the midpoint of the $\$436 \frac{3}{4}$ high ($436 \div 2$). The $240 \frac{3}{8}$ price is the extreme low of 44 + the high of $436 \frac{3}{4}$ divided by 2 ($436 \frac{3}{4} + 44 = 480 \frac{3}{4} \div 2$). The $251 \frac{7}{8}$ price is the low price at 67 + the high price $436 \frac{3}{4}$ divided by 2. Finally, the $319 \frac{1}{8}$ price is the low at $201 \frac{1}{2}$ + $436 \frac{3}{4}$ high divided by 2. Gann then starts adding proportionate parts of the circle to these prices. He is also directly referencing the corresponding zodiac longitude to the price levels. What he is doing is creating an Astrological aspectarian of price. He is trying to calculate where the Trines and squares are to different price levels. Gann said that these were the "mathematical reasons" why

Soybeans made a bottom at \$201½. The extreme low at $44 + 360 = 404$. Both of these prices would equal 14 Taurus. The midpoint or one-half point is 202 ($404/2$) = 22 Libra. The low of 201½ occurred February 9th, 1949. This price is equal to 15 Leo on the "Circle Chart". On this date, you will find Pluto at 15 Leo (conjunct the price), Neptune at 15 Libra (60° to price), Jupiter is 150° to price, etc.

The next part of the letter talks about active or live angles. "By live or active angles is meant Prices and Time Periods where the longitude of the major planets are or where the squares, triangles, oppositions are to these planets". "Examples of live, active angles: At the present writing, Jan. 18th, 1954, Saturn Geocentric is 8° to 9°-degrees Scorpio. Add the square or 90-degrees gives 8° to 9°-degrees Aquarius and equals the price \$309 for May Beans". He then does the same aspect procedure for Jupiter at 21 Gemini - 135°-degrees = 306 or 6° degrees Aquarius. In this paragraph, Gann is using pure longitude to price conversions. In the next paragraph Titled 24 Revolutions of Time and Price Gann is using the Circle Chart. An interesting thing about Gann's first example of an "active angle" is related to the first trade horoscope (natal) for Soybeans. Gann said "October 5th, 1936, futures trading started in Soybeans. The first sale was 120". If we cast a horoscope for this date, we can actually see why Saturn's position is "active". From the above example, you can see that Saturn was "active" because it was transiting natal Venus and was in opposition to natal Uranus. Jupiter was "active" by a conjunction with the natal moon at 17° Gemini. Gann says, "The power of Saturn and Jupiter aspects, working out time to these Price Resistance Levels, is what halts the advance in Soybeans".

Gann then moves on to his circle chart. He instructs his student to circle all the low prices with a red circle and all the highs with a green circle. Then note the Angles of 45, 60, 90, 120, 135, 180, 225, 240, 270, 300, 315, and 360 from each high and low. These are not geometric angles! These are astrological angles on the circle chart or "Master Chart". Gann then gives an example for December 2, 1953, May Soybeans high \$31 1/4. If you place the planets on the circle chart, you will get the prices Gann is talking about. When he mentions that this price is 180°-degrees of the averages he is talking about the MOF Formula, CE Average or the Average of Six that I discussed earlier. He says "Having before you all the information outlined above, you would certainly have gone short of May Soybeans on December 2nd, 1953 and covered your shorts on December 17th at 296 because the price was down to the 45-degree angle from 44 on the monthly high low chart". The Monthly high low chart indicates Gann was referring to a Saturn geometric angle from the low price of 44. When you read Gann, you can almost always substitute the word month for Saturn. The 44 low occurred December 1932. From this time period to December 1953, h₂-heliocentric moved 267.067 degrees. The 11 1/4 angle @ 0.9375/degree would = \$250.375 + 44 low = \$294.375 which is within 1 5/8th cents of Gann's 296 price. Using Gann's shortcut method, December 1932 to 1953 = 21 years times 12 (months) = \$252 + 44 = \$296. The low of the move was \$295. The shortcut method was within a 1 full point or \$50.00 per contract the astrological method was only off by 5/8ths of a cent or \$31.25 per contract.

Another interesting point about the 296 price that Gann said to cover the short position is that on this particular day, Pluto's first trade longitude was activated by a transiting square of Mars, i.e. Mars square natal Pluto. Pluto's natal position is 28 Cancer which =

\$296 on the circle chart. The \$296 price is also square to Neptune and conjunct the lunar node geocentrically on Dec. 17, 1953.

Getting back to some of the earlier parts of the Soybean Letter, the $\$218 \frac{3}{8}$ price would equal 8° Scorpio, which is opposite natal Uranus and conjunct natal Venus. Gann also tells us that the low price on the day of the first trade chart was \$1.20. Because the minimum fluctuation of beans at this time was one $\frac{1}{8}$ of a cent, then 0.45 cents would equal 360° , because $\frac{1}{8}$ of 360 is 45° or cents. Then, $\$1.20 / 0.125$ is 9.60 and $9.60/45$ (a complete price circle) is 0.21333333 and 0.21333333 multiplied by a zodiac circle of 360° is 76.80 or $76^\circ 48'$ of a circle or $16^\circ 48'$ Gemini, which is equal to the longitude of the natal moon for beans. 17° is a hot longitude degree for soybeans because you have the natal moon at 17° Gemini, Saturn at 17° Pisces and Neptune at 17° Virgo. Moving Jupiter, 77 degrees, which is equal to 17° Gemini or the natal moon produces interesting turning points in beans. The highest price up to the date that this letter was written was $\$436 \frac{3}{4}$, in January of 1948. On this date, you had Geocentric Jupiter return to his natal position, Saturn conjunct the Natal Midheaven, and Neptune Conjunct the natal Sun. Also, Mars had made 5 complete round trips and returned to his natal position on this date. The price of $\$436 \frac{3}{4} - 90$ is 346.75 or 17 Pisces, which is the longitude of natal Saturn. $\$436 \frac{3}{4}$ is also within 2 cents from exact opposition of Jupiter' natal return longitude of $18^\circ 55'$ Sagittarius. In the letter, Gann also illustrates that $\$436 \frac{3}{4} - 360$ is $\$76 \frac{3}{4}$, which is equal to $16^\circ 45'$ Gemini within $\frac{1}{4}$ cent of 17° Gemini. The low price of \$67 occurred on July 24, 1939. From the first trade date of October 5, 1936, heliocentric Jupiter moved $88^\circ 48'$ and heliocentric Saturn moved $34^\circ 44'$. The $22\frac{1}{2}^\circ$ angle of Jupiter

would be $12 \times 0.125 = 1.5$ and $1.5/45^\circ = 0.0333333$ and this multiplied by 22.5 is 0.75 points per degree. Therefore, 88.81×0.75 equals \$66.61 for the Jupiter angle. The Saturn $22\frac{1}{2}^\circ$ angle would cross at \$65.14 at 1.875 points per degree. Both of these calculations are the angles up from "0" from the first trade date of the futures contract. The sun also moved 1008.63 degrees, which is 33.3 months and $2 \times 33.3 = 66.6$, same as Jupiter. 1008 reject two circles or 720 is 288, which is the second square of twelve, i.e. $12 \times 12 = 144$ and $144 \times 2 = 288$. The June 15, 1930 cash high price at 216, which Gann says "Equals 6 Scorpio"; we have on this day, heliocentric Saturn at $8^\circ 06'$ Capricorn or 278.1 and heliocentric Neptune at $2^\circ 53'$ Virgo or 152.88. Therefore, the midpoint is $278.1 + 152.88 = 430.98$ and this divided by 2 is 215½ very close to 216 price. Also, heliocentric Venus is at 15° Virgo, sextile the price, Mercury is at 16° Aquarius, square or 90° to the price, Neptune is 45° to the price in Virgo, etc. Mercury and Venus are not typically important in this type of calculation, but when you have them supporting the numbers generated by the large outer planets namely Jupiter, Saturn, Uranus and Neptune, it is a very nice bonus.

Market Volume

In the back of this work, you will find a two-page course that I believe was written by Gann Titled: MARKET TRENDS. This course deals with market trends and the correlation between transactional volume. Gann says, "The trend of stock prices is influenced by supply and demand. The object of following a chart is to determine whether stocks are in a period of accumulation, or distribution; whether stocks are being marked up or down, or whether they are being left alone". It is suggested that you read this small course at this time and also Gann's course on Form Reading, which is also included with this document.

Now that you have had time to review the above referenced material, I will discuss my opinion of volume and it's significance in forming an opinion on the markets. Because there is always a seller for every buyer, and a buyer for every seller on any given day with the exception of Limit moves and Panics, volume simply records the total number of units that resulted in a transaction between the buyer and seller. The volume data in this format has limited uses but there are several books that have been written by various authors discussing how to interpret market movements from this raw form of the data. My personal view is that volume needs to be converted so that it more accurately represents buying pressure and selling pressure or supply vs. demand. Because we already know that you must have a buyer for every seller and vice versa before any volume can be recorded, we need to determine how this number of units relates to price fluctuations. If we take the fundamental view that the only reason prices advance is simply because there was a greater demand to purchase the stock or commodity than

there was a willingness to sell the item at the current existing price. Prices decline because there is a greater willingness to sell the stock or commodity verses the demand for the item at the current price. Selling panics occur when there are no buyers for the item at the current price or even substantially lower prices (zero demand driven). Buying panics occur when there are no sellers at the current price and is obviously a supply driven scenario.

Many analysts have argued that the supply of any stock or commodity is definable simply because you can not instantly create more coffee, corn, oil or shares of a particular stock. These analysts believe that the available supply of the item is what creates the demand that will eventually show up as volume or a transaction between buyer and seller. However, this point of view is not entirely accurate because short selling creates fictional or phantom units of the particular item and therefore distorts or enlarges the available supply of the item. This means that to make volume understandable, we must look at it, as it relates to the desire to buy or sell any item at a given moment and we must also quantify this desire.

What I like to do to determine this desire is to look at the percent change up or down on a given day and multiply this fraction by the volume on that day. For example: lets say that Cisco Systems, Inc. (CSCO) is up \$1.00 today on 55 million shares from 77% the day before. The % change is 1 divided by 77% or 0.0128. This fraction multiplied by the volume of 55 million shares is 707,395 and represents the lack of selling desire or excess demand. If the price had declined \$1.00 on this volume it would represent the lack of

buying desire or a greater supply. So based on the advancing scenario, we would assume that of the 55 million shares traded this day, that desire to buy exceeded the willingness to sell by 707,395 shares. This number will now let us separate the Supply & Demand forces as it relates to whole volume. $55,000,000$ divided by 2 is $27,500,000$ and this represents perfect balance of desire to sell and willingness to buy. $27,500,000 - 707,395 = 26,792,605$ (supply) and $27,500,000 + 707,395 = 28,207,395$ (demand). You can plot these numbers on graph paper or have a spread sheet program like Microsoft Excel maintain the chart and data. This will give you a much better visual picture of the psychology in the market as it pertains to Gann's rules on volume. Another valuable calculation is to take the supply volume numbers and divide them by the total number of shares that the company actually has issued (for a commodity, use the Government estimates of supply). This ratio will show the percentage of turnover in the stock or how actively the stock is changing hands. When a stock is in "strong" hands, this number is low because the big block share holders have little if any willingness to sell. When distribution is taking place (Form reading + Cycle Timing), this number will show a pretty dramatic increase compared to it's relative history during the up trend or "Mark-up" phase. The market is churning as the inventory is moving from strong to weak hands. This information should help you with volume analysis and as a useful tool for interpreting the action of the markets.

Conclusion

This concludes my personal analysis of Gann techniques. This work was not intended to teach you how to trade or to provide trading advice. It was intended to give you a deeper insight into the more veiled and esoteric aspects of Gann's work. This was obviously intended for more serious students of Gann's methods. I do not claim to be a "Gann expert" to do so would be foolish. Most of the so-called "Gann experts" have absolutely no idea how Gann accomplished what he did. I believe that I have some valuable pieces of the puzzle but I have much too much respect for the man to call myself anything more than a student. I realize that this material has been crammed full of information and that it seems overwhelming. However, if you put in the study time, I think that it will come to you and you will be rewarded for your efforts. After all, I did not have the luxury of this material when I started! If you were looking for some practical trading advice, I would suggest that you read each of the following Gann courses at least 3 times each:

- (1) Speculation A Profitable Profession – A course of instructions on Grains.
- (2) Mechanical Method & Trend Indicator For Trading in Grains
- (3) Form Reading
- (4) Forecasting

By studying this material along with the work I have provided, I believe that you will be rewarded! I should also point out that Gann's rules on Time or Space overbalance, volume of sales and form reading are meant to be tools to help his students find the underlying cycles in the market.

Wishing you a Success,

Daniel T. Ferrera

Market Example of Gann Techniques

The March 24th S&P500 High was 1553 on 3/24/00

Jupiter moved 1532 from January 1950 low 16.65

Saturn moved 604.9 from January 1950 low 16.65

603 months, 2621 weeks, 18345 days from January 1950 low 16.65

I have calculated the angles based on the minimum fluctuation of 0.05 (based on the futures contract). Both Saturn's and Jupiter's 75-deg angle would cross near the high of 1553. The reason I would do it this way is that I want to know which Gann angle is being hit per the instructions given in the course material.

Saturn's 75-deg angle would move up at $(0.05 \times 50) = 2.5 \text{ points/deg} \times 604.9 = 1512.25 + 16.65 = 1528.90$

Jupiter's 75-deg angle would move up at $(0.05 \times 20) = 1.0 \text{ points/deg} \times 1532 = 1532 + 16.65 = 1548.65$

Uranus's 75-deg angle would move up at $(0.05 \times 140) = 7.0 \text{ pts/deg} \times 224.28 = 1569.96 + 16.65 = 1586.61$

As I mention in my Gann Unveiled course, these are my 3 favorite planets to work with. The average of the 3 is 4664.16 divided by 3 = 1554.72. The actual top was 1552.87

The other things I looked at, was Jupiter was exactly 540-deg from the 8/12/82 Low, and the conjunction of Mars & Saturn took place on the "hot" degree of 18-40 Taurus.

The 56.25 degree angle up from "0" from the July 7th, 1932 low has the following calculations:

Jupiter $(0.05 \times 15) = 0.75/\text{deg}$. Jupiter moved 2054.38 so this gives 1540.78

Saturn $(0.05 \times 37.50) = 1.875/\text{deg}$. Saturn moved 827.78 so this gives 1552.08

Uranus $(0.05 \times 105) = 5.25/\text{deg}$. Uranus moved 297 so this gives 1559.25

The average of the 3 is 1550.70. So we have two important Time/price angles within 3 points of each other at this major top. If you take the average of 1554.72 + 1150.7, you get 1552.71

Gann's "UNEQUAL 8ths" are the Diatonic Musical Scale.

	<u>Reciprocal</u>	<u>1 - Reciprocal</u>	<u>Note</u>
1/8 = 1.1223	0.8910	0.109	D
2/8 = 1.2596	0.7939	0.2061	E
3/8 = 1.3344	0.7494	0.2506	F
4/8 = 1.4976	0.6677	0.3323	G
5/8 = 1.6808	0.5949	0.4051	A
6/8 = 1.8865	0.5301	0.4699	B
7/8 = 2.0	0.5	0.50	C (octave)
8/8 = 2.2446	0.4455	0.5544	D

The October 18th 1999 Low of 1232 x 1.2596 (2/8) = 1551.83 (3/24/00 top) Also, the 10/8/98 low of 923.30 x 1.6808 (5/8th) = 1551.88.

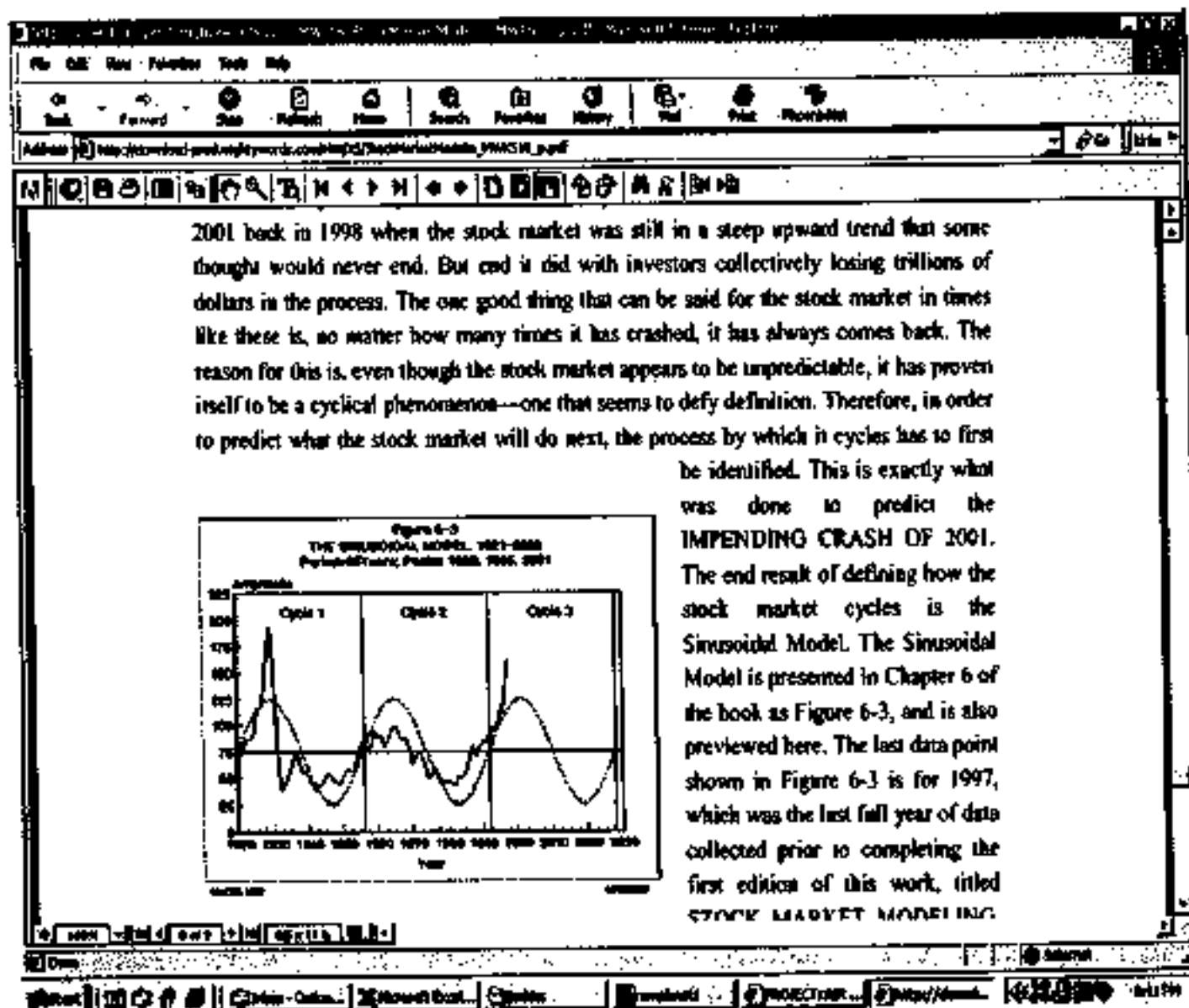
Square of 9. Last major low was 923 on 10/8/98. Root of 923 = 30.38 + 9 (4 & 1/2 cycles) = 39.38 squared = 1550.86
1550.86 is in opposition to 923.

The Low on 10/18/99 was 1233.66. Root = 35.123 + 4.25 = 39.375 (4 & 1/8 cycles) squared = 1550.39.
1550.39 is on a 45-deg angle to 1233.66.

The low just prior to the top on 2/28/00 was 1328.88 Root = 36.453 + 3 (1 & 1/2 cycles) 39.453 squared = 1556.60 (1556.60 is in opposition to 1328.88).

All (3) calculations added together = 4657.85. Divide by 3 and the average is 1552.62
actual was 1552.87

18 Year Sinusoidal Cycle



In his book "Impending Crash of 2001", author Richard Mazze shows the dominance of the 18-year Sinusoidal wave cycle in the U.S. Stock Market. This cycle is related to the 180-degree synodic cycle of Saturn and Neptune. For those interested in long range forecasting, the angle between Saturn & Neptune was 118.31 or 118-deg 18-min (before opposition) at the September 3rd, 1929 Top (see graph above). This angle moved 180-deg (18-yrs) to an angular separation of 61.69 or 61-deg 41-min (before conjunction) on March 30th, 1947. Eighteen years later, the angle between Saturn & Neptune return to the

beginning value of 118.31 on January 29th, 1966. Adding another 180-deg or 18 years takes us to October 2nd, 1982 when the planets are separated by the angle of 61.69 again. On June 24th, 2001, the angle will once again be 118.31 completing an 18-year up-cycle that began in 1982. Each 18-year cycle is $1/8^{\text{th}}$ of Gann's 144-year period or the square of 12. The 120-deg harmonics of this cycle are also strong. For example, adding 120-deg to the 9/3/29 top gives an angle of 121.69 (after opposition) or 121-deg 41-min. This occurred on December 22nd, 1941 at a major stock market low 2 weeks after the Japanese bombed Pearl Harbor and effectively ending the Great Depression. When you combine this cycle with the 90-deg Uranus $1/4$ cycle, which is also sinusoidal, you can immediately see why the Bull Market was so strong from 1982 to present. The Uranus $1/4$ cycle basically runs from 0-deg 41-minutes of Gemini, Virgo, Sagittarius, and Pisces (II, VI, X and XII), with tops occurring in VI and XII and bottoms in II and X. Both of these dominant sinusoidal cycles bottomed in 1982, giving rise to the greatest bull market advance in U.S. history. Another sinusoidal cycle that has a strong market effect is 90-deg of the Lunar North Node. For example, at the July 8th, 1932 Low, the Node was at 20-deg XII, subtract 90 because the node is always retrograde, gives 20-deg X, which occurred on March 8th, 1937 two days before the exact high. Subtract another 90-deg gives 20-deg VI, which happened on November 2nd, 1941, etc. Last but not least, there is a sinusoidal cycle of 40 to 41 months. According to the Foundation For The Study of Cycles, this cycle was discovered when a New York firm hired a mathematician to discover the secret of the Rothschilds investment success. This cycle shows up as a sinusoidal wave, i.e. up for 20 months, then down for 20 months. This cycle was extremely dominant in stock market data from 1871 to 1946. The Astronomical correlation is 60-deg synodic movements of Jupiter & Saturn Top to Top or Bottom to Bottom or 30-deg up then 30-deg down.

Dan

APPENDIX A

COFFEE LETTER REFERENCES

MAY COFFEE SANTOS D

March 19, 1954

High 8729

Using a scale of one point to one degree, 8729 equals 29° Gemini.
Using a scale of 30 points to one degree, equals 21° Capricorn.
Using the Jupiter scale of 12 points to one degree, equals $7^{\circ} 30'$ Aries.
Using one cent to one degree, equals $27^{\circ} 16'$ Gemini.

The dollar value is \$28,171.00, which equals $11^{\circ} 45'$ Capricorn.
The average price of 5 options on March 19, 1954 was 8663, which equals 28° Aries, or 60° from the Heliocentric Jupiter. Heliocentric Jupiter is $20^{\circ} 35'$ Gemini, which means that the price of 8729 was at this degree. Heliocentric Uranus is $21^{\circ} 52'$ Cancer and the price at 21° Capricorn is opposite to this.

April 16, 1954 is 276 months from April 16, 1931, low 435. Using 50 points per month, the 45° angle crossed 8715 on March 19, 1954, and the Sun has moved 8253 degrees from April 16, 1931. Add this to 435 gives 8688 as the resistance angle.

MARCH COFFEE - October 1, 1936 low 300. Time to April 1, 1954 210 months at 30 points per month, the 45° angle crosses at 5600, and at 40 points per month, it crosses at 8700.

1931, April 16 to March 19, 1954 - Geocentric Saturn moved $285^{\circ} 38'$. This would give a price of 8572. 1936, October 1 to March 19, 1954 - Geocentric Saturn moved 23° , which would equal a price of 7230.

1940, May 15 to March 19, 1954 - Saturn moved $181^{\circ} 35'$, which gives a price of 6990 and using 45 points to one degree would give 8715.

1940, August 19 - Saturn moved $173^{\circ} 23'$. At 45 points to one degree, this equals 8760 price.

HELIOCENTRIC SATURN - 1931, April 16 to March 19, 1954. Saturn moved $287^{\circ} 15'$ which equals $17^{\circ} 13'$ Capricorn, price 8632.

1936, October 1, Saturn moved 225° , which gives a price of 7150.

1940, May 15, Saturn gained $179^{\circ} 44'$, price 9940.

1940, August 19, Saturn gained $176^{\circ} 14'$, price 9842.

HELIOCENTRIC PLANETS, March 19, 1954

Jupiter $89^{\circ} 35'$ equals $29^{\circ} 35'$ Gemini.

Saturn 214.44 equals $4^{\circ} 44'$ Scorpio

Uranus 111.52 equals $21^{\circ} 52'$ Cancer

The average of these
6 planets is 164.17
or $14^{\circ} 17'$ Virgo.

Neptune 204.35 equals 24° 35' Libra
 Pluto 144 equals 24° Leo
 Mars 221 equals 11° Scorpio

The average of the 6
 Geocentric planets is
 173.26 or 25° 26' Virgo

One-half of Jupiter to Saturn Helioc is 152.09 or 2° 9' Virgo.
 The average of Saturn, Jupiter, Uranus and Neptune is 155.10, equals 5° 10' Virgo.
 $\frac{1}{4}$ of this average is 17° 35' Gemini. Jupiter, Uranus $\frac{1}{4}$ is 100.43, equals 10° 43' Cancer Heliocentric. $\frac{1}{4}$ of Geocentric Jupiter to Uranus is 93.48 or 3° 48' Cancer.

IMPORTANT DATES EACH MONTH - 1st, 15th, 18th and 19th. The present market is running close to these dates.

1953, June 19 - May Coffee low 5050
 August 17 - high 5765
 Sept. 21 - high 5710
 Oct. 19 - High 5660

July 17, low 5555
 Sept. 15 - low 5565
 Oct. 9 - Last low 5470
 Dec. 9 - High 6240

1954, Jan. 13 - High 7470
 March 15 - High 5625
 March 18 - Low 5465

Jan. 19 - Low 6560 - This was 7 months from June 19.
 March 19 - High 8729 - This extreme high was 9 months from June 19, low, 2 months from Jan. 19 low, 6 months from Sept. 15, 1953 low and 5 months from Oct. 19 high at 5860.

HELIOCENTRIC & GEOCENTRIC ASPECTS

1954, March 24 - Heliocentric Jupiter enters Cancer.
 June 24 - " " " 120° of Saturn.
 April 12 - Sun 60° of Jupiter Geocentric.
 April 15 - Sun 180 of Neptune "
 April 13 - Jupiter 135° of Saturn "
 April 16 - Jupiter 60 of Pluto "
 April 26 - Jupiter 120 of Neptune "
 April 26 - Sun 180 of Saturn "

The month of April is very important. There should be great activity and wide swings in prices due to these aspects.

GEOCENTRIC MAPS MOVEMENTS from low prices on Coffee--

1931, April 16 to August 7, 1953 - Mars has made 12 round trips.
 1954, October 29 - Mars will be opposite or 180° from its place on April 16, 1931.
 1936, October 1 to September 19, 1953 - Mars made 9 round trips of 360° each.
 1954, Dec. 9 - Mars will be 9½ round trips or opposite its place Oct. 1, 1936.
 1940, May 15 to June 12, 1953 - Mars made 7 round trips or complete cycles.
 1954, Apr. 9 - Mars is 7½ cycles or opposite its place on May 15, 1940. Due to the retrograde position of Mars, it will again be 7½ on July 7 and on Aug. 17, 1954 or the third time in opposition to its own place, which is very important.

1940, August 19 to Sept. 15, 1953 - Mars had completed 7 round trips. Note low on Coffee on that date.

1954, Dec. 4 - Mars 7½ round trips or opposite its own place on Aug. 19, 1940.

If Coffee starts to decline between March 22 and 24, 1954, it should continue down to around April 15, when the adverse aspects of Jupiter to Saturn and the Sun to Neptune are completed. From these dates, you should watch for the possibility of a rally up to April 16, 1954, when Jupiter is 120 of Neptune and the

cline. By studying all of the data outlined above and applying it to Coffee, you will be able to learn more about what causes the changes in trend.

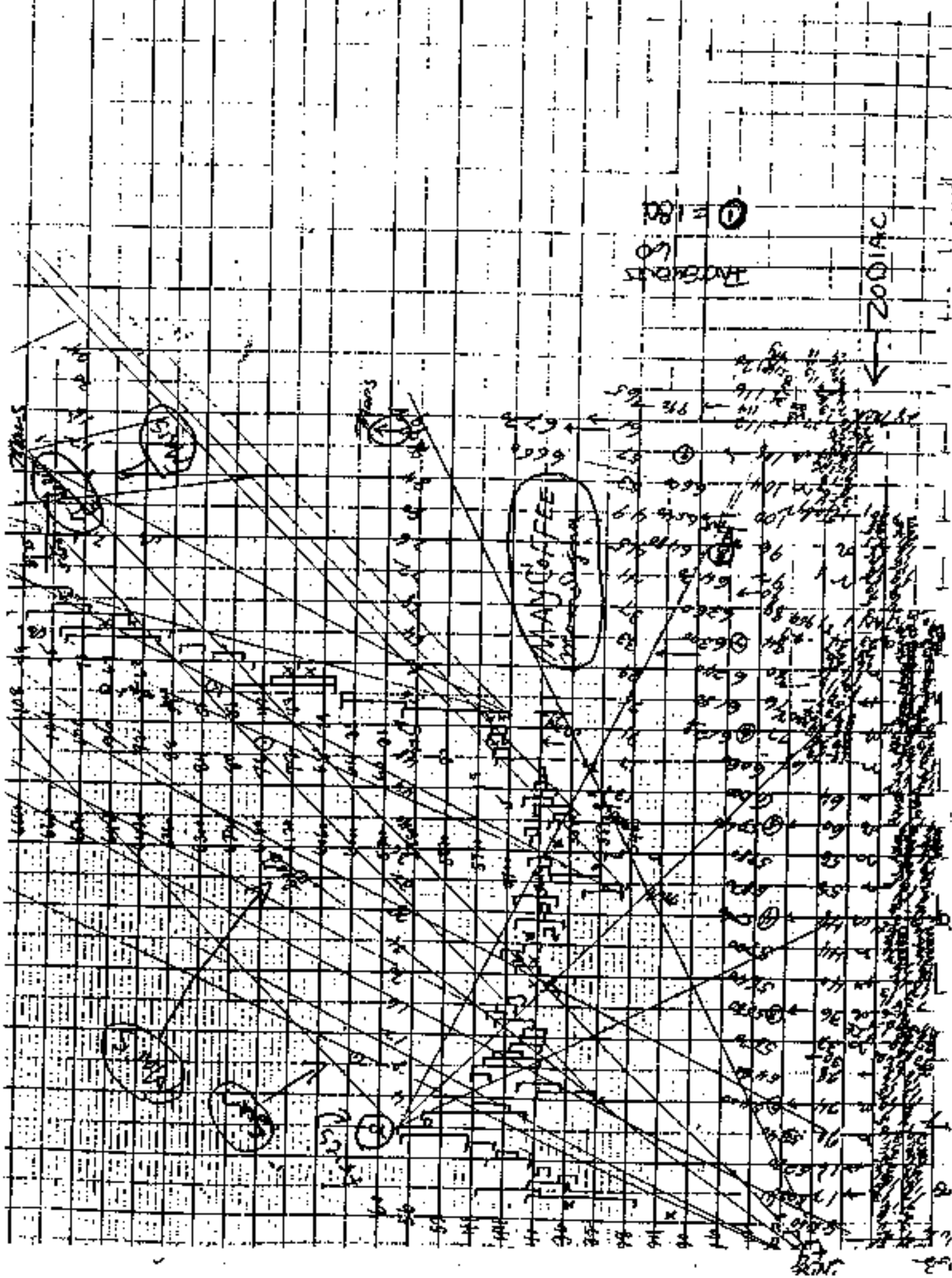
March 20, 1954

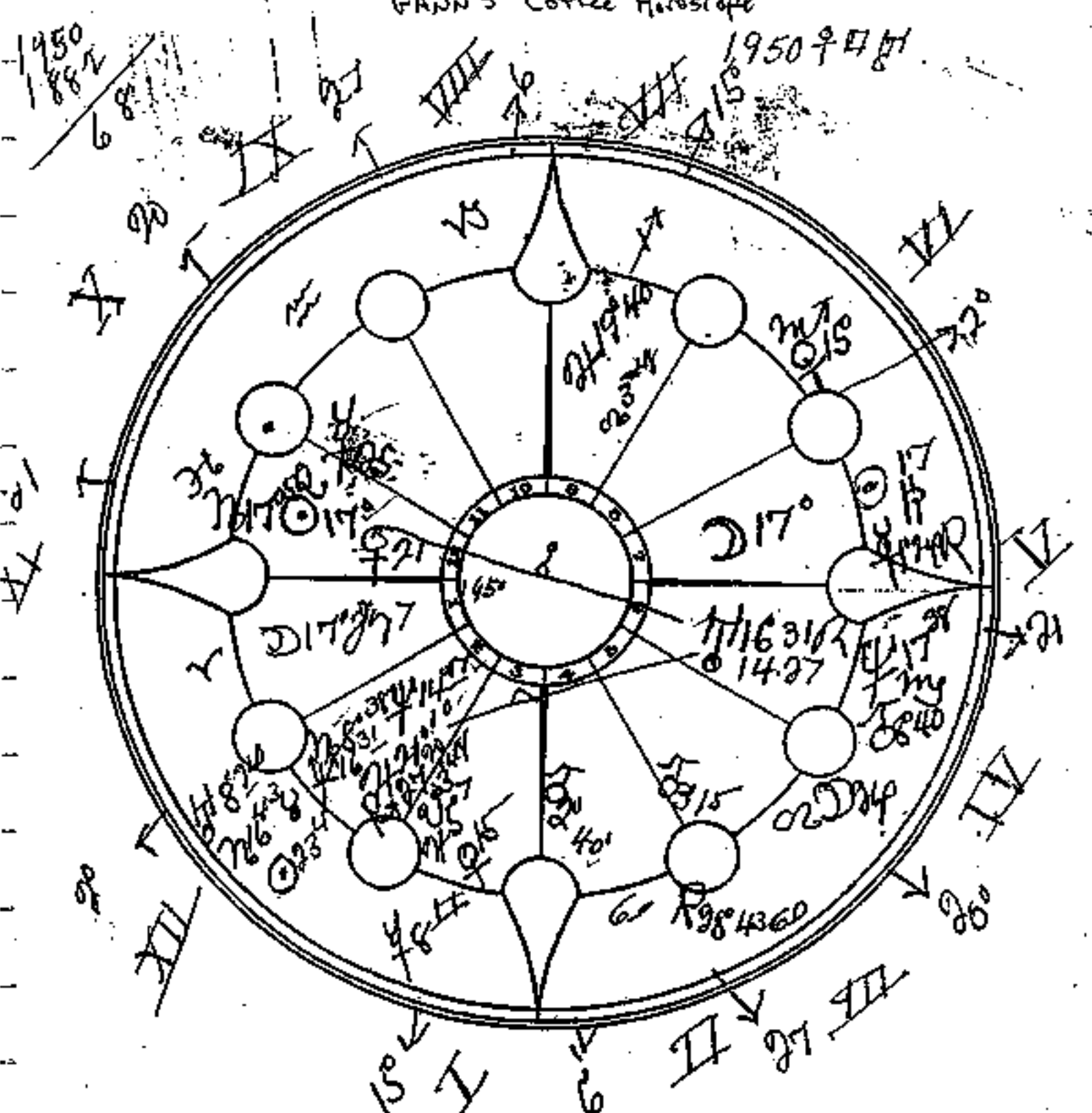
W. D. GANN

DATE 180
60
INDEPENDENT

ZOOIAC
↓

WANDOFFEE





1882 Moh 7 Coffee Future Trading Started
1950- Starts 69 year

Sept 28 1936 172413
404 172413 10 30 AM
172413

55.23.01
1.30
74.31

Oct 10 - 1936
Law 34

SH 1932 June 9 J Helio 31° 30' = 10 30 8

28
12
40

1934 May ①

1936 Apr ②

1938 July ③

1940 Dec ④

1941 Nov ⑤

1943 Oct ⑥

1945 Aug ⑦

1947 July ⑧

1949 June ⑨

1951 Apr ⑩

1952 May ⑪

1938 Apr 10	182 = 2° - 0 = 4500 =
1940 Nov	
1942 Dec	2 - 360 = 12.43
1944 May	2 - 720 =
1946 July	3 - 1080
1948 May	4 - 1440 = 1440
1950 Apr	5 - 1800 = 727
1952 March 8	6 - 2160 = 2167
1954 May 3	7 - 2520
1956 May 3	8 - 2880
1958 May 3	9 - 3240
1960 May 3	10 - 3600
1962 May 3	11 - 3960
1964 May 3	12 - 4320
1966 May 3	13 - 4680
1968 May 3	14 - 5040
1970 May 3	15 - 5400
1972 May 3	16 - 5760
1974 May 3	17 - 6120
1976 May 3	18 - 6480
1978 May 3	19 - 6840
1980 May 3	20 - 7200
1982 May 3	21 - 7560
1984 May 3	22 - 7920
1986 May 3	23 - 8280
1988 May 3	24 - 8640
1990 May 3	25 - 9000
1992 May 3	26 - 9360
1994 May 3	27 - 9720
1996 May 3	28 - 10080
1998 May 3	29 - 10440
2000 May 3	30 - 10800
2002 May 3	31 - 11160
2004 May 3	32 - 11520
2006 May 3	33 - 11880
2008 May 3	34 - 12240
2010 May 3	35 - 12600
2012 May 3	36 - 12960
2014 May 3	37 - 13320
2016 May 3	38 - 13680
2018 May 3	39 - 14040
2020 May 3	40 - 14400

May 3 Helio at 3790

1951 mch 8 $\delta 17^{\circ} 30' 92 = 330 \text{ at } 12^{\circ} \text{ north}$
 $\begin{array}{r} 17^{\circ} 30' 92 \\ - 130 \\ \hline 16^{\circ} 00' \end{array}$ $\begin{array}{r} 330 \\ + 16 \\ \hline 346 \end{array}$ $\begin{array}{r} 3240 \\ + 346 \\ \hline 3586 \end{array}$ $16^{\circ} 42'$

mch 8 - 1951 - δ at $\rightarrow$

1951 Nov 9 $\delta 142^{\circ} = 22^{\circ} 26' \text{ or } 3600 + 90 = 3690$

0° 2'

35 727 + 40 8

1000
100

coffee calculations -
 tobacco 8 mch

* {

100	3500	5000	100
90	3500	5000	100
80	3500	5000	100
70	3500	5000	100
60	3500	5000	100
50	3500	5000	100
40	3500	5000	100
30	3500	5000	100
20	3500	5000	100
10	3500	5000	100
0	3500	5000	100
100	3500	5000	100
90	3500	5000	100
80	3500	5000	100
70	3500	5000	100
60	3500	5000	100
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40	3500	5000	100
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100	3500	5000	100
90	3500	5000	100
80	3500	5000	100
70	3500	5000	100
60	3500	5000	100
50	3500	5000	100
40	3500	5000	100
30	3500	5000	100
20	3500	5000	100
10	3500	5000	100
0	3500	5000	100

Note how BANK is using the minimum fluctuation to set up a zodiac relationship with price. This is what he is doing in the coffee market.

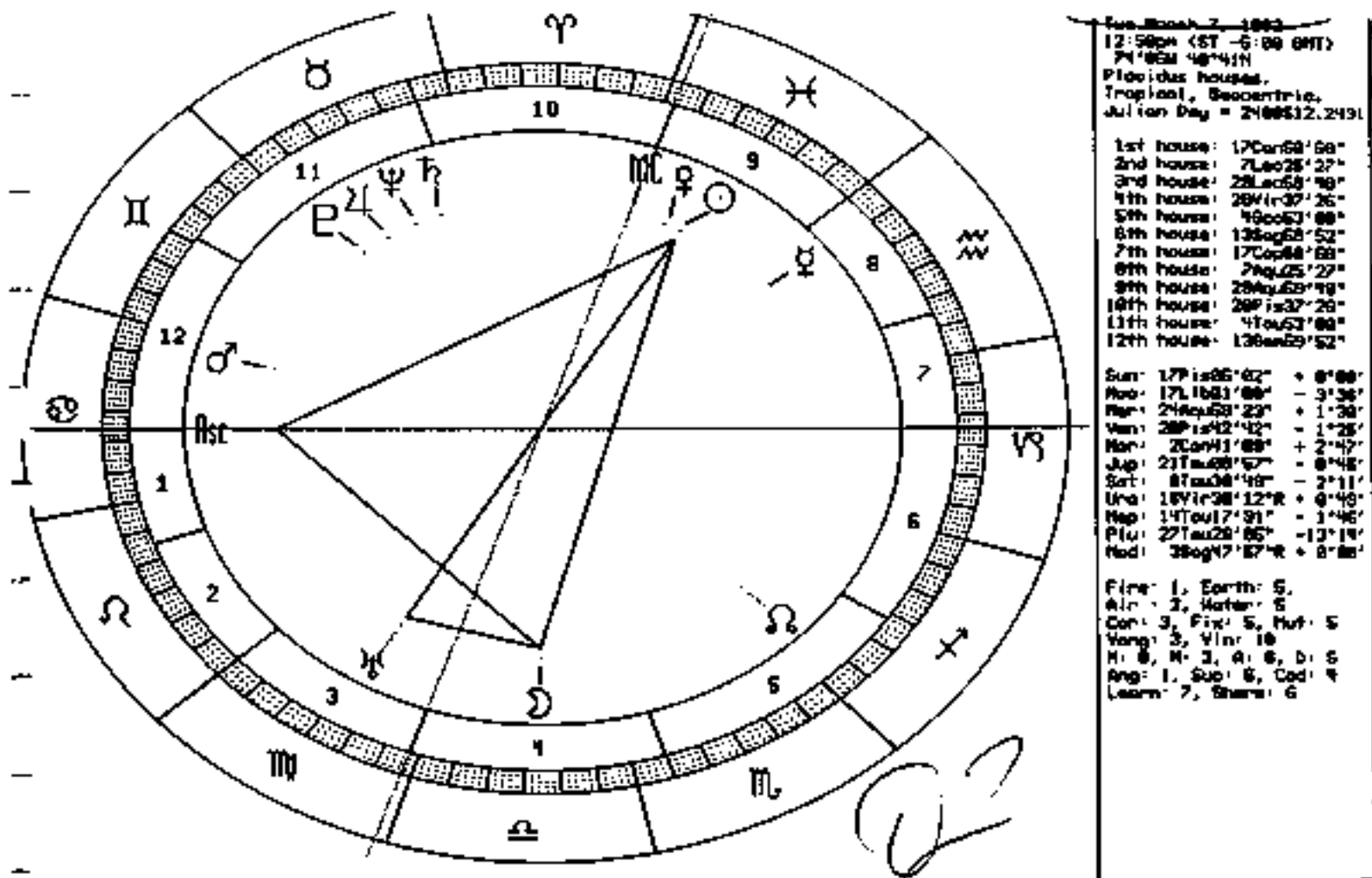
Double numbered Squares

The large units are price
The smaller units are time

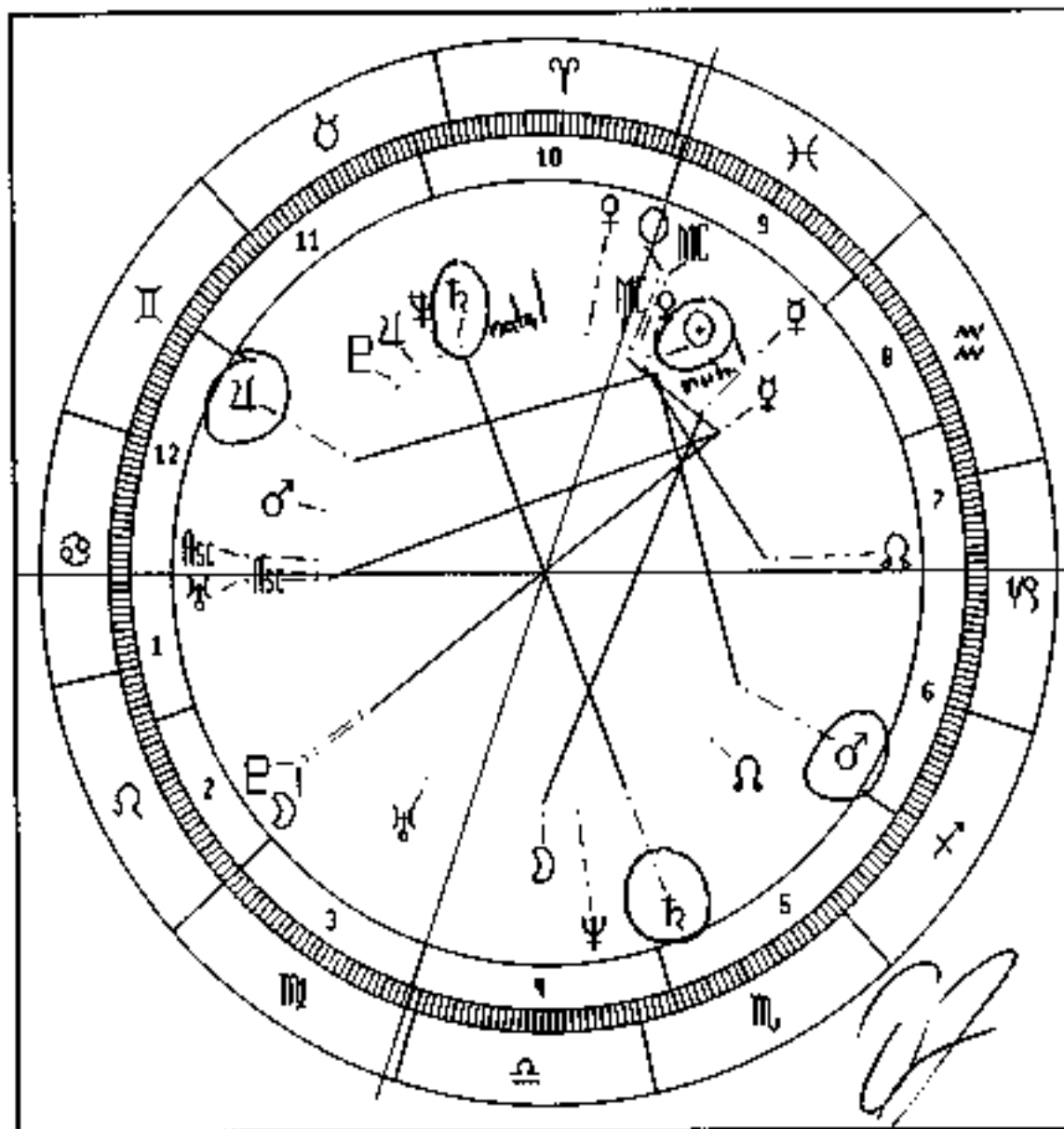
700 to 6400	\$1008.00	7.2160	Gain	\$8.708.00
1500	\$2.160.00	to 5855	Gain	\$6.283.60
Every 70 points to 1 month				
70 points = \$100.80				
5855	\$8.446.60			
6400	\$9.216.00	70	\$1008.00	

\$5460 →
78 time units →
(2) 70pts/month

5460	5390	5320	5250	5180	5110	5040	4970	4900	4830
78	77	76	75	74	73	72	71	70	69
5330	5260	5190	5120	5050	4980	4910	4840	4770	4700
79	78	77	76	75	74	73	72	71	70
5660	5590	5520	5450	5380	5310	5240	5170	5100	5030
80	79	78	77	76	75	74	73	72	71
5770	5700	5630	5560	5490	5420	5350	5280	5210	5140
81	80	79	78	77	76	75	74	73	72
5880	5810	5740	5670	5600	5530	5460	5390	5320	5250
82	81	80	79	78	77	76	75	74	73
5990	5920	5850	5780	5710	5640	5570	5500	5430	5360
83	82	81	80	79	78	77	76	75	74
6100	6030	5960	5890	5820	5750	5680	5610	5540	5470
84	83	82	81	80	79	78	77	76	75
6210	6140	6070	6000	5930	5860	5790	5720	5650	5580
85	84	83	82	81	80	79	78	77	76
6320	6250	6180	6110	6040	5970	5900	5830	5760	5690
86	85	84	83	82	81	80	79	78	77
6430	6360	6290	6220	6150	6080	6010	5940	5870	5800
87	86	85	84	83	82	81	80	79	78



GANN'S Natal Coffee Horoscope



Astrolog 5.48
Gann's Coffee Dr
Tue March 2, 1954
12:58pm (ST -5:1)
71°06'N 48°41'W
Placidus Houses
Tropical, Geocentric
Julian Day = 241

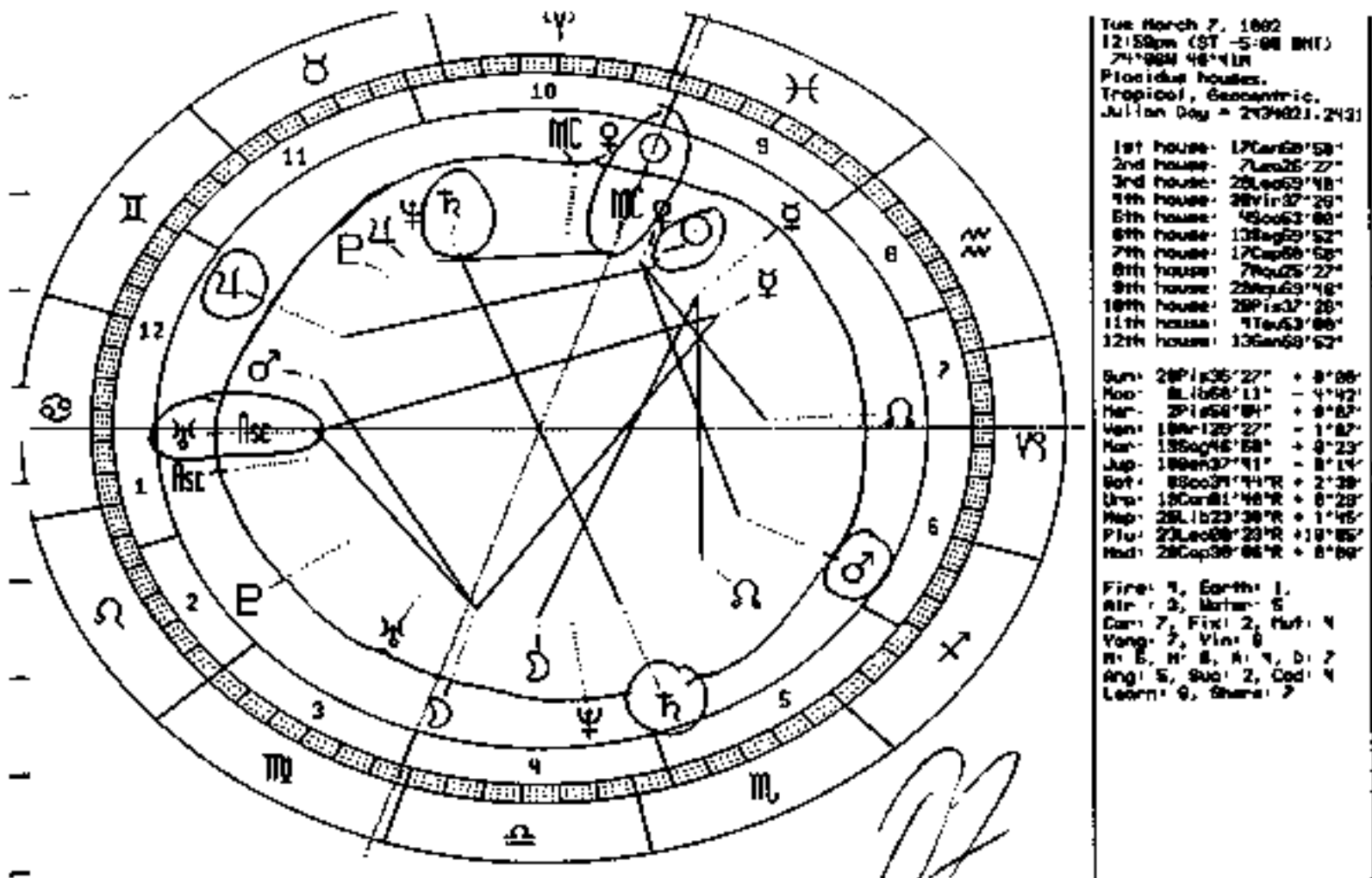
1st house 17Cen64
2nd house 7Leo2
3rd house 28Lib
4th house 20Vir
5th house 4Scor
6th house 13Sag5
7th house 17Cap6
8th house 7Aq
9th house 29Vir
10th house 28Pis
11th house 11Aur
12th house 13Gem5

Sun: 25Pis24'26"
Moon: 24Leo12'36"
Mer: 18Pis55'06"
Ven: 6Ari193'14"
Mars: 18Sag19'48"
Jup: 18Gem17'48"
Sat: 8Scor13'14"
Uran: 18Cen80'13"
Nept: 25Lib27'44"
Pluto: 23Leo43'52"
Node: 28Cap17'44"

Fire: 4, Earth: 1,
Air: 2, Water: 5
Con: 6, Fix: 3,
Yang: 6, Yin: 2
M: 5, M: 6, A: 5,
Ang: 4, Suc: 3, Co:
Learn: 6, Share:

March 19, 1954
VS
Natal

♂ and ♀ are square to the natal Sun ☉
♂ returns to eclipse longitude
♂ is opposite (180°) natal ♀



Inner circle is GANN's Natal Coffee Horoscope
 Outer circle is March 19th, 1954

♄ returns to eclipse longitude ☐ to natal ☉

♂ ☐ natal ☉

♄ is ♂ natal ♄

☉ is ♂ the Midheaven MC

♄ is on the Ascendant AC

APPENDIX B

GEOMETRICAL ANGLES

81 WALL STREET
NEW YORK N. Y.

THE UNIVERSITY OF CHICAGO

Mathematics is the only exact science. All power under heaven and on earth is given unto the man who masters the simple science of mathematics. Emerson said: "God does indeed geometrize." Another wise man said: "There is nothing in the universe but mathematical points." Pythagoras, one of the greatest mathematicians that ever lived, after experimenting with numbers and finding the proofs of all natural laws, said: "Before God was numbers." He believed that the vibration of numbers created God and the Diety. It has been said, "Figures don't lie." Men have been convinced that numbers tell the truth and that all problems can be solved by them. The chemist, engineer, astronomer would be lost without the science of mathematics.

It is so simple and easy to solve problems and get correct answers and results with figures that it seems strange so few people rely on them to forecast the future of business, stocks and commodity markets. The basic principles are easy to learn and understand. No matter whether you use geometry, trigonometry, or calculus, you use the simple rules of arithmetic. You do only two things: you increase or decrease.

There are two kinds of numbers, odd and even. We add numbers together, which is increasing. We multiply, which is a shorter way to increase. We subtract, which decreases, and we divide, which also decreases. With the use of higher mathematics, we find a quicker and easier way to divide, subtract, add and multiply, yet very simple when you understand it.

Everything in nature is male and female, white and black, harmony or disharmony, right and left. The market moves only two ways, up and down. There are three dimensions which we know how to prove--width, length and height. We use three figures in geometry--the circle, the square, and the triangle. We get the square and triangle points of a circle to determine points of time, price and space resistance. We use the circle of 360 degrees to measure Time and Price.

There are three kinds of angles—the vertical, the horizontal, and the diagonal, which we use for measuring time and price movements. We use the square of odd and even numbers to get not only the proof of market movements, but the use.

HOW TO MAKE CHARTS

Charts and records of past market movements. The future is but a repetition of the past. There is nothing new. As the Bible says—"The thing that hath been, it is that which shall be." History repeats and with charts and rules we determine when and how it is going to repeat. Therefore, the first and most important point to learn is how to make charts correctly because if you make an error in the chart, you will make an error in applying the rules to your trading.

YEARLY CHART: You should keep a yearly high and low chart, that is, recording the extreme low and the extreme high price made during the calendar year on one line. The spacing for the price can be used one point to each $1/8$ inch or two points or more, according to the activity and range of the stock.

MONTHLY CHART: You must always keep up a monthly high and low chart, which is the most important chart of all in determining the main trend. This chart records the extreme high and extreme low price for the calendar month on one line, and each space or $1/8$ inch on the cross-section chart paper should represent one point or \$1 per share.

WEEKLY CHART: The next and one of the very important charts to keep is a weekly high and low chart. Where stocks are selling below 50, it usually pays to make this chart up using each $1/8$ inch to represent one-half point, or two spaces to represent one full point, or four points for each one-inch space. When stocks become very active, especially when they are selling above \$100 per share, then you can make up the weekly chart using each space or $1/8$ inch on the chart paper to represent one point or \$1 per share.

SEMI-WEEKLY OR 3-DAY CHART: The next chart of importance to the Weekly Chart is a 3-day chart, that is, taking the extreme high and extreme low price made from the opening of the market on Monday morning until the close on Wednesday night, closing the chart on Wednesday night—then from the opening on Thursday to the close on Saturday, taking the extreme high and low and closing the chart on Saturday. This gives you a time period showing one-half of the week. This chart is very important as will be explained later on in the instructions. The spacing for this chart can be the same as for the weekly high and low chart.

WEEKLY MOVING-AVERAGE OR MEAN POINT: To get a Weekly Moving-Average, we take the extreme high for the week and divide by 2, getting the half-way or mean point for the week. This can be recorded on the weekly high and low chart or on a separate chart, recording the Weekly Moving-Average with a dot and using one line on the chart for each week. Importance of this Weekly Mean Point will be explained later.

DAILY CHART: When you are trading in a stock that is active, you should always keep up a daily high and low chart, but for study purposes it is enough to keep up the Weekly and Monthly Charts, which give you the main trend. The Daily Chart shows the minor trend and shows a change in trend much oftener than any of the other charts, but the indication does not last as long or run so far. This chart should be kept up the same as the others, except when stocks are selling below 50 or when they are in an inactive trading range—then the spacing should be $1/2$ -point to each $1/8$ -inch on the chart paper, allowing two spaces to represent one full point or \$1 per share. When stocks are active and advancing very fast, making a wide range each day, then you can make the Daily Chart the same as the Weekly or Monthly, that is, using one point for each $1/8$ -inch on the chart paper. This spacing cuts the chart down and keeps it in a range where it is easy to see and read when fluctuations are wide.

No spaces are skipped on the Daily Chart for holidays or Sundays, therefore the time period is for actual market days and not calendar days. However, you should carry the calendar days along at least every two weeks, as later, under rules for Time Periods for change in trend, you will find that it is necessary to check up and know when the stock is 30, 60, 80, 120, 135, etc. days from a top

or a bottom, which means calendar days, the exact measurement of Time for the daily chart. Often the Daily Chart on actual daily movements comes out on an exact mathematical angle of time measurement at the same time the calendar days come out on exact time measurement, making it a doubly important point for change in trend.

GEOMETRICAL ANGLES

.....

After long years of practical experience, I have discovered that Geometrical Angles measure accurately Space, Time, Volume and Price.

Mathematics is the only exact science, as I have said before. Every nation on the face of the earth agrees that 2 and 2 make 4, no matter what language it speaks. Yet all other sciences are not in accord as mathematical science. We find different men in different professions along scientific lines disagreeing on problems, but there can be no disagreement in mathematical calculation.

There are 360 degrees in a circle, no matter how large or how small the circle may be. Certain numbers of these degrees and angles are of vast importance and indicate when important tops and bottoms occur on stocks, as well as denote important Resistance Levels. When once you have thoroughly mastered the Geometrical Angles, you will be able to solve any problem and determine the trend of any stock.

After 35 years of research, tests and practical applications, I have perfected and proved the most important angles to be used in determining the trend of the stock market. Therefore, concentrate on these angles until you thoroughly understand them. Study and experiment with each rule I give you, and you will make a success.

We use geometrical angles to measure Space and Time periods because it is a shorter and quicker method than addition or multiplication, provided you follow the rules and draw the angles or lines accurately from tops and bottoms or extreme highs and lows. You may make a mistake in addition or multiplication, but the geometrical angles accurately drawn will correct this mistake. For example: If you should count across the bottom of your chart 120 spaces, which represents 120 days, weeks, or months, then you begin at "C" and number vertically on your chart up to 120—then from this top point at 120 draw a 45-degree angle moving down, this will come out at "C" on 120 points over from the beginning. If you have made a mistake in numbering, this will correct it.

Angles drawn on a chart always keep before you the position of the stock and its trend whereas if you had a resistance point on time written down, you might mislay it or forget it but these angles are always on the chart in front of you.

These angles or moving-average trend lines correctly drawn will keep you from making mistakes or misjudging the trend. If you wait and follow your rules, these angles will show you when the trend changes.

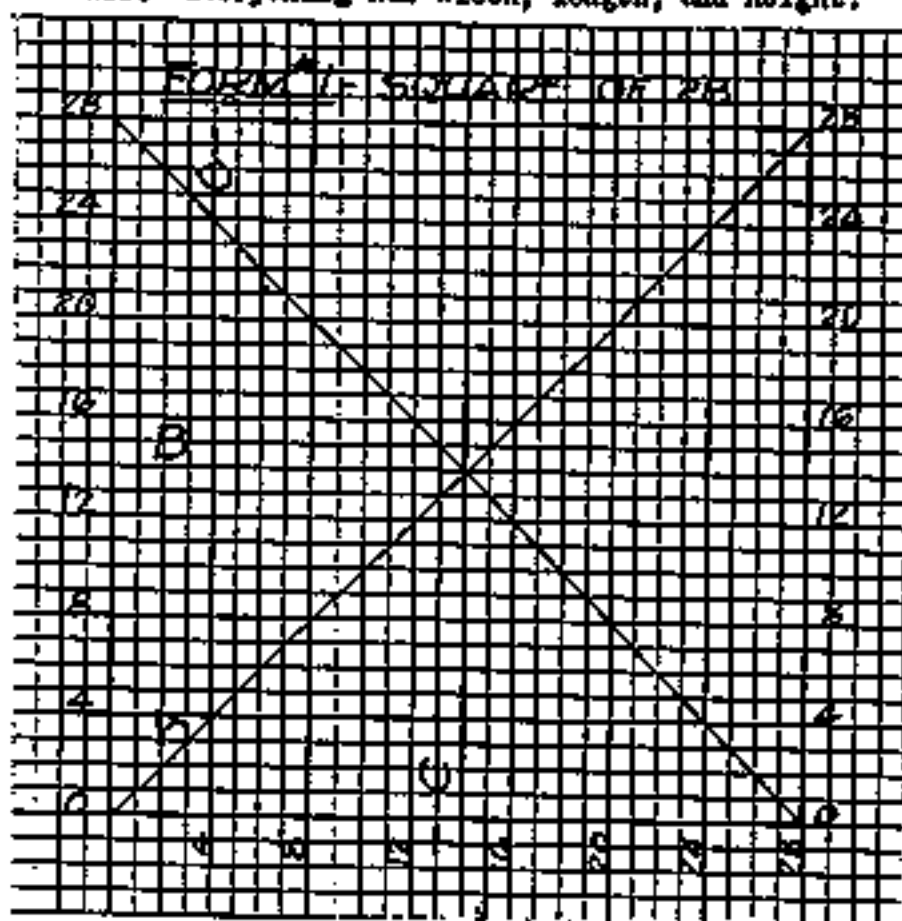
The moving-average as commonly used is obtained by taking the extreme low price and the extreme high price of the calendar day, week or month, and dividing it by two to get the mean or average price for the day, week or month, and continuing this at the end of each time period. This is an irregular movement in spaces or points per week because at one time it may move up 2 points per week

and at another 5 points per week, while the time period is a regular unit. Therefore geometrical angles, which are really moving-averages, move up or down at an uniform rate from any bottom or top on a daily, weekly or monthly chart.

HOW TO DRAW GEOMETRICAL ANGLES

There are three important points that we can prove with mathematics or geometry: the Circle, the Square, and the Triangle. After I have made the Square I can draw a Circle in it using the same diameter, and thereby produce the Triangle, the Square and the Circle.

The Angles or moving-trend-line averages measure and divide Time and Price into proportionate parts. Refer to Form "1", where I have drawn the square of 28. You will note that this is 28 high and 28 wide- in other words, 28 up and 28 across. It is the same as a square room which has a bottom or floor, a top or ceiling, and side walls. Everything has width, length, and height.



To get the strongest and most important points in this Square, I divide it into two equal parts by drawing a horizontal and a vertical line. Note angle "A", which divides each of the smaller squares into two equal parts and runs from "0" to "28" diagonally. This is a diagonal line moving on a 45° angle and divides the large Square into two equal parts. Then note angle "B" at "14" running horizontally across. This divides the Square into two equal parts. Note angle "C", which is a vertical line, running up from "14", which is one-half of "28". This crosses at the center or half-way point at 14, where the other angles cross, dividing the Square into two equal parts. Then note angle "D", which forms another 45° angle moving from the N. W. corner to the S. E. corner, crossing "14" at the exact half-way point. You see by this that if we draw the first line thru the center of the square, we divide it into two equal parts- then when

we draw lines from the other directions, we divide it into four equal parts—then by drawing the two lines from each corner, we divide the square into 8 equal parts and produce 8 triangles.

As you look at this Square, it should be easy for you to tell with your eye where the strongest support point is or resistance point is. It is at the center where all the angles cross. Four angles cross at this point, so naturally this would be a stronger support point than a place where only one angle crosses. I could divide each one of these smaller squares into four or eight equal parts by drawing angles in the same way. Later, when I give you the rules and examples, I will explain how to square the Range of a stock, that is, the difference between the extreme low and the extreme high prices, or the difference between any low point and any high point, and also how to square the bottom price. For example: If the top of a stock is 28, this Square of 28 x 28 would represent squaring the Price by Time, because if we have 28 points up in Price, and we move over 28 spaces in Time, we square the Price with Time. Therefore, when the stock has moved over 28 days, 28 weeks, or 28 months, it will be squaring its price range of 28.

PATTERN CHART FOR GEOMETRICAL ANGLES

The Square of 90, or the Pattern Chart, shows all the measured angles that are important to use in determining the position of a stock. These angles are as follows: 38, 71, 15, 182, 254, 30, 332, 371, 45, 521, 564, 60, 638, 714, 75, 824, 864, and 90 degrees.

It is not necessary to measure these angles with a protractor. All you have to do to get the angles correct is to count the spaces on the chart paper, using 8 x 8 to the inch, and draw the lines or angles accordingly.

On the square of 90, which you will receive with these instructions, note how equal angles drawn from the top and from the bottom prove themselves by the point at which they cross. For example:

The angle of 8 x 1 drawn from "0" and the angle of 8 x 1 drawn from "90" down both cross at 45, 5-5/8 points over from "0" counting to the right. Then, the angle of 4 x 1 from "0" and the angle of 4 x 1 down from "90", you will notice, cross at 111 on 45, equidistant from the other angle and twice the measure. The reason why these angles prove this way is because the 45° angle or 45 points or degrees from "0" to 45 is one-half of 90. Therefore, parallel angles beginning at "0" going up and at 90 coming down, must cross on a 45° angle or at the gravity center.

HOW TO DRAW ANGLES FROM A LOW POINT RECORDED BY A STOCK

An example marked "Form 2" shows you the most important angles to use when a stock is working higher and advancing. (See page 61)

FIRST IMPORTANT GEOMETRICAL ANGLE:

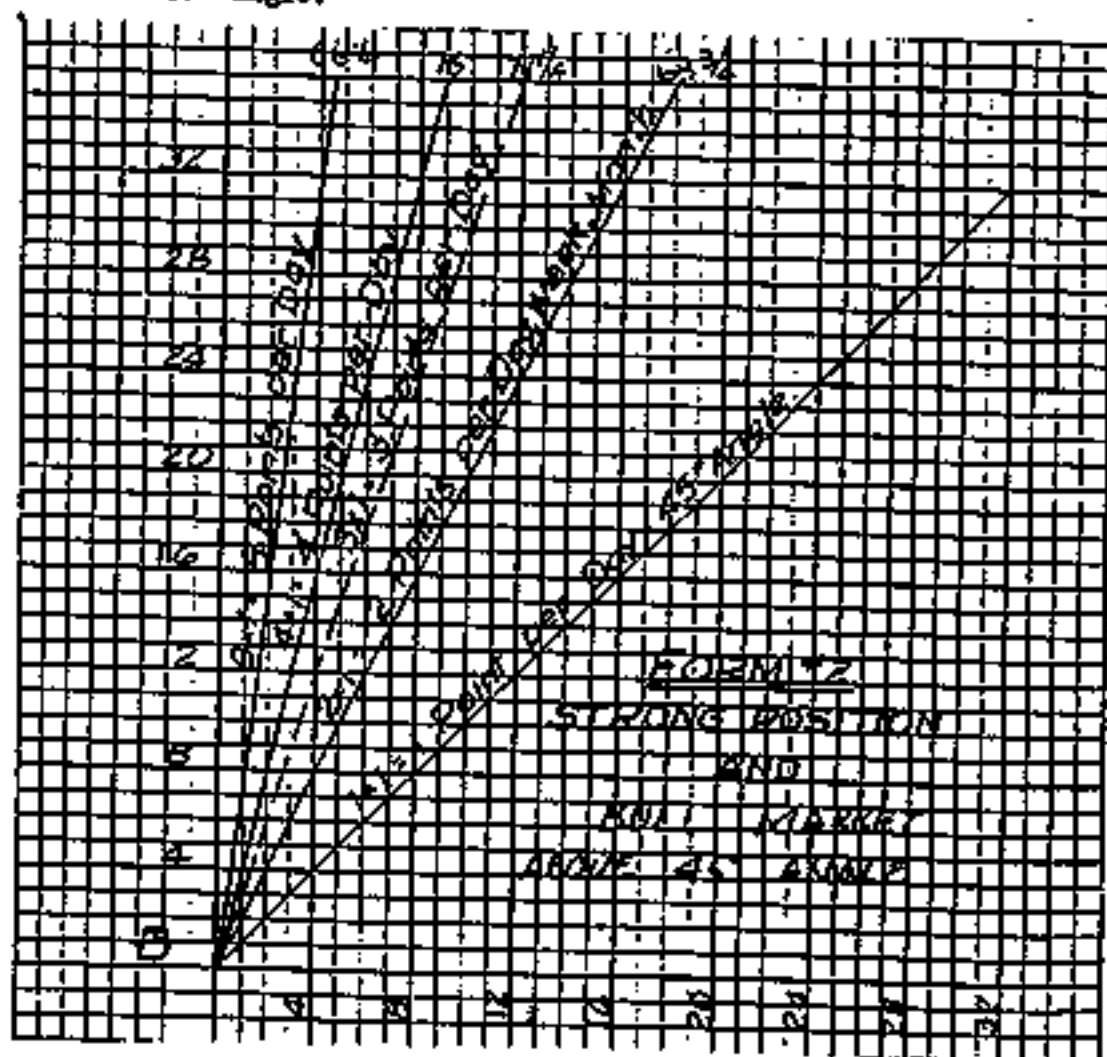
45° or 1 x 1

The first and always most important angle to draw is a 45-degree angle or a moving-average that moves up one point per day, one point

per week or one point per month. This is a 45° angle because it divides the Space and Time Periods into two equal parts. As long as the market or a stock stays above the 45° angle, it is in a strong position and indicates higher prices. You can buy every time a stock rests on the 45° angle with a stop loss order one, two or three points under the 45° angle, but remember the rule—never use a stop

loss order more than 3 points away. Unless stocks are near the low levels or just starting in a bull market or selling at very low prices, I always use a stop loss order one point under the 45° angle. If this angle is broken by one point, you will usually find that the trend has changed at least temporarily and the stock will go lower.

An easy way to calculate accurately how to put on this 45° angle is: For example: If the time is 28 days, 28 weeks, or 28 months from the point where the stock was bottom, then the angle of 45° must be 28 points up from the bottom and would cross at 28. This is one of the easiest angles to put on and one of the simplest to learn. You can beat the market by trading against the 45° angle alone if you stick to the rule—wait to buy a stock on the 45° angle or wait to sell it against the 45° angle.



NEXT IMPORTANT ANGLE is the angle of 2×1 , or the moving-average which moves $\frac{1}{2}$ at the rate of 2 points per day, week or month. It divides the space between the 45° -degree angle and the vertical angle into two equal parts and measures $63\frac{3}{4}^\circ$. That is why it is the next strongest and most important angle. As long as a stock holds above this angle, it is in a stronger position than when it is resting on a 45° angle because it is a more acute angle. When a stock breaks under this angle of 2×1 , or two points for each time period, then it indicates that it will go lower and reach the 45° angle. Remember the rule of all angles: No matter what angle the stock breaks under, it indicates a decline to the next angle below it.

THIRD IMPORTANT ANGLE, $\frac{4 \times 1}{1}$ which is still stronger as long as a stock holds above it, is the angle which moves up 4 points per day, week, or month. This angle is 4×1 , or 4 points of Space equal one period of Time. It measures 75° and divides the space between the angle of 2×1 and the 90° angle into two equal parts. Any stock that continues to advance 4 points per day, 4 points per week, or 4 points per month, and remains above this angle is in a very strong position as long as it stays above it, but when it breaks under, it indicates the next angle or next support point according to the position of the stock on Time.

FOURTH IMPORTANT ANGLE $\frac{8 \times 1}{1}$ is the angle of 8×1 or the one that moves up 8 points per day, week or month. This angle measures $82\frac{1}{2}^\circ$. As long as a stock can hold above this angle on daily, weekly or monthly chart, it is in the strongest possible position, but when it reverses trend and declines below this angle, then it indicates a decline to the next angle.

NEXT ANGLE: $\frac{16 \times 1}{1}$ It is possible to use an angle of 16×1 , or 16 points of Price to one period of Time, which measures $86\frac{1}{4}^\circ$, but this angle is only used in fast, advancing markets, like 1929, when stocks are moving up or down 16 points or more per week or per month. There are very few stocks that will move up 16 points per day, week or month, and very seldom.

You will note that with these four important angles we show the strong or bullish side of the market. All the time by dividing the Space with angles we are getting the half-way point or the gravity center of Time and Price.

3×1 ANGLE: Note the angle drawn in Green, marked " 3×1 ", which moves up at the rate of 3 points per day, week or month, measuring $71\frac{1}{2}^\circ$. This angle is important at times after markets have had a prolonged advance and are a long distance up from the bottom. It is an important angle to use on Monthly and Weekly charts.

These are all the angles you need as long as a stock continues to advance and work up and stays above the angle of 45° or the moving-average of one point per day, week or month.

While there are 360 degrees in a circle and angles can form at any of those degrees, all of the important angles form between " 0° " and " 90° " because 90° is straight up and down and the most acute angle on which a stock can rise. For example: The 45° angle divides the space from " 0° " to " 90° " in half. The angle of 135° is simply another angle of 45° because it is one-half of the next quadrant between 90° and 180° . 225° and 315° in a circle are also 45° angles. Therefore all of the angles valuable in determining the trend of a stock are found between " 0° " and " 90° " degrees. When we divide 90° by 2 we get the most important angles to use—then divide it by 3 we get 30° and 60° angles, which are important to use for Time and Resistance Points.

WHAT KIND OF BOTTOMS TO DRAW ANGLES OR MOVING-AVERAGE LINES FROM

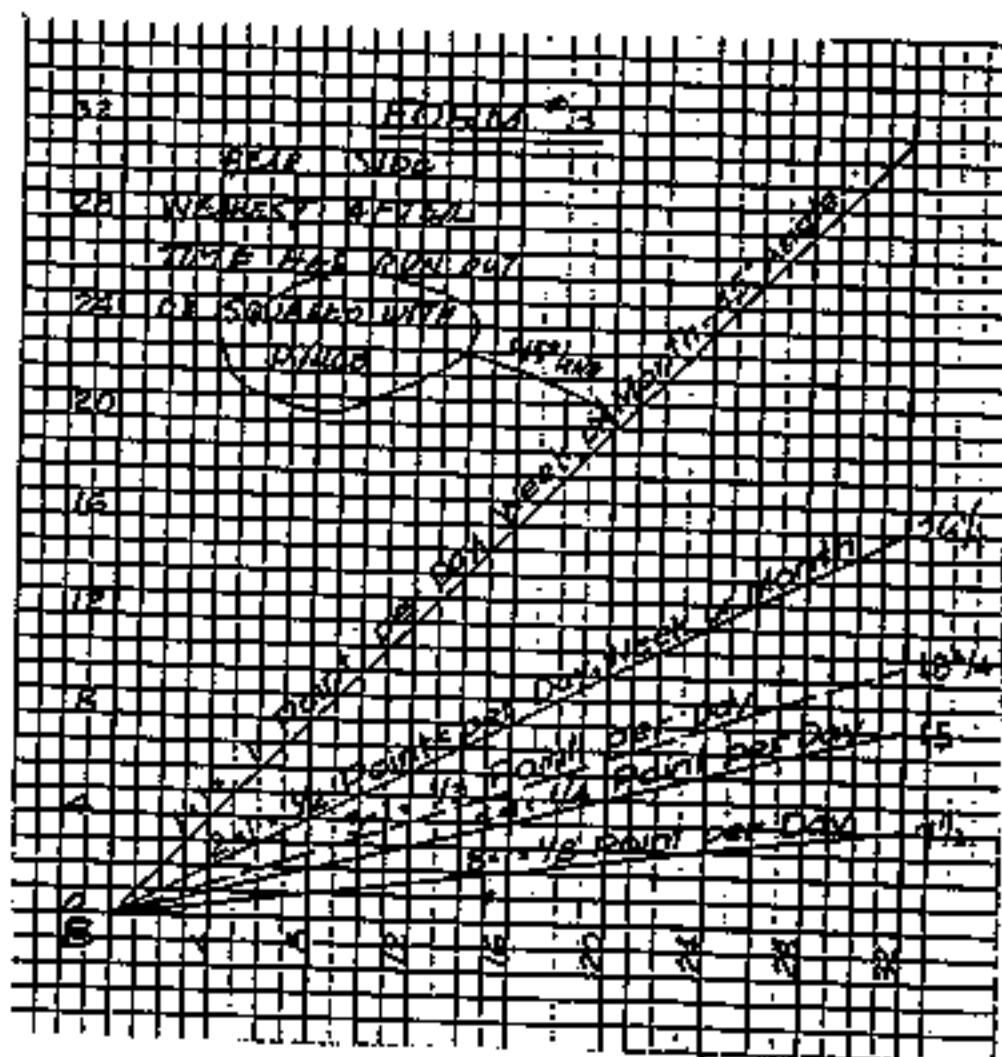
DAILY CHART: If a stock has been declining for some time—then starts to rally (by rallying from a bottom it must make higher bottoms every day and higher tops)—then after a 3-day rally on the daily high and low chart, you can put on the 45° angle and the angle of 2×1 from the bottom or low point. As a rule, it will only be necessary to put on these two angles at first. If this

bottom holds and is not broken, then you can put on the other angles from the bottom.

WEEKLY CHART: If a stock is declining and reacts for more than one week and continues down, we will say, for three weeks or more, then starts to rally and advances two weeks or more, you would start to put the angles on from the low point of the decline, only using the angles above the 45° angle until the stock again breaks under the 45° angle—after that you would use the other angles on the lower or bearish side of the Square.

WHAT TO DO AFTER THE 45° ANGLE FROM BOTTOM IS BROKEN

After a stock makes top, either temporary or otherwise, and breaks under the 45° angle and starts moving down, then the first thing you do is to draw angles below the 45° angle, starting from the bottom or low point. Note example marked "Form #3":



FIRST ANGLE ON BEAR SIDE OF THE SQUARE:

$$\frac{2 \times 1}{1}$$

The first angle that you draw on the bear side of the Square is the angle of 2×1 or 2 points over and one point up, which moves at the rate of one-half point per day, week or month and measures $26\frac{1}{2}^\circ$. This is the first support angle which the stock should reach after it breaks under the 45° angle. As a general rule, when the stock reaches this angle, it

will receive support and rally. Sometimes it will rest on it for a long period of time, holding on this angle and making higher bottoms. But when this angle of 2×1 , or moving-average of one-half point per day, week or month is broken, then you must draw the next angle of 4×1 .

NEXT IMPORTANT ANGLE: $\frac{4 \times 1}{4 \times 1}$ The next important angle on the bear side of the Square, which moves up at the rate of $1/4$ -point per day, is the angle of 4×1 , measuring 15° . It will be the next strong support angle which the stock should get support on and rally from.

NEXT ANGLE 8×1 : Then after the 4×1 angle is broken, the next important angle that you will put on your chart is the angle of 8×1 , which moves at the rate of $1/8$ -point per day, week or month and measures $7\frac{1}{2}^\circ$. This is often a very strong support angle. After a stock has had a big decline, it will often rest on this angle several times or may make final bottom and start up from this angle, crossing other angles and getting back into a strong position again. Therefore this angle is important to use on a monthly or weekly chart after a prolonged decline.

ANGLE 16×1 : This angle can be used on a monthly chart after a long period of time has elapsed from an important bottom. It moves at the rate of $1/16$ point per month and measures $3\frac{3}{4}^\circ$.

ANGLE OF 3×1 : This angle, drawn in red ink, is a very important angle, measuring $18\frac{3}{4}^\circ$. I strongly advise using it at all times and keeping it up on monthly charts from any important bottom. It can also be used to advantage at times on weekly charts, but is seldom of much value on a daily chart. It moves at the rate of one-third point per day, week, or month. By drawing this on the monthly chart for a long period of years, you will soon be convinced of its value and also by testing it on a weekly chart, will find it valuable.

This completes all of the angles that you will need to use from any bottom at any time.

HOW TO DRAW ANGLES FROM TOPS ON DAILY, WEEKLY OR MONTHLY CHARTS

POSITION UNDER 45° ANGLE DRAWN FROM TOP: After a stock has made top and declined for a reasonable length of time, say, three days, three weeks or three months, breaking previous bottoms, then you start to draw angles down from the top. Note example marked "Form #4", which is the pattern for drawing angles from the top under the 45° angle. (see page 10)

45° ANGLE FROM TOP: The first angle you draw is the angle of 45° or a moving-average which indicates a decline of one point per day, week or month. As long as the stock is below this angle, it is in the weakest position and in a bear market.

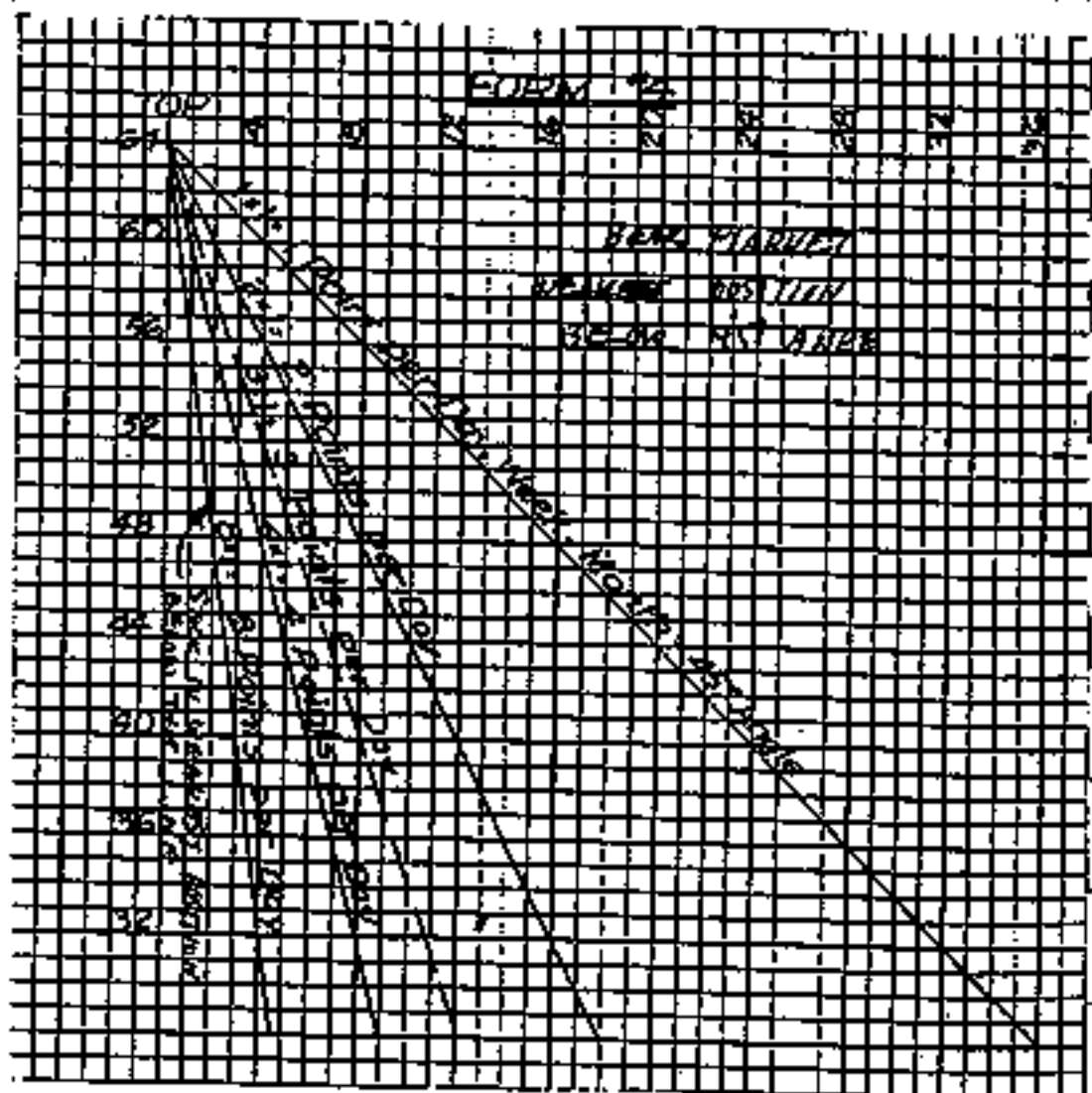
OTHER ANGLES: In many cases a stock will start declining an average of 8 points per day, week or month, or 4 points per day, week or month, or 2 points per day, week or month, therefore you should put on all of these angles from the top, which move down faster than the angle of 45° .

WEAKEST POSITION: The stock is in the weakest possible position when it declines and keeps under the angle of 8×1 . It is in the next weak-

est position when it is dropping down at the rate of 4 points per day, week or month, or under the angle of 4×1 . It is in its next weakest position when it is dropping down under the angle of 2×1 .

STRONGEST POSITION: The stock is in a stronger position and indicates a better rally when it crosses the angle of 2×1 , but this depends on how far it is down from the top and how far the angles are apart, as will be explained later under the rules.

CHANGING TREND: As long as a stock is declining one point per day, week or month, or falling below or under the 45° angle, it is still in a bear market and in a very weak position. When a stock rallies and crosses the angle of 45° after a prolonged decline, then you are ready to put on the angle on the other side of the 45° angle, which shows that the stock is in a stronger position in a bear market and may be getting ready to change into a bull market.



POSITION ABOVE 45° ANGLE DRAWN FROM TOP

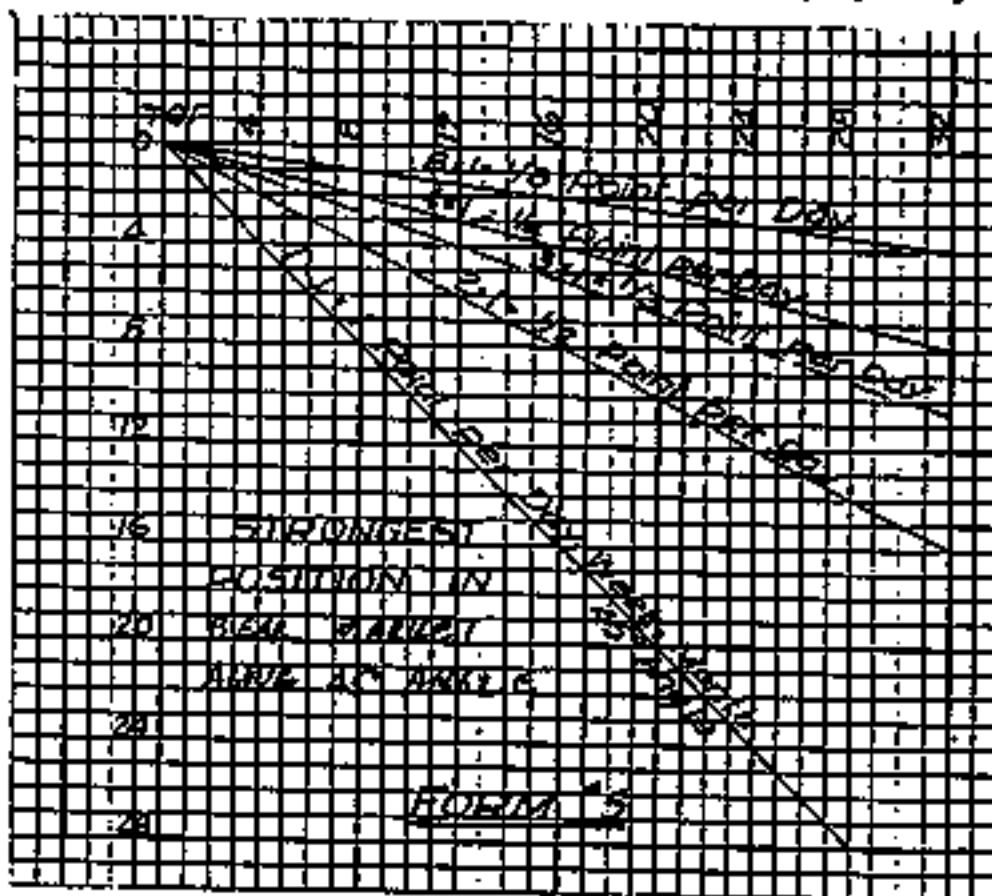
Refer to Form #8, which is the pattern for drawing angles above the 45° angle from the top. (See page 11)

2×1 ANGLE FROM TOP: The first angle or moving-average you draw after the 45° angle from the top is crossed and after the stock indi-

states that it has made a temporary bottom is the angle of 2 x 1, moving over 2 points and down one point, or 1/2-point per unit of Time. This is moving down at the rate of 1/2-point per month, week or day.

4 x 1 ANGLE: The next is the angle of 4 x 1 which moves down at the rate of 1/4-point per day, week or month.

8 x 1 ANGLE: The next angle is the angle of 8 x 1, which moves down at the rate of one point every 8 days, 8 weeks or 8 months, or 1/8-point per time period.



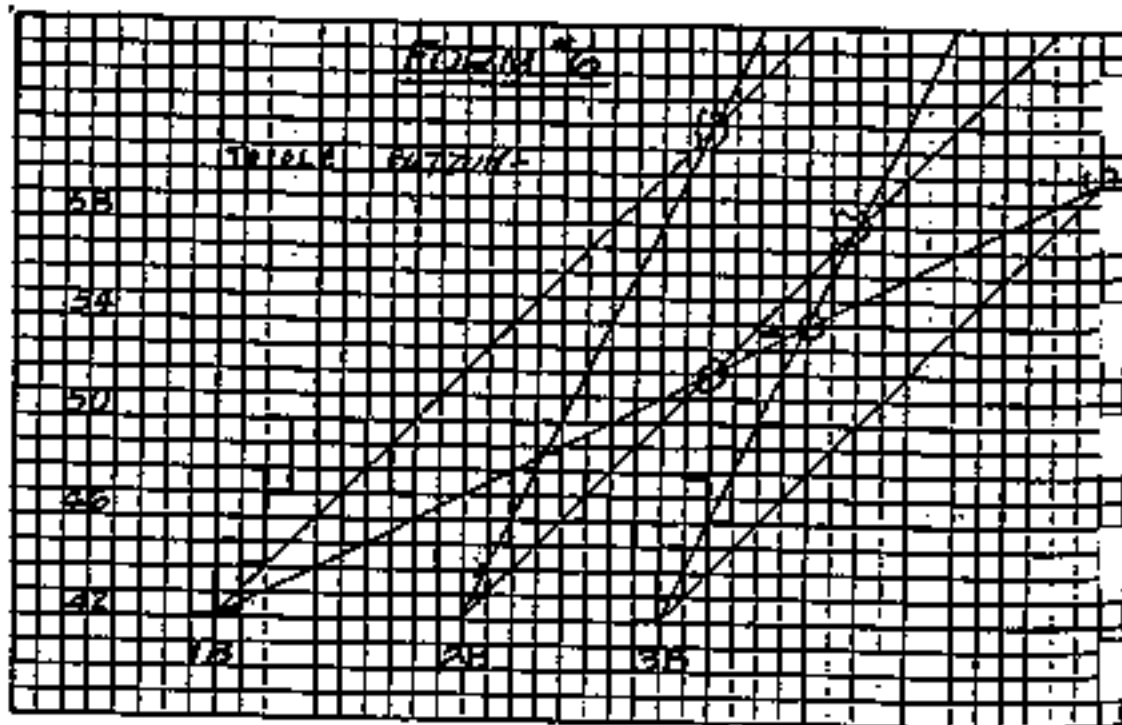
absolutely accurate. Then you can begin to study the rules for determining the trend according to the position of the stock on angles.

DOUBLE AND TRIPLE TOPS OR BOTTOMS

ANGLES CROSSING EACH OTHER: When there is a double bottom several days, weeks or months apart, you draw angles from these bottom which are near the same price levels. For example: From the first bottom draw a 45° angle and from the second bottom draw an angle of 2×1 - then when these angles cross each other, it will be an important point for a change in trend. Note on chart marked Form #6 that I have drawn the 45° angle from the first bottom "1B" and the angle of 2×1 on the right hand side of the 45° angle. Then, from the second bottom "2B" I have drawn a 45° angle and the angle of 2×1 , which gains 2 points per day, week or month, on the left hand or bull side of the 45° angle. You will note that the angle of 2×1 from the second bottom crosses the angle of 2×1 on the bear side from the first bottom at 48, and that when the stock breaks under these angles, a change in trend takes place and it goes lower.

Note that the angle of 2×1 from the third bottom "3B" crosses the angle of 2×1 on the bear side from the first bottom at $53\frac{1}{2}$ and crosses the 45° angle from the second bottom at 58. This would be a point to watch for change in trend. I have placed a circle where these angles from the different bottoms come together.

Apply this rule to double tops and triple tops in the same way. It is not necessary for the tops or bottoms to be exactly at the same price level, but near the same level. Remember, always draw 45° angles from all important tops and bottoms.



PARALLEL ANGLES

Parallel angles or lines run from important tops and bottoms. As previously explained, the 45° angle is the most important and should be drawn from all important tops and bottoms. If a stock starts advancing, we draw a 45° angle from the bottom - then if the stock makes top, declines and makes a higher bottom - then

advances and makes a higher top, we draw a 45° angle from the first top, running the line up. This will give the oscillation or width of fluctuation in a parallel between the 45° angle from the bottom and the 45° angle running up from the top. Often a stock will advance to the 45° angle from the first top, fail to cross it, then decline and rest on the 45° angle from the first bottom—then advance again, working up for a prolonged bull campaign between these parallel angles.

When the angles are very far apart, you can draw another 45° angle equidistant between them, which is often a strong support angle from which a stock will rally, but when it breaks under, it declines to the bottom parallel.

Parallels can form between the angles of 2 x 1 or 4 x 1 just the same as between 45° angles, which often occurs in slow-moving stocks.

GEOMETRICAL ANGLES OR MOVING-AVERAGE LINES DRAWN FROM "0"

When a stock reaches bottom and starts up, you have been instructed to draw angles from this exact low point, which shows the support in time periods, but there are other angles that later on will be just as important and sometimes more important than the angles drawn from the bottom of a stock. These are the angles that begin at "0" or zero and move up at the same rate that they move up from the bottom. The starting point must be on the same line that the bottom is made on as the time period begins from this bottom, but the angles move up from "0". These angles should be started every time a stock makes a bottom, especially on weekly and monthly charts, and should also be carried up on important movements on the daily chart. Example: See chart marked Form #7 on page 14.

If a stock makes low at 20, as shown on the chart, starting the 45° angle from "0", when will this angle reach 20? Answer: It will reach 20 in 20 days, 20 weeks or 20 months from the bottom or its starting point. In other words, in 20 days, 20 weeks, or 20 months, it will be up 20 from "0" and at the price where the stock made bottom. Then the angle will continue on up at the same rate, and later, when the stock breaks under the 45° angle from the actual bottom made at 20 and breaks the other support angles drawn from the actual bottom at 20, the next important point for support will be the angle of 45° moving up from "0". When this angle is broken, it is in the weakest possible position and indicates much lower prices, but this depends on how high the stock is selling and how much it has declined at the time it breaks the 45° angle from "0". These angles drawn from "0", especially the 45° angle, proves when Price and Time are balancing or when the stock is squaring out from its bottom.

"0" ANGLES STARTING AT THE TIME TOP IS MADE

When a stock reaches extreme top on a daily, weekly or monthly chart and the trend turns down, you should start an angle of 45° from "0" moving up from the exact space and date that the top is made. This will prove the square of the time period. It is very important when this angle is reached and indicates a change in trend. It is the last strong support and when broken, it will indicate much lower prices.

I have instructed you in each case to first draw the 45° angle from bottom, top and from "0" at bottom and top, but this does not mean that you must not use the other angles. All of the other angles can be used from "0", but the 45° angle is the first and most important. After this angle is broken, then you can use the other angles. It is not necessary to carry all of them along until you

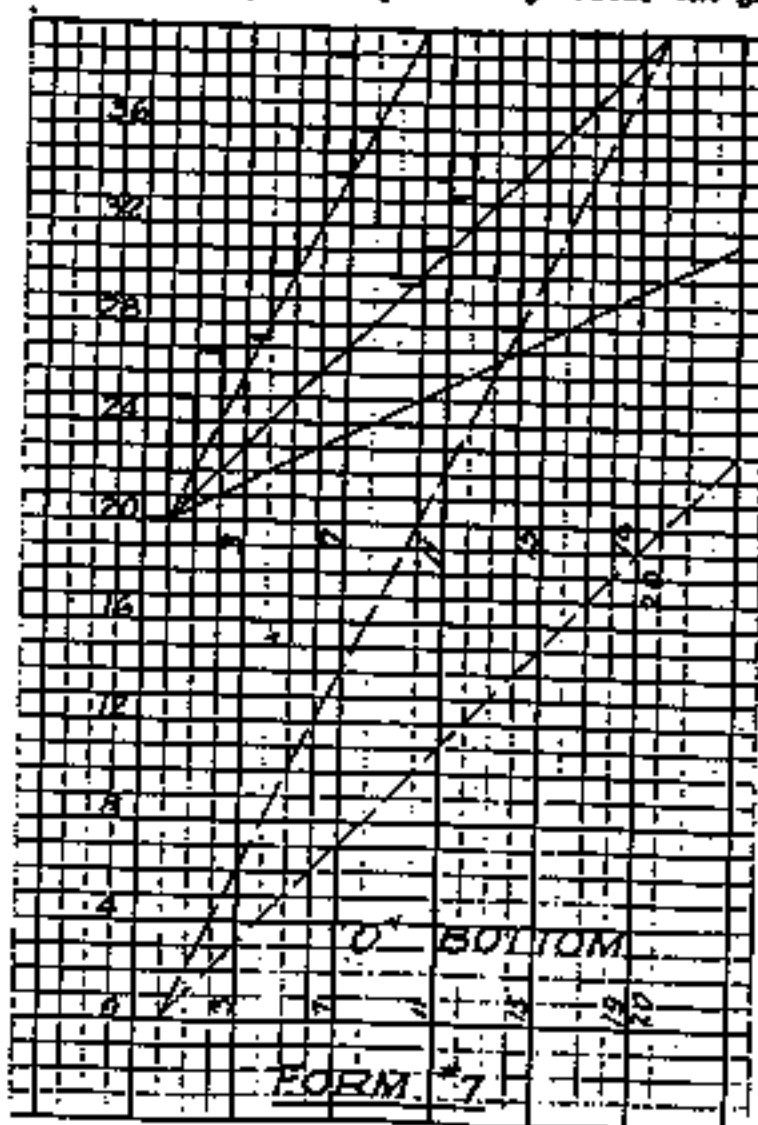
need them, but on the monthly chart, after a long series of years, these other angles should be carried along when the stock begins to approach the levels where they would be broken or where the stock would rest on them and receive support.

45° ANGLE FROM "0" TO TOP AND BOTTOM: When a 45° angle moving up from "0" reaches the line or price of the bottom it is very important- then again when it reaches the point of the extreme high price, it is very important for a change in trend.

You should carry 45° angles and other angles up from "0" from all important first, second, and third higher bottoms, especially those where very much time has elapsed between these bottoms. You should also start the angle of 45° up from "0" from the first, second, and third lower tops, especially those which show much time period elapsed. These angles are the most important to be carried on the weekly and monthly charts.

Never overlook keeping up the angles from "0" because they will tell you when Time is squaring out with Price from tops and bottoms and will locate support angles or moving-average lines at a point on the bear side after the first 45° angle from a bottom is broken. You could not locate this support point in any other way except by the angles from "0".

You should go back over past records and bring up these angles and square out different tops and bottoms so that you can prove to yourself the great value of using these angles.



ANGLES FROM TOPS DOWN TO "0" AND UP AGAIN

A 45° angle starting down from any important top on a monthly or weekly chart should be continued down until it reaches "0" and then started up again at the same rate. After a long number of years between important tops and bottoms, this angle coming down and going up again is important. A 45° angle can also be continued down from any important bottom to "0" and then started up again. This will show the squaring of Price with Time from either top or bottom.

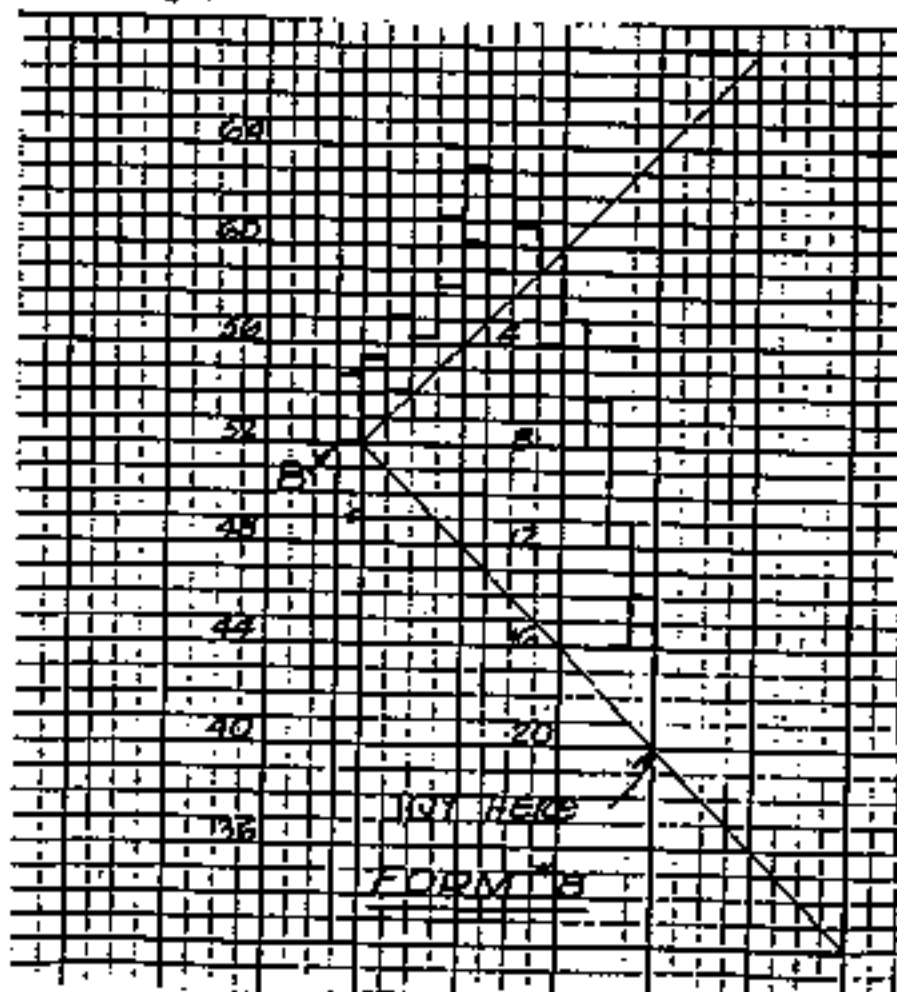
Angles can be started from "0" at the time any important time cycle runs out. For example: U. S. Steel made extreme low in 1904. May, 1924 would be the end of a 20-year cycle or 120 months. In May, 1924 Steel made top at 109 against a 45° angle beginning at "0" at the time bottom was made at 38 in February, 1915. On account of the importance of this top and a 20-year cycle running out here, we would start a 45° angle and other angles, if we need them, from "0" in May, 1924.

May, 1931 would end a 7-year cycle of 84 months from 1924. The 45° angle running up from "0" in May, 1924 crossed at 84 in May, 1931. Note that Steel made low at 83½ in June, 1931. In June, 1924 Steel made the last low at 94½, showing the importance of the end of the 20-year cycle. The 45° angle moving up from "0" in June, 1924 crossed at 84 in June, 1931, and Steel declined and rested on this angle.

TWO 45° ANGLES FROM THE SAME BOTTOM

As we have previously explained, the 45° angle moves up at the rate of one point per month and moves down at the rate of one point per month.

Refer to example on Chart 48:



You will note that the low on this chart is shown as 52 and the stock moves up to a high of 63. A 45° angle is drawn up from the bottom, and after the stock reaches top and starts to work down, it breaks the 45° angle, getting under it, a price of 59. You will note that I have drawn another 45° angle down from the bottom at 52. At the point where the stock breaks under the 45° angle moving up from 52 to the 45° angle moving down from 52, the distance in points is 16, therefore the angles have widened until the stock could decline 16 points, if it went straight down, before it reached the 45° angle moving down from the bottom.

Note that I have shown on the chart that the stock continues down until it reaches 40, where it rests on the 45° angle from the bottom at 52. This would indicate the strongest support point and at least a temporary rally, especially as the stock is down 23 points from the top. Later you will find under "Resistance Levels" that $22\frac{1}{2}$ to 24 points is a strong support point.

U. S. STEEL: Take the extreme low point of U. S. Steel at $111\frac{1}{2}$ in January, 1927. Start a 45° angle on the monthly chart moving up at the rate of one point per month—then start a 45° angle moving down at the same rate. This shows the spreading of the angles and what can happen when Steel breaks under the 45° angle coming up from that bottom and the point where it can decline in extreme panicky markets, like 1931 and 1932.

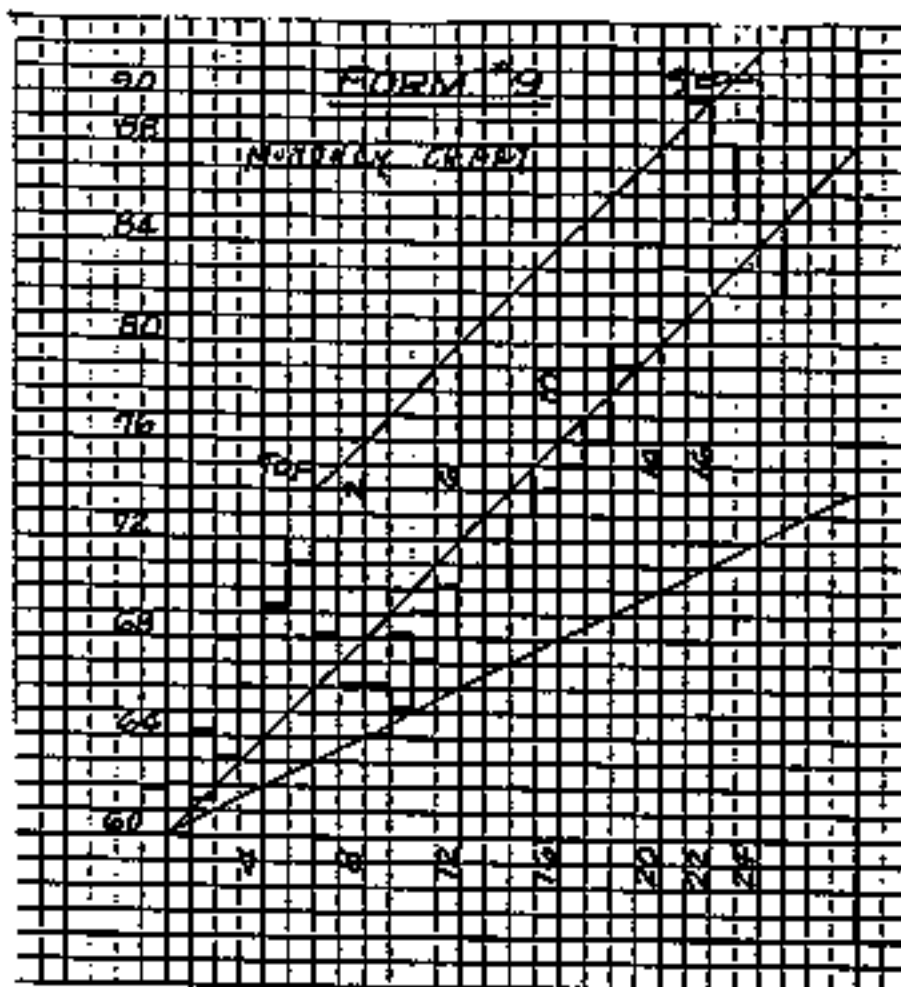
The 45° angle coming up from the low of January, 1927 crossed at 156 in October, 1930. When U. S. Steel broke this angle it went right on down to $134\frac{1}{2}$ in December, 1930, where it rested on the angle of 2×1 from this bottom of January, 1927—then rallied to February, 1931, and at the time it broke under the angle of 45° from $111\frac{1}{2}$, it was on the 45th month, another indication of a sharp, severe decline. Here we look at the 45° angle moving down from $111\frac{1}{2}$ and find it is 90 points down from the 45° angle moving up from $111\frac{1}{2}$. These angles separate at the rate of two points per month and being 45 months from the bottom, the stock would have to decline 90 points to strike the 45° angle moving down from the bottom. The angles being so wide apart indicated that the stock could have a wide-open break. This happened in December, 1931, when Steel broke under the 45° angle moving down from $111\frac{1}{2}$, putting it into a very weak position—in fact, in the weakest position that a stock can get in until it can recover this angle. In June, 1932, when U. S. Steel declined to $21\frac{1}{2}$, it had dropped under the 45° angle moving down from the last low of $113\frac{1}{2}$ made in March, 1925, and closed two months below this angle before it started to recover angle.

This shows that when a stock gets into a very weak position by dropping under important angles moving down from bottoms, after having broken strong angles moving up from bottoms, it can decline to very low levels. These extreme fluctuations and declines have happened in the past and will happen again in the future. This proves the squaring out of Time on the down side or the balancing up of Price and Time.

Here is another illustration of the balancing of Price with Time: The angle of 45° moving up from "C" from the bottom at $21\frac{1}{2}$ from October, 1907 crossed at 262 in September, 1929 and Steel advanced to $281\frac{1}{2}$, which shows that in 262 months from the bottom in 1907 Steel had advanced an equivalent of one point per month. By striking the 45° angle and failing to cross it, it indicated that the Time was up and that the stock was turning downtrend for a prolonged bear market.

ANGLES OR MOVING-AVERAGE LINES FROM ONE TOP TO THE NEXT TOP

Refer to example on Chart #9 on page 17.



GA-17

You will see that we have started the bottom at 60. The stock advances six months to 74, to a point marked "T" and makes top—rests for three months to 84, breaking the 45° angle but resting on the angle of 2 x 1 from the bottom—then starts advancing and finally crosses again the 45° angle from 60, getting into a stronger position, having regained this angle. In order to determine where it might meet resistance, as it is in new high territory, we draw a 45° angle from the top at 74. The stock advances to 90 on the 22nd month from the bottom, striking the 45° angle from the first top at 74, on the 18th month from the first top. Being 18 points up above the first top, the Time equals the advance in the Price above the first top. The 45° angle shows that this is a strong Resistance point and a place to go short with stop one to three points above the 45° angle. A decline starts and in the third month the stock again breaks under the 45° angle from the bottom (at 60) at a very high level. In other words, it is 24 points up from the bottom and is now in a much weaker position, because it is so far from the base of support, and indicates a decline again to the angle of 2 x 1 (marked in green).

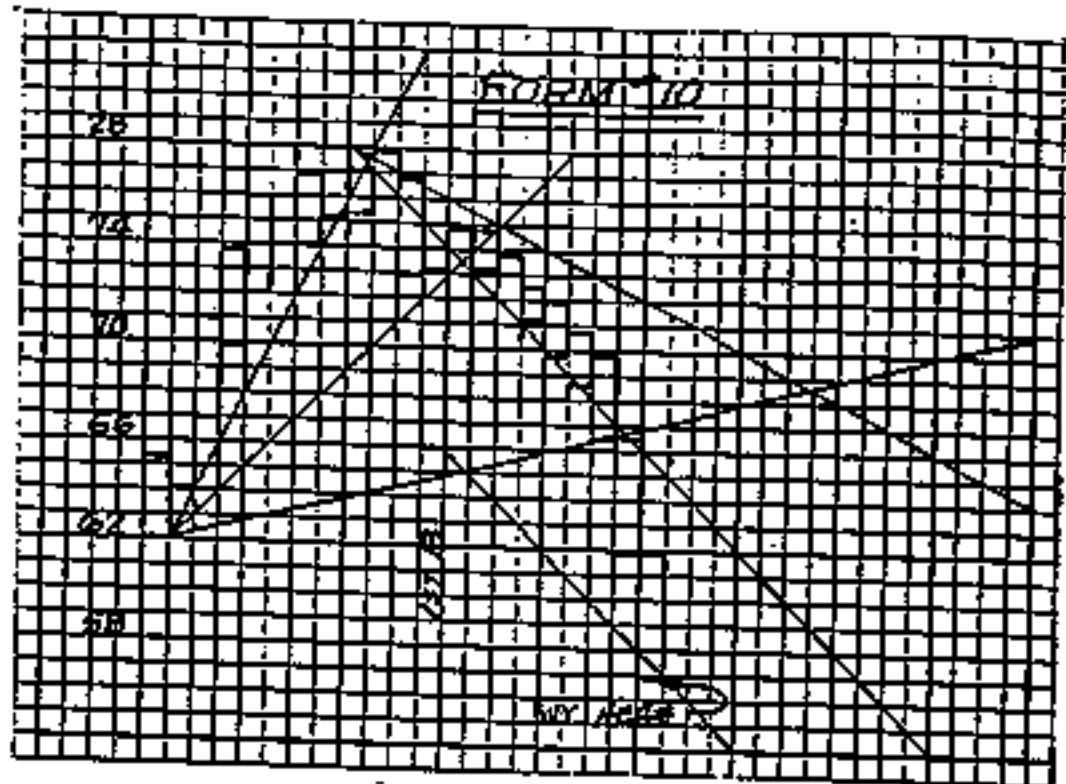
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Don't overlook this rule: After a stock has advanced to a new high level, then declines to the old top at 74, this may be a support point unless it breaks 3 points under it. If it does and also breaks the angle of 2 x 1, it will be in a weaker position and the next point to watch for support and a rally would be the next bottom at 64.

ANGLES FROM BOTTOM OF FIRST SHARP DECLINE

When a stock that has been advancing for some time, makes top and holds for several days, several weeks or several months, then turns the trend down and has a sharp, severe decline, there is always a rally after this first decline. It usually makes a lower top on this secondary rally and then starts to work lower again. The bottom of the first decline is a very important point to draw angles

from, especially the 45° angle moving down, as I have done on the chart marked Form #10.



This chart shows the stock rallying up to around 75, where the 45° angle coming up from the last bottom crosses the angle of 2×1 coming down from the top. Then the decline started and at 66 the stock broke back under the angle of 45° from the top, which put it in a very weak position. It declined to the angle of 45° coming down from the bottom of the first sharp decline. This would be the squaring out of time from the bottom and would be a place to buy for a rally. A stock will often decline and drop a little below this angle from the bottom—then if it holds for several days or weeks under this angle or on it, it is a place to buy for a rally.

On a Monthly Chart always carry this angle down from the bottom of the first sharp decline, as it often becomes very important later on in a campaign.

After a stock has been advancing for some time and then has a sharp break lasting 2 to 3 days, 2 to 3 weeks, or 2 to 3 months—then rallies and afterward breaks under the lows of this first sharp break, it indicates that the main trend has turned down and that it is going lower.

Apply the same rule when a stock has been declining for a long time and then makes a sharp, quick recovery for 2 to 3 days, 2 to 3 weeks, or 2 to 3 months, then reacts and then crosses this first rally point that it made, an indication of higher prices.

LAST SWING IN A BULL OR BEAR MARKET

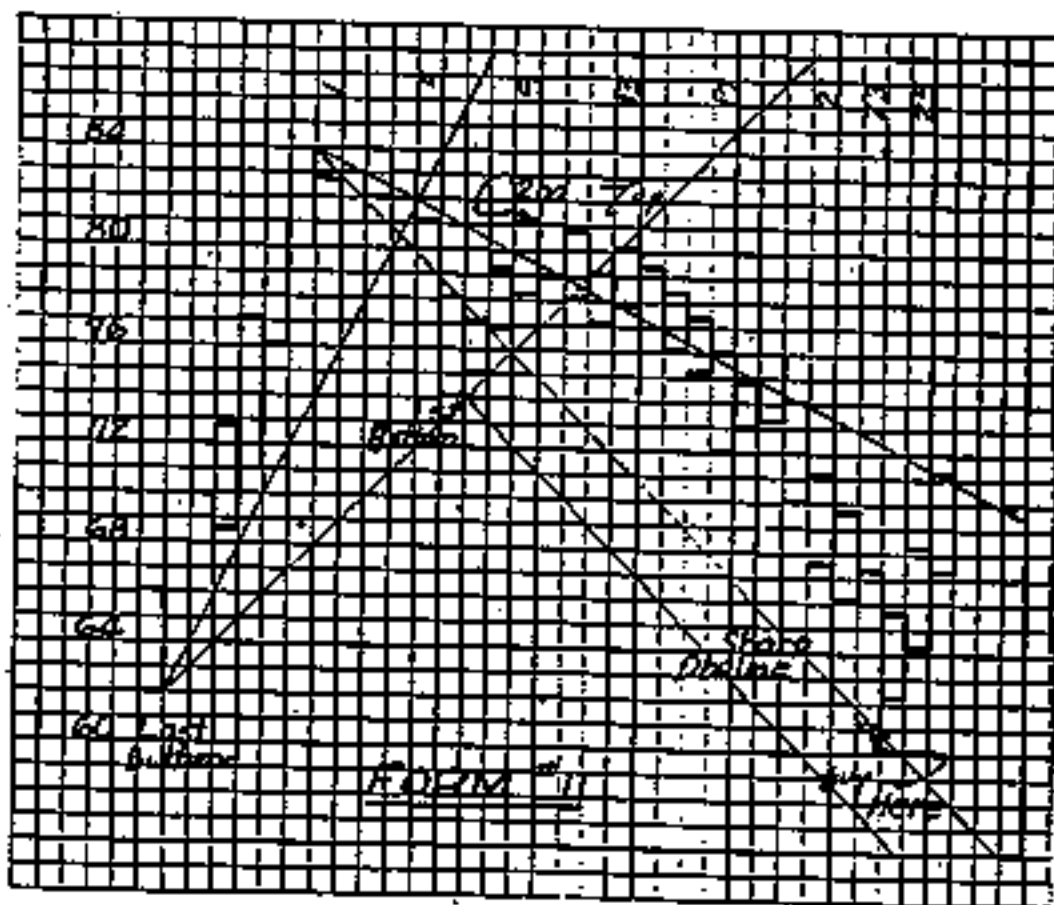
It is important to draw angles from the point where the market starts up and makes its last run in a bull market. Refer to Chart #11 on page 19.

In this example note point marked "last bottom". In the last stage of the bull market a fast advance follows to a price of 84. We have drawn the angle

of 2 x 1 (a gain of 2 points per day, week or month) and the 45° angle from this bottom. When the angle of 2 x 1 was broken, it indicated that the trend had turned down. The stock declined and rested on the 45° angle—then rallied and made a second lower top—then broke the 45° angle—declined sharply and rested on the 45° angle drawn from the top at 84, which indicated that Time and Price had squared out or were equal. This would be a point to buy, with a stop loss order 2 to 3 points under this angle, for a rally back to the angle of 2 x 1 from the top, as shown on the chart.

In very active fast-moving markets a stock may stay above the angle of 4×1 or the angle of 8×1 from the "last bottom," but on the daily or weekly chart after this first acute angle is broken, it indicates that the trend has turned down.

Always remember that after a prolonged advance, when the main trend turns down, it is safer to wait for rallies and sell short than to buy against the trend.



All of these rules are reversed at the end of a bear market or sharp decline. It is important to note when the market starts down from the last top or rally and makes its last run to bottom. Draw angles from this last top and watch when the market reaches these important angles and crosses them. For example:

On March 9, 1932 the Dow-Jones 30 Industrial Averages made last top at 90, from which a decline followed, with very small rallies, reaching bottom at 41 on July 8, 1932. Note on the weekly chart that the angle of 2×1 from the top at 90 crossed at 50 in the week ending July 30, 1932 and after they crossed this level they never declined to 50 again, and advanced to 81 in September, 1932. The cross-

ing of this angle was the first definite indication that the main trend had turned up.

It is also important to review the major swing from November 9, 1931, when the Averages reached a high of 119½, to the low at 40½ in July, 1932. This was the last big swing of the bear market, a decline of 79 points. The half-way point of this was at 80. In September, 1932 the Averages rallied to 81-- then after they reacted to 50 and advanced, getting above the half-way point, and crossed 81, they indicated an advance to 119 anyway. After they crossed the half-way point the second time and advanced above 81, they never sold down to it again until they advanced to 149½ in November, 1933.

After a stock has been advancing for a long time, in the last run when there is a lot of momentum, it may cross angles from previous tops or bottoms, then fall back under them, which is an indication of weakness. When a stock has a sharp decline and is making bottom, it will drop under important angles and then recover quickly, getting above them, which shows that it is getting into a strong position and changing trend.

ANGLES FROM HIGHER BOTTOMS AND LOWER TOPS

What rule should be followed when stocks make higher bottoms and lower tops?

As stocks advance and make higher bottoms on the monthly, weekly or daily chart, you should always draw angles from higher bottoms. Then, in the last section of a bull market, if these important angles are broken from the last bottom, you know that the trend has turned down.

Apply this same rule as a market declines. Draw the angles from each lower top and watch the angles until the stocks again cross the 45° angle from a second, third, or fourth lower top. The second lower top or second higher bottom is always very important to draw angles from and to measure time from as well.

Example: Dow-Jones Industrial Averages--

September 3, 1929 - Extreme high-

November 13, 1929 - Bottom of first sharp decline-

April 17, 1930 - Big rally in bear market, second lower top-

July 8, 1932 - Extreme low, final bottom-

September 8, 1932 - Top of first sharp advance after bear market ended-

February and March, 1933 - Second higher bottom from which bull market was resumed.

These are the most important tops and bottoms to draw angles from.

SECTIONS OF MARKET CAMPAIGNS:

All market campaigns, up or down, move in 3 to 4 sections. When an advance starts, the market runs for several weeks or several months and then halts for several weeks or months, moving up and down over a range of 5 to 20 points, according to the price of the stock--then the advance is resumed and the stock crosses the high level of the first section, moves higher, halts again, and reacts for a period of time-- then crosses the top of the second section and moves up again for another period of time and halts for the third time, which is a very important point to watch as markets often culminate at the end of the third section and a bigger decline follows.

Most markets run out in three important sections or campaigns. However, after resting and reacting, if a stock crosses the third top, it will then move up to the top of the fourth section. This fourth advance may be a shorter period of time than the previous sections, or in some cases may consume a greater period of time, especially if the stock is very active and high-priced. This fourth top is very important and generally marks a culmination and a reversal for a bigger decline.

For example: On March 12, 1935 CHRYSLER MOTORS declined and made low at 31—

FIRST SECTION of the advance carried the stock up to 49½ on May 15—then the stock declined to 41½.

SECOND SECTION— On June 27 the advance started and the stock advanced to new high levels, reaching 62½ on August 10, which was top of the second section— then there was a reaction to 57½ and a resting period.

THIRD SECTION— Then there was another advance which started August 28. The stock crossed the top of the second section and reached high at 74 on September 11, top of the third section. Then followed a reaction to 68 on September 21.

FOURTH SECTION— In October the top of the third section was crossed and on November 18 Chrysler reached 90, top of the fourth section, where it held for 5 weeks in a 6-point range while distribution was taking place. This was a most important point to watch for a final top and a change in trend. Then the trend turned down.

Reverse this rule in a bear market. Watch the action of the market when it makes the third and fourth decline. But, remember, in a bear market when rallies come, they may make only one section or one move or in extreme cases only make the second section— then reverse and follow the main trend down.

You will find it very helpful to study and watch these various sections of a campaign and by applying the angles from tops and bottoms you can detect the first minor and major changes in trend.

STRENGTH OR WEAKNESS DENOTED BY POSITION ON ANGLES

The angles on the Monthly and Weekly Charts are of greater importance than those on the Daily Chart because the daily trend can change quite often, while only the major changes are shown according to the angles on the Monthly and Weekly Charts.

Always consider the distance a stock is from its beginning point when it breaks any important angle or crosses any important angle. The further away from the beginning point, the more important the change in trend, whether this is crossing an angle from the top or breaking under an angle from the bottom.

WHEN A STOCK IS IN THE WEAKEST POSITION:

A stock is in the weakest position when it has completed distribution and broken under a 45° angle from an important bottom on the weekly or monthly chart. It is also in the weakest position when it has broken under the half-way point between any important top or bottom. The longer the time period has run and the higher the price, the weaker the position. For example:

If a stock has advanced to 150 and has only moved down 25 points when the 45° angle from an extreme low on a weekly or a monthly chart is broken, then it is in a very weak position because it is so far above the half-way point on its price movement, already having squared out the time period with price.

Weakness in a stock develops when it breaks the 3/4-point, the 2/3-point, the 1/2-point, etc., but the position on the timing angles from the bottom tells you still more about the weak position. A stock shows its first weakness when it breaks the first important angle coming up from the last bottom in the final run in a bull market.

WHEN A STOCK IS IN THE STRONGEST POSITION:

A stock is always in the strongest position coming up from a bottom when it is holding above the very acute angles on the daily, weekly or monthly charts, especially on monthly and weekly charts.

As long as a stock holds above the angle of 2 x 1 (a gain of 2 points per day) on the daily chart, it is in a very strong position as far as the bottom is concerned. In fact, it is always in a strong position on the daily as long as it holds above the 45° angle. The same applies to weekly and monthly charts, which are the most important trend indicators.

I have found that the stocks which have the biggest advances are those that always hold above the angle of 2 x 1 on the monthly chart or gain 2 points per month for a long period of time. I have seen stocks rest 10 or 15 times on the angle of 2 x 1 and never break it until they have advanced 100 points or more. In this way a stock stays ahead of time and stays within the square of time by being far above the angle of 45°, and therefore is in a very strong position. But the time must come when the cycle has run out and the main trend begins to change from a bull market to a bear market—then the breaking of the angle from the last bottom shows a change in trend.

Another indication that a stock is in a strong position is when it advances and moves up above the half-way point of the previous price movement and then holds the half-way point, that is, advances above it and then reacts and fails to break under it. This is just the same as resting on a 45° angle and indicates a very strong position.

STRONGEST BUYING AND SELLING POINTS:

The cinch buying point is when a stock rests on a 45° angle, placing a stop loss order below it.

Another point to buy is on the half-way point of the price movement, placing a stop loss order under the half-way point.

When the main trend is up, it is also safe to buy when a stock reacts to the angle of 2 x 1 (a gain of 2 points per time period) on the weekly or monthly chart.

REGARDING ANGLES OR CROSSING LINES:

Remember, when any stock breaks under the 45° angle from the extreme low point of a move on the daily, weekly or monthly chart, it is then in a very weak position and indicates a decline to the next angle. However, when a stock can

regain the 45° angle, it is in a stronger position.

The same rule applies to a 45° angle drawn up from any top. When a stock crosses the angle on the daily, weekly or monthly and stays above the 45° angle or any other angle to the left of the 45° angle, it is in a very strong position.

After a stock once drops below or gets above any important angle and then reverses its position by getting back above the angle or dropping back below it, it changes the trend again.

WHEN A STOCK IS IN STRONG POSITION FROM BOTTOM AND IN WEAK POSITION FROM TOP:

A stock is in a strong position from the bottom when it is keeping above the angle of 45° or the angle of 2×1 , but at the same time it can be in a weak position when it rallies up and strikes against a 45° angle or the angle of 2×1 coming down from the top—then it is a short sale until it can cross these angles or cross previous tops. When it breaks the angles from the bottom, it is in a weak position and indicates lower.

A stock can be in a strong position from the top and in a weak position from the bottom, that is, it may cross some important angles from the top after a long period of time, but at the same time may break under the 2×1 angle or 45° angle from the bottom, which would indicate that it is in a weak position and getting ready to go lower.

WHEN ANGLES FROM EXTREME TOP ARE CROSSED:

The 45° angle drawn from the extreme high point of a stock is most important and when it is crossed, a major move may be expected. For example:

On the weekly chart of the Dow-Jones Industrial Averages, note the 45° angle moving down from 386, the high of September 3, 1929. January 12, 1935 was 279 weeks from the 1929 top. Taking 279 from 386, we get 107, the price at which the angle of 45° would cross. These Averages advanced to 108½ in the week ending January 12, 1935—then reacted to 100 in the week ending February 9. This was the first time that they had held within one-half point of this angle and the first time that they had ever reached it since the top was made. During the week ending February 16, 1935, the Averages crossed the 45° angle at 103 for the first time, and during the week ending February 23, 1935 advanced to 108, where they hit the angle of 45° moving up from the low of 86½ in September, 1934, and also hit the angle of 2×1 coming up from the low of July 8, 1932. This was a strong resistance point and the Averages reacted to 96 in the week ending March 18, 1935, where they rested on the 45° angle from the 1929 top and also where the 3×1 angle (a gain of 1/3 point per week), from September, 1929 coming up from "0" crossed the angle of 45° coming down from the 1929 top. This was a strong support point for a change in trend. The advance started and the Averages moved up to new high levels. This proves the importance of angles, especially the 45° angle drawn from any extreme top, and the point at which any other angle crosses the 45° angle.

Watch the 45° angle from 1929 top when it reaches "0" or when it is 386 weeks down from the top. This will be in the latter part of January, 1937. Note what happens at that time.

ANGLES FOR SEMI-WEEKLY CHART

The semi-weekly chart is a great help at the end of extreme advances or

extremes declines. By applying all of the rules and using the geometrical angles from tops and bottoms on the semi-weekly chart, you will often get an indication of a change in trend two to three days before a change in trend is shown on the weekly chart.

A change in trend on the semi-weekly chart is of greater importance than a change in trend on the daily chart. It is much better to rely upon this chart than on the daily chart when markets are in a narrow trading range.

ANGLES FOR NEW LISTED STOCKS

Years of experience and research, which has cost me a large amount of money have enabled me to develop a method that will account for all market movements and give rules to determine the trend from any top or bottom.

It is important to know how to determine the trend when a stock is first listed on any exchange. When a stock has never fluctuated before, we have no top or bottom to draw angles from. Therefore, in order to determine the trend, we use the square of 90, which is 90 up and 90 across, and put all the natural angles on, like the Pattern Chart. As we have said before, the square of 90 is very important because it is one-quarter of a circle of 360° , and as 90° or the vertical angle is the greatest angle that can be used, all of the other angles are found between "0" and "90".

If a new stock opens at 18 or any point below 22½, then you could make out a square of 22½ to determine the position of the stock on angles. If the stock opened at 36 or any point between 22½ and 45, you could make up a square of 45. If it opened at 50 or between 45 and 67, you could make up a square of 67½. However, you could place any stock opening at any price below 90 in the square of 90 and get its proper position and strength or weakness on angles. If the stock opened at 100 or above 90 and under 135, you could make up a square of 135, or could make another square of 90 numbering from 90 to 180.

You could start a monthly chart on a square of 90 at the price where the stock opens or trading begins, as shown on U. S. Steel. (Refer to Special Analysis of U. S. Steel.) After the stock breaks any of these natural angles drawn from "0", it is just the same as breaking under an angle drawn from a bottom. When it crosses any of the angles drawn down from "90", it is just the same as crossing an angle from a top, as you can see by experimenting with U. S. Steel or any other stock, but always consider price resistance levels and how much the stock is up or down from the bottom or top. You can determine the first change in trend by the 3-day or semi-weekly Chart, daily chart, and weekly chart by bringing up the important Geometrical Angles from any higher or lower bottom as the market movements develop.

QUICK CALCULATION OF ANGLES

It is not necessary to draw these angles from a point a long way back. You can make the calculation and determine where they cross. For example: Suppose in 1900, in the month of January, a stock made bottom at 15, and we wish to calculate where the 45° angle will cross 10 years later in January, 1910. The 45° angle rises at the rate of one point per month-- then 10 years would be 120 points or months-- add this to 15 at the bottom-- then the 45° angle would cross at 135 in January, 1910. All of the other angles may be calculated a long period back in the same way.

ANGLES SELDOM USED

3 x 2 ANGLE: This angle of 3 x 2 on the left side of the 45° angle rises at the rate of 6 points in 12 months. A stock must show a gain of 3/4-point per month in order to keep above this angle. This angle can be used when other important angles from the bottom have spread far apart, as it will show the position and resistance or support point between the other angles.

LATITUDE AND LONGITUDE

On all charts—daily, weekly or monthly—the price must move up or down on the vertical angles. Therefore, the price movement is the same as latitude. You should begin with zero or "0" on any chart—daily, weekly or monthly—and draw the important angles and resistance levels across, which measure latitude.

Next, number the time points in days, weeks or months across, and draw the horizontal angle at each important natural angle, such as, 11½, 22½, 33½, 45, 56½, 67½, 78½, 90, 101½, 112½, 120, etc. Then you will know when price reaches these important angles and meets resistance.

Longitude measures the time running across the chart, as it moves over each day, week or month. Therefore, you must keep your chart numbered from each important top and bottom in order to get the time measurements according to angles. These important angles, such as, 11½, 22½, 33½, 45, 56½, 60, 67½, 78½, 90, etc. from each bottom and top will show you where the strongest resistance in price and time takes place. These angles prove the parallel or crossing point. Study past records and see what has happened when prices on monthly charts reached these important angles or time periods.

For example: 90 points up in price from "0" we draw an angle horizontally across the chart. Then 90 days, weeks or months, going to the right across the chart, we draw a vertical angle up, which will cross the horizontal angle at 90 and prove the square. By keeping all these angles up and understanding them on your charts, you will know when important time cycles are running out.

If the price of a stock at 90 comes out on the 90th day, week or month, it will meet strong resistance because it has reached the square of price with Time. It is at the same latitude or price and the same longitude or time period. You can always put the square of 90 on a chart—either daily, weekly or monthly—and use the natural angles, but I advise only using this on the weekly and monthly. You can begin this square of 90 from any bottom or top, that is, going up 90 points, or from the natural points, which are 90, 135, 180, but you must not fail to square the extreme low and high price as well as the second and third lower tops and higher bottoms with Time.

RULE FOR KEEPING TIME PERIODS ON CHARTS

It is very important that you keep the time periods on all of your charts, carrying them across from the bottom and top of each important move in order to check up and know that you have your angles or moving-averages at the correct point and to see where major and minor cycles indicate changes in trend.

TIME PERIODS FROM BOTTOMS: When a stock makes bottom one month and then the following month makes a higher bottom and a higher top, or anyway, after it makes a higher bottom and rallies for one month or more, you can start numbering from that bottom. The month that it makes the low be-

longs to the old or downward movement and is the last move down. You count the first month up as one and then number across on the 1/2-inch squares, running them across, adding four each time.

For example: If a stock has made bottom and advanced 50 points, you look down at the bottom of the chart and find that you are on the 25th month- then the angle of 2×1 , moving up 2 points per month, would cross at 50, while the 45° angle, moving up one point per month, would be at 25, and if the stock broke back under 50 the following month, it would be falling under the angle of 2×1 and indicate a further decline. Now, if you had an error on the chart in the timing or numbering across from the bottom, then the moving-average line or angle would not come out correctly.

TIME PERIODS FROM TOPS:

After a stock has advanced and made an extreme high and reacted for a few days, a few weeks, or a few months, and you start putting on the angles from the top down, you must then begin to number the time periods across from the top. Apply the same rule for the top: The month, week or day that a stock makes extreme high finishes the upward movement and is not to be counted. You can count the number of days, weeks or months moving across after that, allowing the top month to be "0", the next month, week or day over to be "1", adding 4 across on the squares to get the correct position. If this Time Period is carried across on all the charts correctly, then you can always check up and find out if you have made any mistake in bringing down the angles or moving-average lines.

For example: After a stock has declined 75 points, either on a weekly or monthly chart, the angles move down the same, except where the spacing is different. Assuming that the spacing is one point per one-eighth inch, after it has moved down 75 points and all the angles are drawn down from the top, there may be an error in the angle of 2×1 because your ruler may have slipped and you may not have placed it correctly after it is down a distance from the top. Now, in order to prove exactly where the angle of 2×1 comes out, you determine the number of time periods there are. If 40 days, weeks or months have been required to decline the 75 points, the angle of 2×1 moving down 2 points per unit of time, would be down 80 from the top. If you find that this angle does not cross at 80, then you know that you made an error and should correct it.

This is a simple way to always know when the angles or moving-lines are correct because you simply add the movement to the bottom and subtract it from the top. Suppose the price referred to above, when the stock has declined 75 points, was 150, then subtracting 80 from the top at 150, the angle would cross at 70, and the price of the stock down 75 points would be at 75, therefore it would be above the angle of 2×1 from the top and in position for a rally if the time cycle indicated it.

POINTS FROM WHICH TO NUMBER TIME PERIODS

The most important point on the monthly high and low chart to carry the time period from is from the extreme low of the life of a stock and also from the date of incorporation or from the date trading in the stock began on the New York Stock Exchange. From the extreme low point the time period should always be carried across on the chart just the same as the important angles should be continued right along for years.

The next important point to number from is a second or third higher bottom,

but you should not consider a bottom established until the market has held up or advanced three to four months, then commence numbering from that bottom if it appears to be important. For example:

U. S. STEEL was incorporated February 23, 1901. Numbering the months across you will note that February, 1931, was 360 months, or 30 years, from the date of incorporation. Then start a new cycle and begin numbering across from "0". This will be working out the second cycle or circle of 360°.

The next important point is the extreme low of $8\frac{1}{2}$ made May 14, 1904. On the monthly chart carry the numbers across from this bottom, because it is the lowest bottom and therefore the most important. Note this 30-year cycle or 360 months ended May, 1934.

The next important point to number from and draw the angles from, is the low of $21\frac{1}{2}$ in October, 1907, the first higher bottom. Then, the next important is the third higher bottom made in February, 1915. Always draw the angles and number the months across from any other important bottoms where campaigns start.

Use this same rule at tops. After top is reached and the trend turns down, then carry the time numbers across from the top, but after any top is crossed or bottom is broken that you are numbering from, then do not count that top or bottom of importance to number from; except to determine a time period on another cycle 3, 5, 7, 10 or 20 years ahead. Tops that stay for a long time without being crossed are always the most important to carry the Time Periods from. The extreme high reached by a stock is always most important until that high is crossed- then the next high point made on a secondary rally, which is always a lower top; is the next most important top to number from. For example:

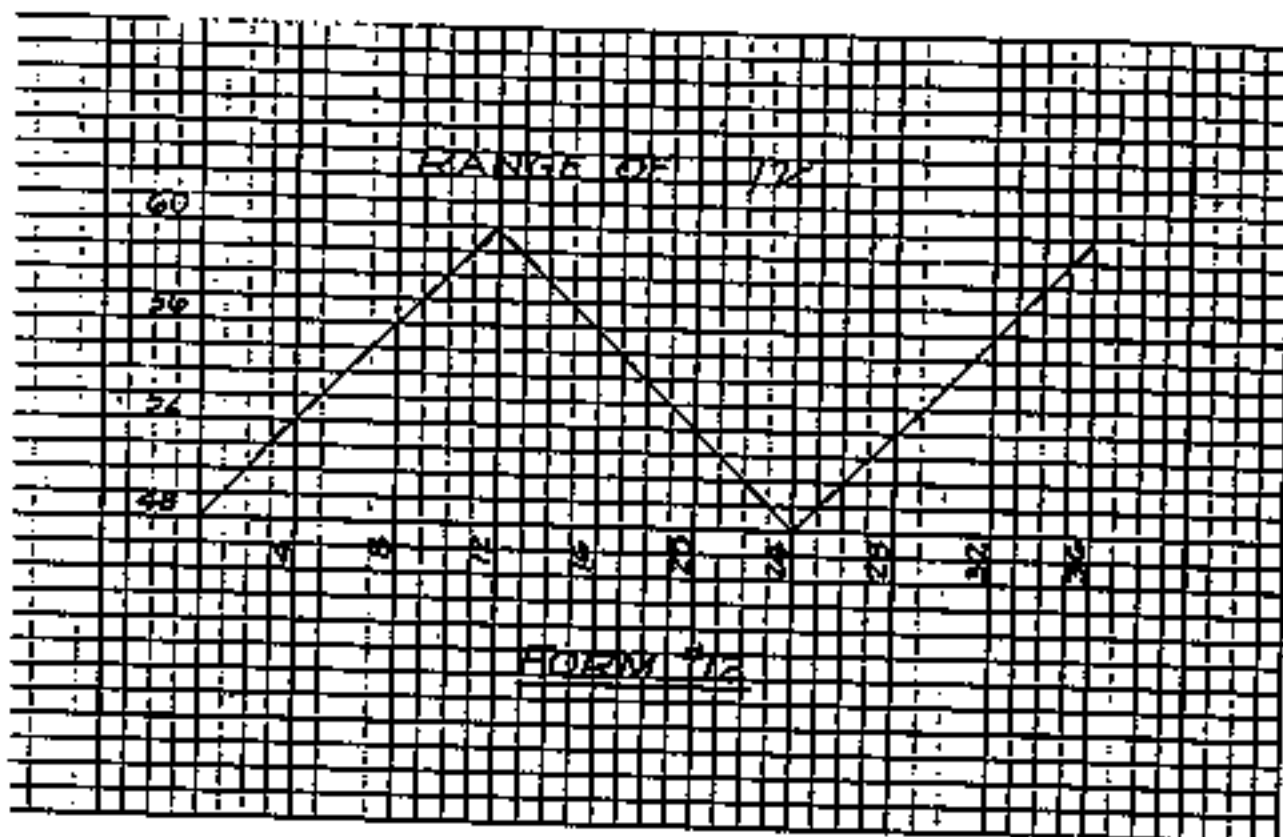
On U. S. Steel you would carry the monthly measurement across first from the high in April 1901- then from the extreme high in October, 1909, and next from the high in May, 1917- then from the final high in September, 1929, being the most important to measure from, and also number from the April, 1930 top.

INDUSTRIAL AVERAGES: The Dow-Jones 30 Industrial Averages reached extreme high on September 3, 1929-- then declined sharply in the panic, reaching low in November 1929-- from this low there was a rally to April, 1930, which was the last high and very important to number from because it was a secondary top, the last rally in a bull market. After final low of the bear market was reached on July 8, 1932, a sharp rally followed to September, 1932, when top was reached-- then a slow decline followed, reaching bottom in late February and early 1933, making this a secondary higher bottom, from which stocks advanced to new high levels. The bottom in 1932 is the most important to number from and the next bottom of March, 1933 is next in importance.

Apply this same rule to weekly and daily bottoms and tops. Discontinue the time periods when any minor top or bottom is exceeded and carry only the main figures on time periods from important tops and bottoms as long as they remain unbroken.

The rule for discontinuing the use of tops and bottoms for Time Periods is: When a bottom or top is exceeded by three points, then discontinue the time period from that bottom or top.

Always note the number of months between extreme high and between extreme low points and note what angle the tops and bottoms come out on.



SQUARING THE PRICE RANGE WITH TIME

This is one of the most important and valuable discoveries that I have ever made, and if you stick strictly to the rule, and always watch a stock when Price is squared by Time, or when Time and Price come together, you will be able to forecast the important changes in trend with greater accuracy.

The squaring of Price with Time means an equal number of points up or down balancing an equal number of time periods—either days, weeks or months. For example: If a stock has advanced 24 points in 24 days, then moving the 45° angle or moving-average-line up at the rate of one point per day, the timing line or time period and the price of the stock are at the same level and the stock is resting on a 45° angle and you should watch for an important change in trend at this point. If a stock is to continue uptrend and remain in a strong position, it must continue to advance and keep above the angle of 45°. If it breaks back under this angle, then it is out of its square on the bear side of the 45° angle and in a weaker position. When you are squaring out Time on a daily chart, look at the weekly high and low chart and monthly high and low chart and see if the stock is in a strong position and has yet to run out the time periods, because on a daily chart it has to react and then recover a position, squaring its price many times, as long as the weekly and monthly point up. Market corrections or reactions are simply the squaring out of minor time periods and later the big declines or big advances are the squaring out of major time periods.

SQUARING THE RANGE: Refer to Form #12, where a range of 12 points is shown from 48 low to 60 high. Now, suppose a stock remains for several weeks or several months, moving up or down, in this range, never getting more than 12 points up from the bottom and not breaking the bottom. We start the 45° angle from the bottom of 48 and move it up to the top of the range to 60, then when we see the stock is holding this range and not going higher, we move the 45° angle back to the bottom; then back to the top of the range again, moving it up or down over this range until the stock breaks out into new low levels or new high levels. You will find that every time the 45° angle reaches the top of this range or the bottom of this range, there is some important change in trend of the stock.

You can also use the angles of 2 x 1 to the right of the 45° angle and the 2 x 1 to the left as they again divide the Time Period into two equal parts and are of some value.

If a stock finally moves out of this range on the up side, then the angles would begin at the new and higher bottom and move up, but from the point where the stock went into new high, or from any important bottom made while it was in the range, especially the last bottom that it made, which would be most important, you should then begin an angle at that bottom and continue on up again; watch when this angle is broken or when Time is squared out again with Price, which would be important for another change in trend, either major or minor.

THREE WAYS TO SQUARE TIME AND PRICE:

We can square the Range, that is, the number of points from extreme low to extreme high, with Time— then square the extreme low point with Time— and square the extreme high point with Time. When the market passes out of these squares and breaks important angles, the trend changes up or down.

- 1 - The range that a stock makes between extreme high and extreme low can be squared so long as it remains in the same price range. If the range is 25 points, it squares with 25 periods of Time— days, weeks or months. Continue to use this time period as long as it stays in the same range.
- 2 - SQUARING TIME WITH BOTTOM OR EXTREME LOW PRICE:
The next important Price to square with Time is the lowest price or bottom of any important decline. For example: If the bottom of a stock is 25, then at the end of 25 days, 25 weeks or 25 months, Time and Price are equal. The watch for a change in trend is based on its bottom or lowest selling price. As long as a stock continues to hold one bottom and advances, you can always use this time period running across and continuing the time period, noting every time it passes out of the squares. Watch especially when the stock reaches the third square, the fourth square, and again the seventh and ninth squares of its time period. These squares only occur frequently on the daily or weekly charts, as the monthly, in most cases, would move out of a range, up or down, before it squared a bottom as many as 7 or 9 times. However, this does sometimes happen when a stock is in a narrow range for many years.
- 3 - SQUARING TIME WITH TOP OR EXTREME HIGH PRICE:
The other important point to square Time with is the extreme high price of a stock. The Time period must be carried across from the high of the daily, weekly or monthly, and the square of the top price in Time must be noted and watched for a change in trend. If the top of a stock is 50, then when it has moved over 50 days, 50 weeks or 50 months, it has reached its square in Time and an important change is indicated. This can be determined by the position of the angles from top and bottom. For example:

Dow-Jones Industrial Averages— The high of 386 on September 3, 1929 would require 386 calendar days to equal the Price in Time. This occurred on September 23, 1930. Look at the chart and note how the trend changed and turned down around that time. Then, on October 14, 1931, it run out this period again— and again November 4, 1932, November 25, 1933, December 16, 1934, and January 6, 1935. Look up these dates and you will see that important changes in trend occurred on the Daily Chart when this time period of 386 days balanced the price of 386.

Both major and minor tops and bottoms on all time periods must be watched as they square out right along. Most important of all is the extreme high point on the monthly high and low chart. This may be very high and work out a long time period before it squares the top, in which case you have to divide the price into 8 equal time periods and watch the most important points, like $1/4$, $1/3$, $1/2$, $3/4$, but most important of all is when Time equals Price.

When you are watching the position of a stock after it has squared out from a bottom or a top, always look up the time period and the angles from the opposite direction. If the market is nearing a low point, squaring out a top, see how its relation is to the bottom as it might be in the second or third square

period from the bottom, which would be a double indication for a change in trend.

SQUARING WEEKLY TIME PERIODS:

The year contains 52 weeks and the square of this in Time and Price is 52 by 52. Therefore you can make up a square of 52 wide and 52 high; put on all of the angles from "0"; then chart the weekly high and low prices of any stock in this square. For example: If the low price of a stock is 50, then the top of this weekly square would be 52 added to 50, which makes 102 as top of the square. As long as the stock stays above 50 and moves up, it will be working in the weekly square of 52. On the other hand, if the stock makes top and works down, you would make up a weekly square 52 points down from the top and 52 over to get the time period.

You can take the past movement of any stock, put on a square of 52 by 52, and study the movement, noting 13 weeks or one-fourth, 26 weeks or one-half, and 39 weeks or three-fourths points on time, and the changes in trend which take place when the stock reaches these important Resistance Points in Time and Price. You would watch for a change in trend around these time periods.

SQUARING MONTHLY TIME PERIODS:

At the time a stock breaks a 45° angle, if it is selling at 135 on the 135th month, it is breaking a doubly strong Resistance Level—a strong angle and a natural Resistance Level. This would be Time and Space balancing at Resistance Levels or geometrical angles and would indicate a big decline to follow. Reverse this rule at the end of a bear campaign.

On a monthly chart twelve months completes a year, therefore the square of 12 is very important for working out time periods on the monthly chart. The square of 12 is 144 and important changes often occur on even 12 months' periods from a bottom or top of a stock. It will help you if you use the Resistance Levels on prices of the even 12's, noting 24, 36, 48, 60, 72, 84, 96, 108, etc. Watch how the stock acts on angles when it reaches these important Resistance points in Price.

PRICE AHEAD OF TIME

Why do stocks often cross the 45° angle on the daily, weekly or monthly chart, then have an advance for a short period of time, decline and rest on the same 45° angle? Because when the stock crosses the 45° angle the first time, it has not run out or overcome the square of Time with Price. Therefore, on the secondary reaction, when it rests on the 45° angle, it is at a time when the stock has reached the square of distance in Time. After that a greater advance follows.

Reverse this rule at the top of a bull market. When a stock breaks under the 45° angle a long distance from the base or bottom, it is most important. Many times a stock will rest on the 45° angle in the early stages of an advance, then later, on a reaction, rest on it again; then have a prolonged advance, react and rest on the 45° again, and then advance to a higher level; then break the 45° angle the next time, which places it in an extremely weak position because it is so far away from the base and so much time has elapsed since the stock made low. Don't forget—it is most important when angles are broken on the monthly and weekly charts.

This accounts for stocks that have a sharp, quick decline from the top and then advance and make a slightly higher top or a series of slightly lower tops, and work over until they overcome the square of the price range at a comparatively high level and break the 45° angle, then a fast decline follows.

STRONGEST ANGLES FOR MEASURING TIME AND PRICE

- 90° ANGLE: Why is the 90 degree angle the strongest angle of all? Because it is vertical or straight up and straight down.
- 180° ANGLE: What is the next strongest angle to the 90° angle? The 180° angle because it is square to the 90° angle, being 90° from the 90° angle.
- 270° ANGLE: What is the next strongest angle to the 180° angle? The 270° angle because it is in opposition to 90, or 180° from the 90° angle, which equals 1/2 of the circle, the strongest point. 270 months equals 22½ years, which is 1/2 of 45.
- 360° ANGLE: What is the next strongest angle after 270? It is 360°, because it ends the circle and gets back to the beginning point and is opposite 180° or the half-way point, or the angle which equals 1/2 of the circle.
- 120° AND 240° ANGLES What angles are next strongest to 90, 180, 270, and 360°?
 Answer: 120° and 240° angles, because they are 1/3 and 2/3 of the circle. 120° is 90 plus 30, which is 1/3 of 90. 240 is 180 plus 1/3 or 60, which makes these strong angles, especially strong for measurements of time.
- 45° - 135° - 225° - 315°: What angles are next in strength?
 Answer: 45° angle, because it is 1/2 of 90,
 135° angle, because it is 90 plus 45,
 225° angle, because it is 45 plus 180, and
 315° angle, because it is 45° from 270.
 The angle of 225° is 180 from 45 and the angle of 315° is 180 from 135.
- CARDINAL & FIXED CROSS: The angles of 90, 180, 270, and 360 form the first important cross, known as the Cardinal Cross. The angles of 45, 135, 225, and 315 for the next important cross, which is known as the Fixed Cross. These angles are very important for the measurements of time and space or price, and volume.
- 22½° - 67½° - 78¾°: Why is the angle of 22½° stronger than 11¼°? Because it is twice as much, being the same reason that a 45° angle is stronger than a 22½° angle. Again, the angle of 67½° is 1½ times 45, therefore quite strong when anything is moving up toward 90°. 78¾° is stronger than 67½°, because it is 7/8 of 90, and therefore one of the strongest points before 90 is reached—important to watch both on time, price, and volume. Many stocks have important moves and make tops or bottoms around the 78th to 80th day, week or month, but don't overlook 84 months or 7 years, a strong time cycle.
- DIVISION OF \$1: 1/8-POINTS Why are the angles of 1/8 of a circle most important for time and space measurement? Because we divide \$1 into 1/2, 1/4, and

$1/8$ parts. We use 25 cents or one quarter, 50 cents or half dollar, and long years ago we had $12\frac{1}{2}$ cent pieces. While the most important figures of our basic of money are the four quarters, we do use the $1/8$ part or $12\frac{1}{2}$ cents in all calculations. Stock fluctuations are based on $1/8$, $1/4$, $3/8$, $1/2$, $5/8$, $3/4$, $7/8$ and the whole figure. Therefore, any price measurement as well as time will work out closer to these figures when changed into angles of time than $1/3$ or $2/3$ points for the simple reason that the fluctuations moving in $1/8$ proportion must come out closer to these figures.

Figuring \$100, or par, as a basis for stock prices and changing these prices to degrees, $12\frac{1}{2}$ equals 45° , 25 equals 90° , $37\frac{1}{2}$ equals 135° , 50 equals 180° , $62\frac{1}{2}$ equals 225° , 75 equals 270° , $87\frac{1}{2}$ equals 315° , and 100 equals 360° . For example:

When a stock sells at 50 on the 180th day, week or month, it is on the degree of its time angle.

On February 1, 1915, U. S. Steel made a low at 38, which is closest to a price of $37\frac{1}{2}$, which is $3/8$ of 100 and equals 135° angle. Steel was 14 years or 168 months old on February 25, 1915, and hit the angle of 135° , which showed that Steel was behind time, but that it was in a strong position, holding at 38 above the 135° angle or the price of $37\frac{1}{2}$.

When Steel reached 200, it equalled 2 circles of 360° . When it advanced to $261\frac{1}{2}$, it was closest to $62\frac{1}{2}$ in the third 100 or nearest the 225° angle or $5/8$ point, which is the strongest angle after it crossed the half-way point at 250 or 180° angle.

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November, 1935

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THE BASIS OF MY FORECASTING METHOD FOR GRAIN

Mathematics is the only exact science. All power under heaven and on earth is given unto the man who masters the simple science of mathematics. Emerson said: "God does indeed geometrize." Another wise man said: "There is nothing in the universe but mathematical points." Pythagoras, one of the greatest mathematicians that ever lived, after experimenting with numbers and finding the proofs of all natural laws, said: "Before God was numbers". He believed that the vibration of numbers created God and the Deity. It has been said, "Figures don't lie." Men have been convinced that numbers tell the truth and that all problems can be solved by them. The chemist, engineer and astronomer would be lost without the science of mathematics.

It is so simple and easy to solve problems and get correct answers and results with figures that it seems strange so few people rely on them to forecast the future of business, stocks and commodity markets. The basic principles are easy to learn and understand. No matter whether you use geometry, trigonometry, or calculus, you use the simple rules of arithmetic. You do only two things: You increase or decrease.

There are two kinds of numbers, odd and even. We add numbers together, which is increasing. We multiply, which is a shorter way to increase. We subtract, which decreases, and we divide, which also decreases. With the use of higher mathematics, we find a quicker and easier way to divide, subtract, add and multiply, yet very simple when you understand it.

Everything in nature is male and female, white and black, harmony or in-harmony, right and left. The market moves only two ways, up and down. There are three dimensions which we know how to prove—width, length and height. We use three figures in geometry—the circle, the square, and the triangle. We get the square and triangle points of a circle to determine points of time, price and space resistance. We use the circle of 360 degrees to measure Time and Price.

There are three kinds of angles—the vertical, the horizontal, and the diagonal, which we use for measuring time and price movements. We use the square of odd and even numbers to get not only the proof of market movements, but the cause.

HOW TO MAKE CHARTS

Charts are records of past market movements. The future is but a repetition of the past. There is nothing new. As the Bible says-- "The thing that hath been, it is that which shall be." History repeats and with charts and rules we determine when and how it is going to repeat. Therefore, the first and most important point to learn is how to make charts correctly because if you make an error in the chart, you will make an error in applying

the rules to your trading.

YEARLY CHART

You should keep a yearly high and low chart, that is, recording the extreme high and the extreme low price made during the calendar year on one line. The spacing for the price can be used 1 cent to each $\frac{1}{8}$ inch or $\frac{1}{2}$ cent on daily high and low charts.

MONTHLY CHART

You must always keep up a monthly high and low chart, which is the most important chart of all in determining the main trend. This chart records the extreme high and extreme low price for the calendar month on one line, and each space or $\frac{1}{8}$ inch on the cross-section chart paper should represent one point or one cent per bushel.

WEEKLY CHART

The next and one of the very important charts to keep is a weekly high and low chart. In Wheat, Corn, Rye and Oats you should use $\frac{1}{8}$ inch to represent $\frac{1}{2}$ cent, or 2 spaces to represent 1 full cent, or 4 cents for each one-inch space.

WEEKLY MOVING-AVERAGE OR MEAN POINT

To get a Weekly Moving-Average, we take the extreme low for the week and the extreme high for the week and divide by 2, getting the half-way or mean point for the week. This can be recorded on the weekly high and low chart or on a separate chart, recording the Weekly-Moving Average with a dot and using one line on the chart for each week. Importance of this Weekly Mean Point will be explained later.

DAILY CHART

When you are trading in Grains, you should always keep up a daily high and low chart, but for study purposes it is enough to keep up the Weekly and Monthly Charts, which give you the main trend. The Daily Chart shows the minor trend and shows a change in trend much oftener than any of the other charts, but the indication does not last as long nor run so far. This chart should be kept up the same as the others, except when grains are selling above \$1.50 per bushel the spacing should be 1 cent per bushel for each $\frac{1}{8}$ inch. When Grains are in a normal trading range then the spacing should be $\frac{1}{2}$ cent for each $\frac{1}{8}$ inch.

No spaces are skipped on the Daily Chart for holidays or Sundays, therefore the time period is for actual market days and not calendar days. However, you should carry the calendar days along at least every two weeks, as later, under rules for Time Periods for change in trend, you will find that it is necessary to check up and know when the market is 30, 40, 60, 120, 135, etc. days from a top or a bottom, which means calendar days, the exact measurement of time for the daily chart. Often the Daily Chart on actual daily movements comes out on an exact mathematical angle of time measurement at the same time the calendar days come out on exact time measurement, making it a doubly important point for change in trend.

GEOMETRICAL ANGLES

After long years of practical experience, I have discovered that Geometrical Angles measure accurately Space, Time, Volume and Price.

Mathematics is the only exact science, as I have said before. Every nation on the face of the earth agrees that 2 and 2 make 4, no matter what language it speaks. Yet all other sciences are not in accord as mathematical science. We find different men in different professions along scientific lines disagreeing on problems, but there can be no disagreement in mathematical calculation.

There are 360 degrees in a circle, no matter how large or how small the circle may be. Certain numbers of these degrees and angles are of vast importance and indicate when important tops and bottoms occur on Grains as well as denote important Resistance Levels. When once you have thoroughly mastered the Geometrical Angles, you will be able to solve any problem and determine the trend of Grains.

After over 40 years of research, tests and practical applications, I have perfected and proved the most important angles to be used in determining the trend of the Grain market. Therefore, concentrate on those angles until you thoroughly understand them. Study and experiment with each rule I give you, and you will make a success.

We use geometrical angles to measure Space and Time periods because it is a shorter and quicker method than addition or multiplication, provided you follow the rules and draw the angles or lines accurately from tops and bottoms or extreme highs and lows. You may make a mistake in addition or multiplication, but the geometrical angles accurately drawn will correct this mistake. For example: If you should count across the bottom of your chart 120 spaces, which represents 120 days, weeks, or months, then you begin at 90° and number vertically on your chart up to 120 - then from this top point draw a 45-degree angle moving down, this will come out at 90° on 120 points over from the beginning. If you have made a mistake in numbering, this will correct it.

Angles drawn on a chart always keep before you the position of the option and its trend whereas if you had a resistance point on time written down, you might mislay it or forget it but these angles are always on the chart in front of you.

These angles or moving-average trend lines correctly drawn will keep you from making mistakes or misjudging the trend. If you wait and follow your rules, these angles will show you when the trend changes.

The moving-average as commonly used is obtained by taking the extreme low price and the extreme high price of the calendar day, week or month, and dividing it by two to get the mean or average price for the day, week or month, and continuing this at the end of each time period. This is an irregular movement in spaces or points per week because at one time it may move up 5 cents per week and at another 10 cents per week, while the time period is a regular unit. Therefore, geometrical angles, which are really moving averages, move up or down at an uniform rate from any bottom or top on a daily, weekly or monthly chart.

HOW TO DRAW GEOMETRICAL ANGLES

There are three important points that we can prove with mathematics or geometry: the Circle, the Square, and the Triangle. After I have made the Square, I can draw a Circle in it using the same diameter, and with the different Squares produce the Triangle and the Square in the Circle.

The Angles or moving-trend-line overages measure and divide Time and Price into proportionate parts. Refer to Form "1" where I have drawn the square of 90. You will note that this is 90 high and 90 wide - in other words, 90 up and 90 across. It is the same as a square room which has a bottom or floor, a top or ceiling, and side walls. Everything has width, length, and height.

To get the strongest and most important points in this Square, I divide it into two equal parts by drawing a horizontal and a vertical line. Note angle marked "A", which divides each of the smaller squares into two equal parts and runs from "0" to "90" diagonally. This is a diagonal line moving on a 45° angle and divides the large Square into two equal parts. Then note angle "B" at "45" running horizontally across. This divides the Square into two equal parts. Note angle "C", which is a vertical line, running up from "45", which is one-half of "90". This crosses at the center or half-way point at "45", where the other angles cross, dividing the Square into two equal parts. Then note angle "D", which forms another 45° angle moving from the N.W. corner to the S.E. corner, crossing "45" at the exact half-way point. You see by this that if we draw the first line thru the center of the square, we divide it into two equal parts - then when we draw lines from the other directions, we divide it into four equal parts - then by drawing the two lines from each corner, we divide the Square into eight equal parts and produce 8 triangles.

As you look at this Square, it should be easy for you to tell with your eye where the strongest support point is, or resistance point is. It is at the center where all the angles cross. Four angles cross at this point, so naturally this would be a stronger support point than a place where only one angle crosses. I could divide each one of these smaller squares into four or eight equal parts by drawing angles in the same way. Later, when I give you the rules and examples, I will explain how to square the Range of option, that is, the difference between the extreme low and the extreme high prices, or the difference between any low point and any high point, and also how to square the bottom price. For example: If the top is 90, the square of 90 x 90 would represent squaring the Price by Time, because if we have 90 points up in Price, and we move over 90 spaces in Time, we square the Price and Time. Therefore, when the option has moved over 90 days, 90 weeks, or 90 months, it will be squaring its price range of 90.

PATTERN CHART FOR GEOMETRICAL ANGLES

The square of 90, or the Pattern Chart, shows all the measured angles that are important to use in determining the position of an option. These angles are as follows: 11, 13, 15, 16, 24, 30, 33, 37, 45, 52, 56, 60, 63, 71, 75, 81, 86, and 90 degrees.

It is not necessary to measure these angles with a protractor. All you have to do to get the angles correct is to count the spaces on the chart paper,

using 6 x 6 to the inch, and draw the lines or angles accordingly.

On the square of 90, which you will receive with these instructions, note how equal angles drawn from the top and from the bottom prove themselves by the point at which they cross. For example:

The angle of 8 x 1 drawn from "0" and the angle of 8 x 1 drawn from "90" down both cross at 45, 5-5/8 points over from "0" counting to the right. Then, the angle of 4 x 1 from "0" and the angle of 4 x 1 down from 90, you will notice, cross at 11; an 45, equidistant from the other angle and twice the measure. The reason why these angles prove this way is because the 45° angle or 45 points or degrees from "0" to 45 is one-half of 90. Therefore, parallel angles beginning at "0" going up and at 90 coming down, must cross on a 45° angle or at the gravity center.

HOW TO DRAW ANGLES FROM A LOW POINT RECORDED BY GRAIN

FIRST IMPORTANT GEOMETRICAL ANGLE

The first, and always the most important, angle to draw is a 45° angle or a moving-average that moves up one point per day, one point per week, or one point per month. This is a 45° angle because it divides the Space and Time Periods into two equal parts. As long as the market or option stays above the 45° angle, it is in a strong position and indicates higher prices. You can buy every time it rests on the 45° angle with a stop loss order 1, 2 or 3 cents under the 45° angle, but remember the rule - never use a stop loss order more than 5 cents away. Unless Grain is near the low levels or just starting in a bull market or selling at a very low price, I always use a stop loss order 1 cent under the 45° angle. If this angle is broken by one point, you will usually find that the trend has changed at least temporarily and the option will go lower.

An easy way to calculate accurately how to put on this 45° angle is: For example: If the time is 30 days, 30 weeks, or 30 months from the point where the option was bottom, then the angle or 45° must be 30 cents up from the bottom and would cross 30. This is one of the easiest angles to put on and one of the simplest to learn. You can beat the market by trading against the 45° angle alone if you stick to the rule--wait to buy on the 45° angle or wait to sell it against the 45° angle.

NEXT IMPORTANT ANGLE — 2 x 1

The angle of 2 x 1, or the moving-average which moves up at the rate of 2 cents per day, week, or month divides the space between the 45° angle and the vertical angle into two equal parts and measures 63 1/3°. That is why it is the next strongest and most important angle. As long as the option holds above this angle, it is in a stronger position than when it is resting on a 45° angle because it is a more acute angle. When an option breaks under this angle of 2 x 1, or 2 cents for each time period, then it indicates that it will go lower and reach the 45° angle.

Remember the rule of all angles: No matter what angle the option breaks under, it indicates a decline to the next angle below it.

THIRD IMPORTANT ANGLE - 4×1

This angle - which is still stronger as long as the price of Grain holds above it - is the angle which moves up 4 cents per day, week, or month. This angle is 4×1 , or 4 points of Space equal one period of Time. It measures 75° and divides the space between the angle of 1×1 and the 90° angle into two equal parts. Any option that continues to advance 4 cents per day, 4 cents per week, or 4 cents per month, and remains above this angle is in a very strong position as long as it stays above it, but when it breaks under, it indicates the next angle or next support point according to the position of the option in Time.

FOURTH IMPORTANT ANGLE - 8×1

The angle of 8×1 or the one that moves up 8 points per day, week or month, measures $82\frac{1}{2}^\circ$. As long as an option can hold above this angle on daily, weekly or monthly chart, it is in the strongest possible position, but when it reverses trend and declines below this angle, then it indicates a decline to the next angle.

You will note that with these four important angles we show the strong or bullish side of the market. All the time by dividing the Space with angles we are getting the half-way point or the gravity center of Time and Price.

These are all the angles you need as long as an option continues to advance and work up and stays above the angle of 45° or the moving-average of one point per day, week or month.

While there are 360 degrees in a circle and angles can form at any of these degrees, all of the important angles form between "0" and "90" because 90 is straight up and down and the most acute angle on which an option can rise. For example: The 45° angle divides the space from "0" to "90" in half. The angle of 135° is simply another angle of 45° because it is one-half of the next quadrant between 90 and 180. $22\frac{1}{2}$ and $31\frac{1}{2}$ in a circle are also 45° angles. Therefore all of the angles valuable in determining the trend of an option are found between "0" and "90" degrees. When we divide 90 by 2 we get 45, the most important angles to use - then divide it by 3 we get 30° and 60° angles, which are important to use for Time and Resistance Points.

KIND OF BOTTOMS FROM WHICH ANGLES OR MOVING-AVERAGE LINES ARE TO BE DRAWN

DAILY CHART

If an option has been declining for some time - then starts to rally (by rallying from a bottom it must make higher bottoms every day and higher tops) - then after a 3-day rally on the daily high and low chart, you can put on the 45° angle and the angle of 2×1 from the bottom or low point. As a rule, it will be necessary to put on these two angles at first. If this bottom holds and is not broken, then you can put on the other angles from the bottom.

WEEKLY CHART

If an option is declining and reacts for more than one week and continues down, we will say, for three weeks or more, then starts to rally and advances two weeks or more, you would start to put the angles on from the low point of

The decline, only using the angles above the 45° angle until the option again breaks under the 45° angle- after that you would use the other angles on the lower or bearish side of the Square.

WHAT TO DO AFTER THE 45° ANGLE FROM BOTTOM IS BROKEN

After an option makes top, either temporary or otherwise, and breaks under the 45° angle and starts moving down, then the first thing you do is to draw angles below the 45° angle, starting from the bottom or low point. Note examples: Angles Drawn on Monthly, Weekly and Daily Charts.

FIRST ANGLE ON BEAR SIDE OF THE SQUARE - 2×1

The first angle that you draw on the bear side of the Square is the angle of 1×1 or 2 points over and 1 point up, which moves at the rate of one-half point per day, week or month and measures $26\frac{1}{2}^\circ$. This is the first support angle which the option should reach after it breaks under the 45° angle. As a general rule, when the option reaches this angle, it will receive support and rally. Sometimes it will rest on it for a long period of time, holding on this angle and making higher bottoms. But when this angle of 2×1 , or moving-average of one-half point per day, week or month, is broken, then you must draw the next angle of 4×1 .

NEXT IMPORTANT ANGLE - 4×1

The next important angle on the bear side of the Square, which moves up at the rate of $1/4$ -point per day, is the angle of 4×1 , measuring 15° . It will be the next strong support angle which the option should get support and rally from.

NEXT ANGLE - 8×1

Then after the 4×1 angle is broken, the next important angle that you will place on your chart is the angle of 8×1 , which moves at the rate of $1/8$ -point per day, week, or month, and measures $7\frac{1}{2}^\circ$. This is often a very strong support angle. After an option has had a big decline, it will often rest on this angle several times or may make final bottom and start up from this angle, crossing other angles and getting back into a strong position again. Therefore this angle is important to use on a monthly or weekly chart after a prolonged decline.

This completes all of the angles that you will need to use from any bottom at any time.

HOW TO DRAW ANGLES FROM TOPS ON DAILY, WEEKLY OR MONTHLY CHARTS

POSITION UNDER 45° ANGLE DRAWN FROM TOP

After an option has made top and declined for a reasonable length of time, say, three days, three weeks or three months, breaking previous bottoms, then you start to draw angles from the top. Note Examples: Angles on Monthly Charts.

60° ANGLE FROM TOP

The first angle you draw is the angle of 45° or a moving-average which indicates a decline of one cent per day, week or month. As long as the option is below this angle, it is in the weakest position and in a bear market.

OTHER ANGLES

In many cases an option will start declining an average of one cent or ten cents per day, week or month, or 4 cents per day, week or month, or 1 cent per day, week or month, therefore you should put on all of these angles from the top, which move down faster than the angle of 45°.

WEAKEST POSITION

The option is in the weakest possible position when it declines and keeps under the angle of 8 x 1. It is in the next weakest position when it is dropping down at the rate of 4 cents per day, week or month, or under the angle of 4 x 1. It is in its next weakest position when it is dropping down under the angle of 2 x 1.

STRONGEST POSITION

The option is in a stronger position and indicates a better rally when it crosses the angle of 2 x 1; but this depends on how far it is down from the top and how far the angles are apart, as will be explained later under the rules.

CHANGING TREND

As long as an option is declining one point per day, week or month, or falling below or under the 45° angle, it is still in a bear market and in a very weak position. When an option rallies and crosses the angle of 45° after a prolonged decline, then you are ready to put on the angles on the other side of the 45° angle, which shows that the option is in a stronger position in a bear market and may be getting ready to change into a bull market.

POSITION ABOVE 45° ANGLE DRAWN FROM TOP

2 x 1 ANGLE FROM TOP

The first angle or moving-average you draw after the 45° angle from the top is crossed and after the option indicates that it has made a temporary bottom is the angle of 2 x 1, moving over 2 points and down one point, or 1/2 point per unit of time. This is moving down at the rate of 1/2 point per day, week or month.

4 x 1 ANGLE

The next is the angle of 4 x 1 which moves down at the rate of 1/4 point per day, week or month.

8 x 1 ANGLE

The next angle is the angle of 8 x 1 which moves down at the rate of one point every 8 days, 8 weeks or 8 months, or 1/8 point per time period.

STRONG POSITION

After the option has crossed the angle of 45° and rallied up to the angle of 2×1 , it will meet selling and react to some angle coming up from the bottom of the last move, but it is in a stronger position when it holds above this angle of 2×1 , and it is in the next strongest position when it crosses the angle of 4×1 . Crossing the angle of 8×1 , which is of least importance, it indicates that it is in a very strong position again from the top. You must always consider a movement coming up from the bottom and its position on angles from the bottom to determine its strength. It is important to consider the number of points it has moved up from the bottom and how many points it is down from the top.

This completes the forms of all the angles that you will need to use at any time from tops or bottoms. Practice putting these angles on tops and bottoms until you are thoroughly familiar with them and know that you are getting them absolutely accurate. Then you can begin to study the rules for determining the trend according to the position of the option on angles.

DOUBLE AND TRIPLE TOPS OR BOTTOMS

ANGLES CROSSING EACH OTHER

When there is a double bottom several days, weeks or months apart, you draw angles from these bottoms, which are near the same price levels. For example: From the first bottom draw a 45° angle and from the second bottom draw an angle of 2×1 - then when these angles cross each other, it will be an important point for a change in trend.

Apply this rule to double tops and triple tops in the same way. It is not necessary for the tops or bottoms to be exactly at the same price level, but near the same level. Remember, always draw 45° angles from all important tops and bottoms.

Angles coming down from tops and crossing angles coming up from bottom are very important for a change in trend when they cross each other.

GEOMETRICAL ANGLES OR MOVING-AVERAGE LINES DRAWN FROM "0"

When an option reaches bottom and starts up, you have been instructed to draw angles from this exact low point, which shows the support in time periods, but there are other angles which later on will be just as important, and sometimes more important than the angles drawn from the bottom of an option. These are the angles that begin at "0" or zero, and move up at the same rate that they move up from the bottom. The starting point must be on the same line that the bottom is made on as the time period begins from this bottom, but the angles move up from "0". These angles should be started every time an option makes a bottom, especially on weekly and monthly charts, and should also be carried up on important movements on the daily chart.

If an option makes low at 120 as shown on the chart, starting the 45° angle from "0", when will this angle reach 120? Answer: It will reach 120 in 120 days, weeks or months from the bottom or its starting point. In other words, in 120 days, 120 weeks, or 120 months, it will be up 120 from "0" and

at the price where the option made bottom. Then the angle will continue on up at the same rate, and later, when the option breaks under the 45° angle from the actual bottom made at 120 and breaks the other support angles drawn from the actual bottom at 120 the next important point for support will be the angle of 45° moving up from "Q". When this angle is broken, it is in the weakest possible position and indicates much lower prices, but this depends on how high the option is selling and how much it has declined at the time it breaks the 45° angle from "Q". These angles drawn from "Q", especially the 45° angle, prove when Price and Time are balancing or when the option is squaring out Time from its bottom.

"Q" ANGLES STARTING AT THE TIME TOP IS MADE

When an option reaches extreme top on a daily, weekly or monthly chart and the trend turns down, you should start an angle of 45° from "Q" moving up from the exact space and date that the top is made. This will prove the square of the Time period. It is very important when this angle is reached and indicates a change in trend. It is the last strong support and when broken, it will indicate much lower prices.

I have instructed you in each case to first draw the 45° angle from bottom, top, and from "Q" at bottom and top, but this does not mean that you must not use the other angles. All of the other angles can be used from "Q", but the 45° angle is the first and most important. After this angle is broken, then you can use the other angles. It is not necessary to carry all of them along until you need them, but on the monthly chart, after a long series of years, these other angles should be carried along when the option begins to approach the levels where they would be broken or where the option would rest on them and receive support.

45° ANGLE FROM "Q" TO TOP AND BOTTOM

When a 45° angle moving up from "Q" reaches the line or price of the bottom, it is very important - then again when it reaches the point of the extreme high price, it is very important for a change in trend.

You should carry 45° angles and other angles up from "Q" from all important first, second, and third higher bottoms, especially those where very much time has elapsed between these bottoms. You should also start the angle of 45° up from "Q" from the first, second, and third lower tops, especially those which show much time period elapsed. These angles are the most important to be carried on the weekly and monthly charts.

Never overlook keeping up the angles from "Q" because they will tell you when Time is squaring out with Price from tops and bottoms and will locate support angles or moving-average lines at a point on the bear side after the first 45° angle from a bottom is broken. You could not locate this support point in any other way by the angles from "Q".

You should go back over past records and bring up these angles and square out different tops and bottoms so that you can prove to yourself the great value of using these angles.

A. D. Gamm

GRAINS SCALE OF PRICES

Beans, Corn, Oats, Rye and Wheat

The scale for daily, weekly and monthly high and low charts is always the same. One cent per bushel to $1/8$ of an inch or 8¢ to one inch. In order to balance time or square time with price, you use 1 unit of time for 1¢ in price.

Example: May soy beans extreme low 67¢ July 27, 1939. This will require 67 calendar days or 67 market days to balance the price. Also, 67 weeks or 67 months to square or balance price and time.

Cocoa *OK 4/16*

Daily chart the scale is 15 points to $1/8$ of an inch and some of the charts are 10 points to $1/8$ of an inch on the daily. The weekly chart is 30 points to $1/8$ of an inch. The monthly charts some of them show 20 points to $1/8$ of an inch and some 30 points. To balance or square time divide the high or low price by the scale and you get the time required to balance or square the price.

Coffee

The daily high and low chart scale is 20 points to $1/8$ of an inch. This works best. The weekly charts are 20 and 30 points to $1/8$ of an inch, and the monthly charts are 30 points to $1/8$ of an inch. To get time balanced with price divide the high or low price by the scale.

Cotton

The scale for the daily chart is 10 points to $1/8$ of an inch. The weekly chart 15 points to $1/8$ of an inch. The monthly chart 20 points to $1/8$ of an inch. However, some of the monthly charts have a scale of 30 points to $1/8$ of an inch, and the trend and angles work good on either, because 20 points equals \$100 on one contract of 100 bales and the 30 point equals \$150 on one contract of cotton. Each fluctuation of one point in cotton equals \$5 on a contract. To balance the price with time divide the high or low price by the scale.

Eggs

Daily chart--the scale is 10 points to $1/8$ of an inch. The weekly chart is 25 points to $1/8$ of an inch which equals $1/4$ of 1¢ or 3¢ profit or loss of one contract. You have one chart with a scale of 25 points to $1/8$ of an inch. This equals \$50 on one contract.

Monthly chart--the scale is 50 points to $1/8$ of an inch, which is $1/24$ per cent on one contract. You have one chart with a scale of 50 to

information for eggs is 1 point or \$7.20 on a contract. If per dozen or 100 points equals \$144. To get the time required to balance any high or low price in the range which is the difference between any top or bottom, you divide the price by the scale to get the required time. You can multiply the high price by 4 when you are using a 25 point scale and get the time to balance price, and where the scale is 50 points to 1/8 of an inch, you can multiply by 2, but you are less likely to make an error if you always divide the high or low price by the scale.

Example: Angles

Read over all of the instructions on how to draw angles from high or low prices and how to draw them from the 1/2 price which can be 1/2 of the highest price or 1/2 between the extreme high and the extreme low. The 45 degree angle is the first and most important angle to draw from any top or bottom or from any 1/2 way point.

Place the dates in days, weeks, or months across the top or bottom of your chart as you will know when the angles reach the high or low price and when time and price square or balance. You should note with arrow on the weekly chart 13, 17, 19 1/2, 26, 32, 35, 39, 45 1/2, 52, and continue these markings for the 1st, 2nd, 3rd, year or more—that is marking 52 weeks for 11/8 years and 65 weeks for 1 1/4 years and 78 weeks for 1 1/2 years. You have a table showing 21 of these time periods up to 30 years, and the calculation is made from the high and low points on eggs so you can see when the time is at 1/4, 1/2, 1/3, 2/3, or 3/4 or a year or at the end of any year from the extreme high or low price.

Yearly time periods are divided by 2 and 1/3 to get these time periods.

Resistance points for time and price

These are the proportionate parts of the circle of 360 degrees and they are as follows: 11 1/4, 22 1/2, 33-3/4, 45, 56 1/4, 60, 67 1/2, 75-3/4, 90, 101 1/4, 112 1/2, 120, 123 1/4, 135, 146 1/4, 157 1/2, 168-3/4, 180, 191 1/4, 207 1/2, 213-3/4, 225, 236 1/4, 247 1/2, 258-3/4, 270, 281 1/4, 292 1/2, 300, 303-3/4, 315, 326 1/4, 337 1/2, 348-3/4, and 360 the complete circle.

These points should be marked in time periods as well as price and watch for important changes in trend.

Natural Squares of Digits

These are 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, and 361.

By studying the action of the market in the past at these important time periods or resistance levels for price and watching the action of prices in the future, you will learn the great value of time periods and resistance levels but remember time is always the most important and causes the price changes in trend.

Points from which to draw angles

Draw angles from any important top or bottom on the daily, weekly, or monthly charts. Always draw the 45 degree angle first and then the angle of 1 x 2 to the left and 2 x 1 to the right or below the 45 degree angle. These are the three most important angles, but all of the other angles can be used when you need them.

2. Draw a straight line or a horizontal from any low price across the bottom of the chart. Draw a straight line from any high price across the top of the chart. When a 45 degree angle or any other angle reaches these base lines, it has squared or balanced the price (or, a proportionate part of it such as $1/8$, $1/4$, $1/2$, etc. **EXAMPLE:** October eggs low 3225, October 3, 1950. The line would be drawn across the bottom from 3225. The high price of October eggs was 5855, and this line would be drawn across the top. A 45 degree angle moving up at the rate of 25 points per week from 3225 and crossing the line at 5855 means that price and time have balanced the range from 3225 to 5855 or that the range is equal by time. We use the 45 degree angle to square the range and divide the range by 25 just the same as we do the high and low prices. (See weekly chart of October eggs from September 10, 1947 to date.) (Where I have placed all time periods and angle and marked with arrows the important points, watch for change in trend. I have put everything on this chart that you need to use at any time so that you can use it as a pattern for any other commodity.)

3. Start a 45 degree angle or any other angle from zero on the exact date when bottom or top was made and continue it until it reaches any previous bottom or top. **EXAMPLE:** Start a 45 degree angle from zero at October 3, 1950, and when this angle reaches 3225 it means that the price of 3225 is balanced by time and when the angle reaches 5855 it means that time and price and time have balanced or squared out from zero. To get the required time for the angle to move up from zero to 5855 you divide 5855 by 25. This gives 234 weeks to square or balance 5855. You of course, must divide this time periods up into 8's and $1/3$, and $2/3$, and mark these points the same as I have on the weekly chart of October eggs. When the 45 degree angle on any other angle crosses the price level of any top or bottom, you should watch for change in trend.

When you start the angle of 1×2 on October eggs which moves up at the rate of 50 points per week from 3225 or any other price, when this angle reaches 5855, it means that $\frac{1}{2}$ of the time is balanced or squared with price.

The 1×4 angle moves up 100 points per week, and when it reaches 5855, it means that $\frac{1}{4}$ of the time and price is balanced. Reverse this rule for a declining market and when the 45 degree angle from any top reaches a low level, it indicates that time and price have balanced or squared out.

The angle of 1×2 declines 50 points per week from any top and when it reaches the low price it indicates that $\frac{1}{2}$ of the price is balanced by time.

4. Calculate the time required to balance or square 3225 at 25 points per week. You get 129 weeks. Mark 129 weeks over on the chart from the low and when you reach this date, start a 45 degree angle from the base line 3225 moving back to the left. When this angle crosses the 45 degree angle moving up from 3225, it will be at the half-way point and is very important when the price crosses this angle, it will indicate higher prices. This will, of course, cross at the $\frac{1}{2}$ or 50% point in price. Apply this to any other commodity or to any other price and you will then know when the angles are crossed coming from the end of the square in price that an important change in trend is due and that the price can be in a strong or weak position.

Balancing 5855

Divide this price by 25 and you get 234 weeks. Move over to the price of 5855 on the base angle and start a 45 degree angle down moving back to the left. When this angle crosses the 45 degree angle moving down from 5855 and the price

below it, it indicates a weak position. This time period of proportionate time period, looked at on the weekly chart for the price and time are balancing out and can see the change in trend with the angles cross and the price goes above or below it. The price is the strongest or weakest position when it is above or below the 45 degree angle, but you must also use all of the other angles to determine the changes in trend.

Apply the same rules to grains or any other commodity and by study and practice you will learn how to determine a change in trend from a strong to a weak position to a strong position.

Remember that you have signed an agreement not to reveal these rules and instructions to any one, and by keeping these secret discoveries confidential for your own use, you will later receive the very important CE AVERAGE, and the NEW FORMULA which is only taught to students who have taken the same course as you have and we do not reveal it to students who take the minor courses and pay less money.

Wishing you success, I am

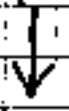
Yours very truly,

November 6, 1954

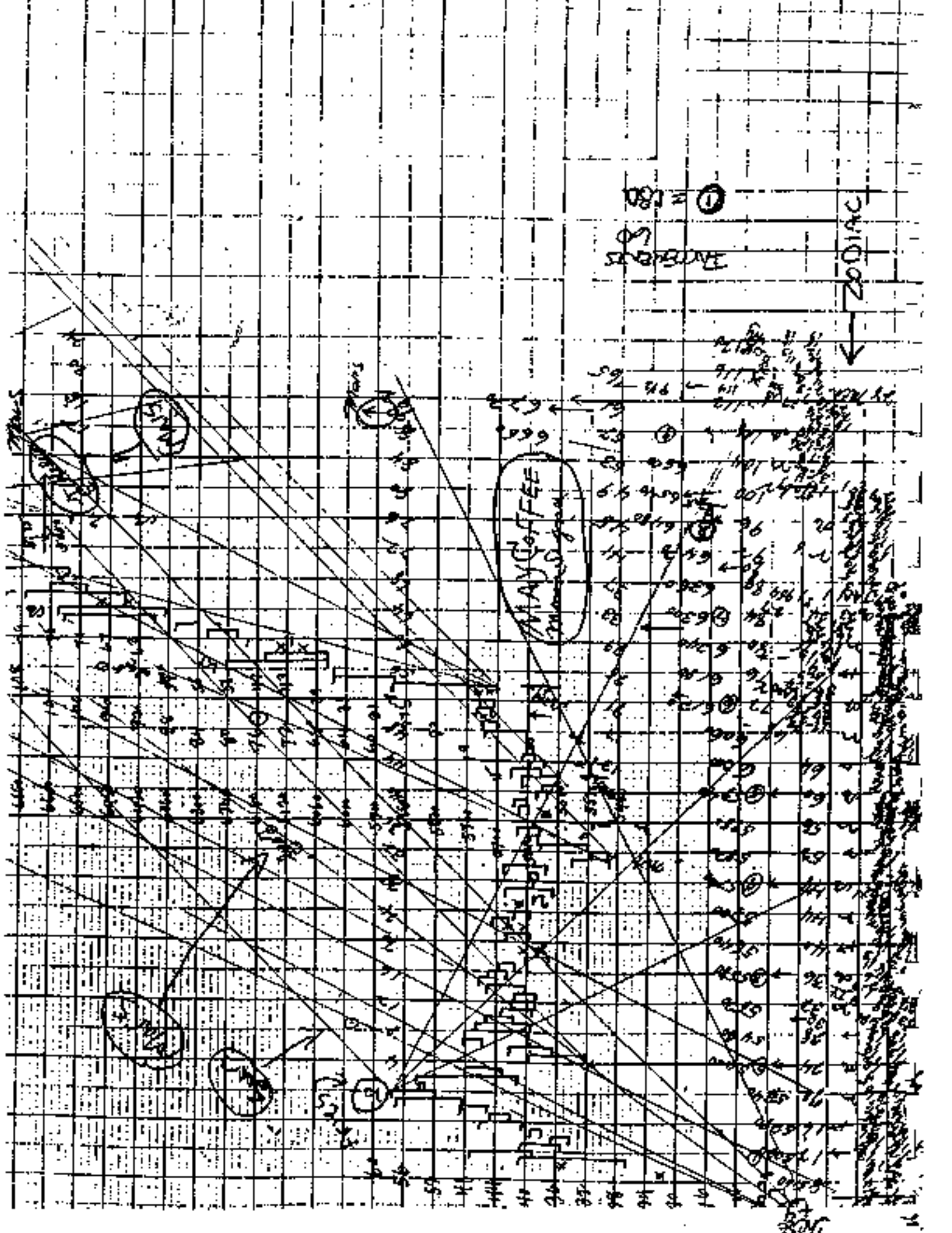
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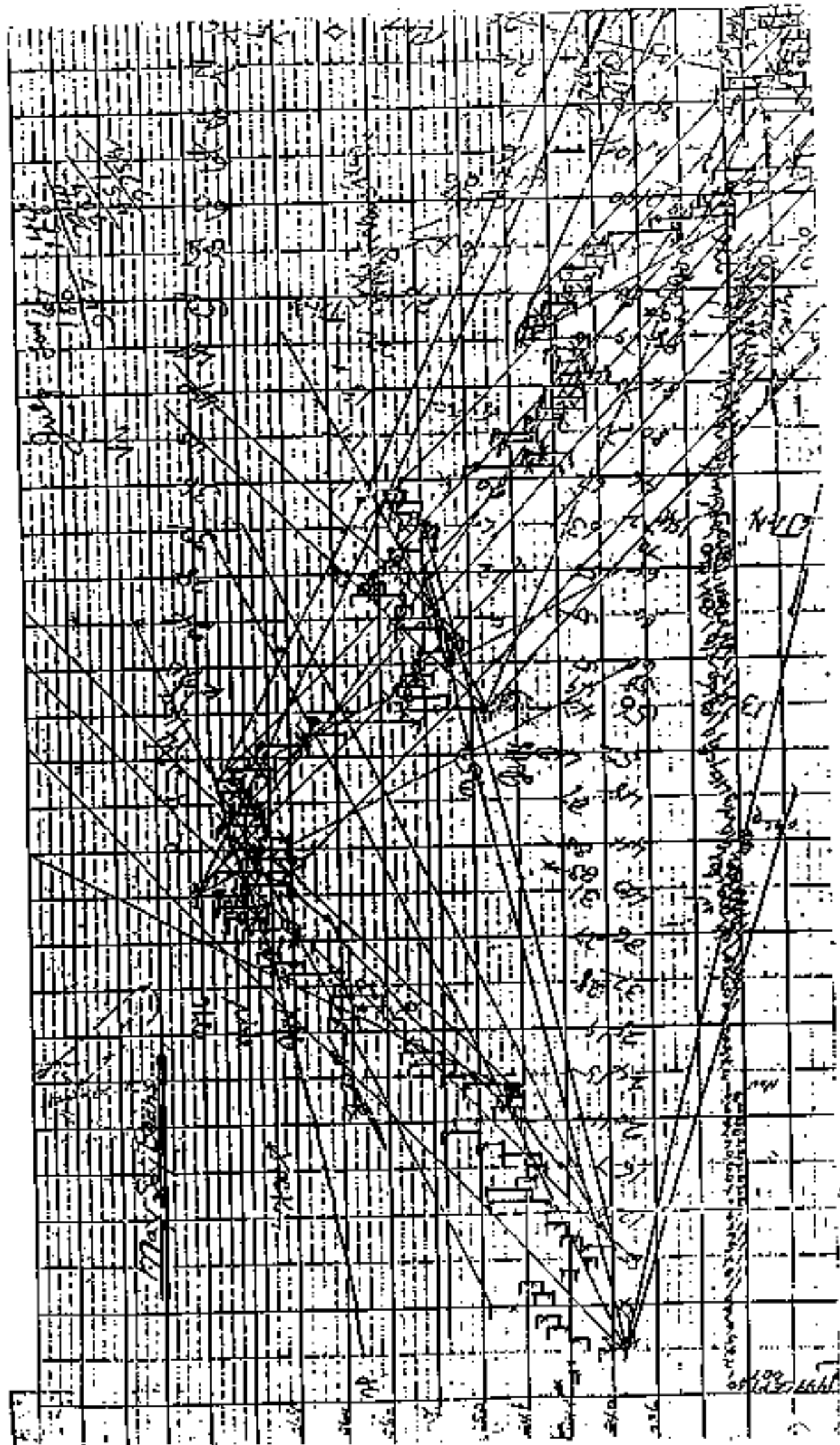
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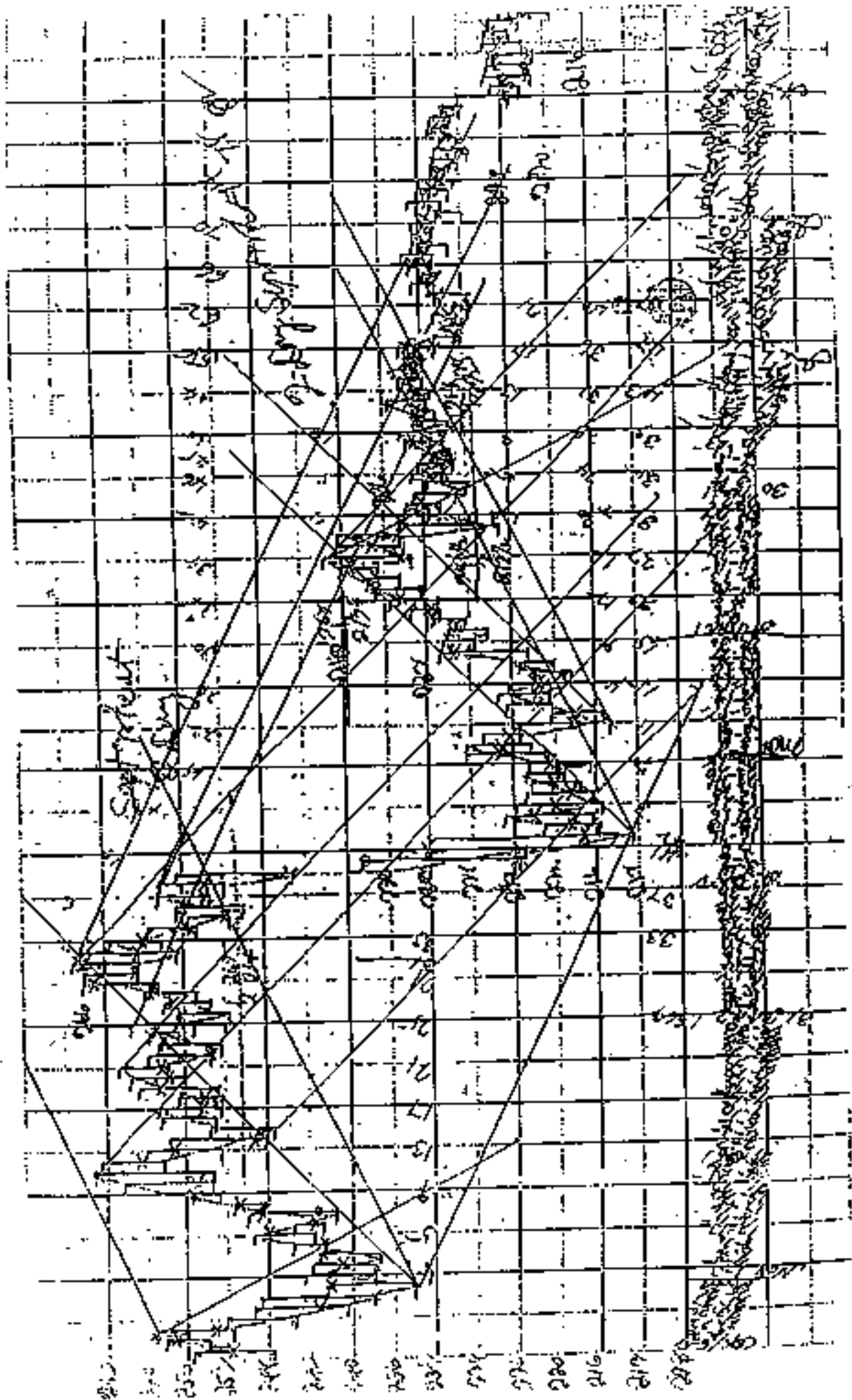
2001AC



WAX COFFEE
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APPENDIX C

TIME CYCLES

W. D. GANN
68 WALL STREET
NEW YORK

Scientific Analysis
and Analytical Reports
of Stocks and Commodities
as "Tales of the Stock Tape"
"Wall Street Stock Selection"
and "The Tunnel That the Air"

Member
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FORECASTING

Every movement in the market is the result of a natural law and a Cause which exists long before the Effect takes place and can be determined years in advance. The future is but a repetition of the past, as the Bible plainly states:

"The thing that hath been, it is that which shall be; and that which is done is that which shall be done, and there is no new thing under the sun." Eccl. 1: 9.

Everything moves in cycles as a result of the natural law of action and reaction. By a study of the past, I have discovered what cycles repeat in the future.

MAJOR TIME CYCLES

There must always be a major and a minor, a greater and a lesser, a positive and a negative. In order to be accurate in forecasting the future, you must know the major cycles. The most money is made when fast moves and extreme fluctuations occur at the end of major cycles.

I have experimented and compared past markets in order to locate the major and minor cycles and determine what years in the cycles repeat in the future. After years of research and practical tests, I have discovered that the following cycles are the most reliable to use:

GREAT CYCLE - MASTER TIME PERIOD - 60 YEARS:

This is the greatest and most important cycle of all, which repeats every 60 years or at the end of the third 20-year Cycle. You will see the importance of this by referring to the war period from 1861 to 1869 and the panic following 1869; also 60 years later—1921 to 1929—the greatest bull market in history and the greatest panic in history followed. This proves the accuracy and value of this great time period.

50-YEAR CYCLE:

A major cycle occurs every 49 to 50 years. A period of "jubilee" years of extreme high or low prices, lasting from 5 to 7 years, occurs at the end of the 50-year cycle. "7" is a fatal number referred to many times in the Bible. It brings about contraction, depression and panics. Seven times "7" equals 49, which is shown as the fatal evil year, causing extreme fluctuations.

30-YEAR CYCLE:

The 30-year cycle is very important because it is one-half of the 60-year cycle or Great Cycle and contains three 10-year cycles. In making up an annual forecast of a stock, you should always make a comparison with the record 30 years back.

20-YEAR CYCLE:

One of the most important Time Cycles is the 20-year cycle or 240 months. Most stocks and the averages work closer to this cycle than to any other. Refer to analysis of "20-year Forecasting Chart" given later.

15-YEAR CYCLE:

Fifteen years is three-fourths of a 20-year cycle and most important because it is 180 months or one-half of a circle.

10-YEAR CYCLE:

The next important major cycle is the 10-year cycle, which is one-half of the 20-year cycle and one-sixth of the 60-year cycle. It is also very important because it is 120 months or one-third of a circle. Fluctuations of the same nature occur which produce extreme high or low every 10 years. Stocks come out remarkably close on each even 10-year cycle.

7-YEAR CYCLE:

This cycle is 84 months. You should watch 7 years from any important top and bottom. 42 months or one-half of this cycle is very important. You will find many culminations around the 42nd month. 21 months or $1/4$ of this cycle is also important. The fact that some stocks make top or bottom 10 to 11 months from the previous top or bottom is due to the fact that this period is $1/8$ of the 7-year cycle.

There is an 84-year Cycle, which is 12 times the 7-year Cycle, that is very important to watch. One-half of this cycle is 42 years— $1/4$ is 21 years, and $1/8$ is 10½ years. This is one of the reasons for the period of nearly 11 years between the bottom of August, 1921 and the bottom of July, 1932. A variation of this kind often occurs at the end of a Great Cycle or 60 years. Bottoms and tops often come out on the angle of 135° or around the 135th month or 11¼-year period from any important top or bottom.

5-YEAR CYCLE:

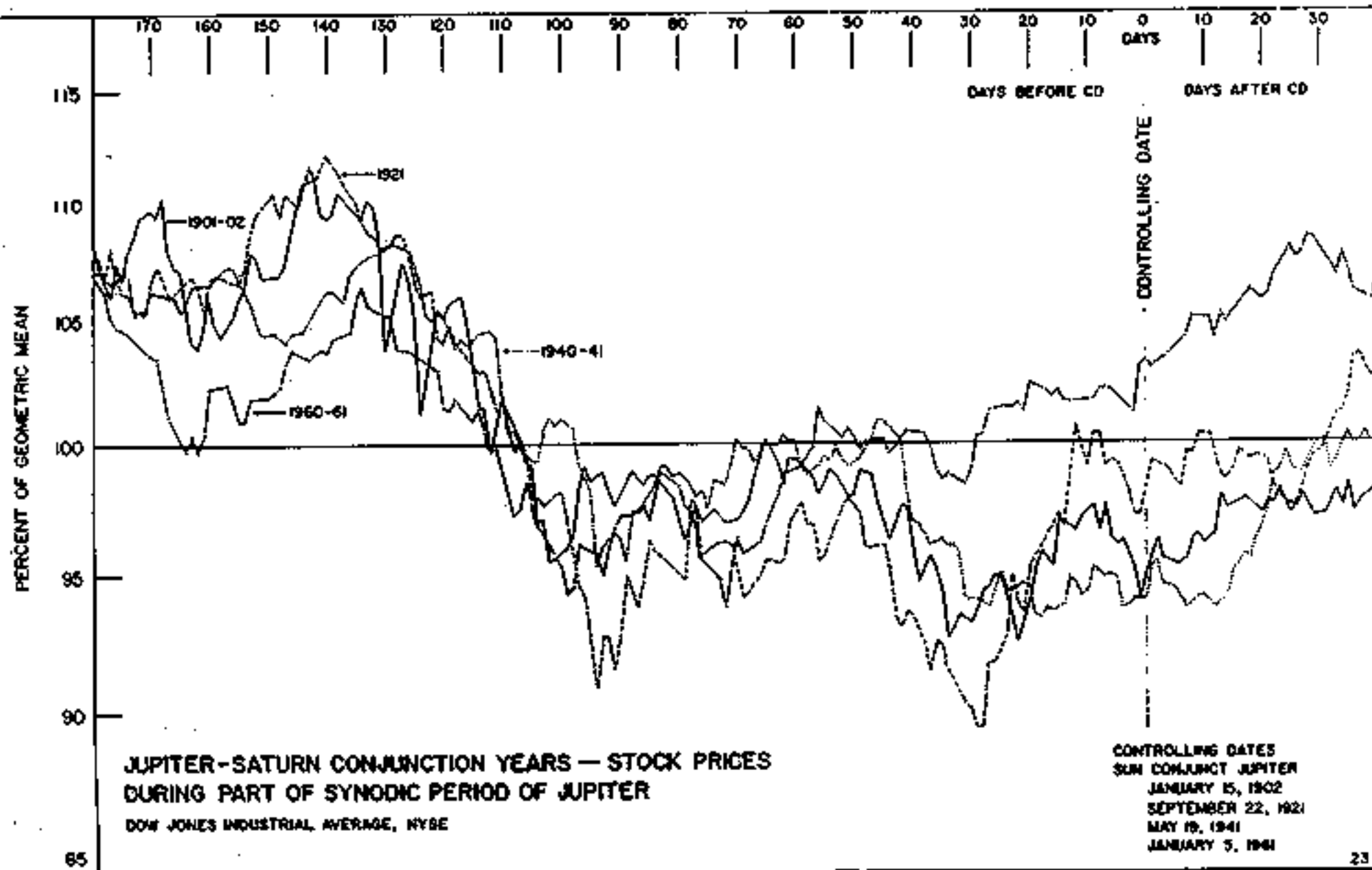
This cycle is very important because it is one-half of the 10-year cycle and $1/4$ of the 20-year cycle. The smallest complete cycle or work-out in a market is 5 years.

MINOR CYCLES:

The minor cycles are 3 years and 6 years. The smallest cycle is one year, which often shows a change in the 10th or 11th month.

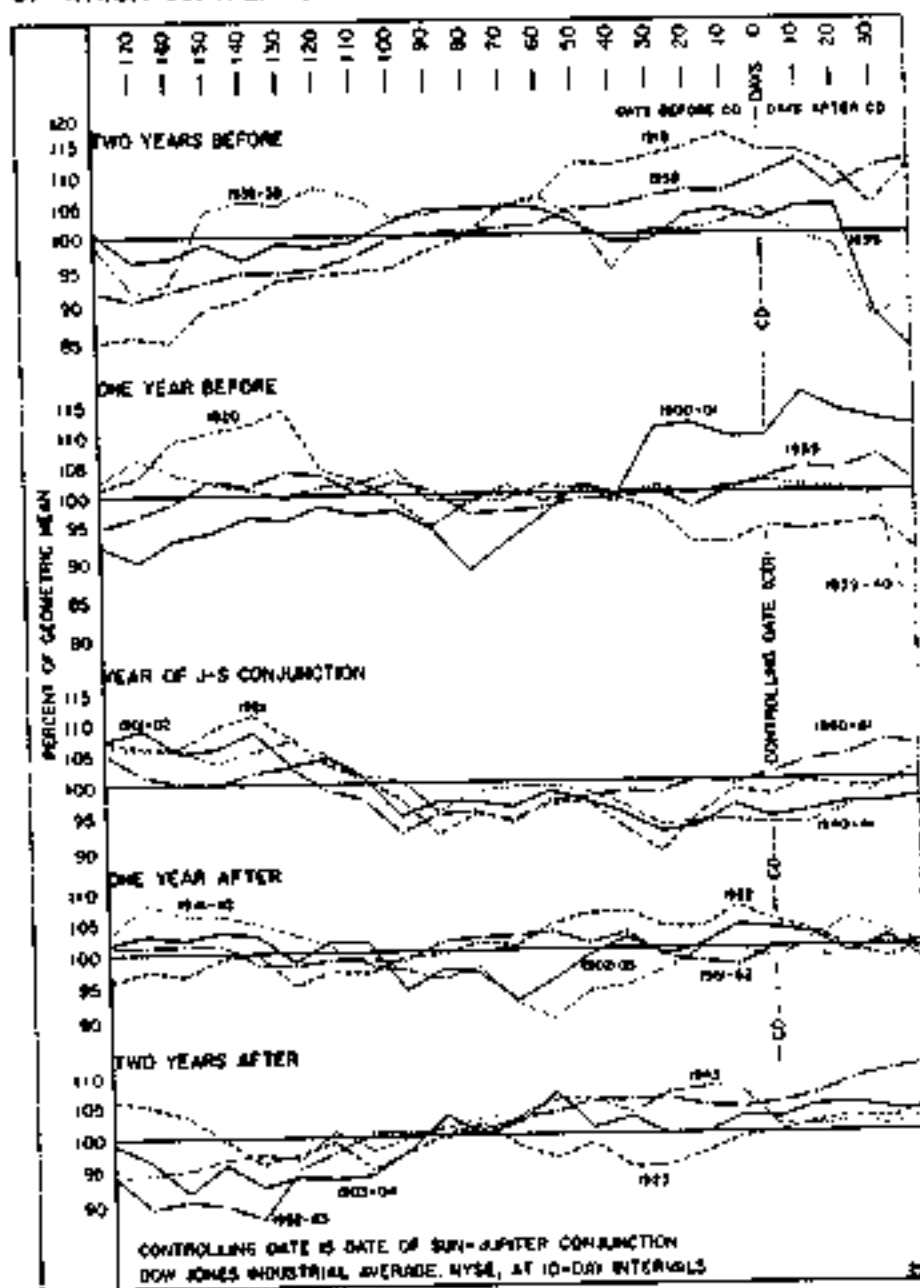
(Book is out of print)

MAN IN COSMOS

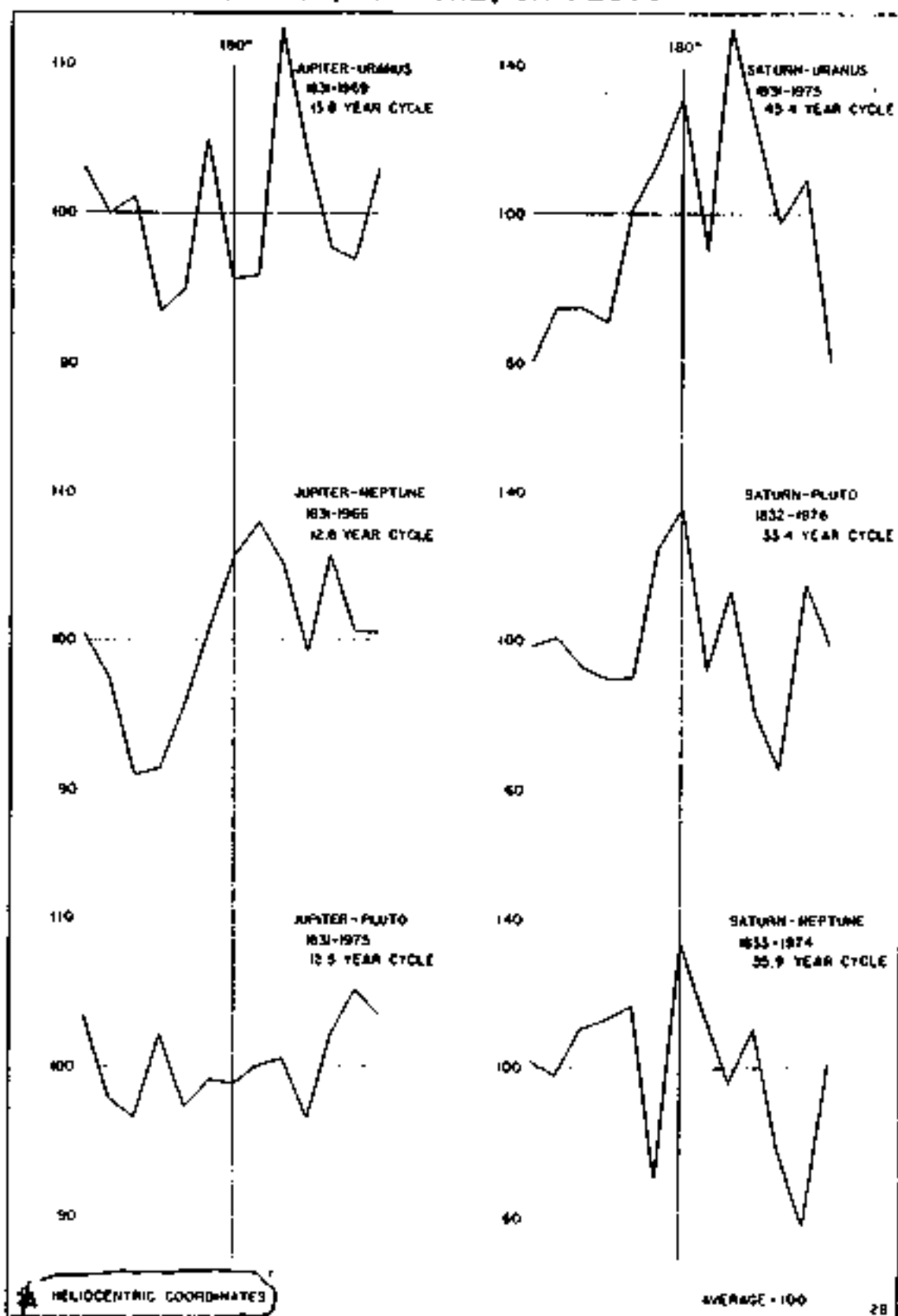


and controlling dates, and by using the above controlling dates to determine whether the data showed the approach and departure of the phenomena to which I have drawn your attention. In order to condense the chart and reduce the statistical burden, I took stock prices at 10-day intervals instead of 1-day intervals backward and forward from the controlling dates. The controlling dates for these additional years were obtained in the same way as in the conjunction year, namely, by ascertaining the date of the conjunction of the sun and Jupiter. This process yielded five charts of four stock market curves each, all of them having one thing in common: vertical alignment by the date of the sun's conjunction with Jupiter. I have put all five of these charts into one grand chart for your examination, the vertical scale being logarithmic as before. The stock price curves in each group are plotted as departures from their own individual averages. If you are not technically inclined, you can ignore all this explanation and simply observe the contours of the charts. They are shown in the chart entitled JUPITER-SATURN CONJUNCTION YEARS — STOCK PRICES DURING PART OF JUPITER SYNODIC PERIOD IN FIVE YEARS OF WHICH JUPITER-SATURN CONJUNCTION YEAR IS CENTER (Chart 25). This comparison shows clearly that the situation of the conjunction year does not exist two years before, begins to appear in the first year before the Jupiter-Saturn conjunction, appears precisely in the year of conjunction, partially departs in the first year after conjunction, and vanishes in the second year after conjunction. You will note that the vertical range of the curves is much less — they are closer together — in the year of conjunction than in the two years before or the two years after the year of conjunction. The stock price pattern in the year of conjunction is unique. The statistical association between stock price tendencies and the sun-Jupiter-Saturn phenomena described is in my estimation a meaningful association. It is a peculiar but important event. Both relative cycles — the Jupiter-Saturn

JUPITER-SATURN CONJUNCTION YEARS — STOCK PRICES
DURING PART OF JUPITER SYNODIC PERIOD IN FIVE YEARS
OF WHICH JUPITER-SATURN CONJUNCTION YEAR IS CENTER



RELATIVE CYCLES — SELECTED INDICES OF STOCK PRICE VARIATION FOR PAIRS OF PLANETS INVOLVING JUPITER OR SATURN WITH URANUS, NEPTUNE, OR PLUTO



17 year "Bull Market" Pattern

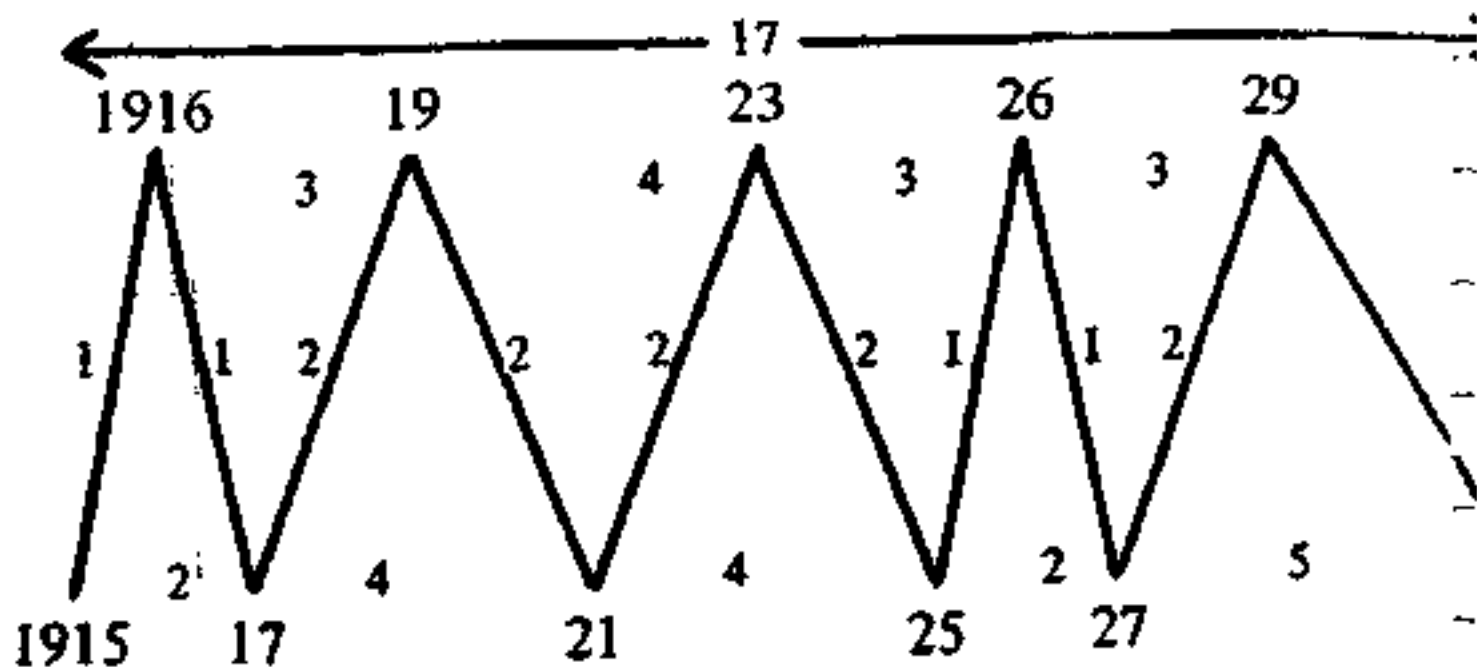


Figure 1. The Ideal Timing Pattern in the Positive Phase, 1915-32

17 year "BEAR MARKET" Pattern

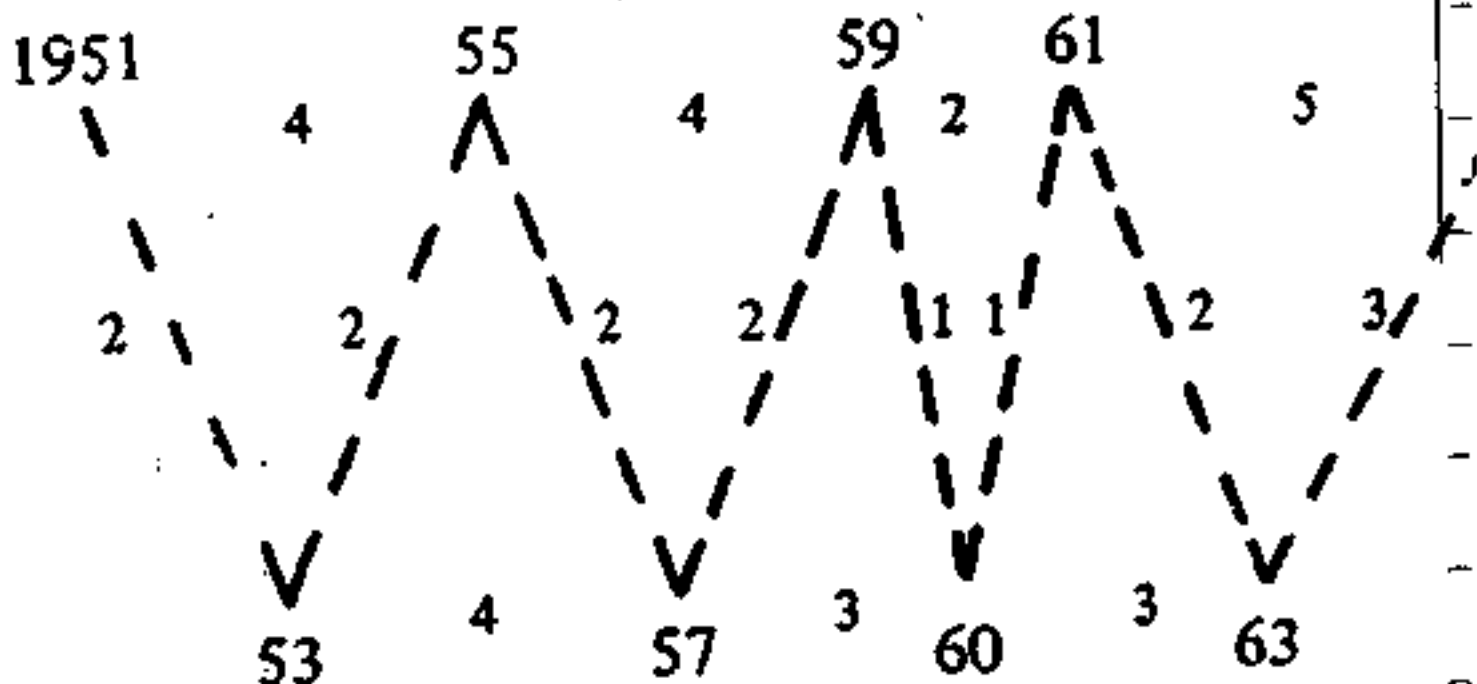


Figure 2. The Ideal Timing Pattern in the Negative Phase, 1951-66

E. D. GANN'S SCIENTIFIC FORECASTS

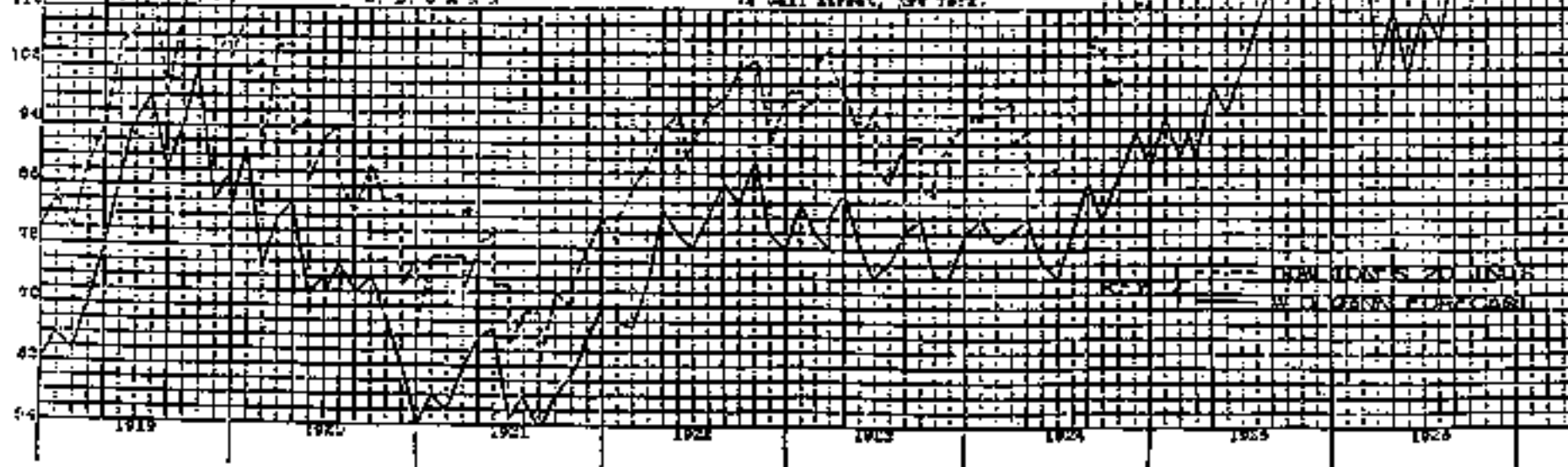
This chart which is a comparison of E. D. Gann's Annual Forecasts (solid black line) with the actual market as shown by the Dow Jones' 20 Industrial Stocks (broken line) proves that his forecasts have been 85 to 90% correct over the period from 1910 to 1926. While the chart is not spaced the same as the original forecasts an account of lack of space, it shows the times when tops and bottoms were forecast by the projected graphs and printed instructions in each year's forecast.

Mr. Gann has been issuing forecasts on stocks and commodities for public and private parties since 1909. The Ticker Magazine (now the Magazine of Wall Street) in December, 1909, first called attention to Mr. Gann's wonderful predictions and trading record. Since that time the New York Herald, New York Sun, New York Evening Telegram, New York Morning Telegraph and various other newspapers and magazines throughout the United States, have commented on the remarkably accurate scientific forecasts made by Mr. Gann. Forecasts of the general stock market and individual stocks, cotton, wheat, corn, coffee, sugar, cacao, rubber, oil and other commodities are issued every year in December, outlining the course of the market during the following year. Write for sample forecasts and prices.

Knowledge is power. Read more and know more. "Truth of the Stock Tape" - a book giving practical rules and methods on stocks and commodities - will enable you to know how to trade successfully and help you make profits instead of losses. Price \$5.00.

E. D. GANN

74 Wall Street, New York.



SAK 1932 June 9 δ Helio 31° 30' = 1° 30' 4

28
12
40

1934 May ①
1936 Apr ②
1938 Feb ③
1940 Dec ④
1941 Nov ⑤
1943 Oct ⑥
1945 Aug ⑦
1947 July ⑧
1949 June ⑨
1951 Apr ⑩
1952 May ⑪

SAK
1938 Apr 10 δ H-182 = 2° - 0 = 430 =
1940 Nov - 2 - 360 = 1243
1942 Oct - 2 - 720 =
1944 Aug - 3 - 1080
1946 May - 4 - 1440 - 1440
1948 Apr - 5 - 1800 - 727
1950 Mch 8 - 6 - 2160 - 7167
1951 Mch 8 - 7 - 2520
1952 May 3 - 11.30 m - 3600
1952 May 3 - 1.30 - 190 - 727
10.00 - 190 - 3790 727 = 3287

May 3 Helio δ at 3790

1951 Mch 8 δ 17° 30' 72 = 330 at 12° noon
- 130 - 16 3240
1600 346 + 346 3586 16° 4'

Mch 8 - 1951 - δ at

1951 Nov 9 δ 142 = 22° v. or 3600 + 90 = 3690

0° 5'

3790 727 + 40 8

40
1080
108

Coffee calculations
helio 2000 of mch 4

Stocks move in 10-year cycles, which are worked out in 5-year cycles — a 5-year cycle up and a 5-year cycle down. Begin with extreme tops and extreme bottoms to figure all cycles, either major or minor.

- Rule 1 - A bull campaign generally runs 5 years — 2 years up, 1 year down, and 2 years up, completing a 5-year cycle. The end of a 5-year campaign comes in the 59th or 60th months. Always watch for the change in the 59th month.
- Rule 2 - A bear cycle often runs 5 years down—the first move 2 years down, then 1 year up, and 2 years down, completing the 5-year downswing.
- Rule 3 - Bull or bear campaigns seldom run more than 3 to 3½ years up or down without a move of 3 to 6 months or one year in the opposite direction, except at the end of Major Cycles, like 1869 and 1929. Many campaigns culminate in the 23rd month, not running out the full two years. Watch the weekly and monthly charts to determine whether the culmination will occur in the 23rd, 24th, 27th or 30th month of the move, or in extreme campaigns in the 34th to 35th or 41st to 42nd month.
- Rule 4 - Adding 10 years to any top, it will give you top of the next 10-year cycle, repeating about the same average fluctuations.
- Rule 5 - Adding 10 years to any bottom, it will give you the bottom of the next 10-year cycle, repeating the same kind of a year and about the same average fluctuations.
- Rule 6 - Bear campaigns often run out in 7-year cycles, or 3 years and 4 years from any complete bottom. From any complete bottom of a cycle, first add 3 years to get the next bottom; then add 4 years to that bottom to get bottom of 7-year cycle. For example: 1914 bottom — add 3 years, gives 1917, low of panic; then add 4 years to 1917, gives 1921, low of another depression.
- Rule 7 - To any final major or minor top, add 3 years to get the next top; then add 3 years to that top, which will give the third top; add 4 years to the third top to get the final top of a 10-year cycle. Sometimes a change in trend from any top occurs before the end of the regular time period, therefore you should begin to watch the 27th, 34th, and 42nd months for a reversal.
- Rule 8 - Adding 5 years to any top, it will give the next bottom of a 5-year cycle. In order to get top of the next 5-year cycle, add 5 years to any bottom. For example: 1917 was bottom of a big bear campaign; add 5 years gives 1922, top of a minor bull campaign. Why do I say, "Top of a minor bull campaign?" Because the major bull campaign was due to end in 1929.
- 1919 was top; adding 5 years to 1919 gives 1924 as bottom of a 5-year bear cycle. Refer to Rules 1 and 2, which tell you that a bull or bear campaign seldom runs more than 2 to 3 years in the same direction. The bear campaign from 1919 was 2 years down—1920 and 1921; therefore, we only expect one-year rally in 1922; then 2 years down—1923 and 1924, which completes the 5-year bear cycle.
- Looking back to 1913 and 1914, you will see that 1923 and 1924 must be bear years to complete the 10-year cycle from the bottoms of 1913-1914.

Then, note 1917 bottom of a bear year; adding 7 years gives 1924 as bottom of a bear cycle. Then, adding 5 years to 1924 gives 1929 top of a cycle.

FORECASTING MONTHLY MOVES

Monthly moves can be determined by the same rules as yearly:

Add three months to an important bottom, then add 4, making 7, to get minor bottoms and reaction points.

In big upswings a reaction will often not last over two months, the third not being up, the same rule as in yearly cycle—2 down and the third up.

In extreme markets, a reaction sometimes only lasts 2 or 3 weeks; then the advance is resumed. In this way a market may continue up for 12 months without breaking a monthly bottom.

In a bull market the minor trend may reverse and run down 3 to 4 months; then turn up and follow the main trend again.

In a bear market, the minor trend may run up 3 to 4 months, then reverse and follow the main trend, altho, as a general rule, stocks never rally more than 2 months in a bear market; then start to break in the 3rd month and follow the main trend down.

FORECASTING WEEKLY MOVES

The weekly movement gives the next important minor change in trend, which may turn out to be a major change in trend.

In a bull market, a stock will often run down 2 to 3 weeks, and possibly 4, then reverse and follow the main trend again. As a rule, the trend will turn up in the middle of the third week and close higher at the end of the third week, the stock only moving 3 weeks against the main trend. In some cases the change in trend will not occur until the fourth week; then the reversal will come and the stock close higher at the end of the fourth week.

Reverse this rule in a bear market.

In rapid markets with big volume, a move will often run 6 to 7 weeks before a minor reversal in trend, and in some cases, like 1929, these fast moves last to 15 weeks or 1/4 of a year. These are culmination moves up or down.

As there are 7 days in a week and seven times seven equals 49 days or 7 weeks, this often marks an important turning point. Therefore you should watch for top or bottom around the 49th to 52nd day, altho at times a change will come on the 42nd to 45th day, because a period of 45 days is 1/8 of a year. Also watch for culminations at the end of 90 to 98 days.

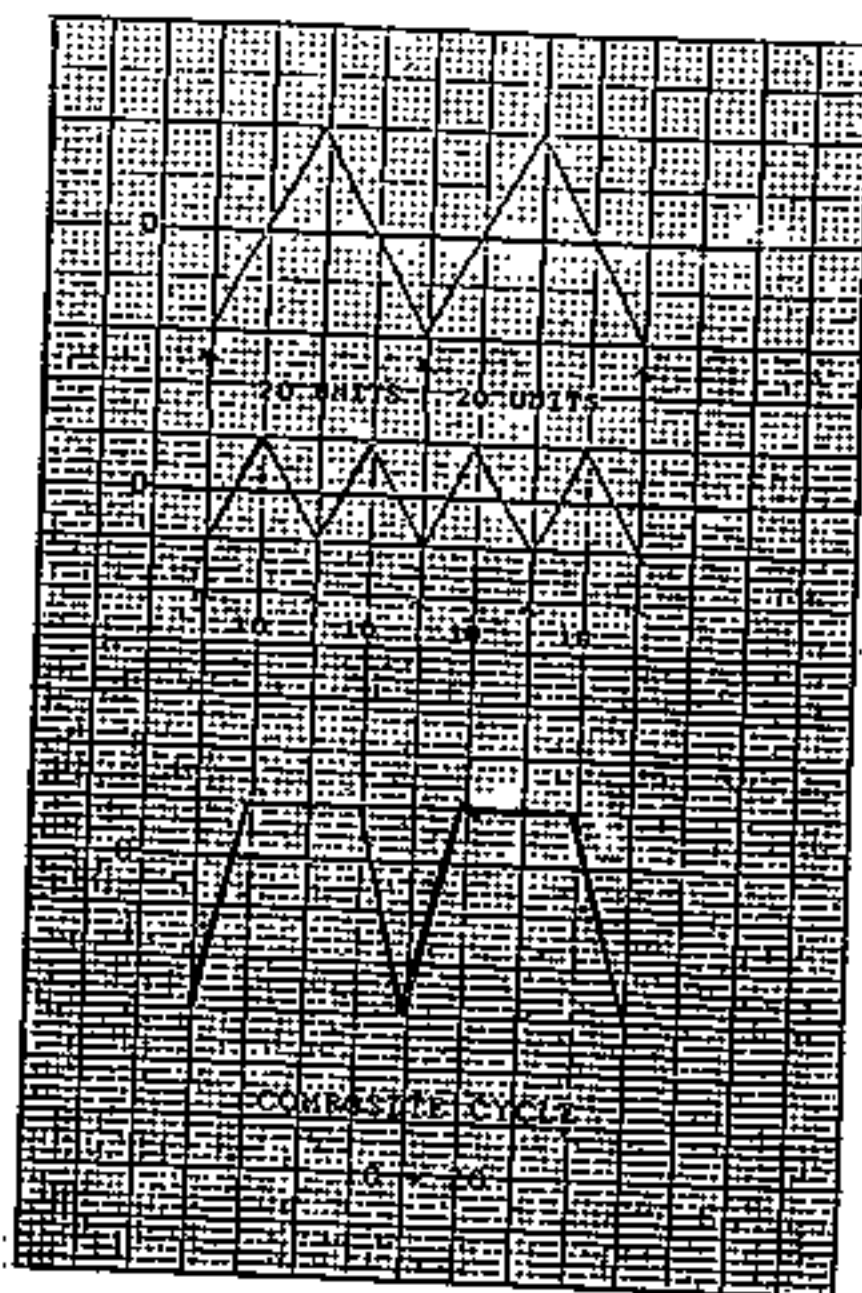
After a market has declined 7 weeks, it may have 2 or 3 short weeks on the decline and then turn up, which agrees with the monthly rule for a change in the trend.

Always watch the annual trend of a stock and consider whether it is in a bull

bear year. In a bull year, with the monthly chart showing up, there are many times that a stock will react 2 or 3 weeks, then rest 3 or 4 weeks, and then go into new territory and advance 6 to 7 weeks more.

After a stock makes top and reacts 2 to 3 weeks, it may then have a rally of 2 to 3 weeks without getting above the first top, then hold in a trading range for several weeks without crossing the highest top or breaking the lowest week of that range. In cases of this kind, you can buy near the low point or sell near the high point of that range and protect with a stop loss order 1 to 3 points away. However, a better plan would be to wait until the stock shows a definite trend before buying or selling; then buy the stock when it crosses the highest point or sell when it breaks the lowest point of that trading range.

HOW TO CREATE A COMPOSITE CYCLE

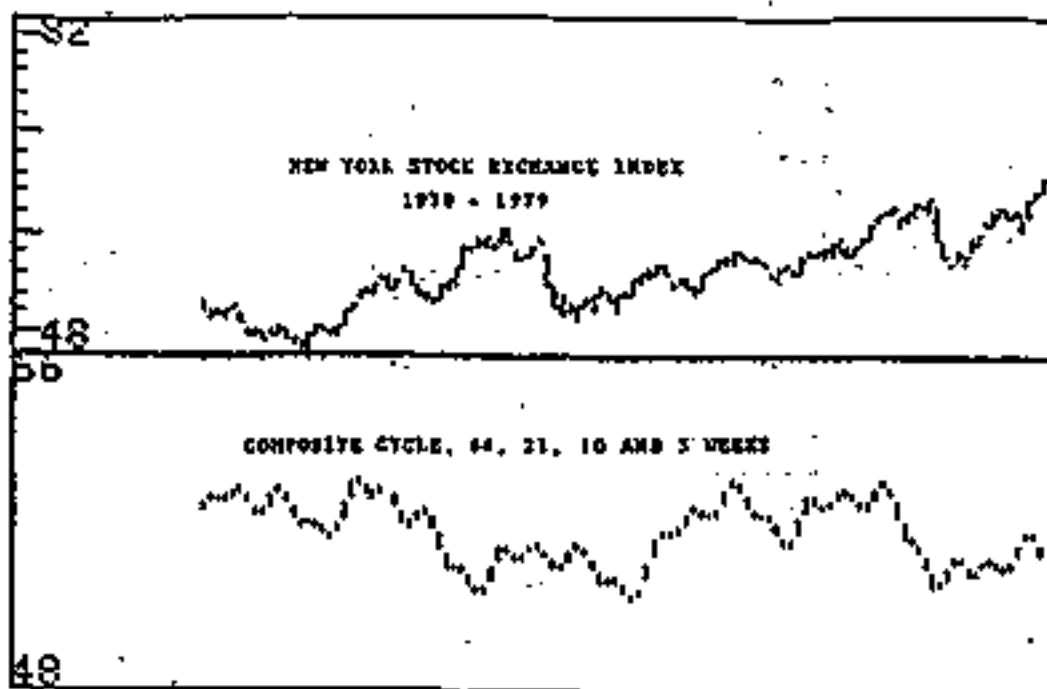


COMPOSITE TRADING CYCLES are cycles which are created by combining the waves of various time cycles that are included within the composite.

At any given time, some cycles will be rising and some cycles will be falling. You have to add the current levels of the various waves together to create the composite, after first laying out your anticipated cyclical configurations.

For example, in the chart above, I have plotted a 10-unit and 20-unit cycle. The composite cycle at the bottom of the chart is secured by adding at each point the positive and negative amplitudes of these two cycles. To illustrate, the 20-unit cycle first peaks at a point 10 units above its "0" Line. At that time, the 10-unit cycle has 5 units below its own "0" Line. We add +10 and -5 and secure a reading of +5 for that entry of the composite cycle.

Notice, that each cycle must be plotted with its proper amplitude (height) and that the phasing of the cycles is very significant. Had I plotted the 10-unit cycle moved say to the left so that its peaks coincided with peaks in the 20-unit cycle, then the composite would be skewed upwards rather than downwards below 0. This procedure is, of course, only approximate, but very useful nonetheless.



The above charts show a Fourier derived composite cycle of the 64, 21, 10 and 5 week market cycles for the 1978-1979 period. The fit of the composite to the actual market is not, of course, perfect, but many of the actual high and low points were well defined by way of this cyclical procedure.

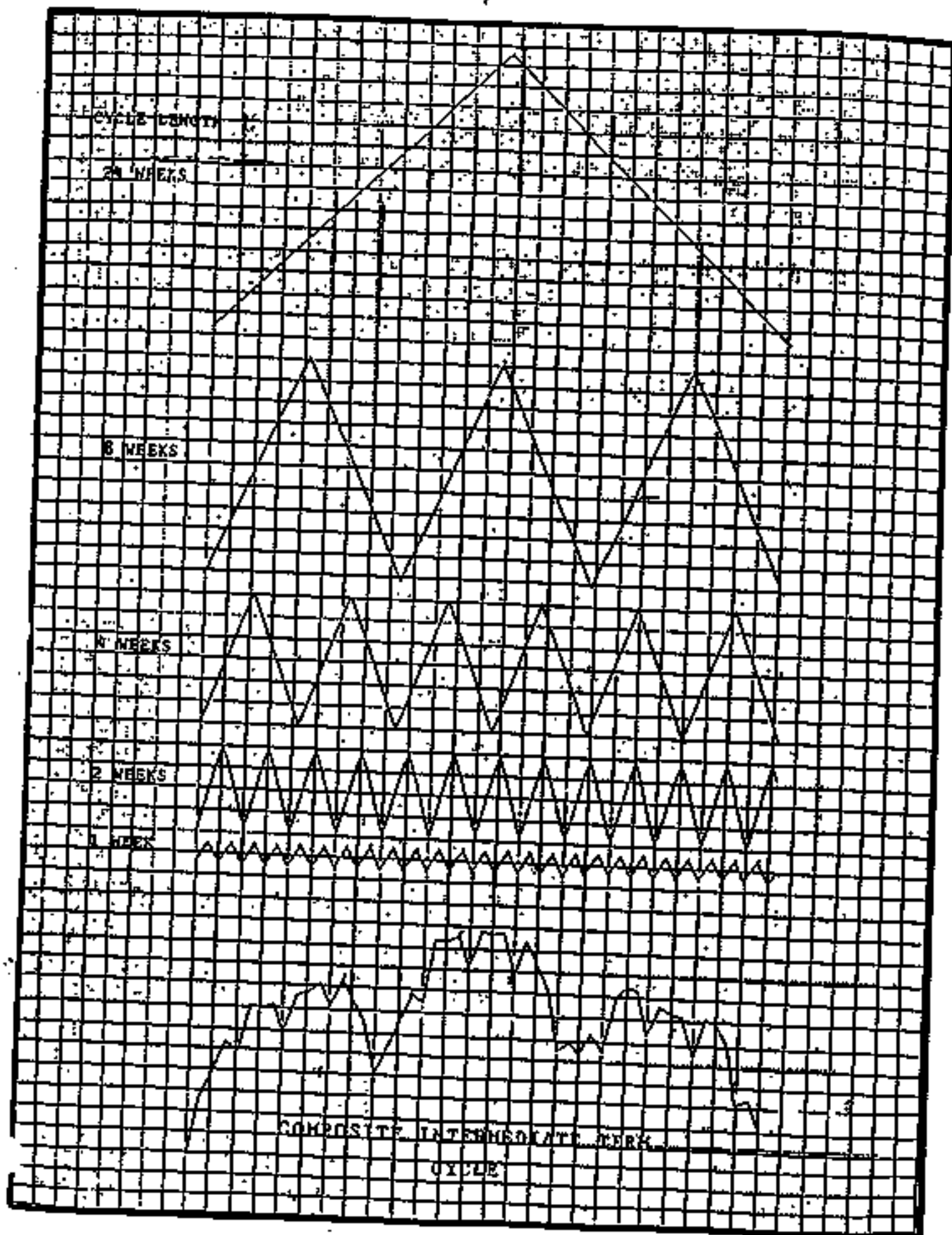
On page 7, I have drawn in a model composite 24-week cycle based upon the 24-week, 8-week, 4-week, 2-week and one-week market cycles. You can use this composite as an approximate road map for intermediate market cycles within a neutral period of the major trend.

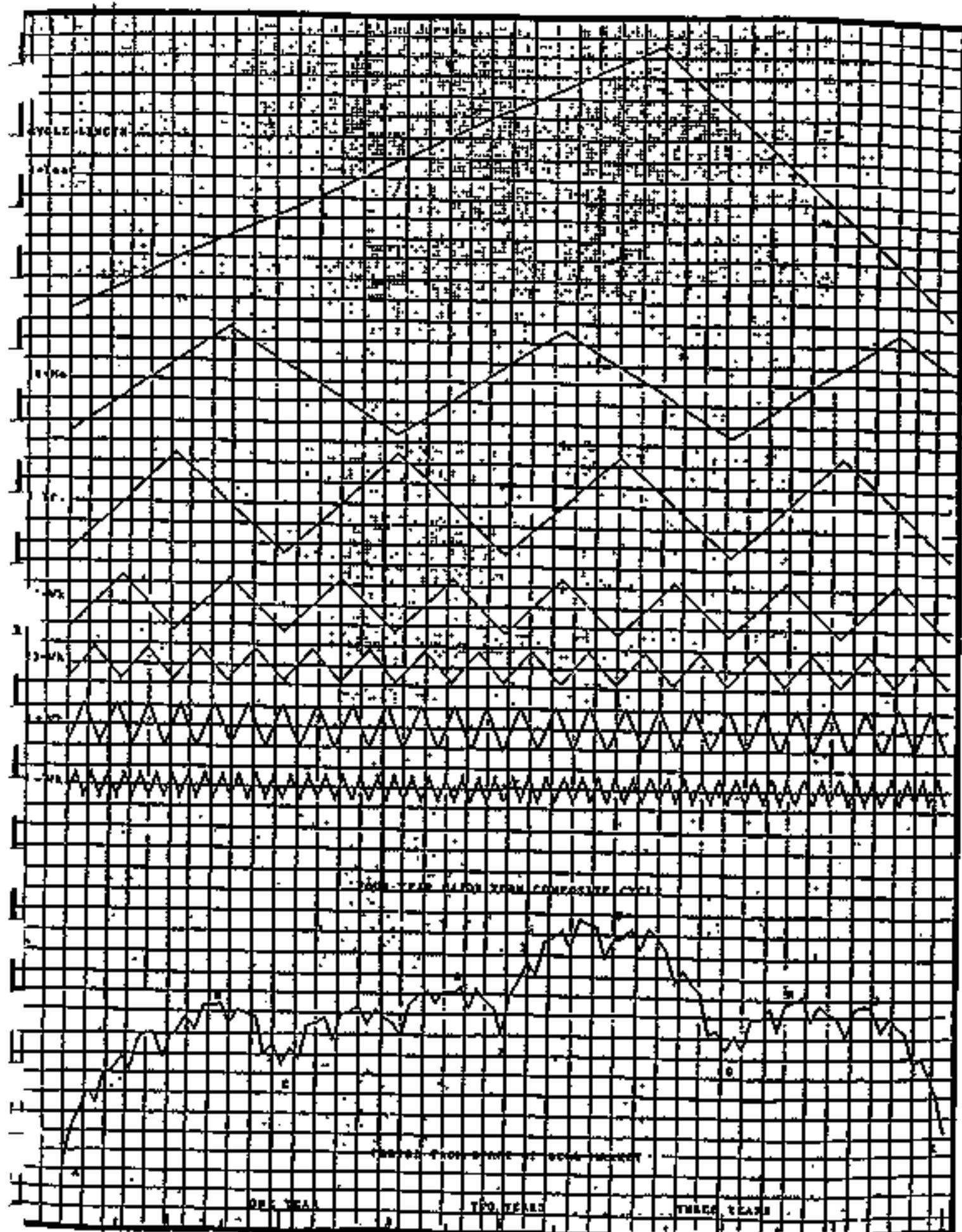
SKewing:

The rendition on page 7 presumes that all of the cycles are equal in amplitude during rising and falling market periods and that they are equal in the lengths of their rising and declining phases. This is not in actuality the case. Stocks spend about 2/3 of the time rising and major bull phases show a rising cycle for about 2/3 of the 4-year major cyclical period. We also find that intermediate cycles show a rising skew during bull markets. Short term cycles show rising skews during rising intermediate periods. Cycles are very influenced by the positions of cycles above them in length.

It is a sign of market strength when cycles show greater periods of rising than falling activity and a sign of market weakness when cycles show longer periods of decline in their phases. During neutral market periods, cycles tend to rise and fall for equal time spans -- often with great precision.

You may notice in the rendition on page 7 how the overall cyclical direction is set by the 24-week intermediate cycle. Shorter term cycles create the fluctuations within the overall intermediate trend with the greatest influence exerted by the 8-week cycle.





Astronomical Cycles

ASTRONOMY is more concerned with cycles than any of the other sciences. For example, the earth and the other planets go around the sun in cycles. The moons go around the planets in cycles. The planets and their moons revolve around their own axes in cycles, and, for that matter, the sun revolves around its own axis too. Sunspots come and go in cycles. Comets have their cycles. The light from many of the stars pulsates in cycles. Even the whole galaxy of which our solar system is a part has a cycle—our part of the galaxy having a cycle or rotation period of about 220 million years.

It is always interesting to compare known cycles on earth with the astronomical cycles to see if the length and timing correspond. If so, we have a hint of a possible interrelationship, even though it may be an indirect one.

Earthly and Celestial Cycles Correspond

In some instances earthly and celestial cycles do correspond. The day and the year are basically astronomical cycles. Tidal cycles and all the various biological cycles associated with them are of course largely based upon the cycles of the moon. Variations in terrestrial magnetism correspond to cycles in sunspots. Many economic phenomena have cycles which correspond to cycles in sunspots with alternate cycles reversed. And there are a few other earthly cycles that rather clearly correspond to celestial ones (see for example the article entitled "A 29½-Day Cycle in the Births of Infants" on pages 15 to 17 of our September 1950 report).

From here on, however, the comparison between the cycles in the heavens and the cycles on the earth offers harder going for those who seek correspondences, but there may be correspondences notwithstanding.

With this possibility in mind, and the more likely possibility of sunspot and planetary interrelationships, I reproduce for reference herewith a tabulation from a book called *Cycles in Weather and Solar Activity* by Maxwell O. Johnson, printed in

1950 by Paradise of the Pacific Press, Honolulu, and purchasable from the Foundation for the Study of Cycles at \$5.00 a copy, postpaid. This table is fully protected by copyright. It is reproduced through the very kind permission of Mr. Johnson.

If you study this table carefully you will note many examples of cycles which do in fact correspond to certain cycles here on earth. For example, the 22.7-month cycle in the sales of Company G, reported upon in the January issue, has the same length as the "synodic period" shown for Mars and Pluto. However these seeming identities of time span do not necessarily indicate any cause and effect relationship; viz solar disturbances or otherwise. The correspondence may be merely a coincidence.

At this point I should perhaps get into matters of terminology.

Sidereal periods

The sidereal period of a planet is the time taken by it to go around the sun with reference to the stars. (Sidereal is derived from a Latin word meaning constellation or star.) These periods are as follows:

Planet	Sidereal Period		
	In Days	In Months	In Years
Mercury	87.9686	2.89	.2408
Venus	224.7007	7.38	.6152
Earth	365.2564	12.00	1.0004
Mars	686.9804	22.57	1.8808
Jupiter	4,332.5870	142.34	11.86
Saturn	10,759.2025	233.48	29.46
Uranus	30,685.93	1,008.14	84.01
Neptune	60,187.64	1,977.38	164.78
Pluto	90,737.	2,981.0	248.42

Synodic Periods

The synodic period of a planet is the length of time required by it to come again into the same position with reference to the earth and the sun.

The synodic periods of the planets, in days and months, are commonly given as follows:

Planet	Synodic Period	
	In Days	In Months
Mercury	116	3.81
Venus	584	19.19
Earth	---	---
Mars	780	25.63
Jupiter	399	13.11
Saturn	378	12.42
Uranus	369	12.12
Neptune	367	12.06
Pluto	366.7	12.05

(The daily figures in both of the above tables are taken from the book *Astronomy* by William T. Skilling and Robert S. Richardson, for sale by the Foundation for the Study of Cycles at \$6.10 a copy postpaid. If you are interested in the subject of *Astronomy* I suggest that you may wish to buy this book.)

To make the difference between sidereal and synodic periods clearer, let me quote a few lines from the same book.

"To make the case simple suppose we think of the planets in their orbits of different sizes all starting together on the same line for a race around the sun. A person at the sun would see all the planets in line toward some certain star.

"Each time a planet would complete a circuit back to the line leading to the same star it would have finished one sidereal period for that planet.

"But when the earth has made one revolution (in one sidereal year) it still has

some distance to go to overtake one of the slower moving outer planets so as to bring planet, earth, and sun again into one line. When this happens, that planet has completed one of its synodic periods. The word synodic is from a Greek word signifying a 'meeting.' From one meeting of earth and planet to the next is a synodic period, the 'meeting' period."

In the table on the following page Johnson has extended the meaning of the term synodic to include the "meeting" of any two of the planets.

Thus when Johnson says that the synodic period of Mercury and Venus is 144.5 days he means that if Mercury and Venus and the sun were lined up so that a straight line drawn through the center of the sun would pass through the center of each (a heliocentric conjunction), it would be 144.5 days before these two planets would again get in line with each other and the sun to create a second heliocentric conjunction.

That is, if Mercury and Venus were racing, at the start of the race, with Mercury at the rail, the faster moving Mercury would pull ahead of Venus and make one complete lap in its sidereal period of 88 days, but in the meantime Venus would have gone ahead only 88/225 of a revolution. For Mercury to catch up would take 56 1/2 days more, or 144.5 days in all. This interval is what Johnson would call the synodic period of Mercury and Venus.

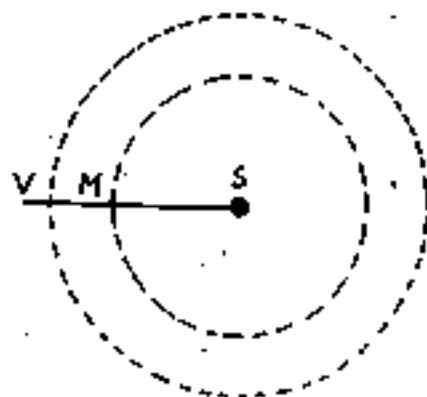


FIG. A

A HELIOCENTRIC CONJUNCTION OF VENUS AND MERCURY. IT WILL TAKE MERCURY EIGHTY-EIGHT DAYS TO MAKE A COMPLETE REVOLUTION AROUND THE SUN AND GET BACK TO THE SAME POSITION. IT WILL TAKE VENUS 225 DAYS.

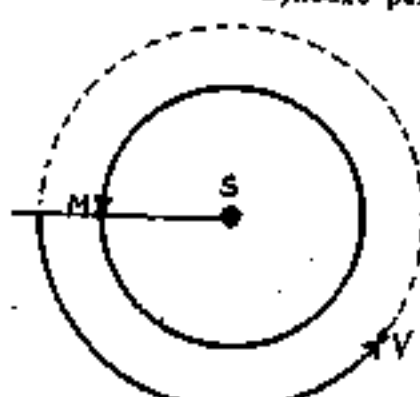


FIG. B

POSITION OF THE PLANETS EIGHTY-EIGHT DAYS AFTER FIG. A. MERCURY HAS MADE ONE COMPLETE REVOLUTION (A SIDEREAL PERIOD) AND VENUS HAS MADE 88/225 (ABOUT 2/10) OF A REVOLUTION.

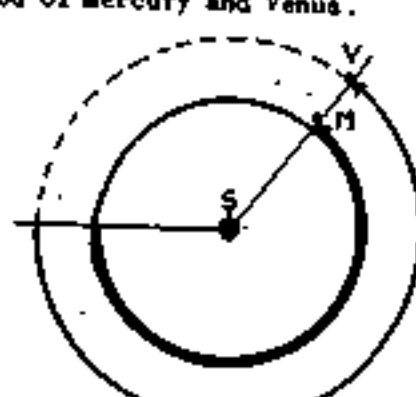


FIG. C

POSITION OF THE PLANETS 144 1/2 DAYS AFTER FIG. A. MERCURY HAS CAUGHT UP WITH VENUS, CREATING ANOTHER HELIOCENTRIC CONJUNCTION. THIS INTERVAL OF 144.5 DAYS, BY EXTENSION OF MEANING, IS CALLED BY JOHNSON THE SYNODIC PERIOD OF VENUS AND MERCURY.

As the planets travel in ellipses instead of circles, successive synodic periods differ slightly. The figures given by Johnson are averages.

It is possible that Johnson's table would be especially useful for comparison in studying the shorter rhythms of sunspot numbers.

Other Astronomical Cycles

I could give you hundreds and hundreds of other astronomical cycle lengths. For example, we have in our library books which require over 400 pages just to list all

the different variable stars, each one of which has its own period of variation. In these lists we could add the periods or cycle lengths of all the comets, and many other cycle lengths as well. However it seems to me that all these other cycles offer such remote possibility of interrelation with any of the terrestrial or solar cycles in which you or I might be interested that a study of them belongs more to the subject matter of astronomy than to the subject matter of cycles. We will therefore leave them to the astronomers.

Calculated Synodic Periods of the Planets

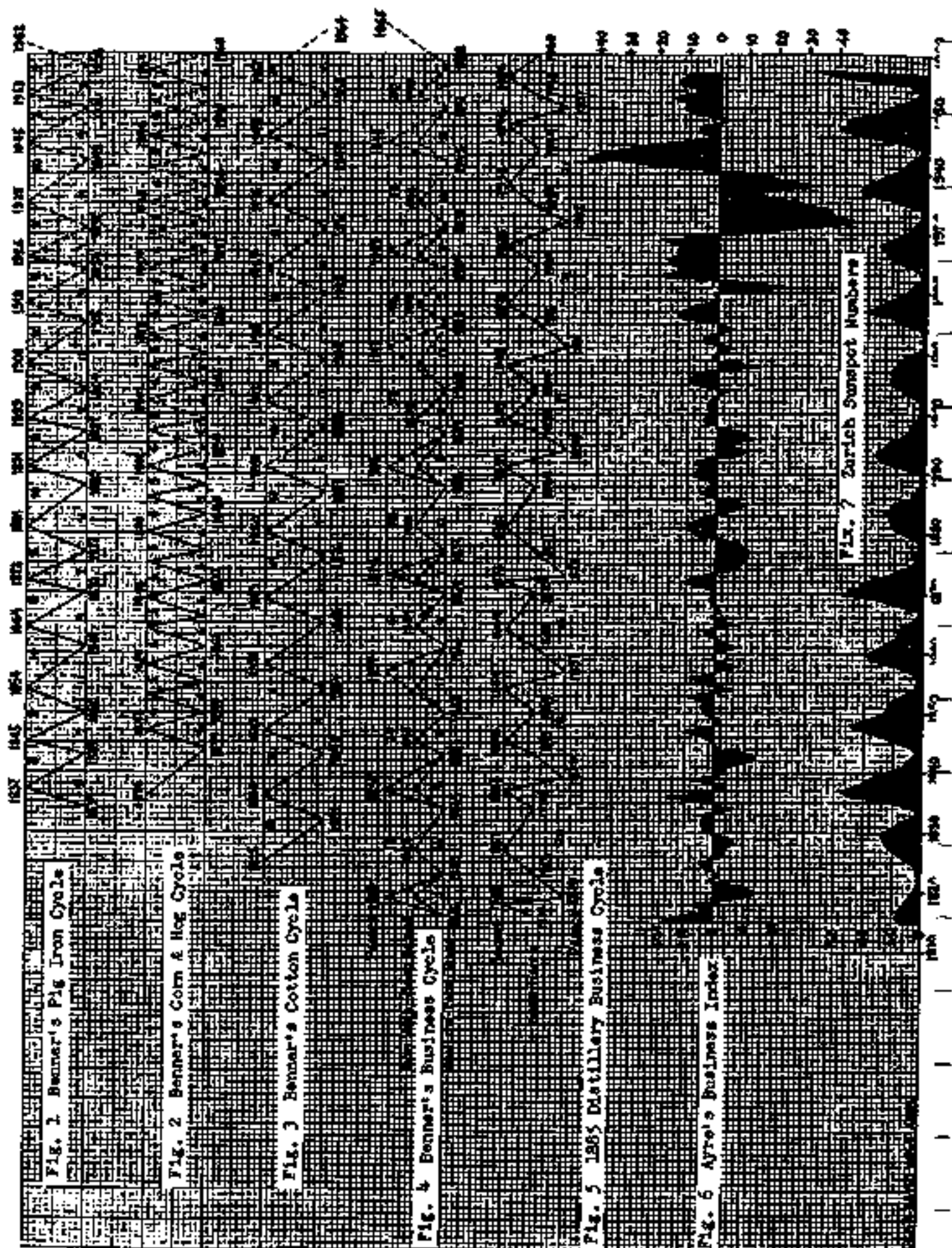
Mercury and	Venus and	Earth and	Mars and	Jupiter and	Saturn and
Venus 0.2855 years 4.35 months 144.5 days 28.9 pentads	Earth 1.5857 years 19.18 months 583.9 days 116.8 pentads	Mars 2.135 years 25.62 months 779.3 days 155.8 pentads	Jupiter 2.235 years 26.82 months 816.1 days 163.2 pentads	Saturn 29.86 years	Uranus 45.36 years
Earth 0.3172 years 3.81 months 115.9 days 23.2 pentads	Mars 0.5142 years 10.87 months 333.9 days 66.8 pentads	Jupiter 1.002 years 12.10 months 368.8 days 73.8 pentads	Saturn 2.009 years 24.1 months 733.5 days 146.8 pentads	Uranus 13.81 years	Neptune 34.87 years
Mars 0.2762 years 3.31 months 100.9 days 20.2 pentads	Jupiter 0.4169 years 7.79 months 237.0 days 47.4 pentads	Saturn 1.033 years 12.42 months 378.8 days 75.8 pentads	Uranus 1.924 years 23.1 months 702.7 days 140.5 pentads	Neptune 13.76 years	Pluto 33.43 years
Jupiter 0.5458 years 6.55 months 60.6 days 13.0 pentads	Saturn 0.6280 years 7.54 months 229.5 days 45.9 pentads	Uranus 3.013 years 36.14 months 369.6 days 73.9 pentads	Neptune 1.903 years 22.8 months 695.1 days 139.0 pentads	Pluto 12.46 years	
Saturn 0.2428 years 2.91 months 83.7 days 16.7 pentads	Uranus 0.5167 years 7.44 months 220.3 days 44.1 pentads	Neptune 1.006 years 12.07 months 367.4 days 73.5 pentads	Pluto 1.895 years 22.7 months 662.1 days 132.4 pentads		
Uranus 0.2415 years 2.90 months 83.2 days 16.6 pentads	Neptune 0.6173 years 7.41 months 225.5 days 45.1 pentads	Pluto 1.004 years 12.05 months 366.7 days 73.3 pentads			
Neptune 0.3412 years 4.09 months 83.1 days 16.6 pentads	Pluto 0.6167 years 7.40 months 225.2 days 45.0 pentads				
Pluto 0.2413 years 2.88 months 83.1 days 16.6 pentads					

Uranus and	Neptune and
Neptune 171.4 years	Pluto 492 years
Pluto 127.1 years	

REPRODUCED FROM CYCLES IN WEATHER AND SOLAR ACTIVITY. BY MAXWELL O. JOHNSON.
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Sun	365.25 days
Moon	29.5 days
Mercury	88 days
Venus	225 days
Mars	687 days
Jupiter	4332 days (11.86 years)
Saturn	10759 days (29.46 years)
Uranus	30706 days (84.07 years)
Neptune	60,199 days (164.82 years)
Pluto	90,801 days (248.6 years)

CHART 6 COMMODITY PRICE, BUSINESS & SUNSPOT CYCLES



CYCLICAL MARKET FORECASTING

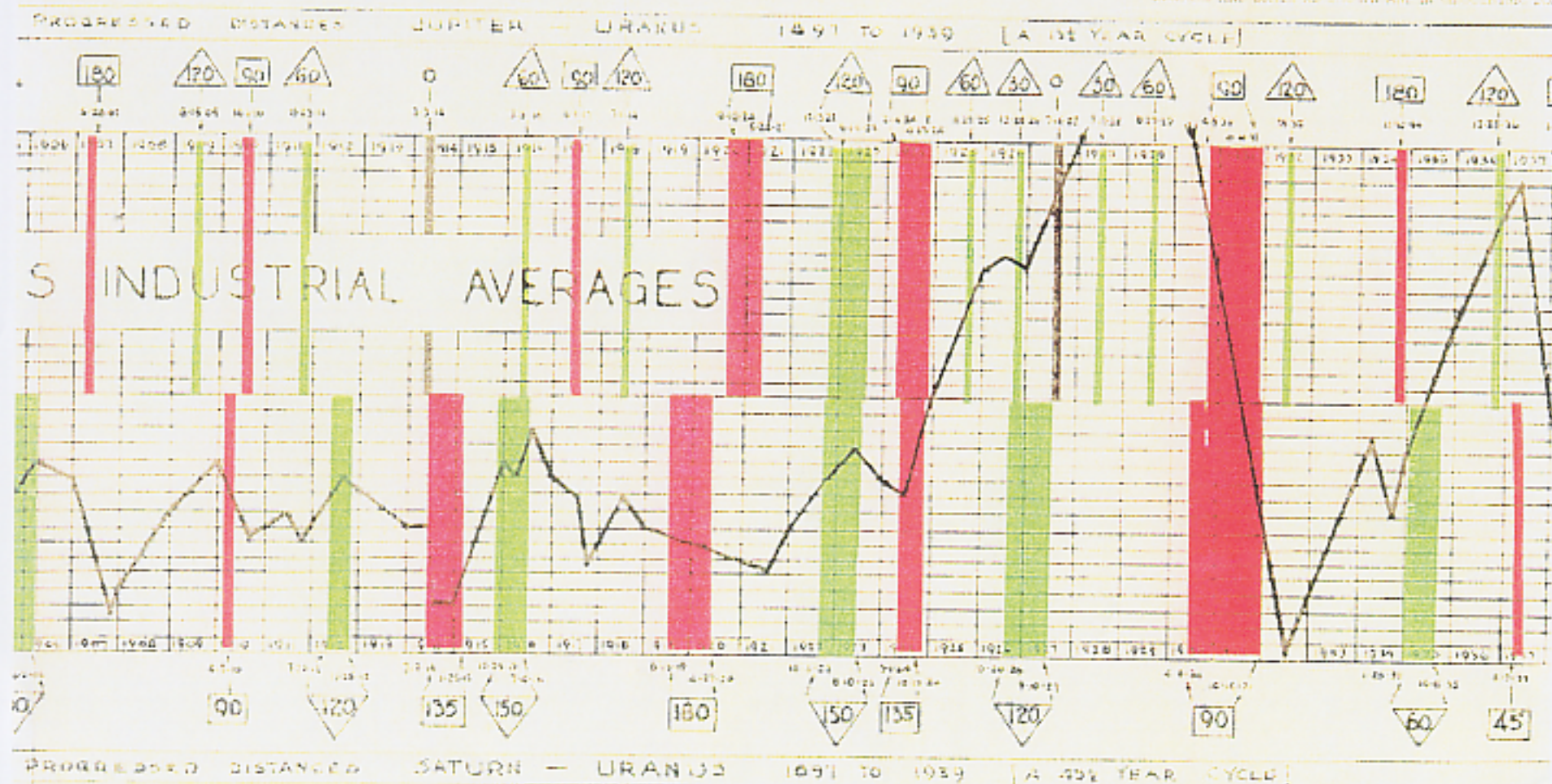
SIMULATIONS

JAMES MARS LANGHAM

Monetary Angles have been an accurate market barometer from 1897 to 1939
Advance positions are available for years ahead.

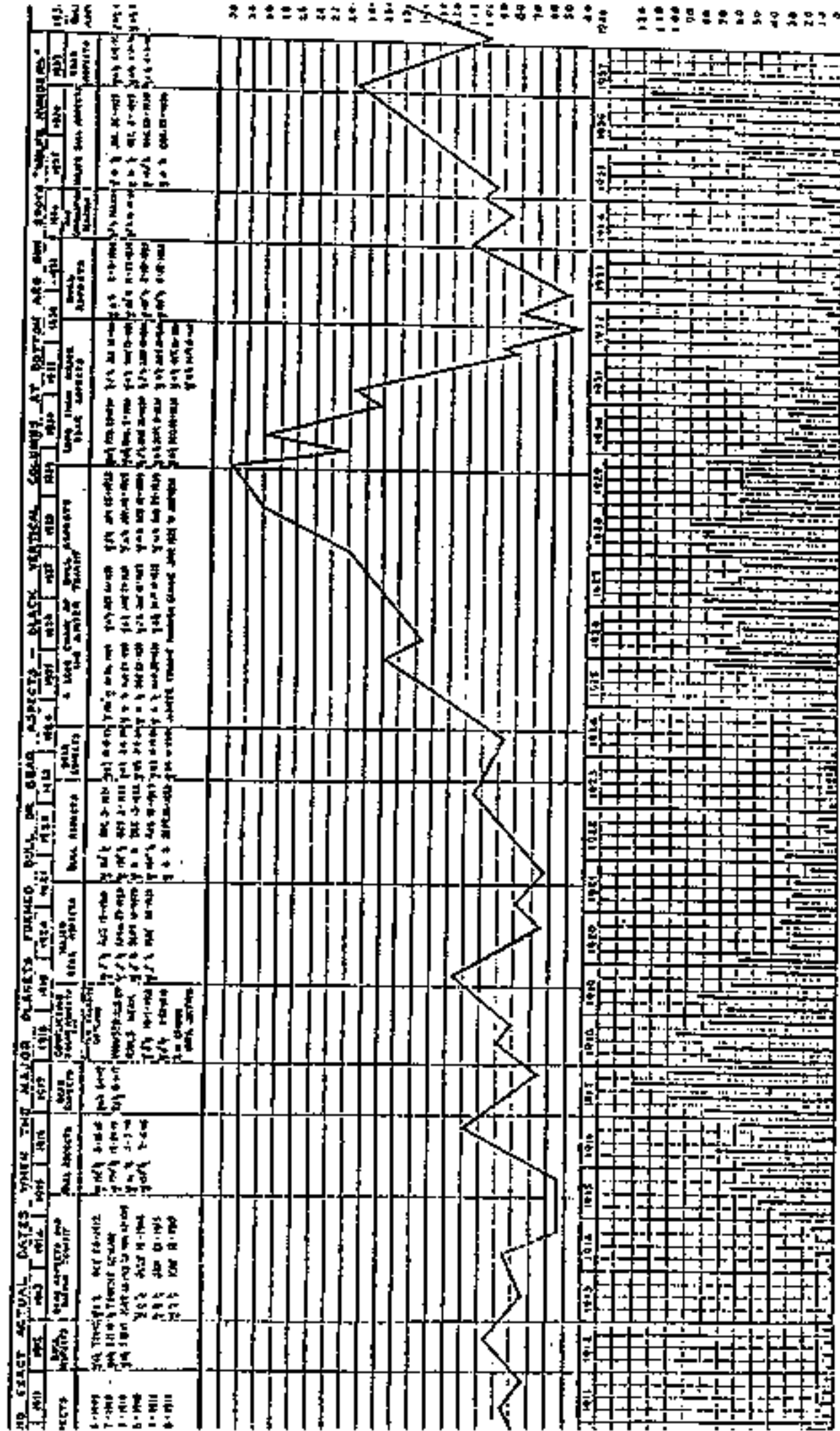
EXPLANATION

NOTES: 1. The 1000- μ m (39.37-in.) diameter plate is shown by the dashed line. The major diameter of the plate is 1000 μ m.



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Newspaper at Montserrat, for the people of Montserrat.

James Moss Langham



YEARS

	1784	1803	1821	1840	1858	1877	1895	1914	1932	1951	1969	1988	2006	2025	2043	2062
A	1785	1804	1822	1841	1860	1878	1896	1915	1933	1952	1970	1989	2007	2026	2044	2063
	1786	1805	1823	1842	1860	1879	1897	1916	1934	1953	1971	1990	2008	2027	2045	2064
	1787	1806	1824	1843	1861	1880	1898	1917	1935	1954	1972	1991	2009	2028	2046	2065
	1788	1807	1825	1844	1862	1881	1899	1918	1936	1955	1973	1992	2010	2029	2047	2066
B	1789	1808	1826	1845	1863	1882	1900	1919	1937	1956	1974	1993	2011	2030	2048	2067
	1790	1809	1827	1846	1864	1883	1901	1920	1938	1957	1975	1994	2012	2031	2049	2068
	1791	1810	1828	1847	1865	1884	1902	1921	1939	1958	1976	1995	2013	2032	2050	2069
C	1792	1811	1829	1848	1866	1885	1903	1922	1940	1959	1977	1996	2014	2033	2051	2070
D	1793	1812	1830	1849	1867	1886	1904	1923	1941	1960	1978	1997	2015	2034	2052	2071
	1794	1813	1831	1850	1868	1887	1905	1924	1942	1961	1979	1998	2016	2035	2053	2072
E	1795	1814	1832	1851	1869	1888	1906	1925	1943	1962	1980	1999	2017	2036	2054	2073
F	1796	1815	1833	1852	1870	1889	1907	1926	1944	1963	1981	2000	2018	2037	2055	2074
	1797	1816	1834	1853	1871	1890	1908	1927	1945	1964	1982	2001	2019	2038	2056	2075
G	1798	1817	1835	1854	1872	1891	1909	1928	1946	1965	1983	2002	2020	2039	2057	2076
H	1799	1818	1836	1855	1873	1892	1910	1929	1947	1966	1984	2003	2021	2040	2058	2077
	1800	1819	1837	1856	1874	1893	1911	1930	1948	1967	1985	2004	2022	2041	2059	2078
I	1801	1820	1838	1857	1875	1894	1912	1931	1949	1968	1986	2005	2023	2042	2060	2079
	1802	1821	1839	1858	1876	1895	1913	1932	1950	1969	1987	2006	2024	2043	2061	2080
J	1803	1822	1840	1859	1877	1896	1914	1933	1951	1970	1988	2007	2025	2044	2062	2081
K	1804	1823	1841	1860	1878	1897	1915	1934	1952	1971	1989	2008	2026	2045	2063	2082

LEGEND

- A Extreme low stock prices, strikes, repression, despair
Beginning of a new business generation of 16 1/2 years
About four years of rising stock prices
About four years of improving business
Markets bare of goods
Young men becoming prominent
- B High stock prices
- C Panic
- D Low stock prices
- E High stock prices
- F Panic
- G Low stock prices
- H Very high stock prices, Most prosperous year. Waste, over speculation, extravagance, No more poverty ever. Most paper money in circulation. Everybody sees stocks higher and its
- J Major Panic - CRASH
About four years of falling prices, also business stagnated, breadlines, soup kitchens
Despair and unemployment
- K Extreme low stock prices, depression, despair
Same as A) strikes, unemployment
Many prominent deaths

18.5 year cycle timetable
for the stock market developed
by W.D. GANN in 1909. Based on the
North NODE of the MOON 00.

These are estimates of what years
to expect the market conditions in
the Legend.

The Stock Market's 37 year "Williams" Cycle

Phase One : This is the beginning of the 37 year stock market cycle. It consists of 4 Bull market waves in a long 14 year uptrend. In the 14th year, a 4 year Bear market begins, ending the life of this phase (18 years total).

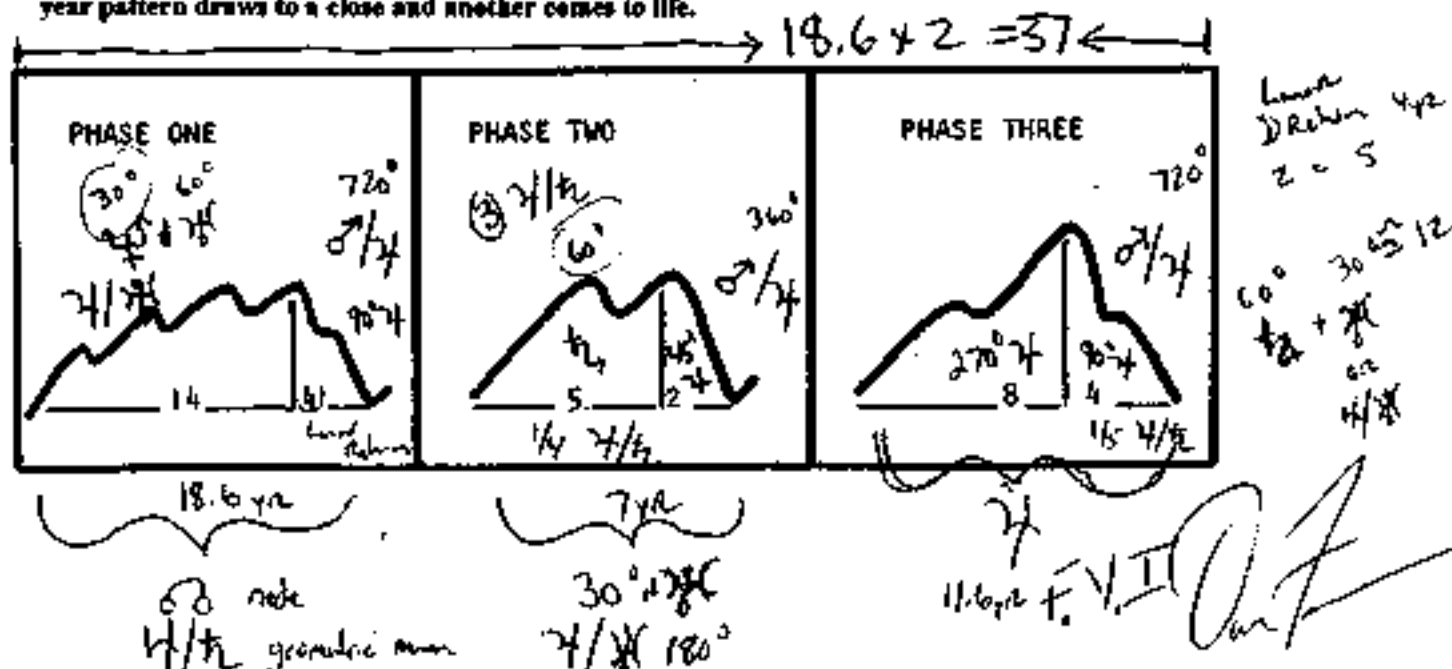
Phase Two : This is the shortest of the three phases of the 37 year cycle. This phase hits its high 5 years from it's start. It makes it's ultimate low point 7 years from the start. This pattern calls for a 5 year Bull move, which consists of two up legs with one intervening Bear market or sideways market. From this five year high, an approximate two year Bear market or sideways market begins. (7 years total)

Phase Three : This is the final phase of the 37 year cycle. The time span from the beginning of this phase to its ultimate market high is 8 years. From this peak, a 4 year Bear market begins, at which point the entire 37 year cycle starts all over again. (12 years total)

Phase 1	Lows	Highs
	1859 - 1896	1872 - 1909
	1896 - 1933	1909 - 1946
	1933 - 1970	1946 - 1983
	1970 - 2007	1983 - 2020
Phase 2	1877 - 1914	1882 - 1919
	1914 - 1951	1919 - 1956
	1951 - 1988	1956 - 1993
	1988 - 2025	1993 - 2030
Phase 3	1884 - 1921	1892 - 1929
	1921 - 1958	1929 - 1966
	1958 - 1995	1966 - 2003
	1995 - 2032	2003 - 2040

Also see
Time Movers
Lower of Business cycle
18.6 yrs

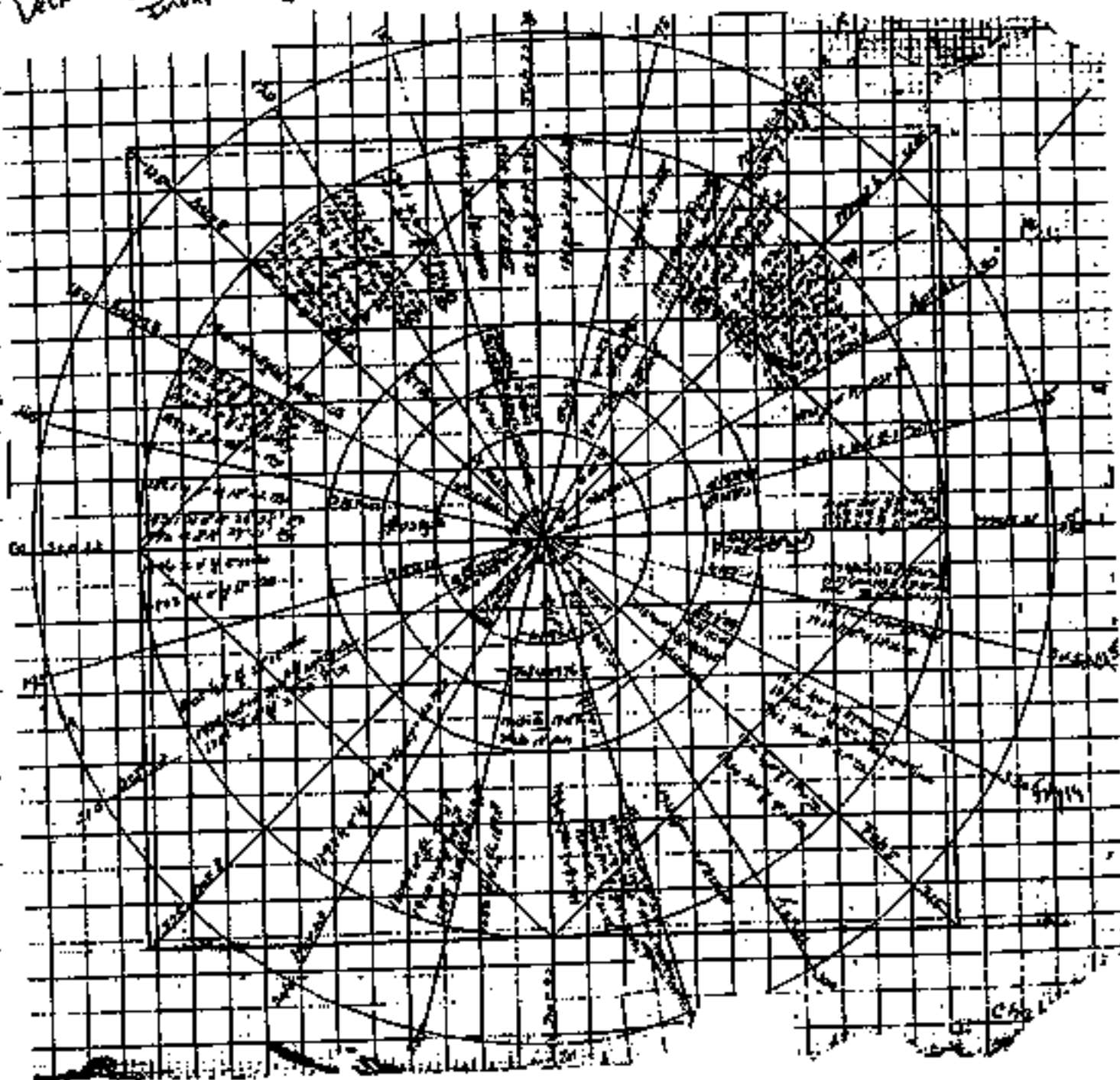
This cycle has predicted all of the markets major bottoms and tops with an incredible 95% accuracy rate. The few times it missed, the cycle was only off by a few months. At the time of this writing, we are about to end phase two for 1995. If this cycle continues to hold true, we should make a significant market low in 1995 similar to the one in 1958 i.e., a 20% correction. From this point we should enter phase three and experience a 8 year long Bull market. This major peak should be reached in the year 2003 (1929 style) with a major low occurring in 2007. From this point the 37 year pattern draws to a close and another comes to life.



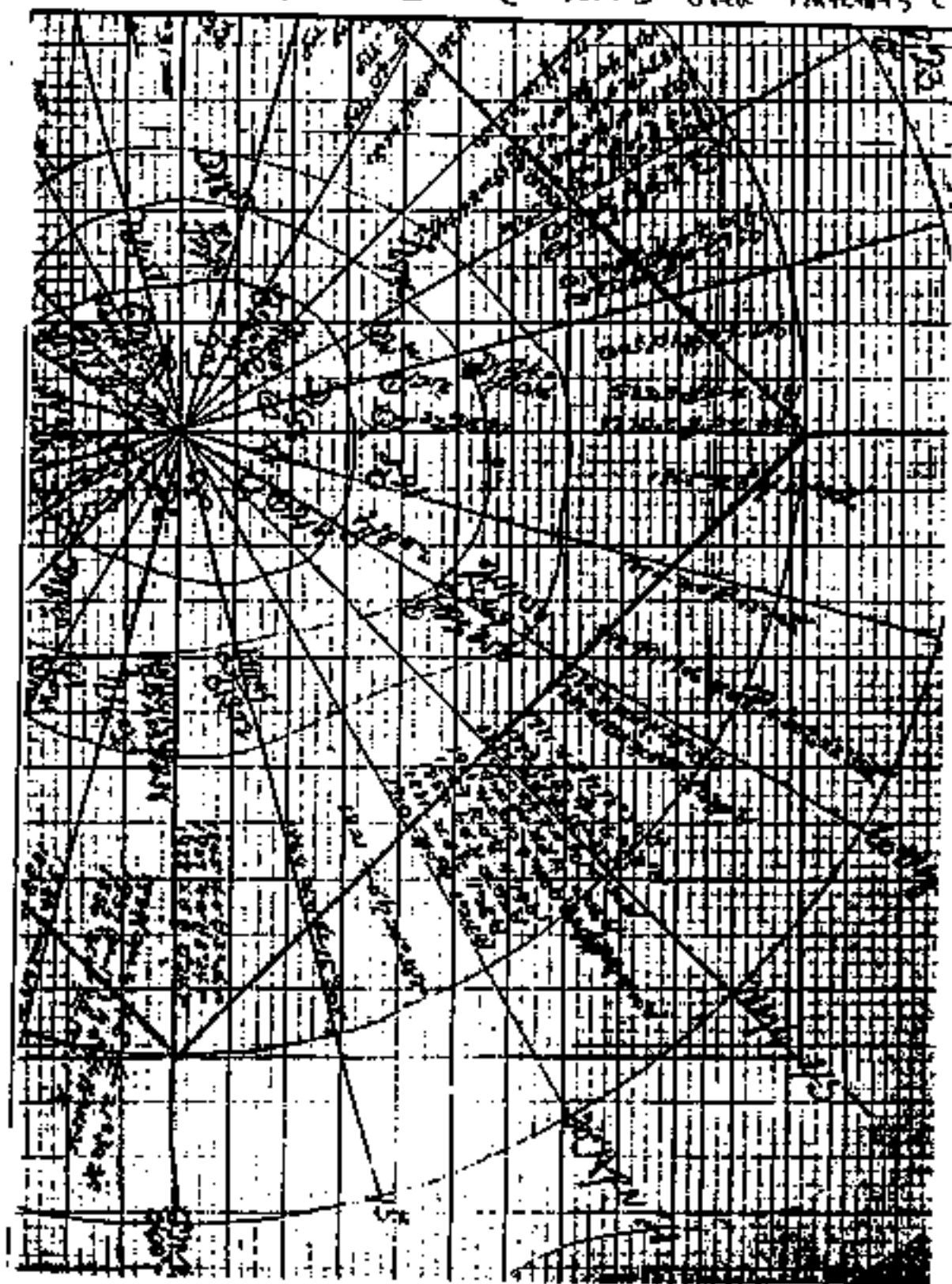
37 Year Table								
1792	1829	1866	1903	1940	1977	2014	2051	2088
1793	1830	1867	1904	1941	1978	2015	2052	2089
1794	1831	1868	1905	1942	1979	2016	2053	2090
1795	1832	1869	1906	1943	1980	2017	2054	2091
1796	1833	1870	1907	1944	1981	2018	2055	2092
1797	1834	1871	1908	1945	1982	2019	2056	2093
1798	1835	1872	1909	1946	1983	2020	2057	2094
1799	1836	1873	1910	1947	1984	2021	2058	2095
1800	1837	1874	1911	1948	1985	2022	2059	2096
1801	1838	1875	1912	1949	1986	2023	2060	2097
1802	1839	1876	1913	1950	1987	2024	2061	2098
1803	1840	1877	1914	1951	1988	2025	2062	2099
1804	1841	1878	1915	1952	1989	2026	2063	2100
1805	1842	1879	1916	1953	1990	2027	2064	2101
1806	1843	1880	1917	1954	1991	2028	2065	2102
1807	1844	1881	1918	1955	1992	2029	2066	2103
1808	1845	1882	1919	1956	1993	2030	2067	2104
1809	1846	1883	1920	1957	1994	2031	2068	2105
1810	1847	1884	1921	1958	1995	2032	2069	2106
1811	1848	1885	1922	1959	1996	2033	2070	2107
1812	1849	1886	1923	1960	1997	2034	2071	2108
1813	1850	1887	1924	1961	1998	2035	2072	2109
1814	1851	1888	1925	1962	1999	2036	2073	2110
1815	1852	1889	1926	1963	2000	2037	2074	2111
1816	1853	1890	1927	1964	2001	2038	2075	2112
1817	1854	1891	1928	1965	2002	2039	2076	2113
1818	1855	1892	1929	1966	2003	2040	2077	2114
1819	1856	1893	1930	1967	2004	2041	2078	2115
1820	1857	1894	1931	1968	2005	2042	2079	2116
1821	1858	1895	1932	1969	2006	2043	2080	2117
1822	1859	1896	1933	1970	2007	2044	2081	2118
1823	1860	1897	1934	1971	2008	2045	2082	2119
1824	1861	1898	1935	1972	2009	2046	2083	2120
1825	1862	1899	1936	1973	2010	2047	2084	2121
1826	1863	1900	1937	1974	2011	2048	2085	2122
1827	1864	1901	1938	1975	2012	2049	2086	2123
1828	1865	1902	1939	1976	2013	2050	2087	2124

USA (GANN'S GREAT Mutations Chart)

Declaration of Independence & Columbus Discovers America.



Soybeans (GANN'S Great Matchless Chart)



PART 5

THE GODS

CHAPTER 17

AGES AND CYCLES OF TIME

The ages given in ancient literature were calculated from the travels of the sun and earth in the heaven. It is written that God (the sun) regulates time into days, seasons, years and ages according to his wisdom and that his laws are unchangeable. The dividing of time into years and seasons is a well known science, but the making of time into ages is not understood. It simply means the same as making days into months and seasons, and a given number of years into ages.

Before the beginning of the historic era (1500 B.C.), the center of population and civilization was in Asia and Africa. From the ruins found on these continents it has been learned that all years and ages were calculated in zodiacal degrees. In the tombs and in the different temples are found inscriptions containing the signs of the zodiac, showing the signs to represent the ages. The ancient zodiac and calendars, which have been preserved, begin the year with the constellation of the Bull (Taurus). The zodiac known as the Elephanta Pagoda (Sallsette) has the sign of the sacred Bull (the ox Apis), as the first ruler of the twelve constellations. The ceiling of a sepulchral chamber at Thebes shows the Bull as the first month of the summer season. The zodiac of Esne, the astronomical picture discovered in the Ramesseum of Thebes, illustrates the Bull to be the leading sign of the age, calculated from the time when the Vernal equinox passed through the Hyades, located in the head of the Bull. The Chinese have referred to the beginning of the cycles for the sun and stars to Taurus and Chinese records relate the Pleiades as marking the beginning of time. Hesiod sings of the Pleiades as beginning the labors of the year and the Roman myths refer to them, in connection with the Virgilia, as the time for the beginning of the cycle.

The Egyptian title of "Pharaohs" and "Dynasties" was applied to what the Hebrew race called "Logos" and the ages of the "Patriarchs." In the different ages names of men and women were given to the cycles, or dynasties to record history in ages. The ages were calculated from the moving of the equinoctial points, within the twelve signs of the zodiac, in 28, 30 and 32 degree cycles of 72 years to each degree. In all ages of the past, and by all races up to the present time, the system described in dividing the cycles of each sign of the zodiac into minor divisions has been used.

We take it for granted that the Christian Age simply represents a cycle of time, and that this age corresponds to the dynasties and patriarchal ages. The system used to divide ages into smaller cycles is very simple. The great

cycle of our solar system contains 25,920 years of 360 degrees each, which makes 72 years to each degree, or day. We next divide 25,920 into twelve ages, which equals 2,160 degrees of 72 years each. The cycle of 2,160 is divided into 30 days and equals 72 degrees to each day, which is a very common system and easily understood.

The Bible contains the statement that at the time of the end, called the judgment day (Doomsday), the world would be destroyed, and we now realize that the end of the world means the end of an age, but not the end of time. The universe is operated in cycles, whereby the minor ages repeat itself into periods of time. Nature gives the best illustration of what an age is and how long it lasts, as it may be a day, season, year, or thousands of years.

The sun appears to rise in the east and goes over the meridian at noon, and disappears in the west to reappear again as a new cycle the following day. An age or a period of time is calculated by the same system, and in as many hours divided into minor divisions as the sun's cycle is divided within the 24 hours of its circuit. A year is divided into seasons, and the condition the sun produces on the earth is according to the formation, revolution and location of the sun. Nature provides that at the end of each year a destruction takes place and we call it fall or autumn. It is so also, with the ages of the earth, the sun and man, and nature gives signs and warnings of the approach of the termination of the ages. The Bible shows that the most common names among the Jewish people were used to describe days, years and cycles. When an age was described in the name of a man, his children and grandchildren represented the smaller cycles in the same sense as minutes and seconds do to the hours of time. "A year is as a day and a day as a thousand years with the Lord," which means, that an age is calculated from the sun and is complete whether it is an hour, day or year and that all are calculated by the same method.

In order to get to the fundamental principle of this study, we will state that there is no difference between degrees and hours, representing space and time. A circle contains 360 degrees for both space and time, and is the basic principle to calculate from. There are 60 seconds in one minute and 60 minutes in one hour, making 360 seconds in a cycle. There are 60 minutes in one hour and 12 hours makes 720 minutes for one day and 1,440 minutes in 24 hours. The 12 hours of a day correspond to the 12 signs of the zodiac, which contain 30 degrees each, making 360 degrees for 12 signs and 720 degrees to equal 24 hours, or one month of 30 degrees. All time and space are calculated as containing 360 degrees in 12 hours, which has to be considered when dividing a day of 24 hours into space.

Time and space are calculated from the earth's diurnal rotation on its axis in 24 hours and from its revolution around the sun in one year. The ecliptic circumference of the earth is 25,920 miles and as the earth revolves on its axis in 24 hours, the rate of motion, therefore, is 1,080 miles per hour. We divide the 1,080 miles per hour into minutes in dividing it by 60, and the rate of the earth's rotation on its axis will be 18 miles per minute. The

earth makes its circuit around the sun in one year, which corresponds to 24 hours and to 25,920 degrees, at the same rate of motion as it rotates on its axis. From this it will be seen that the earth rotates on its axis at the rate of 18 miles per minute and that it travels through space, around the sun, at the rate of 18 miles per minute, which makes the earth's actual motion 36 miles per minute. Astronomers calculate that our solar system travels at the rate of 18 miles per second. We state that the motion of the sun and earth, among the stars, is 36 miles per minute, making 2,160 miles per hour, 25,920 miles in 12 hours and 51,840 miles in 24 hours, which makes the motion of the earth 72 miles per minute, without adding the motion of the polar fall.

A quarter circle is called one week and contains one fourth of 24 hours, which is six days or degrees. Let it be distinctly understood that one of God's quarter divisions, called a week, contains six days, or degrees and corresponds to a quarter of the daily circuit of 24 hours. A day is as one hour and contains 60 degrees, therefore, making six days contain 360 degrees; 12 days 720 degrees; 24 days 1,440 degrees; and 30 days 1,800 degrees, hours or minutes. A month of 30 days is to 12 hours, as a day of 60 degrees is to 15 hours, which makes 15 degrees to one hour or distance. A year of 12 months of 30 degrees each contains 360 degrees and is to the distance around the earth, as 24 hours is to a day, and also as a month of 30 degrees is to 24 hours. From this it will be seen that 90 is to 360 as 6 is to 24, and represents one fourth of space and time.

It is written that in the beginning God created the world in six days, which means that the first six degrees represent 90 degrees in space from the beginning, or Vernal equinox. A solar year contains 360 degrees (within twelve months), therefore making six months contain 180 days and three months 90 days, which again represent degrees of longitude. There are 24 hours in a day and 30 days makes 720 hours, which are divided into deacons by three and equal 240 hours to each deacon of 10 degrees, making 8 hours to one deacon.

The circumference of the earth contains 25,920 miles and as the earth revolves on its axis in 24 hours, the motion of the earth is 1,080 miles per hour. The motion of the earth in 25,920 miles is divided into 12 hours, or months and equals 2,160 miles, which is again divided into 30 days and equals 72 miles to each degree. One hour, or day equals 72 miles; 10 hours 720 miles; 30 hours 2,160 miles; and 12 hours, or months contain 25,920 hours, miles, or degrees. One revolution of the earth is called a generation in the Bible, and is calculated from the polar meridian degree until it returns to the same meridian again in precession.

God, the sun, as father time, created the heaven and the earth in six days and on the seventh day rested, from which we learn that time is divided into six and seven days. It does not say, in the story about creation, that a day contains any given number of hours, but that time is divided into day and night, which is again described as divided into seven and five tribes.

A week of six days represents six signs, or hours, and a week of seven days contains seven signs, or hours. A week of seven days, months, or seasons

consist of the seven months from the Vernal equinox to the Autumn equinox, which is from February 19th to September 23rd. A day and the summer season contain seven hours and months; the nights and the winter season contain five hours and months. The seven months of summer have also been described and calculated from the Bull sign, in April, to the Scorpion in November, and the winter months from November to April. (See plate 6).

A week of six days, or hours, begins at twelve o'clock and extends to six o'clock, which correspond to the space from the first degree Pisces to Virgo, and if calculated for seven days, it includes Virgo as the seventh day, or sign. The second week is calculated from six o'clock to twelve o'clock and contains the Southern six signs from August to February. It will be seen that a week of six days contains six signs and that a week of seven days, seven signs, making nineteen precessional weeks terminate the age.

The six day week when applied to the signs of the zodiac and to the twelve hours of the day is in the following order: Monday is the first day of the week and corresponds to the sign Pisces and the hour between 12 and 1 o'clock. Tuesday corresponds to the time between 1 and 2 o'clock in Aries; Wednesday to the time between 2 and 3 o'clock in Taurus; Thursday to the hour between 3 and 4 o'clock in Gemini; Friday to the hour between 4 and 5 o'clock in Cancer; Saturday to the hour between 5 and 6 o'clock in Leo. The second six day week begins with Monday in Virgo and corresponds to the hour between 6 and 7 o'clock; Tuesday to between 7 and 8 o'clock in Libra; Wednesday to between 8 and 9 o'clock in Scorpio; Thursday to between 9 and 10 o'clock in Sagittarius; Friday to between 10 and 11 o'clock in Capricorn; Saturday to between 11 and 12 o'clock in Aquarius.

It will be observed that the opposite hours and days of the signs, represent the same day in the week, in both the Northern and Southern Hemisphere. This is best understood by the signs of Cancer and Capricorn for June and December, which represents Friday. In ancient writings Friday was given as hangman's day and the day when time of all description is killed. Time is killed on the cross formed by the solstices and the equator. The sun is furthest north and south of the equator in June and December respectively, which forms a cross with the equator, making the earth's four cardinal points, on which time is crucified. The earth remains stationary about three days, when it reaches its furthest points north and south. It has been referred to that the age was dead and buried for three days and when this was applied to the weeks, it is described to take place from Friday to Sunday. It now becomes evident that one week contains six hours, days, months, or years. Two weeks correspond to twelve hours and four weeks of six days each to twenty-four hours. Six days, or hours of 60 minutes to the hour, makes 360 minutes, which corresponds to hours and days. Two weeks of 12 hours, make 720 minutes and four weeks or 24 hours 1,440 minutes, or degrees, thus representing an age.

Six days of six hours each, correspond to 36 degrees, twelve days of six hours to 72 degrees, and twenty-four days represent 144 degrees. Six hours, days, or a week of six days, correspond to 36 degrees or one-tenth of the

cycle of 360, which makes 36 one-tenth of all calculations. A year of 360 degrees was divided into twelve months of 30 days each. A cycle of 360 degrees was also divided into ten hours, days and months of 36 days each, making five days equal 180 degrees, or hours. From this it will be seen what the will of Father Time means, and that his laws are unchangeable. It is written that God creates and destroys according to his will, which can be readily seen from the motion of the earth. A day is as a year and represents a generation, which describes what is meant by the beginning and ending of the different generations. A day of creation begins at sunrise in the morning and terminates at sunset, which means that the day is destroyed when night comes on. The seasons of the year and ages are destroyed when Father Time changes the seasons and years according to his will.

The past age described in the Bible contained 32 years, and is given in the house of Pisces in the name of Jesus. This age terminated in the destruction of the Christian Age, when the North Magnetic Pole passed the solar meridian at the 96th degree west longitude, in the year 1920 A.D. From this we may learn that the age of Jesus, containing 32 years, represents 60 degrees in space as an age.

One of the comparative ages described in the Bible contain 5,924 years, which have been so divided that 4,004 years are termed B.C. and 1920 A.D., up to the time of the crucifying of Christ, which terminates the age. The cycle described after the age of Jesus, is in the name of the Apostle St. Paul, containing 84 years. This age is divided into three cycles, or months of 28 years each, and represents the polar distance from the 96th degree west longitude to the 180th meridian.

B.C.—A.D. AND A.M.—P.M.

The dividing point between B.C. and A.D. corresponds to 12 o'clock noon (the zero degree and the Vernal equinox), which means the zero meridian degree. It corresponds to our present method of using A.M. for the forenoon hours and P.M. for the afternoon hours. The term B.C. means before zero and A.D. after date. When time is described as B.C. it was counted backward from zero, or 12 o'clock, and means that the time is lacking so many minutes, years, or degrees before reaching zero time. If the age is given at 60 B.C. it corresponds to ten o'clock, because it is 60 degrees before 12 o'clock. The term 60 B.C. when applied to the distances in the heaven and also on the earth in degrees, means 60 degrees east, or before the zero and first meridian degree.

The term equinox means node, or pole, and relates to the two polar points called spring, or vernal equinox, and the fall, or autumn equinox, which is calculated from the first and seventh meridians and the time when the sun enters the first degree of Aries and Libra every year. The zero meridian equinox takes place about February 19th; the Autumn equinox about September 23rd; and when this method is applied to the hours of clock time, it corresponds to 12 and 7 o'clock. It now becomes evident that the Zero degree and vernal equinox represent 12 o'clock and February 19th as well as the north polar degree for a day, year, and all cycles of time; and that

the precession of the equinoxes are calculated from both the Zero degree and the first meridian.

The principle of describing the meridian degree, for the earth, is from the sun's motion in the 12 signs of the zodiac. The sun rises in the east, which corresponds to A.M. or B.C., and reaches the meridian at 12 o'clock, and then sets in the west, which again corresponds to P.M. or A.D. From this we learn that the signs of the zodiac from Virgo to Pisces, represent A.M. and forenoons in the east, and from Aries to Virgo the afternoons as P.M. in the west.

The 12 signs of the zodiac must be located on earth as they are in the heaven, for the reason that the rotation and revolution of the earth corresponds to the motion of the sun. It has been proven that when the sun enters Aries on March 21st, it corresponds to what is known as the Greenwich meridian, which crosses England and France in longitude. From this degree the signs and hours can be calculated westward around the earth, in 30 degree periods, to correspond to 360 degrees. The Zero meridian degree on earth corresponds to 30 degrees in longitude east of the Greenwich degree, which crosses Alexandria, Egypt, and Leningrad, Russia. This meridian represents the Vernal equinox, or 12 o'clock on earth, and the first degree of Pisces in the heaven. It is an old saying that the pyramids are located in the center of the land surface of the earth, and that it is the place where the East meets the West. It is evident that it was known anciently where the Zero meridian was located, and that the pyramids were purposely built at this locality.

In order to describe the location on earth in degrees, which corresponds to the ages of ancient time, we will describe the location of the year and degree of 600 B.C. to show the system used. The year 600 B.C. is given in nearly all ancient records as the termination of an age and the beginning of a new cycle. The story about Noah and the flood is given as having taken place at his 600th year, and this age corresponds to the beginning of all Asiatic religions, from Confucius to Buddha, Nebuchadnezzar, Solon and many others. The method employed, to locate places on earth, is the same as used to locate polarities in the heaven. We can say that 720 represents the first meridian and that it crosses England in longitude, and as 600 B.C. is 120 degrees before or less than 720 degrees, we deduct 120 degrees east from England, and we find this location to be on the East Coast of Asia. This corresponds to the sign of the zodiac of Sagittarius and R.A. 270, or 9 o'clock. From this it will be seen that Sagittarius is located on the East Coast of China, and that the beginning of a day, or a year, is at sunrise, when the sun rises above the horizon in the East and in Sagittarius.

In order to show that the ancients described the law and that the year 600 is the actual termination of a cycle, we will copy the days and hours which describe this calculation scientifically. If we take the lunar month to be 29 days, 12 hours, 44 minutes and 3 seconds, we find that 219,146½ days, make 7,421 lunar months and that this number of days gives 600 solar years, each containing 365 days, 5 hours, 51 minutes, 31 seconds. As this is nearly the same calculation as used at the present time, it will be seen that the ancient

Bible writers and astronomers knew the law, but used a different method of reaching their conclusions.

In all the sacred books of the Asiatic races, as well as the Bible, the year 600 B.C. is given as the beginning of their historic records, as an age. We have explained elsewhere that the first meridian degree represents the poles and the 720th degree, as 600 B.C. means 120 degrees east of Greenwich and the beginning of Sagittarius. This location is on the East Coast of Asia and represents the place and hour of one-third of the earth's polar fall of 32 degrees and is a turning point. It represents the beginning of a day containing seven signs or hours, which is from November 22nd to June 21st; and from 9 A.M. to 4 P.M., and means a day of 7 hours of 360 seconds, containing 2,520 seconds, or degrees to the age. The beginning is 600 B.C. and if we add 1,920 A.D. we obtain 2,520 years. From this it will be seen that the age which began at 600 B.C. terminated in the year 1,920 A.D., and is the termination of a cycle of seven generations of 2,520 years.

We have referred to the fact that cycles of time were called by the name of man to represent ages, which we will explain. The days of the week and months of the year, are given individual names and represent cycles or ages. The day of Wednesday is named after the Norse God Odin, or Woodin; Thursday after Thor; Friday after Freiva; Saturday after Kronos (Saturn), and Sunday after the sun. It simply means to give names to distinguish one from the other, whether it is a day, year, or age.

We have described in detail that the ages given in the Bible are calculated from 4004 B.C., and that the last age terminated in the year 1920 A.D., after a period of 5,924 years. The Bible also contains a description that the period of time, from Adam to the end of the age, contained 6,048 years, and that all ages terminated in the year 1920 A.D. Christian Bible students have calculated the time described as the creation and beginning of time, in the Garden of Eden, to October 1st, 4004 B.C., and that the age of 6,048 years terminated September 30th, 1920 A.D. The two ages of 5,924 and 6,048 years each, simply represent cycles, which were calculated from the earth's polar motion. The two ages do not represent the described creation of the earth, but the beginning of time as an age; the termination of time is not the destruction of the earth, but the ending of the age.

The ancients calculated time and ages from the earth's motion around the sun, as already described. It is written that all ages begin, or as it also is called resurrected, four days, or years after the termination of the previous one. Bible history dates from 4004 B.C.; the Christian Age began at 4 B.C., but is calculated from 1 A.D. and terminated in 1920. The Millennium Age began in 1921, but is calculated from 1925, making the four years referred to, which is added to the age. This can best be understood from the changing of the earth's polar motion in December every year. The beginning of the age takes place when the earth enters the sign of Capricorn on December 21st, but the Christian Age is described from midnight December 25th, and that represents the four days which is comparatively described from 4 B.C. to 1 A.D. It is these four days as years, when applied to the present age,

which corresponds to from the year 1921 to 1925, when the Millennium actually begins.

The name for the day of rest, or Sabbath, is derived from the fact that at every seventh hour, day, year and age, the earth's polar motion becomes stationary, which can readily be understood from the motion of the moon and the tides in the ocean. The day of rest is called the Lord's, or sun's day, and in B.C. time calculated from the new and full moon. This makes the four quarter periods of the moon's time identical to four weeks and to the four tides at which time the motion of the sun, earth, moon and the tides becomes stationary. The sun's day when God rests, therefore, corresponds to the time when the sun and all polar motion, including the tides, becomes stationary, that is at rest. It is this period of rest which was called the Sabbath day in ancient time.

We will mention here, that there is a disagreement regarding the location of the Vernal equinox in the celestial heaven, when this degree is located from the zodiac. There are astronomers who claim that the Vernal equinox was on the 13th degree of Pisces in the year 1896, and others claim that it entered Aquarius in 1883. We state that the sun's vernal equinox passed over the very point of the Zero meridian in the year 1920, as we have learned from the description given in the Bible that this is the year when the change in ages takes place. The Millennium, or Aquarian Age begins when the Vernal equinox passes from Pisces and enters Aquarius. The Bible and other books contain descriptions of the change in the ages. It states that the termination of the Christian Age is the time when the Millennium Age begins. If astronomers are correct, the Millennium Age, consequently, will not begin before 1,200 years in the future. The Bible, however, declares that the Millennium Age begins at the termination of the Christian Age. It is given for the time when the old heaven and earth are destroyed, after which time a new heaven and earth are to be created. We base all of our astronomical calculations, regarding ages, from the description given in the Bible, and have found it necessary, in several instances, to differ with the present method of astronomical calculations. We are informed that astronomers have been unable to outline the location of the sign of Pisces, and it is, therefore, possible that a mistake has been made. It is also possible that astronomers in 1896 located the Vernal equinox from the first meridian and calculated space from the Zero meridian on the zodiac. It is evident that a change in polarity took place in the year 1920, and that this polar motion caused the termination of the Christian Age, and also that 1925 is the time for the beginning of the Millenium Age.

Europe is in the Bible described by the name of "Egypt" and the location of Europe corresponds to the time between the zero and first meridian in the sign of Pisces. In the description of this age, we are told that the people fled into Egypt for safety, as the destroyer could not kill anyone there. The reason that Europe is described as a safety zone is that its location corresponds to time between twelve and one o'clock. Time is destroyed at twelve o'clock and the beginning of time is calculated from one o'clock, which

makes this space and locality exempt from the killing by Pharaoh, the father and destroyer of time.

The story that Moses killed an Egyptian in the 15th century B.C., represents the destruction of Europe at the time. Moses is described as having led the population of Egypt out of bondage into the promised land. He led the children of Israel across the Atlantic Ocean, described as the Red Sea, and over the flooded wilderness of the eastern part of the present location of the United States. He brought the children of Israel to the promised land, at the 120th degree west longitude, and died as an age on the mountain when he was 120 years of age.

When it is realized that Europe was called "Egypt," and that it was flooded at the same time the eastern part of the United States was under water, the reason for the emigration described in the name of Moses will then be understood. In this gradual flooding of the lowlands of Europe the population continued to emigrate to America and the trouble described by Moses in the wilderness represents the rising water. It states that the people complained to Moses, saying that they might as well have stayed in Egypt and died there, as to come to this wilderness, but they finally arrived at the mountain of the promised land.

The Books of Moses describe the law, given to the children of Israel in the mountains of what is now known as the Pacific Coast States, and this locality is the only place on earth, where the law is written in the formation of the country, because the law of Moses and the locating of the tribes described, contains the astronomical law. The tribes named represent the twelve signs of the zodiac, and the rediscovery and understanding of this mystery can only be solved by someone who knows the law and the formation of the Pacific Coast States.

The following periods represent ages, which have been calculated by Bible students and are supposed to be correct. The first age described in the Bible is from Adam to the termination of the flood and consists of 1,656 years. The second age is known as the covenant to Abraham and consists of 427 years. The period of the Exodus contains 430 years. The time in the wilderness represents 40 years and the division of the land by Joshua 6 years. The ages of the Judges consist of 450 years and that of the Kings 513 years. The time of Desolation is given as 70 years and from this time to 1 A.D. 536 years. The Hebrew Bible describes an age of 4,128 years. The Christian Age contains 1,920 years and all these ages, as years, added together consist of 6,048 years. From this it will be seen that this age of 6,048 years terminated in 1920 A.D., which is the time described as the end of the world, when Jesus would return to earth and establish his Kingdom for another age of one thousand years.

The ancient Sacred Books of the Hindu Aryans, now preserved in India, relates history in cycles, and called the ages "yuga" and "kalpa." A yuga is given as containing 432,000 years. The great cycle of 518,400 years divided into 12 yuga equals 432,000 years of 72 periods of 6, 60 and 600 to a minor yuga. From this it will be seen the Hindu-Brahmans also em-

played the zodiacal system of the Egyptians to describe history in ages.

The four ages referred to in ancient writings as the Gold, Silver, Bronze, or Brass and Iron Ages, correspond to the four seasons of the year and to the four periods of a day of 6 hours each, as well as to the dividing of hours, minutes and seconds into four periods of 15 minutes and seconds to each quarter division. The earth's age of 6,000 years is divided into four periods of 1,500 years each and corresponds to the dividing of an hour into quarters of 15 minutes. These four ages also correspond to the four tides in the ocean of six hours each, and to the four periods of 90 degrees each of the circle of 360. All divisions are calculated by the same method and corresponds to the moon's circuit of four weeks, which begins when the moon crosses the equator going north as the Golden Age and terminates with the Iron Age as the fourth week.

The Iron Age begins on December 21st; the Golden on March 21st; the Silver on June 21st; and the Brass, or Bronze on September 21st every year. The 24 hours of the day are divided into four six hour ages, in the following order. The iron period begins at midnight; the golden represents the forenoon; the silver the afternoon and the brass the evening hours until midnight.

The Christian Age contained 1,920 years, which made the four ages contain 480 years each, but as the spiritual and philosophical religion will continue 84 years in the future, it makes the age contain 2,004 years. Bible history contains a description of 6,000 years, divided into four ages of 1,500 years each, and also into twelve ages of 500 years to each age. The birth of the Christian Age took place on Christmas, December 24th, which corresponds to the beginning of the Iron Age containing 500 years. The Golden Age extends between the years 500 to 1000 A.D. The Silver Age terminated at 1,500 A.D., and the Bronze, or Brass Age, which is the present age, will terminate in 2004 A.D. The termination of the Iron Age in 2504 A.D. corresponds to the ending of the year at midwinter, in December, when a new year is born again.

The age containing 6,000 years is a quarter of the sun's cycle of 24,000 years, which is again a quarter of 96,000 years. There are six ages of 96,000 years, making 576,000 years, which is to the solar system as $2/3$ is to the earth. The 576,000 year age is divided into quarters of 144,000 years to each quarter, and corresponds to the distance of 240 degrees on earth. There are six ages of 144,000 years, making 864,000 years of 120 degrees, or years, and as 864,000 represents one-third of an age, a complete age will, therefore, contain 2,592,000 years. The circumference of the sun contains 2,592,000 miles.

SIGNS OF THE TIMES

The Bible says "There is a time for everything." All the laws of Nature teach this. There is a time to sow and a time to reap. The four seasons of the year teach us that there is a reaping time and a sowing time, and that we can not reverse this order of Nature's laws. Man does not try to grow oranges on Greenland's icy mountains; neither does he expect to cut ice from the tropical rivers in Florida, because it is out of season, time, and place. It is the same with the stock market. There is a time to buy and a time to sell, and when this time comes, neither bunches of bears nor bevys of bulls with hot air, hope, optimism, extreme pessimism, depression or bad reports, can force prices above or below the zones of Supply and Demand, out of season. You must learn to go with the tide, and not against it. Discern the signs of the times, and do not get caught in the undertow when the tide is flowing out. Those who hesitate and are late in buying or selling in the last stage invariably have to take losses.

GEORGE BAYER

Chart readers have a subconscious idea that in charting stock movements they discover similarities with previous movements, hoping for exact repetitions of movements one or two months later. This is a very fine idea if it would do so. We merely look to the Bible for help. Genesis 10 shows us the big cycles of stock movements or any other movements; so does Genesis chapter 5, wherein we note that the Adam cycle is 930 years long, equalling in our language 930° of the motion of one certain planet. After the Adam cycle is in operation for 130°, the given age of Adam at the time he beget Seth, another cycle makes its appearance and therefore is in operation already 800° before the former dies out. Jupiter happens to be the planet that is meant by Adam. Looking up the time it requires Jupiter to make a motion of about 90°, we find it to be about one year measured in our time. Dividing 930 by 30 gives you the answer of how long the Adam cycle must be. Thus, we have to do with large time spaces to go through the entire family of Adam to arrive at Noah. Noah, being again the same as Adam, i.e. Jupiter, another cycle starts that ends with Abraham. Abraham again is Jupiter. Let us now turn to the first book of the New Testament, St. Matthew. In chapter 1 we find the subsequent cycles of Jupiter beginning with Abraham. Verse 17 of that chapter tells us that we have 14 Jupiter cycles up to David, fourteen more up to the Babylonian time and fourteen more up to Christ, making a total of 42 cycles. The repetition therefore runs in fourteen cycles of Jupiter, at which time likewise we may look for similarities of repetition of movements. You may not be able "to tie" this statement with anything you heard before; but you are free to do such research yourself and convince yourself of these facts.

It has taken me years to shake off concrete facts. The idea that Washington or France move the market, but once discarded, a new foundation had to be built. The Bible was used as foundation and it proved to be a solid one. For those who want further proof that the Bible is nothing but an astrological book and proof that the writers of the Bible knew more about astronomy than we ever will know and especially of the influences of heavenly bodies upon human beings, I bring chapter 220 of Harshel's Outlines of Astronomy, from my copy, dated: Phila., Pa., 1849. "Now, although, it is true that the mean motion of no planets are exactly commensurate, yet cases are not wanting in which there exists an approach to this adjustment. For instance in the case of Jupiter and Saturn, a cycle composed of five periods of Jupiter and two of Saturn, although it does not exactly bring about the same configuration, does so pretty nearly. Five periods of Jupiter are 21663 days. (Hold this value for later!! GB), and two periods of Saturn, 21519 days. The difference is only 144 days, in which Jupiter describes, on an average, 12°, and Saturn about 3°; so that after the lapse of the former interval they will only be 7° from a conjunction in the same parts of their orbits as before. If we calculate the time which will exactly bring about, on the average, three conjunctions of the two planets, we shall find it to be 21760 days, their synodic period being 7253.6 days. In this interval Saturn will have described 804° in excess of two sidereal revolutions, and Jupiter the same angle in excess of 5°. Every third conjunction, then, will take place 8°6' in advance of the preceding, which is near enough to establish, not, it is true, an identity with, but still a great approach to the case in question. The excess motion, for several such triple conjunctions (seven or eight) in succession, will lie the same way, and at each of them the elements of P's orbit and its angular motion will be similarly influenced (Note: the author here states clearly that one planet influences the other, two enormous big bodies!!) so as to accumulate the effect upon its longitude; thus giving rise to an irregularity of considerable magnitude and very long period, which is well known to astronomers by the name of the great inequality of Jupiter and Saturn.

The arc 8°6' is contained 44 4/9 times in the whole circumference of 360°; and, accordingly, if we trace round this particular conjunction we shall find that it will return to the same point of the orbit in so many times 21760 days or in 2668 years. But the conjunction we are now to consider is only one out of three. The other two will happen at points of the orbit about 123° and 246° distant and these points will also advance by the same arc of 8°6' in 21760 days. Consequently the period of 2668 years will bring them all around, and in the interval each of them will pass through that point of the orbit from which we commenced; hence a conjunction (one or the other of the three) will happen at that point once in one third of this period, or in 889 years.

(hold this value for later, G2); and this is therefore, the cycle in which the great inequalities would undergo its full compensation, did the elements of the orbits continue all that time invariable."

This is what herchel has to say. A cycle, or rather the circle has a length of $368^{\circ}6'$; one third of that circle is $122^{\circ}42'$.

To prove astrology in the Bible is very difficult, because of the clever substitutes the writers have used. The maze of statements and stories, the individual meanings of each chapter, and especially the fact that everything that is measured therein in degrees, years or days, is meant to be a part of $368^{\circ}6'$ and not a part of 360° .

We find in the Book of Ezra, Chapter 2, verse 2 up, the people who joined in the first expedition under Zerubbabel (Jupiter), the number of the people of Israel (which again is Jupiter!) verse 2 tells us that Zerubbabel came with 10 other leaders which are called by name, making a total of 11. We therefore use only the first 11 tribes enumerated thereafter. Verses 3 to 13 gives us their names and number:

Children of Parosh	2172	(note astronomer's value of 2176011)
Children of Shephatiah	372	
Children of Arub	775	
Children of Pahathmoab	2812	
Children of Elam	1254	
Children of Zattu	945	
Children of Zaccai	760	
Children of Bani	642	
Children of Bebai	623	
Children of Azgad	1222	
Children of Adonikam	656	Grand total 13243 or $122^{\circ}43'$

This shows the specific cycle of Jupiter divided in its parts, a value which the astronomer found too.

The balance of the count refers to $37^{\circ}10'$ and $82^{\circ}36'$ cycles, which latter value is shown when adding the tribes as 3196 ($31^{\circ}96' = 82^{\circ}36'$). These cycles have not been discussed in this book at all. However, added all together gives us $242^{\circ}29'$. Chapter 2, verse 65 of Ezra states that the total number of men in the community were 42360; omit the 0 and you have $42^{\circ}36'$.

In the book of Numbers, chapter 26, verse 7 up, we also find a count of population.

Tribe of Reuben	43730	people
Tribe of Simeon	22200	
Tribe of Gad	40500	
Tribe of Judah	76500	
Tribe of Issachar	64300	
Tribe of Zebulun	60500	
Tribe of Manasse	52700	
Tribe of Ephraim	32500	
Tribe of Benjamin	45600	
Tribe of Dan	64400	
Tribe of Asaer	53400	
Tribe of Naphtali	45400	Grand Total 601730 people or $601^{\circ}73'$ or $602^{\circ}13'$.

Deduct 360° from this value gives you the same as above, $243^{\circ}13'$.

Going back to the time of the Deluge, Genesis 8, 14, we may find it strange that Noah kept a diary for posterity in all his excitement the Deluge brought him. He noted carefully, that it was in the 2nd month on the 27th day of the month in the 601st year of his life that the earth dried. This value represents nothing but the same cycle as shown above.

How close these ancient astrologers came to find the inequalities between Saturn and Jupiter may be seen in Chronicles, first book, chapter 15-verse 5 up: The number of men that carried the ark:

Uriel	120 men
Asaiah	220 men
Joel	130 men
Shemaiah	200 men
Eliel	80 men
Aminadab	112 men
grand total 862 men.	

We have shown that these men represent degrees of the circle, but the circle containing $368^{\circ}6'$ (actual value is $368^{\circ}10'48''$). Then we turn these degrees of the 360° circle into degrees of the 368° circle, as

120° equal	$122^{\circ}43'36''$	
220	224 59 56	
130	132 57 14	
200	204 32 40	
80	81 49 4	
112	114 32 41	Grand total $882^{\circ}35'11''$

These degrees we have to take as representing years; in that case have the great inequalities of Jupiter and Saturn just as the astrologers found them 4000 years later....

The book of Numbers, chapter 2 gives us another account. The work of this particular book seems to have done the work of previous work all over and found a figure that somewhat deviates from others, and it shows that there was checking and rechecking done of these values.

Tribe of Judah	74600	
Issachar	54400	
Sebulon	57400	
Reuben	46500	
Simeon	59300	
Gad	45650	
Ephraim	40500	
Manasse	32200	
Benjamin	35400	
Dan	62700	
Asser	41500	
Naphthali	53400	Grand total 603750 or 603075'

Chapter 2,32 tells us that there were 603550.

We wonder now what this value of $601^{\circ}23'$ and $603^{\circ}75'$ respectively means. We know from astronomy that the lunar apogee circulates in 3232.57 days. The moon's Node has a rate of retreat each day of $1'$ and is carried around the entire Zodiac in 18.6 years.

When we divide now our 601730 by 3232.57, we obtain 18.615 years. It indicates that these parts of the Bible have to do with the Moon's motion, or with astronomical laws. The stories, however, that follow and precede have to do with effects of the planets. They all denote astrological cycles and even though universities or governments do not recognize a science that is based on the greatest book in existence, the revolutionary action of Neptune at this period may bring about a change of attitude.

As a final remark it may be said, that these ancient prophets had no telescopes, not logarithmic tables or any other facilities at their disposal.

This last few lines may show you that I have gone much deeper into market movements than I can bring forth in this volume and state that I produce very accurate forecasts for stocks or any commodity for any period, even without using any of the laws given to you.

Low Jones
Average

3/29

to
40/37

? to Top

240/37

to
20/39

13/29

to
13/37

240/37

0/20/39

5/31/46

1/13/32

5/31/46

1/31/46

to
1/14/56

Helio Centric MOVES

$$t_2/\gamma \quad t_2 \text{ moved } 85^\circ 51' \quad \gamma \text{ moved } \boxed{29^\circ 45'}$$

Pythagorean

$$\sqrt{85.5^2 + 29.95^2} = \boxed{90^\circ 60' = 90^\circ 36'}$$

Square

$$\gamma \text{ moved } \boxed{29^\circ 57'} \text{ \& the Hypotenuse} = \underline{\underline{90^\circ 36'}} \square \text{ Square}$$

$$t_2 \text{ moved } 84.07 \quad \gamma \text{ moved } 10.654 \quad t_2 \text{ moved } 32.7$$

$$\text{Geometric mean of } t_2/\gamma = \sqrt{84.07 \times 10.654} = \boxed{29^\circ 55'}$$

$$\odot t_2 \text{ moved } \boxed{90^\circ 43'} \text{ Square}$$

$$t_2/\gamma \text{ Geometric Mean } \sqrt{231.633 \times 90.73} = \underline{\underline{144^\circ 58'}} \text{ or } 12^\circ$$

$$t_2/\gamma \text{ Harmonic Mean } \frac{90.73 \times 231.633}{(90.73 + 231.633)/2} = \underline{\underline{46^\circ 57'}}$$

$$t_2 \text{ moved } 121^\circ 55' \quad (2^\circ \text{ past } 120)$$

$$\text{Geometric Mean } t_2/\gamma \quad \sqrt{284.384 \times 39.44} = 104^\circ 33' \quad (27^\circ 105)$$

$$\text{Harmonic Mean } t_2/\gamma \quad \frac{121.55 \times 39.44}{(121.55 + 39.44)/2} = 58^\circ 24' \quad (1^\circ 36' \pm 6)$$

$$\text{Square \& Geometric Mean } t_2/\gamma \quad \sqrt{416,216 \times 175,516} = \boxed{270^\circ 16'}$$

$$\text{Hypotenuse } t_2/\gamma \quad \sqrt{416,216^2 + 57,10^2} = 420^\circ 06' - 360 = \underline{\underline{60^\circ}}$$

$$\text{Square Geometric Mean } t_2/t_2 \quad \sqrt{304,116 \times 238.5} = \boxed{269^\circ 19'} \quad (41^\circ)$$

$$\text{Harmonic Mean } t_2/\gamma \quad \frac{304,116 \times 43.75}{(304,116 + 43.75)/2} = \boxed{76^\circ 29'} \quad (1^\circ 29' \text{ } ^\circ)$$

5/4/42
to
4/4/56

γ/δ moved 61° (1' past $\star$) Sextile
Geometric Mean γ/δ $\sqrt{421.25 \times 176.56} = 272.57$ (9' 42")

Note
Arithmetic
Mean

γ/δ Harmonic Mean $\frac{421.25 \times 61}{(241.125)} = 106^\circ 34$ (1' 34 past
Time $240^\circ \rightarrow$ Arithmetic Mean $\rightarrow$ (241.125) Time
 δ/γ Harmonic Mean $\frac{176.56 \times 61}{118.933} = 90^\circ 42'$ (42' past
Time $120^\circ \rightarrow$ Arithmetic Mean $\rightarrow$ 118.933 Squall

July 13
1932
to
5/4/42

δ moved $120^\circ 09'$ Δ TRINE
Geometric Mean γ/δ $\sqrt{299.683 \times 120.15} = 189$ nothing Sext
Hypotenuse of γ/δ $\sqrt{299.683^2 + 39.85^2} = 301.43$ (1' 43" off
Sextile
Harmonic Mean of δ/γ $\frac{120.15 \times 39.85}{(120.15 + 39.85)/2} = 59^\circ 51'$ (9' off Sextile

8/7/82
to
10/15/99

Geometric Mean of δ/γ $\sqrt{72.033 \times 200.06} = 120^\circ 02'$ (2' past
TRINE

1/30/94
to
1/3/2000

~~γ/δ $\sqrt{183.52 \times 73.65} = 115.11$ (1' 11" off)~~
 ~~δ/γ Harmonic Mean $\frac{183.52 \times 73.65}{(183.52 + 73.65)/2} = 105.11$~~
 γ/δ Harmonic Mean $\frac{183.52 \times 73.65}{(183.52 + 73.65)/2} = 105.11$

* Three Term Proportion
Example's from
my personal notes

CASH AND MAY SOY BEAN FUTURES

by

W. D. GANN

TIME PERIODS FROM FEBRUARY 15, 1930 to NOVEMBER 4, 1934

These periods run from 1 to 36. Suppose you want to get the $1/2$ points or 26 weeks from any of these important highs and lows, you move down to the column marked " $1/2$ " and run across to get all the time periods which are $1/2$ year or 26 weeks from any important high or low. Running from 1 to 36, these are:

Aug 19,	Jun 29,	Apr 2,	Oct 14,	Apr 17,	Jan 25,	Jun 19,	Feb 6,
Mar 11,	Apr 14,	Jul 31,	Apr 26,	Apr 26,	Jul 19,	Aug 18,	Nov 22,
Apr 9,	May 24,	Aug 12,	Sep 26,	Feb 20,	May 24,	Jun 29,	Nov 10,
Apr 12,	Aug 12,	Jan 7,	Jun 7,	Oct 25,	Feb 12,	Aug 17,	Oct 20,
Feb 19,	Oct 31,	Nov 20,	Jan 25,	Mar 25,	May 2,		

All of these dates are $1/2$ of the yearly time periods.

MOST IMPORTANT TIME PERIODS FROM ANNIVERSARY DATES

These are the actual dates of extreme highs and lows from 1 to 36 on the Time Table and are as follows:

Feb 15,	Dec 20,	Oct 8,	Apr 10,	Oct 20,	Jul 27,	Dec 20,	Aug 10,
Sep 13,	Oct 17,	Jan 27,	Oct 28,	Oct 28,	Jan 15,	Feb 14,	May 23,
Oct 9,	Nov 26,	Feb 6,	Mar 21,	Aug 24,	Nov 18,	Dec 28,	May 8,
Oct 16,	Feb 8,	Jul 8,	Dec 10,	Apr 21,	Aug 14,	Feb 13,	Apr 16,
Aug 20,	Apr 27,	May 10,	Jul 27,	Sep 27,	Nov 4,		

Study these periods in years that follow to see when high and low prices repeat around these dates. Examples:

Suppose you want to look up November, 1934 for May Soy Beans; first you look for the actual dates of highs or lows. You find November 26, 1930, high $276-3/4$ and November 16, 1929, low 220. Next you look for the $1/2$ point and find November 22, 1930, November 10, 1930, and November 20, 1934. Then you would look through the column of $1/3$ points in time and find November 19, November 23, November 30, and November 12. Next, the $1/4$ points in time periods are: November 11, November 25, November 18, and November 21. Next, $1/5$ points are: November 27, November 9, and November 27. Next, the $3/8$ points are: November 8, November 24, November 6. Then, the $5/8$ points are: November 28, November 12. Next, the $2/3$ points are: November 26. The $3/4$ points are: November 16, November 17, November 12, November 11, November 16. Next, the $7/8$ points are: November 14, November 5, November 14.

NEARLY-TIME PERIODS FOR SOY BEANS FEBRUARY 15, 1930 to NOVEMBER 4, 1934

All of these time periods from the different highs and lows are shown

HIGH AND LOW PRICES FOR CASH AND MAY BEAN FUTURES

We are giving these prices as far back as there are any records of Cash Soy Beans so that you can check them in working out future yearly and monthly cycles.

1913 Nov 15 low 154	1923 Feb 15 high 214	1930 Jun 15 high 216
1915 Jan 15 high 238	1923 Oct 15 low 209	1931 Jan 15 high 146
1916 Oct 15 low 189	1924 Oct 15 low 216	1931 Nov 15 low 152
1916 Oct 15 low 210	1925 Feb 15 high 264	1932 Mar 15 high 66
1917 Oct 15 high 257	1925 Dec 15 low 216	1932 Dec 20 low 44
1918 Feb 15 high 382	1926 Jan 15 high 238	1933 Jul 15 high 104
1919 Mar 15 low 292	1926 Dec 15 low 192	1933 Oct 15 low 68
1920 Feb 15 high 406	1927 Jun 15 high 222	1934 Jul 20 high 154
1921 Feb 15 low 217	1927 Dec 15 low 160	1934 Nov 15 low 84 1/2
1921 Nov 15 high 222	1928 Jan 15 high 213	1935 Feb 15 high 126
1921 Dec 15 low 216	1928 Nov 15 low 189	1935 Oct 15 low 68
1922 Feb 15 high 216	1929 Jul 15 high 248	1936 Aug 15 high 119
1922 Oct 15 low 189	1929 Nov 15 low 160	1936 Sep 20 low 110

The above figures are taken from Government records. They only reported cash prices on the 15th day of each month.

Futures trading started October 5, 1936. The low of May Beans on that day was 120. From this date on, all of the future high and low prices are given in the printed Time Tables. You can check any time period in weeks from these Tables but use these Cash Prices to get future cycles. Examples: 1915, January 15, a 40-year cycle terminates January 15, 1955. From February 1925, a 30-year cycle ends in February 1955, and the low of a 30-year cycle ends in December 1955.

PRICE RESISTANCE LEVELS FOR CASH AND MAY BEAN FUTURES

When any extreme high or low price is reached, look up the position on this Table for Weekly Periods and Prices; then when the price is down or up from an important level, look to see what resistance level it is at.

The Table below gives (1) the advance or decline from a low or a high price and (2) the position of the high or low price. All of these are covered from 1920 through November 4, 1954. In future, when high or low prices are reached, look up the price resistance in the same way and look up at the same time the Time Periods and Time Resistance Levels, the same as we have shown in the examples. In this way you get the position for Time and Price Resistance and changes in trend which will be shown on the Master Calculator and the Tables for Time and Price Periods.

		cents		
1920	high 406			on 7-3/4 years
1932	low 44	down	361	on 7/8 year and 7 years
1933	high 104	up	60	on 2 years and 1-1/8 year
1935	low 68	down	35	on 1 1/2 years and 2/3 year

5-9 of 52 = 6.5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400
401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500
501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600
601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	691	692	693	694	695	696	697	698	699	700
701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	721	722	723	724	725	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800
801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900
901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000

on the column running down from November 4, 1954. The high on November 4 was 299. Run down the column to 18, then look at the top at the date and you will see that 18 is February 9, 1949, low 201. Therefore, 299 was the 45° angle moving up from "0" from 201.

To get the present position up to the week ending January 22, 1955, you add 11 weeks. Add 11 to 299 gives 310. In the same way, add 11 to all the other figures in the same column and it brings them all up to date. Look in column 11 and you will see the time is 253 weeks to November 4, 1954. This 253 weeks is from December 29, 1949, when the last low was made at 224.

For the week ending January 22, 1955, you add 11 weeks, which gives 264, which means that the 45° angle from "0" on the Weekly Chart crosses at 264. Each week this angle moves up 1/2. Therefore, when the price closes below this angle, it will be in a weaker position and indicate lower prices.

The most important time cycle is from July 27, 1939, low 67. This is under column 6. You move over on column 6 to the line under November 4 and you find the time 797 weeks. Add 11 to this to bring it to January 22, 1955, gives 808 weeks. This can be divided by 2, 3, and 4 to get the angles up to date.

Check all of the other time periods in the same way. Check the Price and Time resistance levels with the Time Table which runs from 1 to 40 years. Example: The last high on May Beane, December 29, 1944, was 288. 288 is 1/2 the range of the option. Look at the table under 1/2 points and you see that 288 is in the 1/2 zone and is 1 1/2 years in time. 288 is in the same zone is 3-1/8 years. You will find all of these shown in the instructions and the resistance levels for Time and Price.

Study this Price and Time Resistance Table in connection with the Master Square of 82 and you will see that it is the most important Price and Time Trend Indicator that I have ever discovered.

MAY BEANE: NOVEMBER 1954 - JANUARY 1955

November 4 High 299, a Signal Day on the daily high and low chart. Look at 299 on the Price Scale and you will find that it comes out at 5-3/4 years, an exact resistance level, and also 299 weeks from February 9, 1949, therefore, Time and Price balanced on the 45° angle from "0", making this a selling level.

Why did May Beane make high on November 4? Look at the Table and you will see that the 5/8 point or Midseason comes out on November 8, which is important for a change in trend. Then, if you lay the bottom of the Calculator at "0" on 201, the last extreme low for May Beane on February 14, 1949, you find that this angle crossed at 299 in the week ending November 8, 1954, which was why the price made top and turned down.

November 16 Low 252. 252 is 17/2 down from 299, which is 1/3 of a year

and an exact resistance level.

August 20, 1953, May Beans low 239 $\frac{1}{2}$. Note 240-3/8 is 1/2 of the range from 44 to 436-3/4. Place "0" of the Calculator at 239 $\frac{1}{2}$ on this date. Adding 62 to this price gives 281 $\frac{1}{2}$, which is the top of the time period for the first year. The week ending November 20, 1954, was 65 weeks or 1 $\frac{1}{4}$ years from the extreme low and the price made 282 in this week and the minor trend turned up.

April 27, 1954, May Beans extreme high 422. This was an advance of \$1.82 $\frac{1}{2}$ per bushel from the low of August 20, 1953. 182 equals 3 $\frac{1}{2}$ years in time and is at the exact resistance level of 3 $\frac{1}{2}$. The reason for the high on April 27 was 1/3 of the year from December 28, 1932, the extreme low of 44 $\frac{1}{2}$ per bushel and 3/4 year from July 27, 1939, low 47 $\frac{1}{2}$. Therefore, both time periods and the price came out on exact resistance levels. If we use the exact high price of 422, it equals 8 years, which is 416 plus 6 $\frac{1}{2}$ = 422 $\frac{1}{2}$ as the 1/8 point and natural resistance level. From April 27, 1954 to the week ending November 6, 1954, the time was 25 weeks or 1/2 year, which was the reason for the top at 280.

From the low of 265-1/4 on September 27, 1954, the week ending December 30, 1954 was 13 weeks or 1/4 of a year, and December 28 was the anniversary date or 27 years from December 28, 1927, making this very important for a change in trend. On December 29, 1954 May Beans made last high of rally at 285 $\frac{1}{2}$ and trend turned down again.

January 16 - The next or most important point to consider is January 15, 1946, the extreme high. Therefore, January 15, 1955 ends 9 years or 364 weeks. Prices can go straight up or straight down at the end of the 7-year period. On January 17, 1955 May Beans sold down to 263 $\frac{1}{2}$, and made bottom for a rally.

HOW TO DETERMINE CHANGE IN TREND

Calculate the total cents per bushel that the price is up from extreme low levels and minor low levels. Calculate how much the actual high or low price is up from "0". Calculate how many cents per bushel the price is down from any extreme high level or minor tops. Then calculate the range between extreme high and low and you will find that the price will come out on the resistance level that is in exact agreement with the Calculator and the Tables for the Price and Time.

Example: February 9, 1949, low 201 $\frac{1}{2}$. This was down 215 $\frac{1}{2}$ from 436-3/4. If you will look at the Table, you will see that 4 years equals 216 weeks, making this an exact resistance or support level. Next, consider the price 201 $\frac{1}{2}$. 182 is 3 $\frac{1}{2}$ years; add 19 $\frac{1}{2}$ to 182, which is 3/8 year, gives exactly 201 $\frac{1}{2}$ as a resistance and buying level. Then consider the circle of 260 $\frac{1}{2}$. 1/2 is 180; add 22 $\frac{1}{2}$ to 180 gives 202 $\frac{1}{2}$. 22 $\frac{1}{2}$ is 1/6 of the circle and 1/2 of 45 $\frac{1}{2}$. The high in 1920 was 405, 1/2 of this is 202 $\frac{1}{2}$. All of this indicated a support and buying level at 202 $\frac{1}{2}$ to 201 $\frac{1}{2}$. Refer to the high of April 10, 1937 and you will find it was 202 $\frac{1}{2}$. Therefore, the price at that time was exactly on a selling level which later became a buying level.

LONG.XLS

Zodiac	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price	Price
Aries	1	361	721	1081	1441	1801	2161	2521	2881	3241	3601	3961	4321	4681	5041
1	1	361	721	1081	1441	1801	2161	2521	2881	3241	3601	3961	4321	4681	5041
2	2	362	722	1082	1442	1802	2162	2522	2882	3242	3602	3962	4322	4682	5042
3	3	363	723	1083	1443	1803	2163	2523	2883	3243	3603	3963	4323	4683	5043
4	4	364	724	1084	1444	1804	2164	2524	2884	3244	3604	3964	4324	4684	5044
5	5	365	725	1085	1445	1805	2165	2525	2885	3245	3605	3965	4325	4685	5045
6	6	366	726	1086	1446	1806	2166	2526	2886	3246	3606	3966	4326	4686	5046
7	7	367	727	1087	1447	1807	2167	2527	2887	3247	3607	3967	4327	4687	5047
8	8	368	728	1088	1448	1808	2168	2528	2888	3248	3608	3968	4328	4688	5048
9	9	369	729	1089	1449	1809	2169	2529	2889	3249	3609	3969	4329	4689	5049
10	10	370	730	1090	1450	1810	2170	2530	2890	3250	3610	3970	4330	4690	5050
11	11	371	731	1091	1451	1811	2171	2531	2891	3251	3611	3971	4331	4691	5051
12	12	372	732	1092	1452	1812	2172	2532	2892	3252	3612	3972	4332	4692	5052
13	13	373	733	1093	1453	1813	2173	2533	2893	3253	3613	3973	4333	4693	5053
14	14	374	734	1094	1454	1814	2174	2534	2894	3254	3614	3974	4334	4694	5054
15	15	375	735	1095	1455	1815	2175	2535	2895	3255	3615	3975	4335	4695	5055
16	16	376	736	1096	1456	1816	2176	2536	2896	3256	3616	3976	4336	4696	5056
17	17	377	737	1097	1457	1817	2177	2537	2897	3257	3617	3977	4337	4697	5057
18	18	378	738	1098	1458	1818	2178	2538	2898	3258	3618	3978	4338	4698	5058
19	19	379	739	1099	1459	1819	2179	2539	2899	3259	3619	3979	4339	4699	5059
20	20	380	740	1100	1460	1820	2180	2540	2900	3260	3620	3980	4340	4700	5060
21	21	381	741	1101	1461	1821	2181	2541	2901	3261	3621	3981	4341	4701	5061
22	22	382	742	1102	1462	1822	2182	2542	2902	3262	3622	3982	4342	4702	5062
23	23	383	743	1103	1463	1823	2183	2543	2903	3263	3623	3983	4343	4703	5063
24	24	384	744	1104	1464	1824	2184	2544	2904	3264	3624	3984	4344	4704	5064
25	25	385	745	1105	1465	1825	2185	2545	2905	3265	3625	3985	4345	4705	5065
26	26	386	746	1106	1466	1826	2186	2546	2906	3266	3626	3986	4346	4706	5066
27	27	387	747	1107	1467	1827	2187	2547	2907	3267	3627	3987	4347	4707	5067
28	28	388	748	1108	1468	1828	2188	2548	2908	3268	3628	3988	4348	4708	5068
29	29	389	749	1109	1469	1829	2189	2549	2909	3269	3629	3989	4349	4709	5069
Taurus	30	390	750	1110	1470	1830	2190	2550	2910	3270	3630	3990	4350	4710	5070
1	31	391	751	1111	1471	1831	2191	2551	2911	3271	3631	3991	4351	4711	5071
2	32	392	752	1112	1472	1832	2192	2552	2912	3272	3632	3992	4352	4712	5072
3	33	393	753	1113	1473	1833	2193	2553	2913	3273	3633	3993	4353	4713	5073
4	34	394	754	1114	1474	1834	2194	2554	2914	3274	3634	3994	4354	4714	5074
5	35	395	755	1115	1475	1835	2195	2555	2915	3275	3635	3995	4355	4715	5075
6	36	396	756	1116	1476	1836	2196	2556	2916	3276	3636	3996	4356	4716	5076
7	37	397	757	1117	1477	1837	2197	2557	2917	3277	3637	3997	4357	4717	5077
8	38	398	758	1118	1478	1838	2198	2558	2918	3278	3638	3998	4358	4718	5078
9	39	399	759	1119	1479	1839	2199	2559	2919	3279	3639	3999	4359	4719	5079
10	40	400	760	1120	1480	1840	2200	2560	2920	3280	3640	4000	4360	4720	5080
11	41	401	761	1121	1481	1841	2201	2561	2921	3281	3641	4001	4361	4721	5081
12	42	402	762	1122	1482	1842	2202	2562	2922	3282	3642	4002	4362	4722	5082
13	43	403	763	1123	1483	1843	2203	2563	2923	3283	3643	4003	4363	4723	5083
14	44	404	764	1124	1484	1844	2204	2564	2924	3284	3644	4004	4364	4724	5084
15	45	405	765	1125	1485	1845	2205	2565	2925	3285	3645	4005	4365	4725	5085
16	46	406	766	1126	1486	1846	2206	2566	2926	3286	3646	4006	4366	4726	5086
17	47	407	767	1127	1487	1847	2207	2567	2927	3287	3647	4007	4367	4727	5087
18	48	408	768	1128	1488	1848	2208	2568	2928	3288	3648	4008	4368	4728	5088
19	49	409	769	1129	1489	1849	2209	2569	2929	3289	3649	4009	4369	4729	5089
20	50	410	770	1130	1490	1850	2210	2570	2930	3290	3650	4010	4370	4730	5090
21	51	411	771	1131	1491	1851	2211	2571	2931	3291	3651	4011	4371	4731	5091
22	52	412	772	1132	1492	1852	2212	2572	2932	3292	3652	4012	4372	4732	5092
23	53	413	773	1133	1493	1853	2213	2573	2933	3293	3653	4013	4373	4733	5093
24	54	414	774	1134	1494	1854	2214	2574	2934	3294	3654	4014	4374	4734	5094
25	55	415	775	1135	1495	1855	2215	2575	2935	3295	3655	4015	4375	4735	5095
26	56	416	776	1136	1496	1856	2216	2576	2936	3296	3656	4016	4376	4736	5096
27	57	417	777	1137	1497	1857	2217	2577	2937	3297	3657	4017	4377	4737	5097
28	58	418	778	1138	1498	1858	2218	2578	2938	3298	3658	4018	4378	4738	5098
29	59	419	779	1139	1499	1859	2219	2579	2939	3299	3659	4019	4379	4739	5099
Gemini	60	420	780	1140	1500	1860	2220	2580	2940	3300	3660	4020	4380	4740	5100
1	61	421	781	1141	1501	1861	2221	2581	2941	3301	3661	4021	4381	4741	5101
2	62	422	782	1142	1502	1862	2222	2582	2942	3302	3662	4022	4382	4742	5102
3	63	423	783	1143	1503	1863	2223	2583	2943	3303	3663	4023	4383	4743	5103
4	64	424	784	1144	1504	1864	2224	2584	2944	3304	3664	4024	4384	4744	5104
5	65	425	785	1145	1505	1865	2225	2585	2945	3305	3665	4025	4385	4745	5105
6	66	426	786	1146	1506	1866	2226	2586	2946	3306	3666	4026	4386	4746	5106
7	67	427	787	1147	1507	1867	2227	2587	2947	3307	3667	4027	4387	4747	5107

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LONG.XLS

19	136	494	458	4210	1176	9009	2358	2468	3010	3336	3734	4084	4436	4800
19	136	494	458	4210	1176	9009	2358	2468	3010	3336	3734	4084	4436	4800
20	140	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
21	141	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
22	142	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
23	143	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
24	144	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
25	145	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
26	146	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
27	147	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
28	148	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
29	149	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
Virgo	150	504	468	4320	1188	9040	2368	2480	3020	3348	3748	4100	4452	4824
1	151	511	471	4351	1191	9051	2371	2491	3031	3351	3751	4111	4461	4831
2	152	512	472	4362	1192	9062	2382	2502	3042	3362	3762	4122	4472	4842
3	153	513	473	4373	1193	9073	2393	2513	3053	3373	3773	4133	4483	4853
4	154	514	474	4384	1194	9084	2404	2524	3064	3384	3784	4144	4494	4864
5	155	515	475	4395	1195	9095	2415	2535	3075	3395	3795	4155	4505	4875
6	156	516	476	4406	1196	9106	2426	2546	3086	3406	3806	4166	4516	4886
7	157	517	477	4417	1197	9117	2437	2557	3097	3417	3817	4177	4527	4897
8	158	518	478	4428	1198	9128	2448	2568	3108	3428	3828	4188	4538	4908
9	159	519	479	4439	1199	9139	2459	2579	3119	3439	3839	4199	4549	4919
10	160	520	480	4450	1200	9150	2470	2590	3130	3450	3850	4210	4560	4930
11	161	521	481	4461	1201	9161	2481	2601	3141	3461	3861	4221	4571	4941
12	162	522	482	4472	1202	9172	2492	2612	3152	3472	3872	4232	4582	4952
13	163	523	483	4483	1203	9183	2503	2623	3163	3483	3883	4243	4593	4963
14	164	524	484	4494	1204	9194	2514	2634	3174	3494	3894	4254	4604	4974
15	165	525	485	4505	1205	9205	2525	2645	3185	3505	3905	4265	4615	4985
16	166	526	486	4516	1206	9216	2536	2656	3196	3516	3916	4276	4626	4996
17	167	527	487	4527	1207	9227	2547	2667	3207	3527	3927	4287	4637	5007
18	168	528	488	4538	1208	9238	2558	2678	3218	3538	3938	4298	4648	5018
19	169	529	489	4549	1209	9249	2569	2689	3229	3549	3949	4309	4659	5029
20	170	530	490	4560	1210	9260	2580	2700	3240	3560	3960	4320	4670	5040
21	171	531	491	4571	1211	9271	2591	2711	3251	3571	3971	4331	4681	5051
22	172	532	492	4582	1212	9282	2602	2722	3262	3582	3982	4342	4692	5062
23	173	533	493	4593	1213	9293	2613	2733	3273	3593	3993	4353	4703	5073
24	174	534	494	4604	1214	9304	2624	2744	3284	3604	4004	4364	4714	5084
25	175	535	495	4615	1215	9315	2635	2755	3295	3615	4015	4375	4725	5095
26	176	536	496	4626	1216	9326	2646	2766	3306	3626	4026	4386	4736	5106
27	177	537	497	4637	1217	9337	2657	2777	3317	3637	4037	4397	4747	5117
28	178	538	498	4648	1218	9348	2668	2788	3328	3648	4048	4408	4758	5128
29	179	539	499	4659	1219	9359	2679	2799	3339	3659	4059	4419	4769	5139
Libra	180	540	500	4670	1220	9370	2690	2810	3350	3670	4070	4430	4780	5150
1	181	541	501	4681	1221	9381	2701	2821	3361	3681	4081	4441	4791	5161
2	182	542	502	4692	1222	9392	2712	2832	3372	3692	4092	4452	4802	5172
3	183	543	503	4703	1223	9403	2723	2843	3383	3703	4103	4463	4813	5183
4	184	544	504	4714	1224	9414	2734	2854	3394	3714	4114	4474	4824	5194
5	185	545	505	4725	1225	9425	2745	2865	3405	3725	4125	4485	4835	5205
6	186	546	506	4736	1226	9436	2756	2876	3416	3736	4136	4496	4846	5216
7	187	547	507	4747	1227	9447	2767	2887	3427	3747	4147	4507	4857	5227
8	188	548	508	4758	1228	9458	2778	2898	3438	3758	4158	4518	4868	5238
9	189	549	509	4769	1229	9469	2789	2909	3449	3769	4169	4529	4879	5249
10	190	550	510	4780	1230	9480	2790	2920	3460	3780	4180	4540	4890	5260
11	191	551	511	4791	1231	9491	2801	2931	3471	3791	4191	4551	4901	5271
12	192	552	512	4802	1232	9502	2812	2942	3482	3802	4202	4562	4912	5282
13	193	553	513	4813	1233	9513	2823	2953	3493	3813	4213	4573	4923	5293
14	194	554	514	4824	1234	9524	2834	2964	3504	3824	4224	4584	4934	5304
15	195	555	515	4835	1235	9535	2845	2975	3515	3835	4235	4595	4945	5315
16	196	556	516	4846	1236	9546	2856	2986	3526	3846	4246	4606	4956	5326
17	197	557	517	4857	1237	9557	2867	2997	3537	3857	4257	4617	4967	5337
18	198	558	518	4868	1238	9568	2878	3008	3548	3868	4268	4628	4978	5348
19	199	559	519	4879	1239	9579	2889	3019	3559	3879	4279	4639	4989	5359
20	200	560	520	4890	1240	9590	2900	3030	3570	3890	4290	4650	4990	5370
21	201	561	521	4901	1241	9601	2911	3041	3581	3901	4301	4661	5001	5381
22	202	562	522	4912	1242	9612	2922	3052	3592	3912	4312	4672	5012	5392
23	203	563	523	4923	1243	9623	2933	3063	3603	3923	4323	4683	5023	5403
24	204	564	524	4934	1244	9634	2944	3074	3614	3934	4334	4694	5034	5414
25	205	565	525	4945	1245	9645	2955	3085	3625	3945	4345	4705	5045	5425
26	206	566	526	4956	1246	9656	2966	3096	3636	3956	4356	4716	5056	5436
27	207	567	527	4967	1247	9667	2977	3107	3647	3967	4367	4727	5067	5447

28	288	888	888	1288	8888	2888	2888	2728	3888	3888	3888	4888	4888	4888
29	289	889	889	1289	8889	2889	2889	2729	3889	3889	3889	4889	4889	4889
Scorpio	219	879	879	1289	8890	2879	2879	2730	3890	3890	3890	4890	4890	4890
1	211	871	871	1281	8891	2871	2871	2731	3891	3891	3891	4891	4891	4891
2	212	872	872	1282	8892	2872	2872	2732	3892	3892	3892	4892	4892	4892
3	213	873	873	1283	8893	2873	2873	2733	3893	3893	3893	4893	4893	4893
4	214	874	874	1284	8894	2874	2874	2734	3894	3894	3894	4894	4894	4894
5	215	875	875	1285	8895	2875	2875	2735	3895	3895	3895	4895	4895	4895
6	216	876	876	1286	8896	2876	2876	2736	3896	3896	3896	4896	4896	4896
7	217	877	877	1287	8897	2877	2877	2737	3897	3897	3897	4897	4897	4897
8	218	878	878	1288	8898	2878	2878	2738	3898	3898	3898	4898	4898	4898
9	219	879	879	1289	8899	2879	2879	2739	3899	3899	3899	4899	4899	4899
10	220	880	880	1290	8900	2880	2880	2740	3900	3900	3900	4900	4900	4900
11	221	881	881	1291	8901	2881	2881	2741	3901	3901	3901	4901	4901	4901
12	222	882	882	1292	8902	2882	2882	2742	3902	3902	3902	4902	4902	4902
13	223	883	883	1293	8903	2883	2883	2743	3903	3903	3903	4903	4903	4903
14	224	884	884	1294	8904	2884	2884	2744	3904	3904	3904	4904	4904	4904
15	225	885	885	1295	8905	2885	2885	2745	3905	3905	3905	4905	4905	4905
16	226	886	886	1296	8906	2886	2886	2746	3906	3906	3906	4906	4906	4906
17	227	887	887	1297	8907	2887	2887	2747	3907	3907	3907	4907	4907	4907
18	228	888	888	1298	8908	2888	2888	2748	3908	3908	3908	4908	4908	4908
19	229	889	889	1299	8909	2889	2889	2749	3909	3909	3909	4909	4909	4909
20	230	890	890	1300	8910	2890	2890	2750	3910	3910	3910	4910	4910	4910
21	231	891	891	1301	8911	2891	2891	2751	3911	3911	3911	4911	4911	4911
22	232	892	892	1302	8912	2892	2892	2752	3912	3912	3912	4912	4912	4912
23	233	893	893	1303	8913	2893	2893	2753	3913	3913	3913	4913	4913	4913
24	234	894	894	1304	8914	2894	2894	2754	3914	3914	3914	4914	4914	4914
25	235	895	895	1305	8915	2895	2895	2755	3915	3915	3915	4915	4915	4915
26	236	896	896	1306	8916	2896	2896	2756	3916	3916	3916	4916	4916	4916
27	237	897	897	1307	8917	2897	2897	2757	3917	3917	3917	4917	4917	4917
28	238	898	898	1308	8918	2898	2898	2758	3918	3918	3918	4918	4918	4918
29	239	899	899	1309	8919	2899	2899	2759	3919	3919	3919	4919	4919	4919
Sagittarius	210	880	880	1289	8890	2879	2879	2730	3890	3890	3890	4890	4890	4890
1	241	891	891	1301	8911	2891	2891	2751	3911	3911	3911	4911	4911	4911
2	242	892	892	1302	8912	2892	2892	2752	3912	3912	3912	4912	4912	4912
3	243	893	893	1303	8913	2893	2893	2753	3913	3913	3913	4913	4913	4913
4	244	894	894	1304	8914	2894	2894	2754	3914	3914	3914	4914	4914	4914
5	245	895	895	1305	8915	2895	2895	2755	3915	3915	3915	4915	4915	4915
6	246	896	896	1306	8916	2896	2896	2756	3916	3916	3916	4916	4916	4916
7	247	897	897	1307	8917	2897	2897	2757	3917	3917	3917	4917	4917	4917
8	248	898	898	1308	8918	2898	2898	2758	3918	3918	3918	4918	4918	4918
9	249	899	899	1309	8919	2899	2899	2759	3919	3919	3919	4919	4919	4919
10	250	900	900	1310	8920	2900	2900	2760	3920	3920	3920	4920	4920	4920
11	251	901	901	1311	8921	2901	2901	2761	3921	3921	3921	4921	4921	4921
12	252	902	902	1312	8922	2902	2902	2762	3922	3922	3922	4922	4922	4922
13	253	903	903	1313	8923	2903	2903	2763	3923	3923	3923	4923	4923	4923
14	254	904	904	1314	8924	2904	2904	2764	3924	3924	3924	4924	4924	4924
15	255	905	905	1315	8925	2905	2905	2765	3925	3925	3925	4925	4925	4925
16	256	906	906	1316	8926	2906	2906	2766	3926	3926	3926	4926	4926	4926
17	257	907	907	1317	8927	2907	2907	2767	3927	3927	3927	4927	4927	4927
18	258	908	908	1318	8928	2908	2908	2768	3928	3928	3928	4928	4928	4928
19	259	909	909	1319	8929	2909	2909	2769	3929	3929	3929	4929	4929	4929
20	260	910	910	1320	8930	2910	2910	2770	3930	3930	3930	4930	4930	4930
21	261	911	911	1321	8931	2911	2911	2771	3931	3931	3931	4931	4931	4931
22	262	912	912	1322	8932	2912	2912	2772	3932	3932	3932	4932	4932	4932
23	263	913	913	1323	8933	2913	2913	2773	3933	3933	3933	4933	4933	4933
24	264	914	914	1324	8934	2914	2914	2774	3934	3934	3934	4934	4934	4934
25	265	915	915	1325	8935	2915	2915	2775	3935	3935	3935	4935	4935	4935
26	266	916	916	1326	8936	2916	2916	2776	3936	3936	3936	4936	4936	4936
27	267	917	917	1327	8937	2917	2917	2777	3937	3937	3937	4937	4937	4937
28	268	918	918	1328	8938	2918	2918	2778	3938	3938	3938	4938	4938	4938
29	269	919	919	1329	8939	2919	2919	2779	3939	3939	3939	4939	4939	4939
Capricorn	270	920	920	1330	8940	2920	2920	2780	3940	3940	3940	4940	4940	4940
1	271	921	921	1331	8941	2921	2921	2781	3941	3941	3941	4941	4941	4941
2	272	922	922	1332	8942	2922	2922	2782	3942	3942	3942	4942	4942	4942
3	273	923	923	1333	8943	2923	2923	2783	3943	3943	3943	4943	4943	4943
4	274	924	924	1334	8944	2924	2924	2784	3944	3944	3944	4944	4944	4944
5	275	925	925	1335	8945	2925	2925	2785	3945	3945	3945	4945	4945	4945
6	276	926	926	1336	8946	2926	2926	2786	3946	3946	3946	4946	4946	4946
7	277	927	927	1337	8947	2927	2927	2787	3947	3947	3947	4947	4947	4947

8	276	636	996	1356	1716	2076	2436	2796	3156	3516	3876	4236	4596	4956
9	279	639	999	1359	1719	2079	2439	2799	3159	3519	3879	4239	4599	4959
10	282	642	1002	1362	1722	2082	2442	2802	3162	3522	3882	4242	4602	4962
11	285	645	1005	1365	1725	2085	2445	2805	3165	3525	3885	4245	4605	4965
12	288	648	1008	1368	1728	2088	2448	2808	3168	3528	3888	4248	4608	4968
13	291	651	1011	1371	1731	2091	2451	2811	3171	3531	3891	4251	4611	4971
14	294	654	1014	1374	1734	2094	2454	2814	3174	3534	3894	4254	4614	4974
15	297	657	1017	1377	1737	2097	2457	2817	3177	3537	3897	4257	4617	4977
16	300	660	1020	1380	1740	2100	2460	2820	3180	3540	3900	4260	4620	4980
17	303	663	1023	1383	1743	2103	2463	2823	3183	3543	3903	4263	4623	4983
18	306	666	1026	1386	1746	2106	2466	2826	3186	3546	3906	4266	4626	4986
19	309	669	1029	1389	1749	2109	2469	2829	3189	3549	3909	4269	4629	4989
20	312	672	1032	1392	1752	2112	2472	2832	3192	3552	3912	4272	4632	4992
21	315	675	1035	1395	1755	2115	2475	2835	3195	3555	3915	4275	4635	4995
22	318	678	1038	1398	1758	2118	2478	2838	3198	3558	3918	4278	4638	4998
23	321	681	1041	1401	1761	2121	2481	2841	3201	3561	3921	4281	4641	5001
24	324	684	1044	1404	1764	2124	2484	2844	3204	3564	3924	4284	4644	5004
25	327	687	1047	1407	1767	2127	2487	2847	3207	3567	3927	4287	4647	5007
26	330	690	1050	1410	1770	2130	2490	2850	3210	3570	3930	4290	4650	5010
27	333	693	1053	1413	1773	2133	2493	2853	3213	3573	3933	4293	4653	5013
28	336	696	1056	1416	1776	2136	2496	2856	3216	3576	3936	4296	4656	5016
29	339	699	1059	1419	1779	2139	2499	2859	3219	3579	3939	4299	4659	5019
30	342	702	1062	1422	1782	2142	2502	2862	3222	3582	3942	4302	4662	5022
31	345	705	1065	1425	1785	2145	2505	2865	3225	3585	3945	4305	4665	5025
32	348	708	1068	1428	1788	2148	2508	2868	3228	3588	3948	4308	4668	5028
33	351	711	1071	1431	1791	2151	2511	2871	3231	3591	3951	4311	4671	5031
34	354	714	1074	1434	1794	2154	2514	2874	3234	3594	3954	4314	4674	5034
35	357	717	1077	1437	1797	2157	2517	2877	3237	3597	3957	4317	4677	5037
36	360	720	1080	1440	1800	2160	2520	2880	3240	3600	3960	4320	4680	5040
37	363	723	1083	1443	1803	2163	2523	2883	3243	3603	3963	4323	4683	5043
38	366	726	1086	1446	1806	2166	2526	2886	3246	3606	3966	4326	4686	5046
39	369	729	1089	1449	1809	2169	2529	2889	3249	3609	3969	4329	4689	5049
40	372	732	1092	1452	1812	2172	2532	2892	3252	3612	3972	4332	4692	5052
41	375	735	1095	1455	1815	2175	2535	2895	3255	3615	3975	4335	4695	5055
42	378	738	1098	1458	1818	2178	2538	2898	3258	3618	3978	4338	4698	5058
43	381	741	1101	1461	1821	2181	2541	2901	3261	3621	3981	4341	4701	5061
44	384	744	1104	1464	1824	2184	2544	2904	3264	3624	3984	4344	4704	5064
45	387	747	1107	1467	1827	2187	2547	2907	3267	3627	3987	4347	4707	5067
46	390	750	1110	1470	1830	2190	2550	2910	3270	3630	3990	4350	4710	5070
47	393	753	1113	1473	1833	2193	2553	2913	3273	3633	3993	4353	4713	5073
48	396	756	1116	1476	1836	2196	2556	2916	3276	3636	3996	4356	4716	5076
49	399	759	1119	1479	1839	2199	2559	2919	3279	3639	3999	4359	4719	5079
50	402	762	1122	1482	1842	2202	2562	2922	3282	3642	4002	4362	4722	5082
51	405	765	1125	1485	1845	2205	2565	2925	3285	3645	4005	4365	4725	5085
52	408	768	1128	1488	1848	2208	2568	2928	3288	3648	4008	4368	4728	5088
53	411	771	1131	1491	1851	2211	2571	2931	3291	3651	4011	4371	4731	5091
54	414	774	1134	1494	1854	2214	2574	2934	3294	3654	4014	4374	4734	5094
55	417	777	1137	1497	1857	2217	2577	2937	3297	3657	4017	4377	4737	5097
56	420	780	1140	1500	1860	2220	2580	2940	3300	3660	4020	4380	4740	5100
57	423	783	1143	1503	1863	2223	2583	2943	3303	3663	4023	4383	4743	5103
58	426	786	1146	1506	1866	2226	2586	2946	3306	3666	4026	4386	4746	5106
59	429	789	1149	1509	1869	2229	2589	2949	3309	3669	4029	4389	4749	5109
60	432	792	1152	1512	1872	2232	2592	2952	3312	3672	4032	4392	4752	5112
61	435	795	1155	1515	1875	2235	2595	2955	3315	3675	4035	4395	4755	5115
62	438	798	1158	1518	1878	2238	2598	2958	3318	3678	4038	4398	4758	5118
63	441	801	1161	1521	1881	2241	2601	2961	3321	3681	4041	4401	4761	5121
64	444	804	1164	1524	1884	2244	2604	2964	3324	3684	4044	4404	4764	5124
65	447	807	1167	1527	1887	2247	2607	2967	3327	3687	4047	4407	4767	5127
66	450	810	1170	1530	1890	2250	2610	2970	3330	3690	4050	4410	4770	5130
67	453	813	1173	1533	1893	2253	2613	2973	3333	3693	4053	4413	4773	5133
68	456	816	1176	1536	1896	2256	2616	2976	3336	3696	4056	4416	4776	5136
69	459	819	1179	1539	1899	2259	2619	2979	3339	3699	4059	4419	4779	5139
70	462	822	1182	1542	1902	2262	2622	2982	3342	3702	4062	4422	4782	5142
71	465	825	1185	1545	1905	2265	2625	2985	3345	3705	4065	4425	4785	5145
72	468	828	1188	1548	1908	2268	2628	2988	3348	3708	4068	4428	4788	5148
73	471	831	1191	1551	1911	2271	2631	2991	3351	3711	4071	4431	4791	5151
74	474	834	1194	1554	1914	2274	2634	2994	3354	3714	4074	4434	4794	5154
75	477	837	1197	1557	1917	2277	2637	2997	3357	3717	4077	4437	4797	5157
76	480	840	1200	1560	1920	2280	2640	3000	3360	3720	4080	4440	4800	5160
77	483	843	1203	1563	1923	2283	2643	3003	3363	3723	4083	4443	4803	5163
78	486	846	1206	1566	1926	2286	2646	3006	3366	3726	4086	4446	4806	5166
79	489	849	1209	1569	1929	2289	2649	3009	3369	3729	4089	4449	4809	5169
80	492	852	1212	1572	1932	2292	2652	3012	3372	3732	4092	4452	4812	5172
81	495	855	1215	1575	1935	2295	2655	3015	3375	3735	4095	4455	4815	5175
82	498	858	1218	1578	1938	2298	2658	3018	3378	3738	4098	4458	4818	5178
83	501	861	1221	1581	1941	2301	2661	3021	3381	3741	4101	4461	4821	5181
84	504	864	1224	1584	1944	2304	2664	3024	3384	3744	4104	4464	4824	5184
85	507	867	1227	1587	1947	2307	2667	3027	3387	3747	4107	4467	4827	5187
86	510	870	1230	1590	1950	2310	2670	3030	3390	3750	4110	4470	4830	5190
87	513	873	1233	1593	1953	2313	2673	3033	3393	3753	4113	4473	4833	5193
88	516	876	1236	1596	1956	2316	2676	3036	3396	3756	4116	4476	4836	5196
89	519	879	1239	1599	1959	2319	2679	3039	3399	3759	4119	4479	4839	5199
90	522	882	1242	1602	1962	2322	2682	3042	3402	3762	4122	4482	4842	5202
91	525	885	1245	1605	1965	2325	2685	3045	3405	3765	4125	4485	4845	5205
92	528	888	1248	1608	1968	2328	2688	3048	3408	3768	4128	4488	4848	5208
93	531	891	1251	1611	1971	2331	2691	3051	3411	3771	4131	4491	4851	5211
94	534	894	1254	1614	1974	2334	2694	3054	3414	3774	4134	4494	4854	5214
95	537	897	1257	1617	1977	2337	2697	3057	3417	3777	4137	4497	4857	5217
96	540	900	1260	1620	1980	2340	2700	3060	3420	3780	4140	4500	4860	5220
97	543	903	1263	1623	1983	2343	2703	3063	3423	3783	4143	4503	4863	5223
98	546	906	1266	1626	1986	2346	2706	3066	3426					

LONG.XLS

18	343	706	8089	1429	8798	2148	2568	2888	3228	3588	3948	4308	4668	5028
19	349	708	8096	1438	8799	2149	2569	2889	3229	3589	3949	4309	4669	5029
20	354	710	8110	1438	8750	2188	2610	2970	3338	3690	4048	4410	4770	5130
21	358	711	8071	1428	8791	2181	2611	2971	3334	3691	4051	4411	4771	5131
22	363	712	8072	1432	8792	2162	2612	2972	3332	3692	4052	4412	4772	5132
23	368	713	8073	1433	8793	2153	2613	2973	3333	3693	4053	4413	4773	5133
24	364	718	8074	1434	8794	2164	2614	2974	3334	3694	4054	4414	4774	5134
25	368	715	8075	1438	8795	2156	2615	2975	3335	3695	4055	4415	4775	5135
26	368	716	8076	1436	8796	2158	2616	2976	3336	3696	4056	4416	4776	5136
27	367	717	8077	1430	8797	2157	2617	2977	3337	3697	4057	4417	4777	5137
28	366	719	8078	1438	8798	2160	2618	2978	3338	3698	4058	4418	4778	5138
29	366	718	8079	1439	8799	2160	2618	2978	3338	3698	4058	4418	4778	5138

STANDARD ASTROLOGICAL SYMBOLS



THE ZODIACAL SIGNS

Symbol	Sign	Symbol	Sign
	Aries		Libra
	Taurus		Scorpio
	Gemini		Sagittarius
	Cancer		Capricorn
	Leo		Aquarius
	Virgo		Pisces

ASPECT (ANGLE) SYMBOL, ASPECT NAME AND KEYWORD - TABLE 7-12

<u>Aspect</u>	<u>Symbol</u>	<u>Name</u>	<u>Keyword</u>
0°	♌	Conjunction	Power, emphasis, reinforcement (strong aspect).
30°	♊	Semi-sextile	Harmony (weak).
45°	∟	Semi-square	Challenge, friction.
60°	♋	Sextile	Opportunity, harmony (stronger than semi-sextile).
90°	☐	Square	Conflict, stress, tension, obstacle.
120°	♍	Trine	Inspiration, flow, ease, luck.
135°	♎	Sesqui-square	Friction, agitation (weak).
150°	♏	Inconjunct	Strain, adjustment, expansion.
180°	♐	Opposition	Separation, conflict, awareness.

ASTROLOGICAL POWERS OR STRENGTHS - TABLE 7-13

Planets in Signs:

Ruler	Powerful, harmonious expression
Detriment	Less powerful, coloration of house more effective than planets.
Exalted	Harmonious, like rulership, but not as forceful.
Fall	Similar to detriment and with disappointments of planet concerns.

Planets in Houses:

Strongest to the weakest:

1, 10, 7, 4, 11, 8, 9, 12, 2, 3, 5, 6.

Aspect (Angle) Strengths:

Strongest to weakest:

<u>Order</u>	<u>Aspect</u>
1.	360°, 0°
2.	180°
3.	120°
4.	90°
5.	72°
6.	60°
7.	51.4°
8.	45°
9.	40°
10.	36°
11.	32.7°
12.	30°

Note: These astrological values are obtained by dividing 360° by 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9, 1/10th.

SIGNS, SYMBOLS, AND KEYWORDS-PHRASES - TABLE 7-11

<u>Sign</u>	<u>Symbol</u>	<u>Keywords</u>	<u>Phrases</u>
Aries	♈	Active, assertive, pioneering.	I am
Taurus	♉	Stable, possessive, patient.	I have
Gemini	♊	Versatile, communicative.	I can
Cancer	♋	Protective, devoted, sensitive.	I feel
Leo	♌	Creative, joyful, magnetism	I want
Virgo	♍	Studies details, critical, practical	I analyze
Libra	♎	United, harmonious, impartial, refined.	I balance
Scorpio	♏	Intense, passionate, secretive, inspires	I desire
Sagittarius	♐	Generous, visualize, straightforward and bold.	I understand
Capricorn	♑	Ambition, prudent, industrious, cool	I can
Aquarius	♒	Imaginative, intuitive, scientific, detached.	I think
Pisces	♓	Strong spiritual aspirations, understanding, sees intangibles.	I believe

TABLE OF PLANETS, SYMBOLS, KEYWORDS - TABLE 7-10

<u>Planet</u>	<u>Symbol</u>	<u>Keywords</u>
Sun	☉	Life, vitality, leadership, authority, power, real self, males.
Moon	☾	Emotions, response, instincts, change, subconscious, females.
Mercury	☿	Expression, communication, thought, children.
Venus	♀	Beauty, harmony, attraction, pleasure, ease, artists.
Mars	♂	Energy, aggression, courage, heat, action, soldiers.
Jupiter	♃	Expansion, devotion, justice, abundance, judges.
Saturn	♄	Limitation, contraction, cold, tradition, teachers.
Uranus	♅	Disruptions, revolutions, sudden changes, originality, sudden events, inventors.
Neptune	♆	Confusion, vagueness, illusion, mystic, inflation, sailors.
Pluto	♇	Regeneration, renewal, transformation, gangsters.

MUNDANE ASTROLOGICAL TERMS

MUNDANE/CORPORATE KEYWORDS FOR PLANETS - TABLE 7-9

Moon:

The masses, popular ideas,
yellow people, crowds,
change, crop harvests.

Mercury:

The media, publishers,
trade, Arab people,
short trips, mind.

Venus:

Art, architecture,
fashion, recreation,
Latin people, music,
soybeans.

Mars:

Violence, war, the army,
epidemics, surgeons,
sports, India and
Indian nationals.

Jupiter:

Abundance, brokers, banking,
investments, courts, religion,
Greenland.

Saturn:

Disaster, depression,
cold, limits, wars,
black people, grains.

Uranus:

Unexpected sudden
change, inventions,
aircraft, strikes,
electronics.

Neptune:

Inflation, drugs,
hospitals,
photography,
actors, oil and gas.

Pluto:

Energy, atomic power,
dictatorships,
rockets, underworld.

The RADIX of any company is the noon of the date and place of registration of that Company, whereby, under the laws of the country the promoters are given powers to form the company and proceed with business.

The RADIX of a Stock Exchange IS THE NOON OF THE DATE OF FOUNDING. The charts for New York and London are attached hereto and will serve as examples of all others. FROM THIS RADIX or root figure of the heavens a variety of influences are by natural process of celestial motions indicated. These are found to depend altogether upon the Sun, which is the controlling center of their interactions. The Sun's MAN MOTION IS 59 minutes and eight seconds per day. The measure of TIME IS ONE DAY FOR A YEAR. Thus the influences arising out of the movement of the bodies in relation to the Sun will influence successive years of the Company, the indications arising in the second day affecting the second year, those formed on the third day influence the third year, and so on continuously.

SOLAR MOTION: THE APPARENT MOTION OF THE SUN AS SEEN FROM THE EARTH is slightly variable, according to the season of the year. This is due to the fact that the apparent orbit of the Sun is not concentric with the

earth but is slightly removed therefrom, thus producing an eccentricity of its apparent motion about the earth, and this may amount to as much as one degree and fifty six minutes either way, as compared with its mean, or average motion, which is 59 minutes 8 seconds per day. It has been found in a great number of tests that the MEAN MOTION OF THE SUN IS THE CORRECT MEASURE TO BE OBSERVED, AND THE SUN REQUIRES 365 1/4 days to complete the circle of 360 degrees we have a mean motion of 59 minutes 8 seconds per day.

This amount is multiplied by the number of years from the radix or date of registration and added to the Sun's place in the radix (called Sun r), also to the Midheaven, and to the Moon r. The other planets are in the same way carried forward through the zodiac at the same rate as the Sun and maintain their radical relations with the Sun thereby, but in the course of their progress from day to day at this uniform rate, they form aspects to the radical places of the Sun, Midheaven, Moon, and ascendant of the radix. These aspects are called DIRECTIONAL ASPECTS and they INFLUENCE THE AFFAIRS OF THE COMPANY OR THE BUSINESS OF ANY EXCHANGE ACCORDING TO THEIR SEVERAL NATURES.

THE PRIME SIGNIFICATOR OF ANY CHART IS THAT PLANET WHICH RULES THE SIGN IN WHICH THE SUN IS FOUND. Thus, in the chart for the New York Stock Exchange, the Sun is in the sign Taurus. This sign is ruled by Venus, and Venus therefore becomes PRIME SIGNIFICATOR.

THE SECONDARY SIGNIFICATOR IS THE PLANET WHICH RULES THE DECANATE IN WHICH THE PRIME SIGNIFICATOR HAPPENS TO BE PASSING THROUGH IN THE COURSE OF ITS DAILY MOTION IN THE HEAVENS as shown in the ephemeris for the current year. The use of these significators will be given further on in this exposition. FIRST, let us look at the method of DIRECTING which has been found to be a guide TO THE GENERAL MARKET OUTLOOK.

The New York Chart is set for noon on the 17th of May, 1792 as shown in the map herewith. Now at the tragic assassination of President Abraham Lincoln we find the years from the above date of founding to the date of that event, APRIL 14TH, 1865, to be a period of 73 YEARS ONE MONTH. This at the rate of 59' 8" per day (year) is 72 degrees 2 minutes and being added to the radical Midheaven of Taurus 28 we come to the 11th degree of the sign LEO, AND, THE ASCENDANT UNDER THIS M.C. is the 5th of SCORPIO. The ascendant was therefore in opposition to Venus radical. Referring to the ephemeris for 1865 we find that on that very day Neptune, the author of plots and schemes, was in Aries 8:40 and therefore exactly on the place of the Moon radical.

AXIOM ONE

Planets in transit over the radical places of the SUN, MOON, MIDHEAVEN, AND ASCENDANT OF THE CHART have in-

fluence on the course of events according to their natures, and so effect the VALUE OF SECURITIES. In November 1857 there was a considerable panic (financial panic) on the Exchange. Time 65 years 6 months. M.C. progress to Leo 2:34, Asc. in Libra 27. The Asc. is opposed to SATURN and in conjunction with NEPTUNE. VENUS was then in the end of LIBRA (CONJ. NEPTUNE r) and SATURN was in CANCER in square aspect to VENUS Midmonth in November 1857. URANUS was in transit over M.C. radical.

AXIOM TWO

DIRECTIONS OF THE GENERAL SIGNIFICATORS, SUN, MOON, M.C. AND ASC., TO AFFLICTING ASPECTS (SQUARE OR OPPOSITION) OF MALEFIC PLANETS WILL CAUSE GREAT DEPRESSION OF SECURITIES.

AXIOM THREE

The ephemeris aspects to the PRIME SIGNIFICATOR WILL DEFINE THE PROBABLE TIME FOR THIS DEPRESSION. The Railway Strike of August 1890 at 98 years 3 months from the foundation of the chart shows M.C. in Virgo 5. Saturn was then in transit through that degree of the Zodiac.

AXIOM FOUR

Saturn IN CONJUNCTION, OR SQUARE OR OPPOSITION-by transit-to ANY OF THE RADICAL PLACES OF THE GENERAL SIGNIFICATORS, OR TO THEIR PROGRESSED PLACES, WILL CAUSE A SERIES DEPRESSION. In August 1914 Saturn was in Gemini 28 in square to Mars in Virgo 28. The MC. of the chart had progressed to Virgo 28. Mars was therefore in transit over

the MC. Progressed and Saturn was in square to it. Mars is here the cause of the excitement and Saturn that of the Depression.

EFFECTS OF PLANETS. NEPTUNE in BAD ASPECT CAUSES BEAR SALES AND HIS ACTION IS GENERALLY DUE TO "RINGS" AND PROFESSIONAL COMBINES MOVING FOR A LOWER BUYING LEVEL. ITS GOOD ASPECTS ADVANCE VALUES FROM THE SAME CAUSES.

URANUS USUALLY BRINGS IN THE GOVERNMENT BROKER AND SHOWS OFFICIAL MANIPULATION OF THE MARKET. In good aspect it makes for buying, and in bad aspect for selling. At such times keep one eye on URANUS and the other on the Government Broker.

SATURN steadies the market when in good aspect, but greatly depresses it when in evil aspect to the SIGNIFICATORS, RADICAL AND PROGRESSED.

JUPITER enhances values by its conjunction or good aspects, and causes heavy selling for profits when in bad aspect. It deflates what it previously had inflated.

MARS stimulates the market and in good aspect causes sharp rises and a brisk market, while in evil aspect it causes sharp depressions. Neither are enduring.

VENUS in good aspect gives equable values and a steady rise in values but is not enduring. In bad aspect it causes a flat and lifeless market.

MERCURY causes rises when in good aspect by 'reports'

and cannot talk and makes for brisk but small operations.
In bad aspect it causes quick in and out buying and selling
with balance against values.

ANIO!

All planets in good aspect enhance values according
to their several natures, the major planets being more
powerful than the minor. Jupiter, Mars and Venus, have
the greatest power for good when in good aspect, but Venus
has not the same power in transit as in Direction.

THE DECANATES

Aries	0 to 10	Mars	11 to 20	Sun	21 to 30	Venus
Taurus	"	Merc	"	Moon	"	Saturn
Gemini	"	Jupiter	"	Mars	"	Sun
Cancer	"	Venus	"	Merc	"	Moon
Leo	"	Saturn	"	Jupiter	"	Merc
Virgo	"	Sun	"	Venus	"	Merc
Libra	"	Moon	"	Saturn	"	Jupiter
Scorpio	"	Mars	"	Sun	"	Venus
Sagit	"	Merc	"	Moon	"	Saturn
Capri	"	Jupiter	"	Mars	"	Sun
Aquarius	"	Venus	"	Merc	"	Moon
Pisces	"	Saturn	"	Jupiter	"	Mars

These decans are parts of signs each ten degrees in extent
and their rulers are set against them. When the PRIME
SIGNIFICATOR is passing through any decan it is the guest
of the ruler of that decan and is subject in some measure
to the aspects affecting that ruler so long as the PRIME
SIGNIFICATOR remains in that Decan. These can be found
in the ephemeris from week to week throughout the year and
FORM THE BASIS OF "FLUCTUATIONS", while being subject to the
general tone of the market indicated by the DIRECTIONS of

the general significations and the major transits over them.

For example:

At the time of the railway strike in 1890, we find Saturn in transit over the M.C. Progressed in Virgo 5 and in square aspect to Mars, then in Sagittarius 5, and near to square of Neptune in Gemini 6, while Uranus was in transit over Jupiter's place in the radix on the cusp of the third house (Railways). But Venus, the PRIME SIGNIFICATOR, was then in the end of Virgo and not in aspect to any planet in the heavens. The Secondary Significator is Mercury, which rules the last decan of Virgo, where Venus is guest. Now we find Mercury was then in conjunction with Saturn in Virgo 5 and therefore square to Mars and square to Neptune in the heavens at the time. Here are adequate causes for predicting a severe set-back.

NATURE OF ASPECTS

The good aspects are the trine of 120 degrees, the sextile of 60 degrees, the conjunction of Jupiter and Venus and the PARALLELS of these also. The bad aspects are the opposition of 180 degrees, the square of 90, the sesq of 135, and the semi-square of 45. Also the conjunctions of SATURN AND ITS PARALLELS. The planets URANUS, NEPTUNE, AND MARS, ACT VARIOUSLY WHEN IN CONJUNCTION AND ARE LARGELY UNDER THE DIRECTION OF PRIMARY INDICATIONS CURRENT AT THE TIME.

ASPECT

ALL PLANETS IN GOOD ASPECT ACT TO INFLUENCE VALUES.

ALL PLANETS IN BAD ASPECT ACT TO DEPRECIATE VALUES.

BUT JUPITER IS THE GREATEST EXPANDER, AND SATURN THE GREATEST DEPRESSOR OF THE MARKETS.

ANALOGY

A PLANET IS JUDGED BY ITS OWN NATURE, AND AN ASPECT BY ITS OWN NATURE.

A BOOM on the market is produced from a succession of good aspects to the PRIME SIGNIFICATOR OR ITS DECAN RULER. WHEN BOTH ARE WELL ASPECTED THE EXTENT IS DOUBLED. It is usually terminated by a change of decan, if not sooner by a break of benefic causes.

A SLUMP is caused by a succession of bad aspects to the Significator and its Decan Ruler

TRANSITION POINTS are formed midway between a good and a bad aspect to the same indicator. Thus:

Venus on the 8th of the month is opposition Jupiter.

On the 11th is sextile Saturn.

On the 20th it is semi-square Mercury.

On the 30th it is semi-square Neptune.

Therefore we should expect a fair amount of selling or realizing of profits. This would be followed, about the 9th or 10th, by a steadying influence under the ap-

proaching sextile of Saturn and Saturn would give a steady upward trend until the 15th-16th, which is the transition point between Saturn and the next aspecting planet Mercury. At this time the market turns and begins to be talked down, and under Venus semi-square Neptune it continues down to the 28th. Here it forms another transition point and proceeds to the sextile of Mercury, which it reaches on August 2nd. These aspects are formed in July 1914. After the 2nd of August the planet Venus meets the conjunction with Mars in Virgo on the 5th, and then goes to the square of Saturn, the change of Decan taking place on August 2nd and changing the good influence of Mercury to that of Mars (war). Venus, being then in the last decan of Virgo, ruled by Mercury, we have further indication of Mercury conjunction Neptune at that time, showing a state of chaos. The direction for 122 years 3 months is 120 degrees 29 minutes and it will be seen that if we add this amount to the M.C. of the radix (Taurus 28) we arrive at just this same position in the zodiac, namely Virgo 28, which is occupied by Mars and Venus in square aspect to Saturn in Gemini at that time. So we have good reason for anticipating a considerable stir in the markets of the world and a great depreciation of values.

DAILY FLUCTUATIONS CAN ONLY BE FOLLOWED WITH SUCCESS by those who are on the spot and watching the clock and ticker. For it will be found that the course of the market follows the MERIDIAN TRANSIT of the planets and the aspects formed to them by the other at the time of their transit. Taking the Sun as indicating noon position, it will be seen that some planets precede the Sun and others follow it. Thus at the beginning of August 1914 we have Sun opposition Uranus, on the 2nd of the month, followed by the Sun semi-square Saturn on the 6th. The heaviest point should therefore be on the 4th, or mid-distance between the aspects. On that date Mercury passed over the MERIDIAN 19 degrees, or 1 hour 16 minutes before noon, in opposition at the time to the Moon. About 24 minutes later Neptune passed the MERIDIAN. Then came the Sun at Noon in Opposition to Uranus and approaching the semi-square of Saturn. Here we have the big depression. It is followed by the transit of Jupiter over the lower Meridian at 17:28 or 28 minutes after noon. Then there is a lot of selling done. Then comes Mars and Venus, in close conjunction in Virgo 23-24 degrees in sextile to Mercury in Cancer 22, but also coming to the square of Saturn in Gemini. This would be about 3:50 P.M. Local Time, and it then became a pressing question as to Peace or War -- Mars or Venus -- question that would serve no market any good.

THE CHART OF ANY PRIVATE COMPANY OR SYNDICATE is to be judged by exactly the same set of indications, and the share values of such company, if offered on the open market, will follow the specific indications belonging to and derived from that chart, subject only to the general indications of the Exchange Chart and the ephemeral transits as already set forth. But each set of indications must be taken account of in THIS ORDER:

1. PRIMARY INDICATIONS, derived from directions of the Midheaven, etc for the current year.
2. TRANSITS OVER THESE PRIMARY POINTS - Sun, Moon, Midheaven, Asc., both radical and progressed, the latter being the more important in relation to the current period.
3. ASPECTS FORMED TO THE PRIME SIGNIFICATOR in the ephemeris from day to day or from week to week, as the case may be.
4. ASPECTS FORMED TO THE RULER OF THE DECAN THEN HELD BY THE PRIME SIGNIFICATOR.

All of these, which being duly noted, will form a concatenation or depending series of influences which will inevitably lead to correct estimate of the TREND OF THE MARKET.

5. Having decided to buy or sell, enter the market WHEN THE MERIDIAN TRANSITS ARE AGREEABLE TO THE CURRENT INDICATIONS FROM WHICH YOUR DECISION WAS MADE. This is the whole business of stock and share trading by chart.

SIXTH RULE

The Sun being in Capricorn or Aquarius - - Saturn is Prime Sig.
The Sun being in Sagittarius or Pisces - Jupiter is Prime Sig.
The Sun being in Libra or Taurus - Venus is P. S.
The Sun being in Virgo or Gemini - Mercury is P. S.
The Sun being in Leo - Sun is P.S.
The Sun being in Cancer - - - - - Moon is P.S.

When the Moon becomes the P.S. through the Sun being in the sign Cancer, as happens from Midsummer Day onwards for a month, it is obvious that we cannot trace primary indications from the ephemeral aspects of so fast and variable a body as the Moon, so what is to be done? Experience says, watch the position and aspect of the ASCENDING NODES of the Moon, which is given for the second and third day continuously throughout the year in any good ephemeris. Merely trace the node as if it were the Moon itself and the rest follows in due sequence.

A few tests of the early July charts will show how this works out. THE NODE OF THE EARTH IS ALWAYS THE VERNAL EQUINOX, AND ALWAYS ZERO ARIES. BUT THE NODE OF THE MOON FROM WHICH ITS COURSE BEGINS, IS ONLY IN THE SAME LONGITUDE AFTER A PERIOD OF 19 YEARS. The node thus becomes a very important indicator. At the beginning of August 1914 it was in Pisces 7 and had no aspects. But it was then in the Decan of Saturn, and Saturn had a succession of bad aspects all through the first three weeks of that month, so that Cancerian Charts were under evil aspects at that time.

It will be observed that there is a line of sympathy between the New York and British Charts in that Venus is

the common PRIME SIGNIFICATOR. In the New York Chart it is strong in its own sign Taurus and this perhaps accounts for the strong Hibernian affinity. In the British Chart (London) Venus is elevated in the Sign Gemini, which is the ruling sign of the USA as a Republic. Consequently there is much sympathy of action between the two markets, and the coincidence of the midheaven degree is remarkable. It is not inappropriate either that the Moon in the NY Chart should be in the ruling sign of England, its P.S. in the ruling sign of Ireland, and Fortuna in the ruling sign of Scotland.

FORTUNA indicates the POSITION OF THE MOON AT THE TIME OF LOCAL SUNRISE FOR ANY SET DATE. IT IS MEASURED BY LONGITUDE, i.e., degrees and minutes of the zodiac.

THE COMPANY. In the case of a company, the M.C. is THE CHAIRMAN, THE 11th HOUSE DENOTES THE DIRECTORS OR BOARD OF CONTROL, THE ASCENDANT DENOTES THE SHAREHOLDERS, THE SIXTH HOUSE SHOWS THE EMPLOYEES.

SPECIFIC HOUSE INDICATION.

THE FIRST HOUSE governs the PUBLIC in which is vested all forms of enterprise and development.

THE SECOND HOUSE relates to PRICE OF MONEY, TRADE, RETURNS, BULLION IMPORTS, BILLS OF EXCHANGE, ETC.

THE THIRD HOUSE rules railroads, tramways, omnibuses, traction of all sorts, locomotion, telephone, aircraft,

canals, bridges and transports as well as postal service and all means of communication.

THE FOURTH HOUSE rules real estate, land explorations, mines, developments, crops, produce of raw material from the soil.

THE FIFTH HOUSE shows educational matters, art, theatres, cinemas, amusements and schools.

THE SIXTH HOUSE governs foodstuffs, clothing, equipment, outfitting supplies, upholstering, furnishings, building and upfitting.

THE SEVENTH HOUSE rules accountancy, banking, corporations, exchanges, contracts, equity, discounting, surveying, valuations, probate, etc.

THE EIGHTH HOUSE controls waste products, conservancy, dredging, petrol, parafine, benzine, medical accessories, chemicals and nitrates.

THE NINTH HOUSE is connected with insurance, cables, publishing, wireless, radiographs, liners and foreign affairs.

THE TENTH HOUSE rules state affairs, the government and political activity generally.

THE ELEVENTH HOUSE rules the exchequer, bonds, government loans, electric and gaslight companies.

THE TWELFTH HOUSE rules laundries, breweries, fisheries, boot mfgs, hosiery and COLD STORAGE.

THE CHART OF THE HEAVENS FOR LONDON.

This is introduced in order to display the influence of ECLIPSES when falling on the places of the significators, either radical or progressed.

On the 12th of March 1914, there was an eclipse of the Moon which fell in Virgo 21. Incidentally, it happened to be in opposition to the ex-Kaiser's radical M.C. in Pisces 21. But what is more pertinent to our present study is the fact that Virgo 21 was the progressed M.C. for the London Exchange for the year 1914. Thus $59^{\circ} 8'$ daily mean motion of the Sun, multiplied by 113, the years elapsed since 1801, gives 111 deg. 26 min., or 3 signs, 21 degrees, 26 min. This being added to the MC of the radix, Taurus 28, gives Virgo 20 on the MC for 1914.

Now Mars passed over the line of the eclipse in Virgo 20 on the ²⁴28-29th of July of that year and ~~was~~ was declared against Serbia by Austria on the 28th. On the 31st a state of war was declared in Germany, who declared war on Russia on the 31st or 1st of August and invaded Luxembourg on the same date, French territory being invaded on August 2nd. This war and the exact date of it was foreseen by "Sepharial" and notified by him in print throughout the country as early as July 1913.

Here we see how the meridian line of the Exchange hereoscope, affecting all securities, was brought into line

with the eclipse which immediately preceded the Great War, and how both were brought into line with the meridian of the ex-Kaiser's horoscope. These considerations determined the prediction of War and the date was defined by the transit of Mars over this triple combine of Virgo Pisces meridian.

The fiscal troubles which have dominated all British politics arose from the eclipse of the meridian degree for the year 1914, and its effects could be traced right through to 1918, when, as early as August 1st, 1914, "Sephariael" fixed the end of the war in these words: "The war will be Titanic and will last until 1918" ---

The Hohenzollerns will bite the dust and gather the DIAD SEA fruit of an inordinate ambition."

When for the last time in the period of 4 years (duration of eclipse) Mars was in transit over the same point, Virgo 20, on March 21st, 1918, Germany launched its last great offensive. America was then well represented at the front and brought new blood into the conflict which was steadily wearing out the reserves of Germany. It was then said: "Hostilities will cease in November of this year." By adding the increment of mean motion for 4 years (1914-1918) to Virgo 20, we reach Virgo 24, and under this meridian in the latitude of London we find Scorpio 29:24 to be rising. This is in 60 degrees aspect to Jupiter, with a following 60 of the Moon in the radix, and this was follow-

ed by the 60 of the Sun to the place of Jupiter and the Moon. The ascendant at this time being past the opposition of the Sun and the trouble done and over, while Fortuna had come up to the sextile of the progressed M.C. at the opening of the WAR.

Fiscal troubles began to arise between America and Great Britain in 1928 on the question of War Debts, and had considerable influence on stock and share values. The total eclipse of May 19¹⁹²⁸ fell on the meridian of the two charts. It will last until 1932, critical dates from the source being November 1929, July 1930 and October-November 1931.

Therefore we have reason to regard the action of eclipses as constituting a major influence upon the VALUE OF SECURITIES.

NEW YORK STOCK EXCHANGE HOROSCOPE

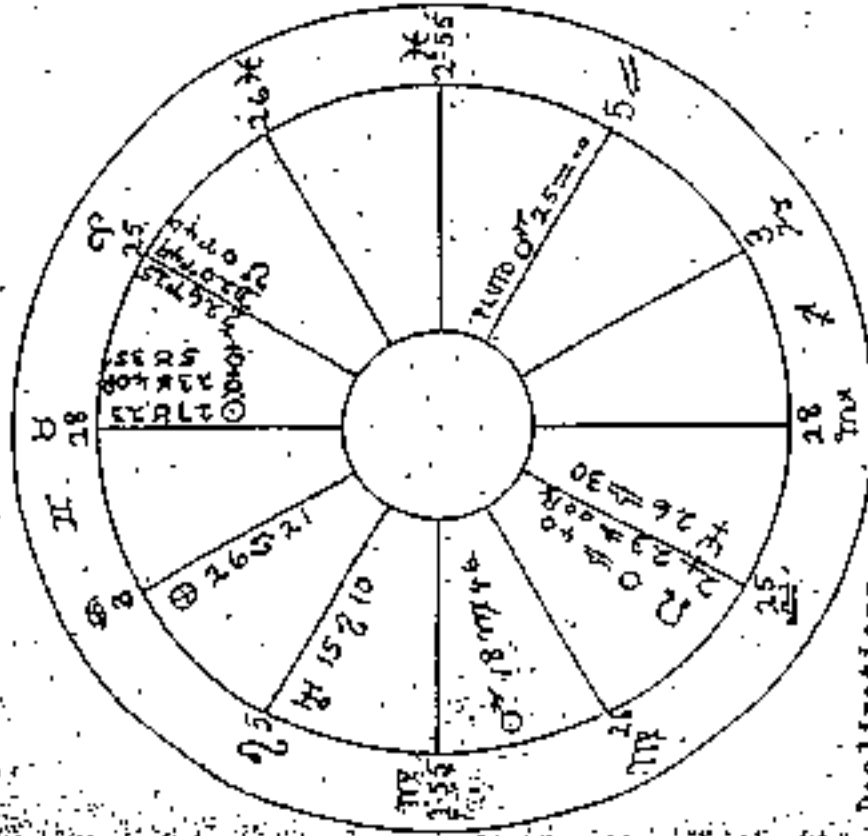
May 17th 1792 @ 12 Noon: New York city

S.T. Or-noon 3-43-11

S.T. at birth 3-42-57

G.M.T. 6 P.M.

Adj. Calo. Date: March 2nd 1792



Declinations

☉ 19-36 n
 ☿ 6-45 n
 ♀ 18-06 n
 ☿ 19-00 n
 ☼ 8-18 n
 ♀ 7-36 n
 ♀ 8-02 n
 ♀ 18-57 n
 ♀ 8-00 n

LONDON STOCK EXCHANGE HOROSCOPE

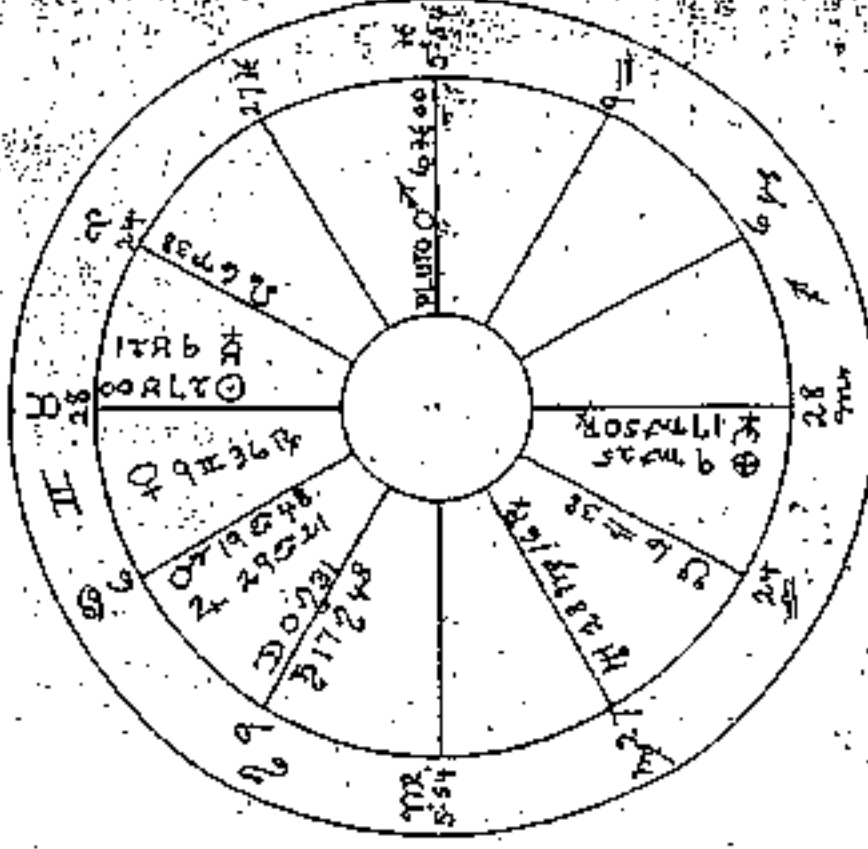
May 18th 1601 @ 12 Noon: London, Eng.

S.T. Or-noon 3-42-45

S.T. at birth 3-42-57

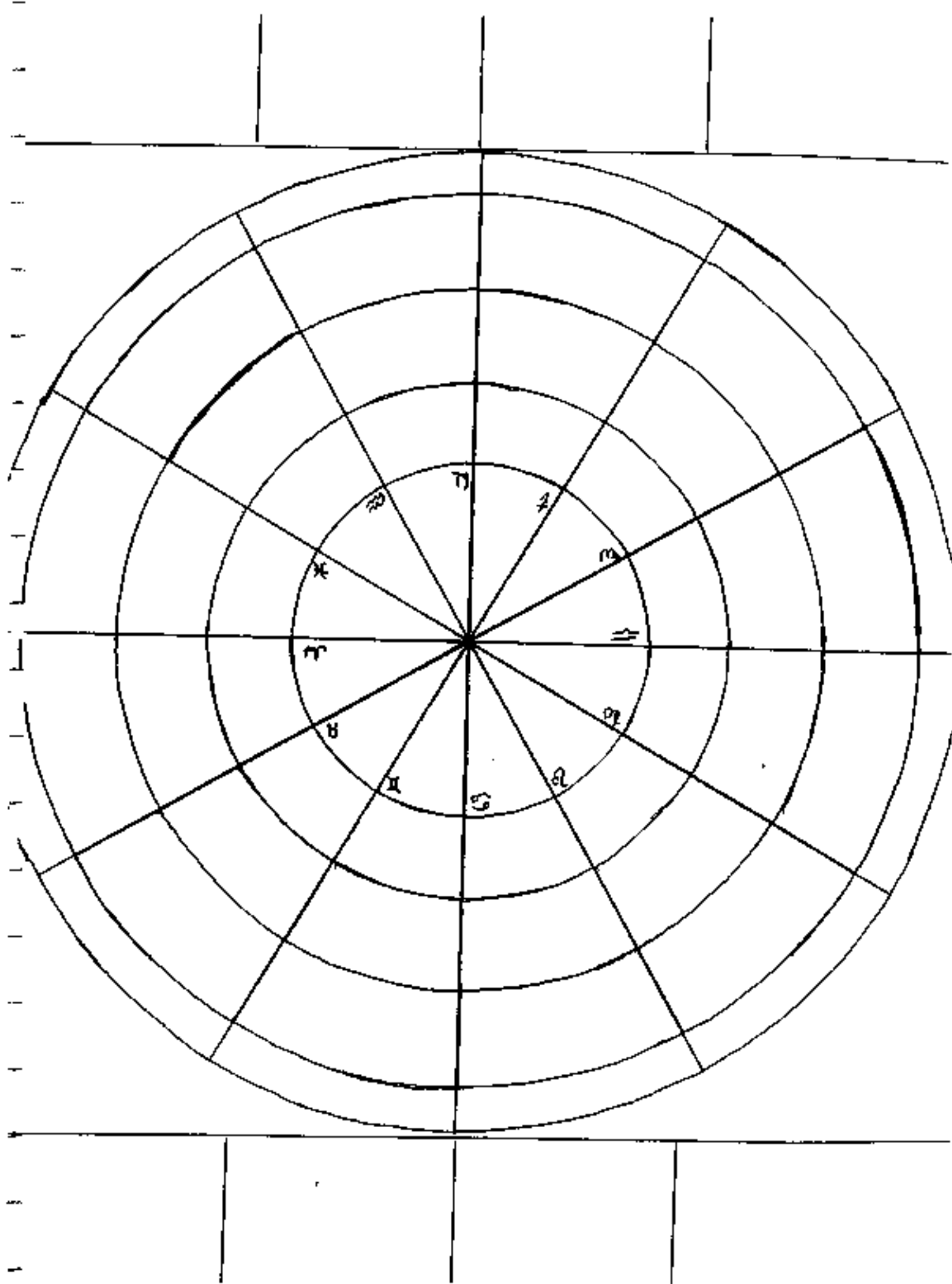
G.M.T. 12 Noon

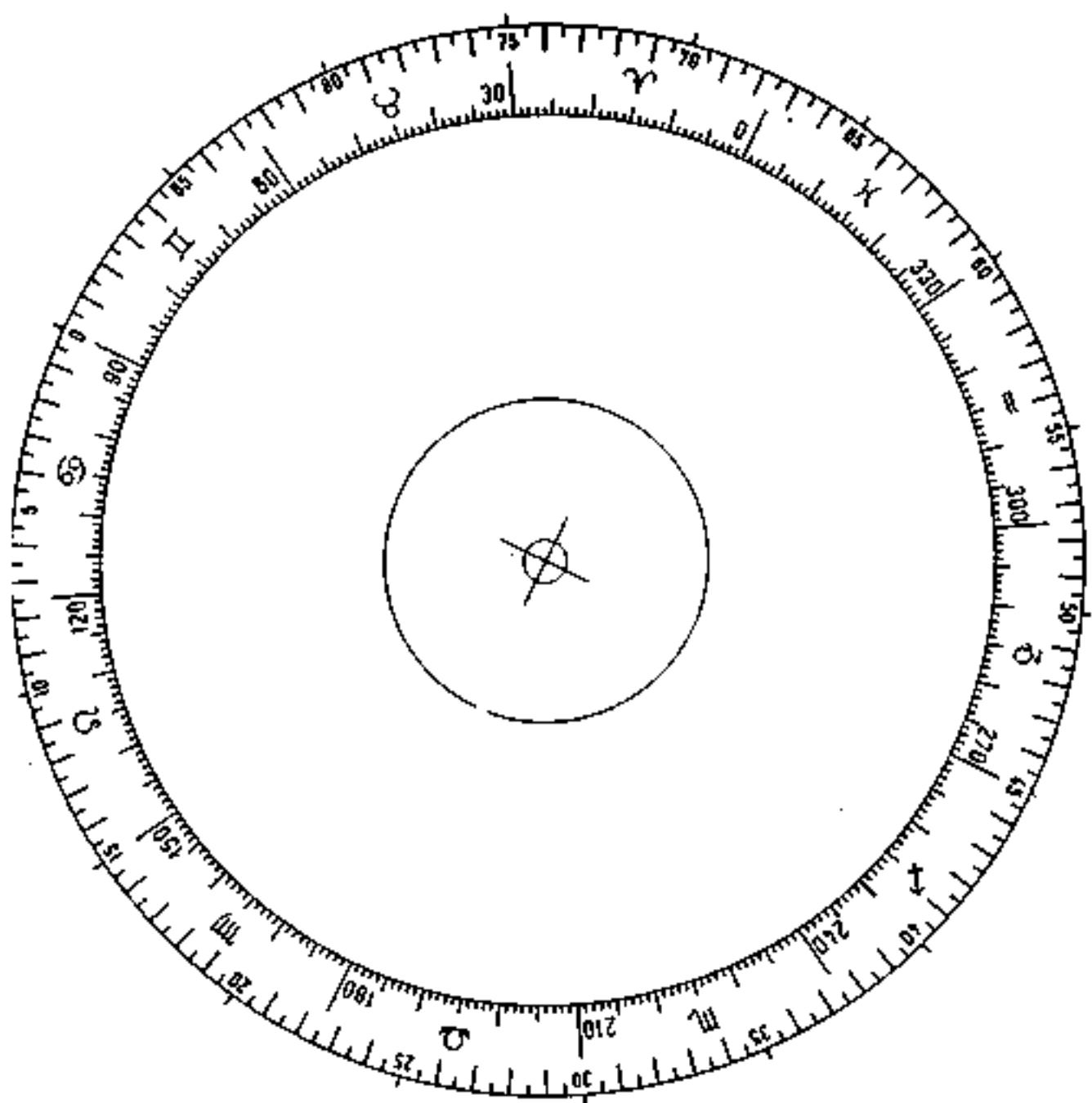
Adj. Calo. Date: May 18th 1801



Declinations

☉ 19-31 n
 ☿ 24-42 n
 ♀ 12-45 n
 ☿ 26-04 n
 ☼ 23-29 n
 ♀ 20-55 n
 ♀ 16-46 n
 ♀ 1-23 n
 ♀ 11-57 n





ASPECTARIAN

Planet	Abbrev.	Sun	Moon	Mercury	Venus	Mars	Jupiter	Saturn	Uranus	Neptune	Pluto	Decln.
Sun	Sa											
Moon	Mo											
Mercury	Me											
Venus	V											
Mars	M											
Jupiter	J											
Saturn	S											
Uranus	U											
Neptune	N											

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APPENDIX D

SOYBEAN LETTER REFERENCES

SOY BEANS
PRICE RESISTANCE LEVELS

The one-half point of the highest selling price and one-half between the high and low are very great because these equal the 45 degree angle or the gravity center. For May Beans these prices are 218 $\frac{3}{8}$, 240 $\frac{3}{8}$, 251 $\frac{7}{8}$ and 319 $\frac{1}{8}$.

On the weekly or monthly chart, use the date when the extreme high or extreme low is made and draw a 45 degree angle up and down starting from the above half-way points. These angles are very important for determining high and lows and a change in trend.

From the extreme lows or extreme highs, you add or subtract the proportionate parts of the circle, as given below.

44 added to 360 gives 404. From 404 subtract 90 gives 314, and the degree in the sign is 14 degrees Aquarius. 90 degrees or square from this is 234 or 16 degrees Scorpio.

436 $\frac{3}{4}$ minus 360 gives 76 $\frac{3}{4}$ or 16 degrees 45' in the sign Gemini. Subtract 120 gives 16 degrees 45' Aquarius or a price of 316 $\frac{3}{4}$. A square or 90 degrees from this is 16 degrees 45' Scorpio, price 226 $\frac{3}{4}$.

The lows of 67, 68 and 69 equal 7, 8 and 9 degrees in Gemini. 120 degrees subtracted gives 307, 308 and 309, the triangle points which are 7, 8 and 9 degrees Aquarius. Subtract 90 degrees from these prices gives 217, 218 and 219 or 7, 8 and 9 degrees in Pisces.

131 high, add 180, gives 311 or 11 degrees Aquarius. Subtract 90 gives 221 or 11 degrees Scorpio.

October 5, 1936, futures trading started in Soy Beans. The first sale was 120. Add 180 gives 300 or 30 degrees in Capricorn.

218 $\frac{3}{8}$ is 8 degrees 22' Scorpio. Add 90 gives 308 $\frac{3}{8}$ or 8 $\frac{1}{2}$ degrees in Aquarius.

240 $\frac{3}{8}$ is 0 degrees 22' in Sagittarius. Add 60 gives 300 $\frac{3}{8}$. Add 90 gives 330 $\frac{3}{8}$.

251 $\frac{7}{8}$ is 12 degrees in Sagittarius. Add 60 gives 311 $\frac{7}{8}$ or 12 degrees Aquarius. Add 90 gives 348 or 12 degrees Pisces.

1930, June high 215. Equals 6 degrees Scorpio. Add 90 gives 306 or 6 degrees Aquarius. Add 225 gives 441, which is the square of 21, and 441 is 21 degrees in the sign Gemini, which is 61 more than 360, the square of 9.

436 $\frac{3}{4}$, subtract 135, gives 201 $\frac{3}{4}$ or 1 degree 45' Aquarius. Subtract 180

gives $256 \frac{3}{4}$ or 16 degrees 45' Sagittarius. Subtract 225 gives $221 \frac{3}{4}$ or 1 degree 45' Sagittarius. Subtract $236 \frac{1}{2}$ gives $200 \frac{1}{2}$ or 20 degrees 15' Min. Libra. The low on May Beans in Feb. 1949 was $201 \frac{1}{2}$.

$344 \frac{1}{2}$, subtract $33 \frac{3}{4}$ gives $310 \frac{3}{4}$ or 10 degrees 45' Aquarius. Subtract 45 gives $299 \frac{1}{2}$ or 29 degrees 30' Capricorn.

120, add 90 gives 210 or 30 degrees Scorpio. Add 125 gives 255 or 15 degrees Sagittarius. Add 180 gives 300 or 30 degrees Capricorn.

67, add 90 gives 157 or 7 degrees Virgo. Add 135 gives 202 or 22 degrees Libra. Add 120 gives 127 or 7 degrees Libra. Add 180 gives 247 or 7 degrees Sagittarius. Add 225 gives 292 or 22 degrees Capricorn. Add 240 gives 307 or 7 degrees Aquarius. Add 270 gives 337 or 7 degrees Pisces. Add 315 gives 382 or 22 degrees Aries. Add 360 gives 427 or 7 degrees Gemini. Add $271 \frac{1}{2}$ gives $438 \frac{1}{2}$. High on May beans was $436 \frac{3}{4}$. After that high the next extreme low was $201 \frac{1}{2}$. Note that 67 plus 125 gives 202, and that one-half of 405 is $202 \frac{1}{2}$, and 180 plus $22 \frac{1}{2}$ is $202 \frac{1}{2}$, which are the mathematical reasons why May Soy Beans made bottom at $201 \frac{1}{2}$.

All of the above price levels can be measured in Time Periods of days, weeks and months, and when the time periods come out at these prices, it is important for a change in trend, especially if confirmed by the geometrical angles from highs and lows.

ACTIVE ANGLES AND DEGREES

By live or active angles in mean Prices and Time Periods where the Longitude of the major planets are or where the squares, triangles, oppositions are to these planets.

The averages of the six major planets Heliocentric and Geocentric are the most powerful points for Time and Price resistance. Also the Geocentric and Heliocentric average of the five major planets with Mars left out, is of great importance and should be watched.

You should also calculate the averages of eight planets which move around the Sun as this is the first most important odd square. The square of "1" is one, and "1" is the Sun. 8 added to "1" gives 9, the square of 3 and completes the first important odd square, which is important for Time and Price.

Examples of live, active angles: At the present writing, Jan. 18, 1954, Saturn Geo. is 8 to 9 degrees Scorpio. Add the square or 90 degrees gives 8 to 9 degrees Aquarius and equals the price 309-309, for May Beans. The planet Jupiter is at 21 degrees Gemini, which is 61 degrees in longitude from "0" the square of 9. Subtract 135 degrees from Jupiter gives 306 or 6 degrees Aquarius. This is why Soy Beans have met resistance so many times between 306 and $311 \frac{1}{2}$. The Price Resistance levels come out strong around these degrees and prices and the Geometrical angles come out on daily, weekly and monthly, but the power of Saturn and Jupiter aspects, working out Time to these Price Resistance Levels, is what halts the advance in Soy Beans.

24 Revolutions of Time and Price The earth makes one revolution on its axis in 24 hours, moving 360 degrees in Longitude. One hour of Time equals 15 degrees in Longitude, and for one hour of Time, we use one cent of Price. This is for daily active markets but can be used for weekly and monthly time periods, as

you can see by the Master Charts.

The Longitude of the Planets and the Longitude of the average of the planets determine the Resistance Levels as the price moves around each cycle of 24 cents per bushel. You mark on the Master Chart all low prices with a red circle around them and place around all high prices a green circle. Then note the Angles of 45, 60, 90, 120, 135, 180, 225, 240, 270, 300, 315 and 260 from each high and low. Then check the Longitude of the Planets and the Longitude of the average of the planets to see when the Price reaches these degrees or aspects and meets resistance.

Example: Dec. 2, 1953, May Soy Beans high $311\frac{1}{4}$. This equalled 18 degrees 45' in Pisces, close square or 90 degrees of Jupiter, 135 degrees to Saturn and 180 degrees of the averages, and 170 degrees of Uranus.

300 price equals 30 degrees Virgo. 302 equals 30 degrees Libra. 304 equals 30 degrees Scorpio. On Jan. 18, 1954, the planet Saturn Con. is 8 degrees 30' Scorpio, and 15 degrees Scorpio gives a price of 303, therefore when May Beans decline to 302, they will be below the body or longitude of Saturn and will indicate lower. At the same time, using the Earth's annual revolution of $365\frac{1}{4}$ days to move around the Sun, a price of 308 is 90 degrees or square to Saturn. As long as the price is below 308 it is within the square and in position to go lower. But by the 24 revolution, when the price breaks below 304, it is in the bear sign Scorpio, a fixed sign and will indicate lower prices.

Study and analyze all options of all commodities in the same way as we have analyzed May Beans. Remember, when these Resistance Points are met you must give the market time to show that it is making tops or bottoms and getting ready to make a change in trend. Do not guess, wait until you get a definite indication of a change in trend before deciding that the main trend has changed. You can buy or sell against these resistance levels and place a stop loss order. Having before you all the information outlined above, you would certainly have gone short of May Soy Beans on Dec. 2, 1953 and cover your shorts on Dec. 17 at 296 because the price was down to the 45 degree angle from 311 on the Monthly high and low chart.

24 CENT MOVES OR MORE

It is very important to watch the action on the daily and weekly chart when the price is up or down 24 cents from any high or low, 48 cents, 72 cents most important because three times 24, 96 cents, 120 or 5 times 24, 120 of great importance because 6 times 24 and the square of 12, 144 which is 7 times 24 - very important. You can also use one-half of 24, which is 12, and watch 36, 60, 84, etc. which equals 180 degrees or half the circle or cycle.

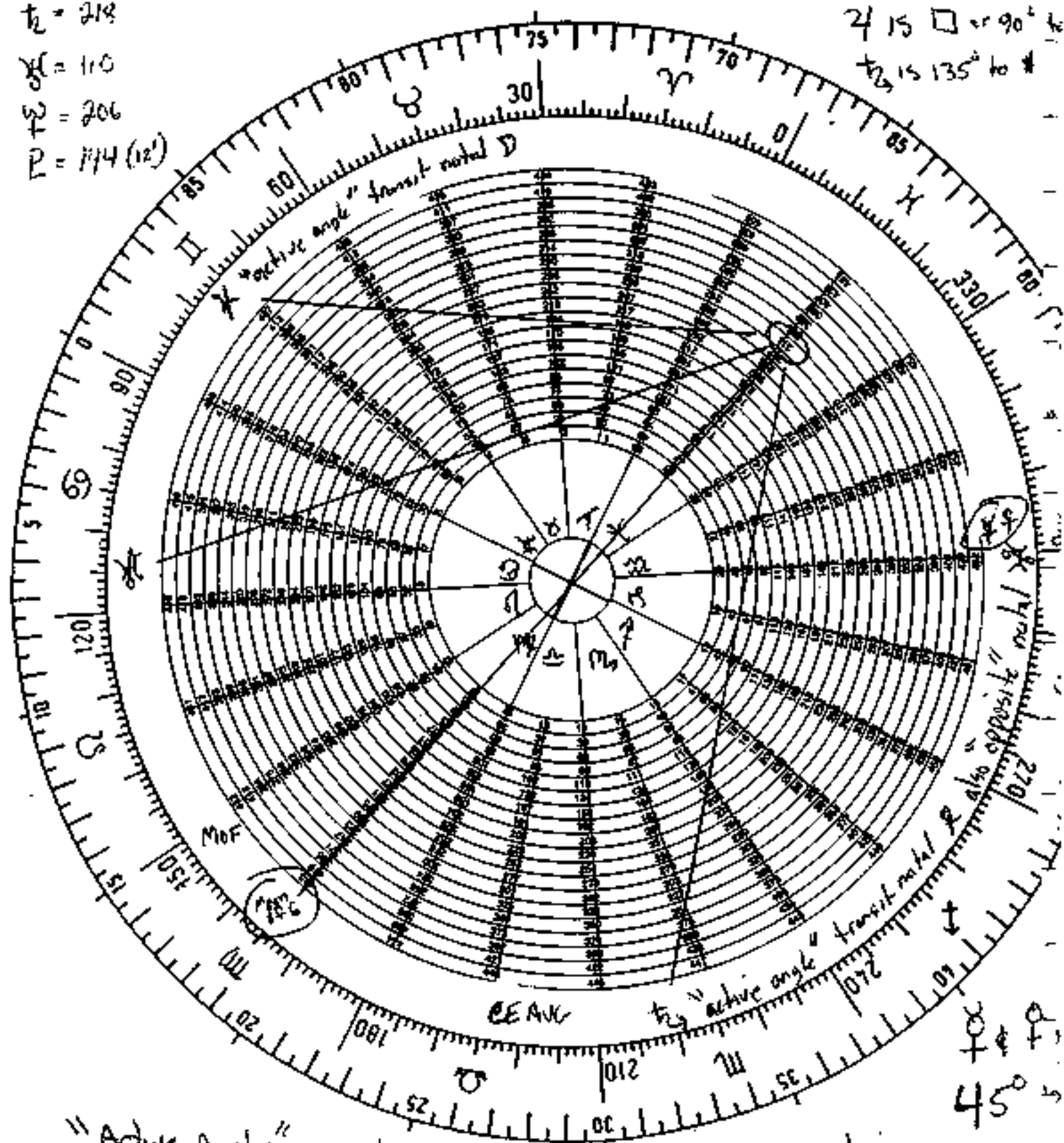
$\phi = 300$
 $\phi = 295$
 $\phi = 227$
 $\phi = 77$
 $t_2 = 218$
 $\phi = 110$
 $\phi = 206$
 $P = 114 (12')$

$CE/MY = 17/10 = 1.7$
MEAN OF 6 = 163 or 13 P2
 $MOF = 151 or 10 P2$

JAN 18, 1954

MEAN of six is of opposi-
Price of #311

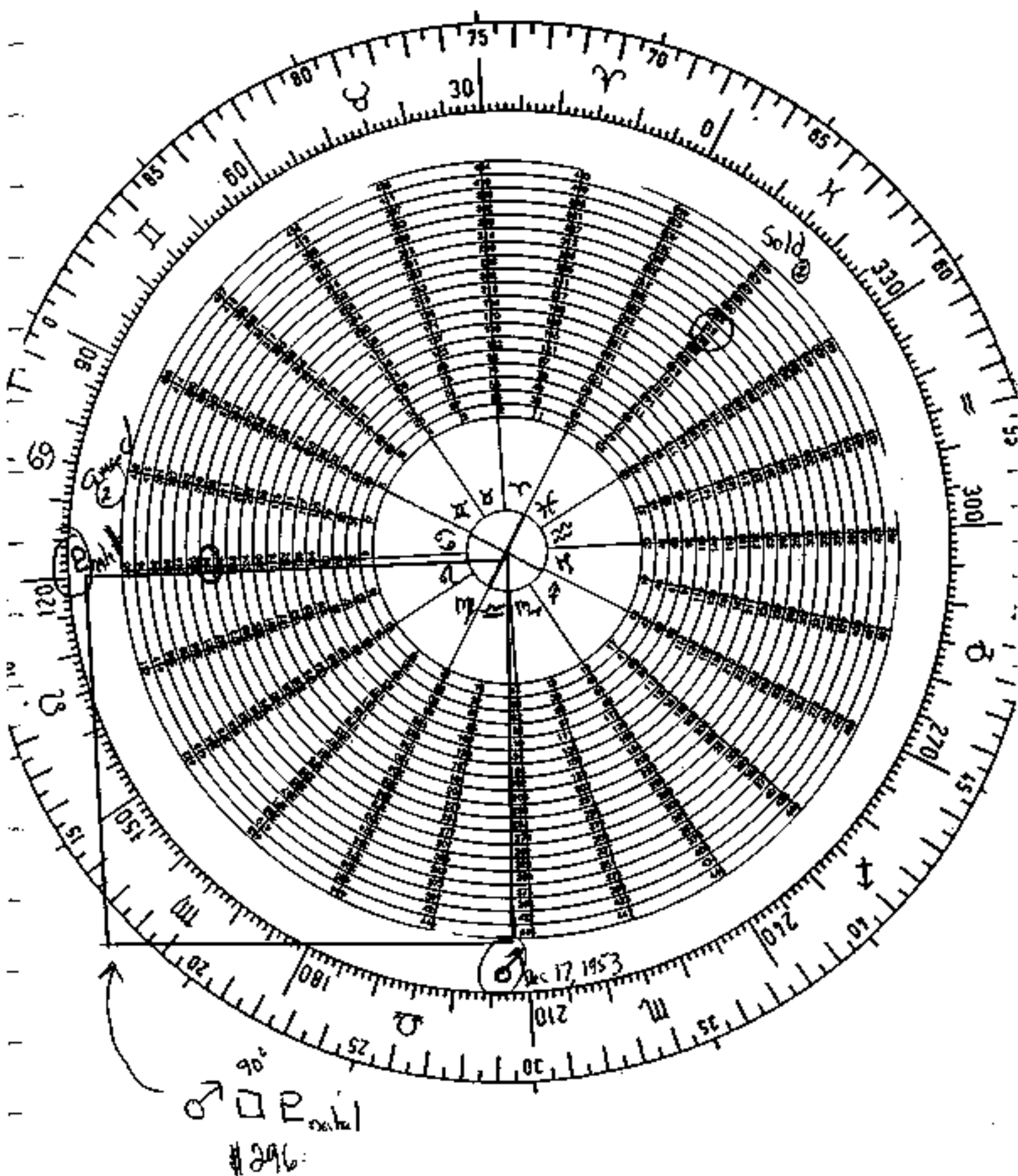
ϕ is $\square$ or 90° to
 t_2 is 135° to #



"Active Angles" are where the squares, triangles, and oppositions are to these planets (I would also include the conjunction)

ϕ ♀
 ϕ ♀
 ϕ ♀
 45°

Also note that Geocentric Saturn's $11\frac{1}{4}^\circ$ angle would have hit #312 early in December 2 1953 & that Uranus' cover was at Helio to Saturn's $11\frac{1}{4}$ angle. The low price 12/17/1953 was 295.



2f returns to ecliptic longitude
2441' 8" & so far decline again.

Aug 11th, 1999
Total Eclipse

← 2f →
Jupiter

November Soybeans
1999

[Signature]

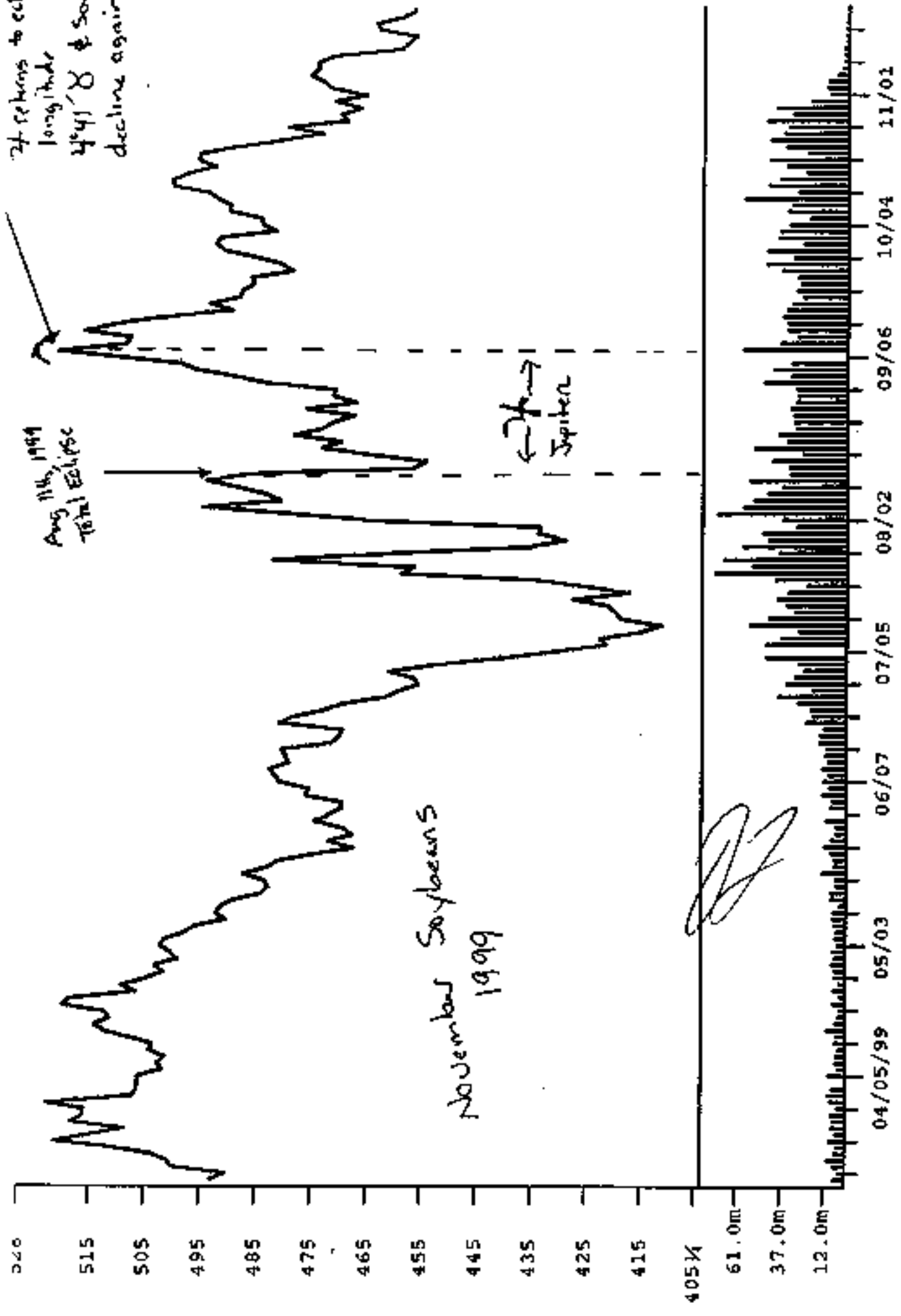


Table for 39V

1st 5 Points
 1st Sign = 150 points
 1800 points = 360

minimum tide

Note how Ganes is using the minimum fluctuations to set up a zodiac relationship with price. This is what he is doing in the coffee letter.

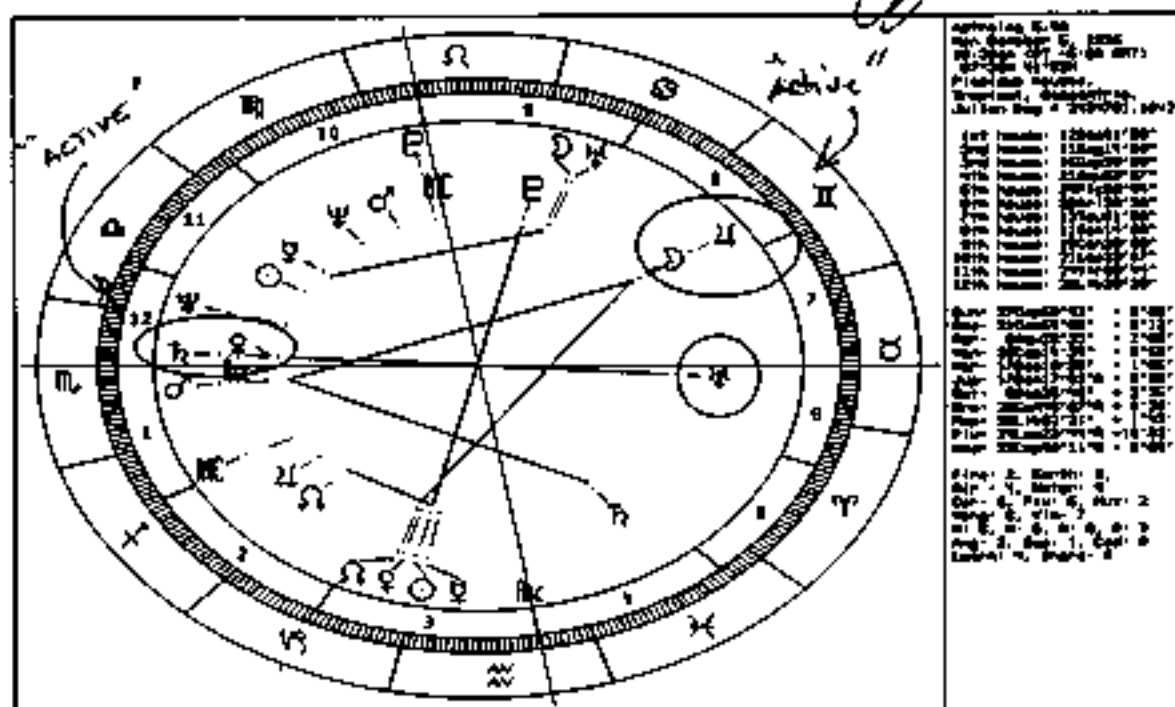
000	300	500	100
30	350	550	110
320	380	580	120
340	410	610	130
360	440	640	140
380	470	670	150
400	500	700	160
420	530	730	170
440	560	760	180
460	590	790	190
480	620	820	200
500	650	850	210
520	680	880	220
540	710	910	230
560	740	940	240
580	770	970	250
600	800	1000	260
620	830	1030	270
640	860	1060	280
660	890	1090	290
680	920	1120	300
700	950	1150	310
720	980	1180	320
740	1010	1210	330
760	1040	1240	340
780	1070	1270	350
800	1100	1300	360
820	1130	1330	370
840	1160	1360	380
860	1190	1390	390
880	1220	1420	400
900	1250	1450	410
920	1280	1480	420
940	1310	1510	430
960	1340	1540	440
980	1370	1570	450
1000	1400	1600	460
1020	1430	1630	470
1040	1460	1660	480
1060	1490	1690	490
1080	1520	1720	500
1100	1550	1750	510
1120	1580	1780	520
1140	1610	1810	530
1160	1640	1840	540
1180	1670	1870	550
1200	1700	1900	560
1220	1730	1930	570
1240	1760	1960	580
1260	1790	1990	590
1280	1820	2020	600
1300	1850	2050	610
1320	1880	2080	620
1340	1910	2110	630
1360	1940	2140	640
1380	1970	2170	650
1400	2000	2200	660
1420	2030	2230	670
1440	2060	2260	680
1460	2090	2290	690
1480	2120	2320	700
1500	2150	2350	710
1520	2180	2380	720
1540	2210	2410	730
1560	2240	2440	740
1580	2270	2470	750
1600	2300	2500	760
1620	2330	2530	770
1640	2360	2560	780
1660	2390	2590	790
1680	2420	2620	800
1700	2450	2650	810
1720	2480	2680	820
1740	2510	2710	830
1760	2540	2740	840
1780	2570	2770	850
1800	2600	2800	860
1820	2630	2830	870
1840	2660	2860	880
1860	2690	2890	890
1880	2720	2920	900
1900	2750	2950	910
1920	2780	2980	920
1940	2810	3010	930
1960	2840	3040	940
1980	2870	3070	950
2000	2900	3100	960
2020	2930	3130	970
2040	2960	3160	980
2060	2990	3190	990
2080	3020	3220	1000
2100	3050	3250	1010
2120	3080	3280	1020
2140	3110	3310	1030
2160	3140	3340	1040
2180	3170	3370	1050
2200	3200	3400	1060
2220	3230	3430	1070
2240	3260	3460	1080
2260	3290	3490	1090
2280	3320	3520	1100
2300	3350	3550	1110
2320	3380	3580	1120
2340	3410	3610	1130
2360	3440	3640	1140
2380	3470	3670	1150
2400	3500	3700	1160
2420	3530	3730	1170
2440	3560	3760	1180
2460	3590	3790	1190
2480	3620	3820	1200
2500	3650	3850	1210
2520	3680	3880	1220
2540	3710	3910	1230
2560	3740	3940	1240
2580	3770	3970	1250
2600	3800	4000	1260
2620	3830	4030	1270
2640	3860	4060	1280
2660	3890	4090	1290
2680	3920	4120	1300
2700	3950	4150	1310
2720	3980	4180	1320
2740	4010	4210	1330
2760	4040	4240	1340
2780	4070	4270	1350
2800	4100	4300	1360
2820	4130	4330	1370
2840	4160	4360	1380
2860	4190	4390	1390
2880	4220	4420	1400
2900	4250	4450	1410
2920	4280	4480	1420
2940	4310	4510	1430
2960	4340	4540	1440
2980	4370	4570	1450
3000	4400	4600	1460
3020	4430	4630	1470
3040	4460	4660	1480
3060	4490	4690	1490
3080	4520	4720	1500
3100	4550	4750	1510
3120	4580	4780	1520
3140	4610	4810	1530
3160	4640	4840	1540
3180	4670	4870	1550
3200	4700	4900	1560
3220	4730	4930	1570
3240	4760	4960	1580
3260	4790	4990	1590
3280	4820	5020	1600
3300	4850	5050	1610
3320	4880	5080	1620
3340	4910	5110	1630
3360	4940	5140	1640
3380	4970	5170	1650
3400	5000	5200	1660
3420	5030	5230	1670
3440	5060	5260	1680
3460	5090	5290	1690
3480	5120	5320	1700
3500	5150	5350	1710
3520	5180	5380	1720
3540	5210	5410	1730
3560	5240	5440	1740
3580	5270	5470	1750
3600	5300	5500	1760
3620	5330	5530	1770
3640	5360	5560	1780
3660	5390	5590	1790
3680	5420	5620	1800
3700	5450	5650	1810
3720	5480	5680	1820
3740	5510	5710	1830
3760	5540	5740	1840
3780	5570	5770	1850
3800	5600	5800	1860
3820	5630	5830	1870
3840	5660	5860	1880
3860	5690	5890	1890
3880	5720	5920	1900
3900	5750	5950	1910
3920	5780	5980	1920
3940	5810	6010	1930
3960	5840	6040	1940
3980	5870	6070	1950
4000	5900	6100	1960
4020	5930	6130	1970
4040	5960	6160	1980
4060	5990	6190	1990
4080	6020	6220	2000
4100	6050	6250	2010
4120	6080	6280	2020
4140	6110	6310	2030
4160	6140	6340	2040
4180	6170	6370	2050
4200	6200	6400	2060
4220	6230	6430	2070
4240	6260	6460	2080
4260	6290	6490	2090
4280	6320	6520	2100
4300	6350	6550	2110
4320	6380	6580	2120
4340	6410	6610	2130
4360	6440	6640	2140
4380	6470	6670	2150
4400	6500	6700	2160
4420	6530	6730	2170
4440	6560	6760	2180
4460	6590	6790	2190
4480	6620	6820	2200
4500	6650	6850	2210
4520	6680	6880	2220
4540	6710	6910	2230
4560	6740	6940	2240
4580	6770	6970	2250
4600	6800	7000	2260
4620	6830	7030	2270
4640	6860	7060	2280
4660	6890	7090	2290
4680	6920	7120	2300
4700	6950	7150	2310
4720	6980	7180	2320
4740	7010	7210	2330
4760	7040	7240	2340
4780	7070	7270	2350
4800	7100	7300	2360
4820	7130	7330	2370
4840	7160	7360	2380
4860	7190	7390	2390
4880	7220	7420	2400
4900	7250	7450	2410
4920	7280	7480	2420
4940	7310	7510	2430
4960	7340	7540	2440
4980	7370	7570	2450
5000	7400	7600	2460
5020	7430	7630	2470
5040	7460	7660	2480
5060	7490	7690	2490
5080	7520	7720	2500
5100	7550	7750	2510
5120	7580	7780	2520
5140	7610	7810	2530
5160	7640	7840	2540
5180	7670	7870	2550
5200	7700	7900	2560
5220	7730	7930	2570
5240	7760	7960	2580
5260	7790	7990	2590
5280	7820	8020	2600
5300	7850	8050	2610
5320	7880	8080	2620
5340	7910	8110	2630
5360	7940	8140	2640
5380	7970	8170	2650
5400	8000	8200	2660
5420	8030	8230	2670
5440	8060	8260	2680
5460	8090	8290	2690
5480	8120	8320	2700
5500	8150	8350	2710
5520	8180	8380	2720
5540	8210	8410	2730
5560	8240	8440	2740
5580	8270	8470	2750
5600	8300	8500	2760
5620	8330	8530	2770
5640	8360	8560	2780
5660	8390	8590	2790
5680	8420	8620	2800
5700	8450	8650	2810
5720	8480	8680	2820
5740	8510	8710	2830
5760	8540	8740	2840
5780	8570	8770	2850
5800	8600	8800	2860
5820	8630	8830	2870
5840	8660	8860	2880
5860	8690	8890	2890
5880	8720	8920	2900
5900	8750	8950	2910
5920	8780	8980	2920
5940	8810	9010	2930
5960	8840	9040	2940
5980	8870	9070	2950
6000	8900	9100	2960
6020	8930	9130	2970
6040	8960	9160	2980
6060	8990	9190	2990
6080	9020	9220	3000
6100	9050	9250	3010
6120	9080	9280	3020

Gann's Personal Soybean Letter

I have provided a copy of this letter in the back of the Book. Please read it over before reading this section.

First, I will show you where Gann is getting some of the prices he is using in this letter that are not listed as Major highs or lows. In the first paragraph, Gann says "The one-half point of the highest selling price and one-half between the high and low are very great because these equal the 45-degree angle or the gravity center. For May Beans these prices are 218 3/8, 240 3/8, 251 7/8, and 319 1/8". On this page you will find four prices that are hard to explain. These are \$218 3/8, \$240 3/8, \$251 7/8 and \$319 1/8. The first price is the midpoint of the \$436 1/2 high (436 / 2). The 240 3/8 price is the extreme low of 44 + the high of 436 1/2 divided by 2 (436 1/2 + 44 = 480 1/2 / 2). The 251 7/8 price is the low price at 67 + the high price 436 1/2 divided by 2. Finally, the 319 1/8 price is the low at 201 1/2 + 436 1/2 high divided by 2. Gann then starts adding proportionate parts of the circle to these prices. He is also directly referencing the corresponding zodiac longitude to the price levels. What he is doing is creating an Astrological aspectarium of price. He is trying to calculate where the Trines and squares are to different price levels. Gann said that these were the "mathematical reasons" why Soybeans made a bottom at \$201 1/2. The extreme low at 44 + 360 = 404. Both of these prices would equal 14 Taurus. The midpoint or one-half point is 202 (404/2) = 22 Libra. The low of 201 1/2 occurred February 9th, 1949. This price is equal to 15 Leo on the "Circle Chart". On this date, you will find Pluto at 15 Leo (conjunct the price), Neptune at 15 Libra (Sextile to price), Jupiter is 150 to price, etc.

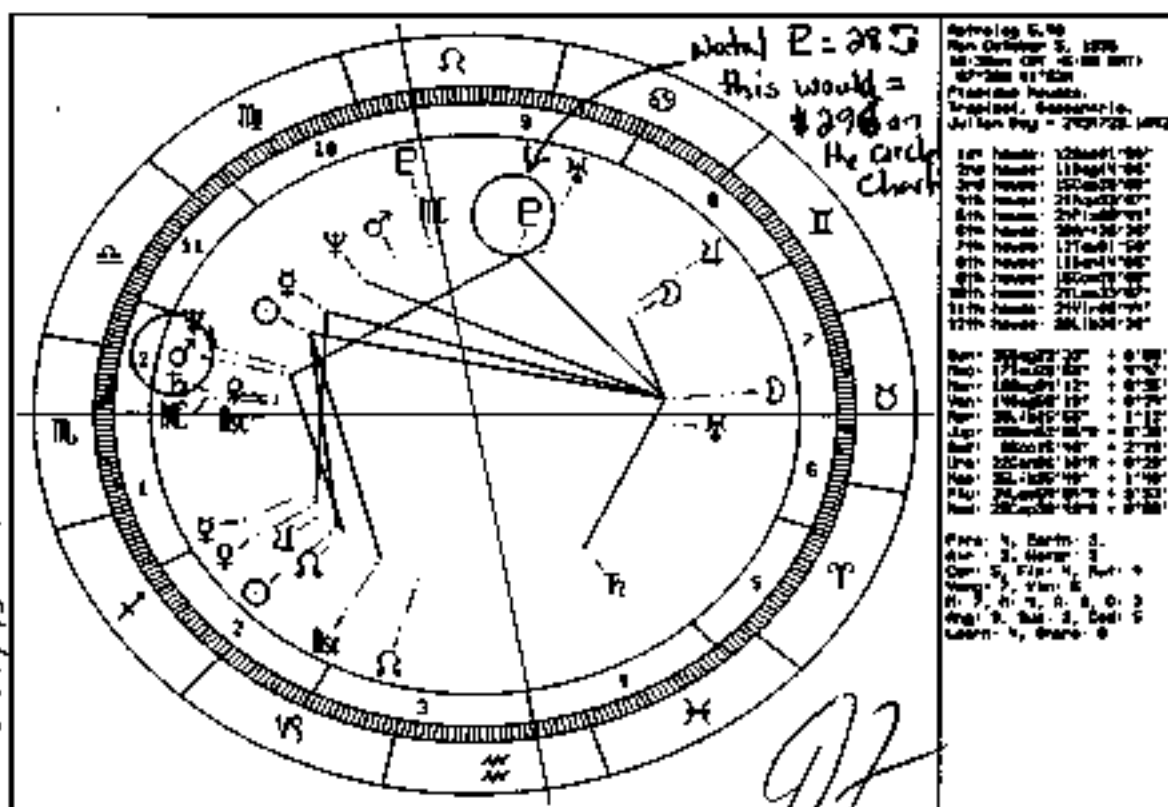
The next part of the letter talks about active or live angles. "By live or active angles is meant Prices and Time Periods where the longitude of the major planets are or where the squares, triangles, oppositions are to these planets". "Examples of live, active angles: At the present writing, Jan. 18th, 1954, Saturn Geocentric is 8 to 9-degrees Scorpio. Add the square or 90-degrees gives 8 to 9-degrees Aquarius and equals the price \$309 for May Beans". He then does the same aspect procedure for Jupiter at 21 Gemini + 135-degrees = 306 or 6 degrees Aquarius. In this paragraph, Gann is using pure longitude to price conversions. In the next paragraph Titled 24 Revolutions of Time and Price Gann is using the Circle Chart. An interesting thing about Gann's first example of an "active angle" is related to the first trade horoscope (natal) for Soybeans. Gann said "October 5th, 1936, futures trading started in Soybeans. The first sale was 120". If we cast a horoscope for this date, we can actually see why Saturn's position is "active".



From the above example, you can see that Saturn was "active" because it was transiting natal Venus and was in opposition to natal Uranus. Jupiter was "active" by a conjunction with the natal moon. Gann says, "The power of Saturn and Jupiter aspects, working out time to these Price Resistance Levels, is what halts the advance in Soybeans".

Gann then moves on to his circle chart. He instructs his student to circle all the low prices with a red circle and all the highs with a green circle. Then note the Angles of 45, 60, 90, 120, 135, 180, 225, 240, 270, 300, 315, and 360 from each high and low. These are not geometric angles! These are astrological angles on the circle chart or "Master Chart". Gann then gives an example for December 2, 1953, May Soybeans high \$31 1/4. If you place the planets on the circle chart, you will get the prices Gann is talking about. When he mentions that this price is 180-degrees of the averages he is talking about the MOF Formula, CE Average or the Average of Six that I discussed earlier. He says "Having before you all the information outlined above, you would certainly have gone short of May Soybeans on December 2nd, 1953 and covered your shorts on December 17th at 296 because the price was down to the 45-degree angle from 44 on the monthly high low chart". The Monthly high low chart indicates Gann was referring to a Saturn geometric angle from the low price of 44. When you read Gann, you can almost always substitute the word month for Saturn. The 44 low occurred December 1932. From this time period to December 1953, $\frac{1}{2}$ helio. Moved 267.067 degrees. The $1\frac{1}{4}$ -angle @ 0.9375/degree would = $\$250.375 + 44 = \294.375 which is within 1.625 cents of Gann's 296 price. Using his shortcut method, December 1932 to 1953 = 21 years times 12 (months) = $\$252 +$ the 44 low = $\$296$. Another interesting point about the 296 price that Gann

Remember the min. fluctuation for Beans was on 1/8 in 1953 so $30 \times 125 = 3.75$ (45°)
 $3.75 / 45^\circ = .08333 \times 11\frac{1}{4}^\circ = .9375$



said to cover the short position is that on this particular day, Pluto's first trade longitude was activated by a transiting square of Mars i.e. Mars square natal Pluto on the circle chart "Master Chart". See the above horoscope:

This concludes my analysis of Gann, if you study and apply the above techniques, I think you will have a greater appreciation of WD Gann and the work that he did. Most people trade the markets as a secondary hobby or interest. Gann made it his profession and worked extremely hard at his craft. If you are not as serious as Gann, then you can not achieve the same results. Fortunately, we have computers today that can do a lot of the tasks. Gann had to do his work without the aid of even a simple calculator. Put the time into your studies and the results should follow.

APPENDIX E

MASTER CALCULATOR

MASTER CALCULATOR FOR WEEKLY TIME PERIODS TO DETERMINE THE TREND OF STOCKS AND COMMODITIES

by W. D. GANN

This Master Calculator shows Weekly Time Periods of 7 days each or a total of 52 weeks in one year. This represents 364 calendar days, therefore at the end of each year there is a gain of one day and at the end of 7 years a gain of 7 days, the time period coming out one week before the date of the important high and low prices. You must also add one day for each leap year. Suppose you want to get the time period for 15 years, you multiply 365 by 15, add the number of leap years and then divide the total days by 7 to get the weekly periods of 7 days each in order to use the Calculator. (See Tables for Price and Time Periods.)

The total square of 52 is 2704, which we can use to measure weeks, days, months, years or hours. In using days, it would, of course, require 2704 days to pass thru the square of 52. This would give 386 weeks and 2 days or approximately 7 years and 5 months, very close to the important cycle of $7\frac{1}{2}$ years, which is 90 months.

If we use hours to balance or square 2704, we get $112\frac{2}{3}$ days by dividing 2704 by 24, the total number of hours in a day during which time the earth makes one complete revolution on its axis.

The Square of 52, which is composed of 7-day periods, is one of the most important for measuring Price and Time. The number 7 is referred to in the Bible more times than any other number, except the number 3. Both of these numbers are very important to use in connection with price and time changes.

You start time periods from the actual dates of important high and important low prices and not from the first day of each month or the first day of each year.

The Calculator is 104 weeks wide, which equals 2 years. The time periods run across the bottom of the Calculator from left to right to 104, which completes 2 years, and to 208, which completes 4 years. At the top of the Calculator, running across to the left, the time periods run to 312 which ends 6 years, and to 416, which ends 8 years, and to 520, which completes a 10-year cycle.

DIVISIONS OF TIME PERIODS

The year is divided by $\frac{1}{8}$, which equals $6\frac{1}{2}$ weeks.

$\frac{1}{4}$	"	"	13	"	
$\frac{1}{5}$	"	"	17	"	
$\frac{3}{8}$	"	"	$19\frac{1}{2}$	"	
$\frac{1}{2}$	"	"	26	"	a most important time and resistance level
$\frac{5}{8}$	"	"	$32\frac{1}{2}$	"	
$\frac{2}{3}$	"	"	35	"	
$\frac{3}{4}$	"	"	39	"	very important for change in trend.
$\frac{7}{8}$	"	"	$45\frac{1}{2}$	"	
1 year, which is 52 weeks.					

The angles run from each of the time periods for Price and Time in order to balance the Square and show resistance levels where price and time periods indicate a change in trend.

THIRD AND FOURTH DIMENSIONS OF TIME AND PRICE

We know of three dimensions--height, width and length--but there is a fourth dimension or element in market movements. We prove the fourth dimension with the Master Calculator or Square of 52 in Time Periods of 7 days each for 7 weeks or more and the same price relation. 7×52 equals 364 or 7 years.

THE CIRCLE, TRIANGLE AND SQUARE

The circle of 360° and the nine digits are the basis of all mathematical calculations. The Square and the Triangle form within the circle but there is an inner circle and an inner square, as well as an outer square and an outer circle which prove the fourth dimension in working out market movements.

PRICE

The most important points to consider are:

1. Lowest price.
2. Highest price.
3. The $1/2$ point, mean or average between the extreme high and the extreme low. We get the fourth dimension, as shown on the Master Calculator, by drawing 45° angles from the $1/2$ point or gravity center, which is the most important for price resistance.
4. Volume of Sales. This is the power which drives the market up or down but remember that TIME is the most essential element and when Time is completed, the volume of sales starts to move the market up or down.

TIME

Time is divided into sections or cycles by which we determine the change in trend.

1. Daily high and low prices
2. Weekly high and low prices
3. Monthly high and low prices
4. Yearly high and low prices

The Weekly and the Yearly Time Periods are most important for trend indications and for changes in trend.

The day is divided into hours, minutes and seconds. The 4 divisions of the day are: Sunrise, Noon, Sunset and Midnight. Of these the most important are: Noon, when the Sun is straight overhead or on a 90° angle, and Midnight, the opposite point or 180° from Noon and 90° from Sunset.

Because we are using 7-day Time Periods with the Calculator, $1/2$ of 7 days or $3\frac{1}{2}$ days is important to watch for change in trend. Always watch the 3rd and 4th day from any important high or low level for a minor change in trend which later may become a major change.

7-day Periods -- The time periods of 7 calendar days from any important high or low level is of great importance. 14 days is the most important and 21 days or 3 weeks is next in importance. Reactions will often run 2 weeks and sometimes 3 weeks and then resume the main trend. Rallies in a Bear Market often run 14 days and sometimes 21 days and then resume the downward or main trend.

Multiples of 7 days -- The square of 7 or 49 days is very important for change in trend. You can start to watch for this change after the 42nd day but the first indication of a change may not occur until the 45th or 46th day, which is $1/8$ of a year or 365 days. $1/16$ of a year is 23 days. Therefore, both the 46-day and 23-day periods are important to watch for change in trend.

Next in importance is 63 to 65 days because 7×9 is 63 and the square of 8 is 64. 81 days or the square of 9 is also of great importance. 90 to 91 days is $1/4$ of a year or 7×13 . This is of very great importance to watch for change in trend. Of course, of next importance is around 182 days or $1/2$ of the year.

YEARS

Later we will refer to the four seasons or the divisions of the year, which are Spring, Summer, Fall and Winter and important to watch for change in trend. However, the divisions of Time are from the date of the actual important high and low prices.

The important yearly cycles are 1, 2, 3, 5, 7, 9, 10, 12, 14, 15, 18, 20, 21, 22, 24, 25, 27, 28, 30, 40, 45, 49, 56, 60, 84, and 90 which is the Great Cycle. We divide the cycles into $1/2$, which is the most important, and also into the periods of $1/8$, $1/3$ and $2/3$, and watch these proportionate parts of the cycles for changes in trend. For example:

- * The Great Cycle of 90 years equals 1080 months;
 - $1/2$ is 45 years or 540 months
 - $1/4$ is 22 $\frac{1}{2}$ years or 270 months
 - $1/8$ is 11 $\frac{1}{4}$ years or 135 months
 - $1/16$ is 5-5/8 years or 67 $\frac{1}{2}$ months

The 30-year Cycle or any other cycle is divided up in the same way.

Multiples of 7 Years -- The multiples of 7 years or 84 months are all important to watch for change in trend. These are 7 years, 14, 21, 28, 35, 42, and 49, which is most important because it is the square of 7. Next 56 and 63 are very important, 63 because it is 7×9 . 81, the square of 9, is very important.

Prices can also be used in sevens. Example: 98, 2×49 . 126, 2×63 . 162, 2×81 , etc.

Yearly Time Periods - Triangles and Squares -- When $1/3$ of a year from any important low comes out at the same time that $1/4$ or $1/2$ of a year from another important top or bottom comes out, it is of great importance for a change in trend. $1/2$ of a yearly time period is always the most important, just the same as $1/2$ of the highest selling price and $1/2$ of the range of the price is important for resistance levels. By practice, study and comparison and by placing the Calculator over the Weekly High and Low Chart, you will see how all these Price and Time Periods work out.

TIME, PRICE, VOLUME, VELOCITY, PITCH OR TREND

When a Time Cycle is completed, Volume increases and the market begins to move up faster or move down faster.

The pitch or trend is determined mostly by the 45° angle, which is the most important, but other angles can be used to determine the trend. The pitch or trend is the 4th dimension and shows whether the market is slow or fast by the angles, whether very acute or above the 45° angle or flat and slow, below the 45° angle, which causes a slow creeping market that may later regain important angles and increase the pitch of the angle and start moving up faster.

All of this is shown on the Master Calculator or Square of 52.

3 WAYS TO SQUARE OR BALANCE TIME AND PRICE

- (1) Balance the lowest price with time measured in weeks and balance the highest price with time.
- (2) Balance the range, which is the total between the extreme high and extreme low.
- (3) Get the 4th dimension by balancing Price and Time in weekly time periods as shown on the Master Square of 52.

PRICE SCALE

The price scale runs up and down to 104, 208, 312, 416, and 520, which balances out with the Time Periods. For GRAINS, this scale is for cents per bushel. For STOCKS, it is \$1.00 per share. The Price Periods are divided up into 1/8 and 1/3, the same as the Time Periods.

Scale for COTTON, COFFEE, COCOA and EGGS -- Each 1/8 cent equals 10 points. Therefore 52 would indicate 520 in price and 104 would indicate 1040 for Cotton or any other commodity trading in 100 points to 1¢.

EGGS--Weekly high and low chart -- Eggs trade at a minimum of 5 points. We use 10 points to 1/8 inch on the daily high and low chart. Experience has proven that a scale of 25 points to 1/8 inch, which represents one week in Time, works best. Therefore, 52 spaces on the Calculator would indicate 1300 or 13¢ for Eggs; 104 would represent 2600; 156 would represent 3900; 208 would represent 5200 or 52¢ per dozen for Eggs, etc. One year Time Period would give a range of 13¢, 1/2 of this would be 6½¢, 1/4 would be 3¼¢ and 1/8 would be 162. Therefore, if you want to get resistance levels for prices above 26¢, you would add 6½ which would give 32½, etc.

All of these are shown on the Table enclosed for prices which run up to the equivalent of 40 years calculated. All you have to do is run across the period marked "1/2" and get the exact date during each year when the Time is at 1/2 or 182 days from an important high or low, and in the same way from the Table you get the price resistance levels based on the Square of 52.

HOW TO USE THE MASTER CALCULATOR

One column on the Calculator can be used for one month or one year but the Calculator is designed to be used on the Weekly high and low Chart for determining the important changes in trend.

HOW TO USE CALCULATOR ON WEEKLY CHART

Place the bottom or "0" of the Calculator on "0" below any price or place it on the low price, then you will see where the angles cross and resistance levels are indicated.

Place the Calculator marked "Top" at the high price on the exact date the important price is made, then you can see the important angles for resistance from the top down.

THE INNER SQUARE OR 1/2 POINT

Place the Calculator on 1/2 of the highest selling price or 1/2 of the range. Place the Calculator where it is marked 1/2 or 26 weeks over the chart on the same line that low or high is made. Placing the Calculator on 26 over the 1/2 point will show the resistance and whether the price is in strong or weak position.

THE INNER SQUARE OF 45° ANGLES

The Inner Square of 45° Angles starts from 26, which is 1/2 of 52, and moves up or down. It crosses at 26, which is 1/2 of 52. The 45° angle moving up from "0" crosses at 52 and 104 and the 45° angle moving down from a top or high level crosses at 52 and terminates at "0". All of these important 45° angles from any important high or low cross on 1/4, 1/2, 3/4, etc., as you can see, and balance out Time and Price.

MOST IMPORTANT TIME PERIODS

The most important Time Periods are the anniversary date of 1, 2, 3 years or more from the dates of the important highs and important lows. Second in importance is 1/2 of the Yearly Time Period; third in importance is 3/4 or 39 weeks in each year; fourth, the 1/3 point or 17 weeks and the 2/3 point or 35 weeks are also very important Time Periods for change in trend.

When you are working out Weekly Time Periods, you must also consider the importance of the 3-year cycle, the 5-year cycle, 7-year cycle, 10-year cycle, 15 years which is 1/2 of the 30-year cycle, and 20 years which is 1/3 of the 60-year cycle, and 30 years which is a complete cycle of 360 months. The longer the Time Period from an important top or bottom, the greater the variation because each year of the time period gains at least one day and in leap years gains an additional day before the end of the complete cycle.

The Table for Time and Price Periods shows the exact time in 52 weeks to a year or 364 calendar days. By figuring the leap years and the time gained at the rate of one day per year, you know how much time to subtract and can make the adjustment for the full period.

SEASONAL TIME PERIODS

In figuring Seasonal Time Periods, we do not start to calculate Time from January 1st but calculate the Time Periods from the date when the Spring Season starts on March 21. These periods are marked on the Table in 1/6, 1/3, etc. They are as follows:

May 5 ends	1/8	or	6 1/2	weeks from March 21
Jun. 21 "	1/4	or	13	"
Jul. 23 "	1/3	or	17	"
Aug. 5 "	3/8	or	19 1/2	"
Sep. 22 "	1/2	or	26	"
Nov. 5 "	5/8	or	32 1/2	"
Nov. 22 "	2/3	or	35	"
Dec. 21 "	3/4	or	39	"
Feb. 4 "	7/8	or	45 1/2	"
Mar. 20 "	1. year	or	52	"

MIDSEASON TIME PERIODS

These are May 5, August 5, November 5, February 4. Important changes in trend occur around these midseason dates but all of the above Time Periods should be watched for important changes in trend.

The Table for Time Periods and important high and low prices is shown with figures at the top running from 1 to 38. The important Time Periods are shown with exact dates and if you want to look up something in the 7th year, you move over to the column marked "7" and move down to the 1/2 point. You will find that 7 1/2 years is 390 weeks, etc. and any price on this same line would be an important resistance level.

By studying, practicing and experimenting with the Master Calculator, you will learn how valuable these Time and Price Periods in multiples of 7 are.

W. D. Gann
W. D. GANN

January 10, 1935

W. D. GANN MATHEMATICAL FORMULA FOR MARKET PREDICTIONS

THE MASTER MATHEMATICAL PRICE TIME AND TREND CALCULATOR

This chart is made on transparent plastic so that you can place it over a daily, weekly, or monthly high or low chart and see at a glance the position on the time and price based on the geometrical angles. It is designed to give QUICK ACCURATE EASY CALCULATIONS: SAVE TIME AND PREVENT ERRORS.

This square of 12 is always important in working out time periods because there are 12 months in a year. The square of 144 is the GREAT SQUARE and works better than any other square both for TIME AND PRICE because it contains all of the squares from 1 to 144. This chart is divided up into sections of 9 both for time and price because 9 is the highest digit. Nine spaces on the daily chart equals 9 days, 9 weeks or 9 months in time periods and 9 equals 9¢ on grains, 9 points on stocks or 90 points on cotton on the daily high and low chart.

One column in the square of 144 contains 144. This would equal \$144 on grain, 144 points on stocks or using a scale of 10 points to 1/8 inch it will equal 1440 points on cotton.

MASTER 144 SQUARE CONTAINS 324 square inches and each square inch contains 64 units which gives 20,736. This is 20,736 weeks or months and the proportionate parts of this are used for the measurement of time and price because this is the great cycle.

THE GREAT CYCLE OF THE SQUARE OF 144:

The time period of this square is 20,736 days, weeks or months. One-half is 10,368 days. One-fourth is 5,184 days. One-eighth is 2,592 days. One-sixteenth is 1,296 days. One-thirty-second is 648 days and the one-sixty-fourth is 324 calendar or market days. 1/128 is 162 days and 1/256 is 81 days or the square of 9.

WEEKLY TIME PERIODS:

THE GREAT CYCLE in weeks is 2962 and 2 days and 1/2 of this period is 1481 weeks and 1 day. One-fourth is 740 weeks. One-eighth is 370 weeks and 2 days. One-sixteenth is 185 weeks, 1 day. One-thirty-second is 92 weeks, 4 days and 1/64 is 41 weeks, 2 days.

MONTHLY TIME PERIODS:

THE GREAT CYCLE in months is 681 months and 23 days. One-half of this is 28 years, 5 months and 23 days. One-fourth is 14 years, 5 months and 8 days. One-eighth is 7 years, 2 1/2 months. One-sixteenth is 43 months and days. The weekly and monthly time periods from any major high and low can be checked to determine the future trend.

THE MASTER NUMBERS:

The Master Numbers are 3, 5, 7, 9 and 12. The No. 9 and its multiple is the most important because 9 digits added together equal 45. The next number of greatest importance is 7, the number mentioned more times in the Bible than any other number. There are 7 days in the week and 7 calendar days as well as 5 market days, and their multiples should be carried on your Daily, Weekly and Monthly Charts. The square of 7 is 49, which is a very important time period. The 2nd square of 7 is 98 and the third square of 7 is 147 and the fourth square of 7 is

196 and 196 is also the square of 14. The next number in importance is No. 5 which is the balancing number between 1 and 9. The square of 5 is 25 and the second square of 5 is 50 which is just 1 over the square of 7 making 49 to 50 very important for a change in trend. Three squares of 5 is 75 and 4 squares of 5 is 100 and 100 is the square of 10, which is also important for changes.

The No. 3 is mentioned in the Bible next to the No. 7 and 3 is important because 3×3 equals 9, the square of 3, and it is the first odd number that forms a square greater than itself. Three must be used in every way possible. 3×7 equals 21, 3×5 equals 15, 3×9 equals 27 and 3×12 equals 36, very important because it is the square of 6. The No. 12 is also spoken of in the Bible many times and is of great importance. Jesus selected 12 disciples. There are 12 months in the year, and 12 signs of the Zodiac. The important 12's in the square of 144 are 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132 and 144. These are all important for both time and price in days, weeks and months.

Referring to the No. 9, 7×9 is 63 and is of great importance because the square of 8 is 64, therefore, around 63 to 64 is very important to watch the change in trend. 7×12 is 84 and this is of very great importance and the number next to this is 90, which is 10×9 and 90 is $1/4$ of the circle, which is very important for time and price changes. Next in importance is 9×12 which is 108 or $3/4$ of 144.

The importance of the circle of 360 degrees must not be overlooked in connection with the square of 144 because the proportionate parts of the circle agree with the parts of 144. $2 \frac{1}{2}$ times 144 equals 360 and $1 \frac{1}{4}$ times 144 equals 180, $1/2$ of the circle, and 90 is $5/8$ of 144. 9 is $1/16$ of 144, 18 is $1/8$, 27 is $3/8$, 36 is $1/4$, 45 is $5/16$ and always very important for time and price changes and for resistance levels. 48 is $1/3$ of 144 and 54 is $3/8$ and 63 is $7/16$, 72 is $1/2$ of 144, 81 the square of 9, is $9/16$ of 144 and 90 is $5/8$, 99 is $11/16$, 108 is $3/4$, 117 is $13/16$, and 126 is $7/8$ and 135 is $15/16$ of 144. These are the most important in the square of 12 and should be watched closely when time periods in days, weeks or months reach these points on the Master Calculator. Remember that you should always watch the Daily Chart for the first indication of the change in trend and at the same time look at the position on the Weekly Chart or 7 day time periods which is next in importance. The Monthly Chart is of the greatest importance for changes in the main trend.

THE IMPORTANCE OF 3 AND 5:

The movement in PRICE and TIME whether on a Daily, Weekly or Monthly chart has three important points, the PRICE, the TIME AND VOLUME of sales, the PITCH or TREND which is the geometrical angle which shows whether time is influencing and driving prices up or down on a slow angle or on acute fast moving angle. There are also four factors that influence prices, PRICE, TIME AND VOLUME AND VELOCITY. Time is the most important factor because when time is up volume increases and the velocity or speed of the market increases and the PITCH or TREND on the angles moves up faster or down faster.

There are three other important points to consider on a Daily, Weekly or Monthly High and Low Chart. These are the LOW PRICE, the HIGHEST PRICE and the RANGE or $1/2$ between the high and the low.

FIVE FACTORS FOR TIME AND PRICE:

These are high, low, halfway point, opening and closing prices. The trend is indicated

by the closing price, especially when the market is very active. If the price closes above the halfway point or near the high, the trend is up. If it closes below the halfway point or near the low, the selling is greater than the buying and the trend is down, at least temporarily. In connection with the Master Time and Trend Calculator apply all of the rules with the Master Time Factor and geometrical angles.

STRONGEST POINTS FOR TIME AND PRICE RESISTANCE:

In using the Master Square of 144, the strongest points are $1/4$, $1/3$, $2/3$, $3/8$, $1/2$, $5/8$, $3/4$, $7/8$ and the complete square of 144.

The points where the most angles cross are the strongest for resistance in PRICE and TIME.

TRIANGLE POINTS:

The triangle points or where the green angles cross are the most important. These are 72, 144, 36, 48, 96, 108 and, of course, 72, and at the end of the square of 144 at the top and the bottom.

SQUARES IN THE SQUARE OF 144:

These squares where the angles cross are of great importance for time and price resistance. These are 36, 48, 54, 63, 72, 90, 108, and at the top end of the square of 144 at the top and the bottom. When the price is at a point equal to 36 and the time periods in days, weeks or months is at 36, TIME and PRICE is SQUARE and it is important to watch for change in trend. With the square of 144 you can get any square from 1 to 144. Suppose you want to get the square of 72; you move across to 72 for time and if the price is at 72 moving up on the chart, PRICE and TIME has balanced, or squared out, and are at the 45 degree angle and at the half-way point on the Master Price, Time and Trend Calculator.

WHERE TO WATCH FOR CHANGES IN TREND:

Most changes in trend occur when the TIME PERIODS are at one-half of the square of 144 and at the end of a square or at the $1/3$, $2/3$, $1/4$, and $3/4$ points in the square of 144; you must always watch the square in time of the HIGHEST PRICE and the MINOR HIGHS and LOWS, also the square in TIME of the LOWEST PRICE and the SECOND OR THIRD higher bottom, and also the time required to square the Range and where this square works out in the Master Square of 144.

EXAMPLE: The lowest price that wheat ever sold was 28¢ per bushel in March, 1852. Therefore, every 28 months would square the lowest price. The highest price that wheat ever sold for was on May 11, 1917, when the May option sold at \$3.25. Therefore, it would require 325 months to square the highest price. The lowest price that the May option ever sold was 44¢. Therefore, it would require 44 months in time to square the low price. The range between 44¢ and \$3.25 is \$2.81 which would require 281 months, 281 weeks or 281 days to square the range. You would look on the Master Chart and see that 2 squares of 144 equal 288, therefore, you would watch for a change in trend between 281 and 288 or near the end of the second square of 144. 7×44 , the extreme low, equals 308, therefore, $6 \frac{1}{2}$ times 44 would equal 286, which is within two points of the end of the square of 144 or the end of the second square making 286 an important time period to watch for a change in trend. In squaring the high of \$3.25, the highest price for May wheat would be 2 squares of 144 and 37 over, therefore, when the time reached 36 days, weeks or months in the Master Square of 144 you

can see that resistance would be met because moving up the time period of 136 you see that the 45 degree angle moving down from 72, which is the INNER SQUARE and the line drawn across from 36 on the price scale crosses at 36. In this way you can see that the Master Chart would indicate a resistance in time and price corresponding with the square of the highest price, the lowest price, and the Range. All other time periods from a high price, low price, or the range of any commodity or of the stock averages or individual stocks should be worked out in this same way.

You will succeed in using the Master Mathematical Price, Time and Trend Calculator by going over the charts and laying the calculator over them and working out past history. In this way you will learn just how it works and prove to yourself the great value of the Master Calculator.

HOURLY TIME PERIODS:

When Markets are very active and making a wide range in price, it is important to keep an hourly High and Low Chart, just the same as you keep the Daily High and Low Chart and the Hourly Chart will give the first change in trend.

There are 24 hours in a day, therefore, 6 days are required to pass through 144 and a total of 864 days to pass through the square of 144.

At the present time, all exchanges are open 5 days a week with the exception of holidays and most of them are open 5 hours each day. Therefore, it will require 28 days and 4 hours to pass through 144 at the rate of 5 hours per day and 5 days per week. Trading time for grains is 225 minutes or 3.75 hours per day or 38 days and 1 1/2 hours to pass through 144 at the rate of 3.75 hours per day and 5 days per week.

GREAT YEARLY TIME CYCLE:

To pass through the square of 144, which equals 20,736, it requires 56 years, 9 months and 23 days, which is a very important time cycle. Next in importance is 1/2 of this time period which is 28 years, 5 months and 8 days and 1/4 which is 14 years, 2 months and 19 days. The 14 year cycle is always very important because it is 2 seven year periods. 14 years equals 168 months and 169 months is the square of 13 making it very important for a change in trend and this is also an important time resistance point.

1/8 of the Great Cycle is 7 years, 1 month and 10 days and is quite important. 1/16 is 42 months and 20 days, 1/32 is 21 months and 10 days. This is an important time period because it is close to 22 1/2 months which is 1/16 of the circle of 360 degrees.

NINE SPACES AND NINE TIME PERIODS:

The chart being divided up into 9 sections gives all the squares of 9 for price resistance of time resistance which, as referred to above is 9¢ on the Daily Chart for grains and the same for Weekly and Monthly Charts; that is, 1¢ to each 1/8 inch. For different commodities, different scales are used. (See Special Instructions for the different commodities.)

The Chart being divided up into 9 sections gives 1/16th squares of 9 for price resistance or time resistance which, as referred to above, is 90 points on the Daily Chart, 135 on the Weekly Chart, and 270 on the Monthly and 2880 on the 20 point scale and 9 spaces equal

180 points.

The angles in red are all drawn on the squares of 9. The Inner Square of 45° is drawn from 72 because 72 is 1/2 of 144. These angles come out at 72, 144 and 72.

The straight lines are in green and are 1/3 of the square of 144 in price or time.

GREEN ANGLES:

The green angles are the angles of 2 x 1 which move up two spaces or 2 points in one period of the time. The other angle below the 45° angle is the angle of 1 x 2 which requires 2 time periods to move up 1 space or 1¢ per bushel on grains for each time period of 1 day, 1 week or 1 month. These angles move down from the top at the rate of 2 points or 2 spaces per time period or at the rate of 1/2 space, 1/2 point, or 1/2¢ per time period. The distance that the green angles and the red angles are apart determines how far prices can advance or decline.

When the market enters the INNER SQUARE it is important for a change in trend and time angles and Position in the square tell at the time it entered whether the price is going up or down. Also, when price breaks below a 45° on the INNER SQUARE it shows weakness in proportion to the time from high or low price.

WHEN TO START A NEW SQUARE:

When the time period of the Daily, Weekly or Monthly Charts has moved 144 you begin a new square. But to get the position you simply move the Master Square over to 144 and place it over the chart to get the position in the next square.

STRONGEST AND WEAKEST POINTS:

Where the greatest number of angles cross or bisect each other in the square of 144 are the points of greatest resistance, such as where the 25 and 2 x 1 angles cross each other.

Study and practice on past action of the market and you will soon learn how to determine the trend very quickly by using the Master Chart.

HOW TO USE THE MASTER SQUARE OF 144:

Follow all rules on angles as given in the Master Forecasting course.

Place this chart at the bottom or 0 on the Daily, Weekly or Monthly Chart or place the bottom of the chart on the low price or SQUARE of the HIGH PRICE, the LOW PRICE, and the square of the RANGE and show where time and price balances.

When you figure the halfway point of the extreme high or the halfway point of the range, place either the top or the bottom of the chart on the halfway point and it will give the correct POSITION and TREND. However, if you will place the CENTER, or 72, on the halfway point in PRICE you will then get the correct position in time and can see how the price is working out with time in the MASTER SQUARE OF 144 and the INNER SQUARE which starts from 72, the gravity center or 1/2 point of the square.

CALENDAR DAYS AND MARKET DAYS:

For any kind of a chart, we move over one space for each unit of time. Therefore, if

would require 144 market days or 144 calendar days to complete the square of 144. When prices pass out of one square into another, a change in trend usually takes place and the periods and geometrical angles on this MASTER CHART tell you which way the trend is going to change.

LEAP YEARS:

In calculating time periods to get exact number of days and weeks the LEAP YEAR must be figures and one extra day added. From 1864 the Leap Years were as follows: 1868; 1872; 1876; 1880; 1884; 1888; 1892; 1896; 1904; 1908; 1912; 1916; 1920; 1924; 1928; 1932; 1936; 1940; 1944; 1948; 1952.

POSITION WITH MASTER SQUARE:

To get the position with the MASTER SQUARE OF 144, lay it over the Daily, Weekly or Monthly and start from extreme low, extreme high, 0 or 1/2 point of the range or 1/2 of the high selling price. You can also place the top of the chart at the high selling price. In this way you can get the correct time and positions or angles at a glance.

Place MASTER CALCULATOR on January to get twelve year periods, each month same way.

You should have all time periods from important highs and lows calculated in days, weeks and month in order that you can look them up quickly on the MASTER CHART.

With all time periods brought up to date in this way, you can get the position on the MASTER CHART and the indicated trend without looking at a daily, weekly or monthly chart. You should have the time periods in months from every important top and bottom.

With the MASTER SQUARE OF 144, your work will be cut down but you must learn to practice and bring up all time periods and study the MASTER SQUARE and learn how to use it to get tops and bottoms accurately. Work and practice will bring PRECISION and PROFITS. I have done my part. It is now up to you to work hard and if you do, your success is assured.

HOW TO PLACE THE MASTER CHART SQUARE OF 144 ON CHARTS

To get correct positions on any of your charts, daily, weekly or monthly, you must place the MASTER CHART OVER THEM correctly. The top of the MASTER CHART has Price and Time Chart by W. D. Gann on it. Always place this on the bottom of your chart unless it is where you move up to a higher price when you place it at 72 or bottom marked "Bottom" 0.

Place the MASTER CHART at 0 on any of your other charts or at a low point on the same side of any high or low point. Place the MASTER CHART at 72 which is 1/2 of 144 or on any one-half point of the price range or 1/2 of the high selling price.

Always place the MASTER CALCULATOR on all previous highs and lows to get changes in trends and price resistance. Never overlook the extreme high and low price. Also, most important 1/2 of high to low and 1/2 of the highest price any commodity or stock ever sold.

TIME PERIODS AND PRICE RESISTANCE

The circle of 360° is most important for time cycles and price resistance. First, we divide the circle by 2 and get 180 which is one-half and is most important for time or price in days, weeks or months. Next we divide the circle by 3 and get the triangle points of 120, 240, and 360. Third, we divide the circle by 4 which gives 90, 180, 270, and 360, which are the squares and most important.

Divide the circle by 8 which gives 8 periods of 45 degree angles which are 45, 90, 135, 180, 225, 270, 315, and 360. Next in importance is to divide the circle by 16 which gives the angles of 22 1/2 degrees. We divide the circle by 32 and get the angles and time periods of 11 1/4 degrees and their multiples. Divide the circle by 64 which gives 5 5/8 degrees and their multiples.

The table below shows each division of 5 5/8 and we run down to 16 in the column which equals 90 or 1/4 of the circle and 16 is 1/4 of 64. This arrangement reading across gives each of the multiples and the figures between the heavy lines are among the ones which are the most important.

Table of 64th of the Circle

1. 5 5/8	17. 95 5/8	33. 185 5/8	49. 275 5/8
2. 11 1/4	18. 101 1/4	34. 191 1/4	50. 281 1/4
3. 16 5/8	19. 106 7/8	35. 196 7/8	51. 286 7/8
4. 22 1/2	20. 112 1/2	36. 202 1/2	52. 292 1/2
5. 28 1/8	21. 118 1/8	37. 208 1/8	53. 298 1/8
6. 33 3/4	22. 123 3/4	38. 213 3/4	54. 303 3/4
7. 39 3/8	23. 129 3/8	39. 219 3/8	55. 309 3/8
8. 45	24. 135	40. 225	56. 315
9. 50 5/8	25. 140 5/8	41. 230 5/8	57. 320 5/8
10. 56 1/4	26. 146 1/4	42. 236 1/4	58. 326 1/4
11. 61 7/8	27. 151 7/8	43. 241 7/8	59. 331 7/8
12. 67 1/2	28. 157 1/2	44. 247 1/2	60. 337 1/2
13. 73 1/8	29. 163 1/8	45. 253 1/8	61. 343 1/8
14. 78 3/4	30. 168 3/4	46. 258 3/4	62. 348 3/4
15. 84 5/8	31. 174 3/8	47. 264 3/8	63. 354 3/8
16. 90	32. 180	48. 270	64. 360

The squares from 1 to 10 are important to watch for time and price resistance as they are important degrees in the circle. These are: 1 - 4 - 9 - 16 - 25 - 36 - 49 - 64 - 81 - 100 - 121 - 144 - 169 - 196 - 225 - 256 - 289 - 324 and 361, which is the square of 19.

Divide the circle by 6 and we get two resistance and time periods which are not contained in the other table. These are 60 and 300.

It is also very important to divide the circle by 12 because there are 12 months in the year and this works out accurately for time periods. The following are the degrees not shown in the above table: 30 - 150 - 210 - 330.

Divide the circle by 24 which gives 15 degree periods in longitude and approximately 15 days in time. Because there are 24 hours in a day and the earth makes 1 revolution on its axis in 24 hours these periods are quite important. The following are not contained in the other table: 15 - 75 - 105 - 165 - 195 - 285 - 345.

Under the table of the 64th Circle reading across from left to right the 2nd column is always 90 degrees from the first. Example: The first is $5 \frac{5}{8}$ and 17 in the next column is $95 \frac{5}{8}$ or a gain of 90. Opposite 33 is $185 \frac{5}{8}$ which is 90 from $95 \frac{5}{8}$, opposite 49 is $275 \frac{5}{8}$ or 90 from $185 \frac{5}{8}$.

The column 1 under 8 between the heavy lines is 45 and the next one is 135, 225, and 315 each 90 degrees apart.

At the bottom of column 1 opposite 16 is 90 and reading across is 180, 270, and 360. These figures are all 45 degrees from the figures in column 8 which are the most important for time and price resistance.

To realize the value and importance of these degrees in the circle you take the price of highs and lows and the time periods, especially in weeks and months and check them over and you will see how well they work out to these important degrees.

Remember you must always figure how many points or cents the prices are up from the extreme low or minor lows and how many cents it is down from the extreme highs or minor highs. Also, how many cents it is above or below the main halfway point or the minor halfway points or gravity centers. You will find that the important halfway points form very close to these natural degrees in the circle.

Example: May Soybeans extreme high $436 \frac{3}{4}$, extreme low 67, gives the halfway point at $251 \frac{7}{8}$. Look in the table and you find the 45th 64th at $263 \frac{1}{8}$. This halfway point is very close. Next $1/2$ of $436 \frac{3}{4}$ is $218 \frac{3}{8}$ and the 39th 64th is $219 \frac{3}{8}$, very close to this natural resistance degree. The extreme low of Cash Beans was 44 and the halfway point between 44 and $436 \frac{3}{4}$ is $240 \frac{3}{8}$. 240 is $2/3$ of a circle or a triangle point and $241 \frac{7}{8}$ is the 43rd 64th of the circle. 44 the extreme low, was only 1 from 45, the important resistance level. 67, the lowest price May futures ever sold, is within $1/2$ point of $67 \frac{1}{2}$ which is $12/64$ or $3/4$ between 0 and 90 and $67 \frac{1}{2}$ is halfway between 45 and 90 which was the cause of May beans making lows for 5 years between 67-69 which indicated the big advance that followed as there were triple bottoms between 67-69 in three different years.

Next consider the time periods December 28, 1932, was low on May Beans and December 28, 1947, was 15 years or 180 months which is $1/2$ of a circle also $1/2$ of the 360 degrees making this a very important time cycle. May beans reached extreme high on January 15, 1948, just 18 days beyond the even 15 year cycle. Check all other time periods and time cycles in the same way and you will find how well they work out in the circle of 360 degrees.



W. D. Gann

September 29, 1953

APPENDIX F

MASTER CHARTS

W. D. GANN

78 WALL STREET
NEW YORK

SCIENTIFIC ADVICE
AND ACCURATE FORECASTS
OF STOCKS AND COMMODITIES
BY "TRENDS ON THE STOCK TAPE"
"THE TUNNEL THROUGH THE AIR"

MEMBER
AMERICAN ECONOMIC ASSN.
ROYAL ECONOMIC SOCIETY
CABLE ADDRESS
"GANNWIDE NEW YORK"

MASTER CHARTS

::::::::::::

The Master Charts are permanent and represent natural angles and permanent resistance points for either price, time or volume. These points do not change and you should study them carefully on each different Master Chart and learn how to apply them.

MASTER "12" CHART

The MASTER CHART is the square of "12" or 12 x 12, making the first square end at 144. The second square of "12" ends at 288, the third square of "12" at 432; and the fourth square at 576, which will cover most anything that you want, but you can make up as many more squares as you need.

This chart may be used and applied to anything--TIME, SPACE, PRICE OR VOLUME, the number of points up or down; days, weeks, months and years.

On Square No. 1, which runs from 1 to 144, I have drawn the finer angles to show the grand-center or strongest resistance point in each minor square. The minor centers, which are the strongest for minor tops and bottoms are 14, 17, 20, 23, 30, 33, 36, 39, 46, 49, 52, 55, 58, 65, 68, 71, 74, 77, 84, 87, 90, 93, 96, 103, 106, 109, 112, 115, 118, 125, 128, 131.

The major center is where the strongest resistance is met. These numbers are 66, 67, 78 and 79. Stocks going up or coming down to these prices will meet with stubborn resistance. The next strong angle is the 45°, and the numbers of greatest resistance on it are 14, 27, 40, 53, 66, 79, 92, 105, 118, 131 and 144. The other diagonal 45° angle from 12 is equally strong. The numbers are 12, 25, 34, 45, 56, 67, 78, 89, 100, 111, 122 and 133.

The numbers which are cut by the 45° angles thru the center of each of the 1/4 squares are next in strength. These numbers are 7, 20, 33, 46, 59, 72, 85, 98, 111, 124, 137, 150, 163, 176, 189, 202, 215, 228, 241, 254, 267, 280, 293, 306, 319, 332, 345, 358, 371, 384, 397, 410, 423, 436, 449, 462, 475, 488, 501, 514, 527, 540, 553, 566, 579, 592, 605, 618, 631, 644, 657, 670, 683, 696, 709, 722, 735, 748, 761, 774, 787, 800, 813, 826, 839, 852, 865, 878, 891, 904, 917, 930, 943, 956, 969, 982, 995, 1008, 1021, 1034, 1047, 1060, 1073, 1086, 1099, 1112, 1125, 1138, 1151, 1164, 1177, 1190, 1203, 1216, 1229, 1242, 1255, 1268, 1281, 1294, 1307, 1320, 1333, 1346, 1359, 1372, 1385, 1398, 1411, 1424, 1437, 1450, 1463, 1476, 1489, 1502, 1515, 1528, 1541, 1554, 1567, 1580, 1593, 1606, 1619, 1632, 1645, 1658, 1671, 1684, 1697, 1710, 1723, 1736, 1749, 1762, 1775, 1788, 1801, 1814, 1827, 1840, 1853, 1866, 1879, 1892, 1905, 1918, 1931, 1944, 1957, 1970, 1983, 1996, 2009, 2022, 2035, 2048, 2061, 2074, 2087, 2100, 2113, 2126, 2139, 2152, 2165, 2178, 2191, 2204, 2217, 2230, 2243, 2256, 2269, 2282, 2295, 2308, 2321, 2334, 2347, 2360, 2373, 2386, 2399, 2412, 2425, 2438, 2451, 2464, 2477, 2490, 2503, 2516, 2529, 2542, 2555, 2568, 2581, 2594, 2607, 2620, 2633, 2646, 2659, 2672, 2685, 2698, 2711, 2724, 2737, 2750, 2763, 2776, 2789, 2802, 2815, 2828, 2841, 2854, 2867, 2880, 2893, 2906, 2919, 2932, 2945, 2958, 2971, 2984, 2997, 3010, 3023, 3036, 3049, 3062, 3075, 3088, 3101, 3114, 3127, 3140, 3153, 3166, 3179, 3192, 3205, 3218, 3231, 3244, 3257, 3270, 3283, 3296, 3309, 3322, 3335, 3348, 3361, 3374, 3387, 3400, 3413, 3426, 3439, 3452, 3465, 3478, 3491, 3504, 3517, 3530, 3543, 3556, 3569, 3582, 3595, 3608, 3621, 3634, 3647, 3660, 3673, 3686, 3699, 3712, 3725, 3738, 3751, 3764, 3777, 3790, 3803, 3816, 3829, 3842, 3855, 3868, 3881, 3894, 3907, 3920, 3933, 3946, 3959, 3972, 3985, 3998, 4011, 4024, 4037, 4050, 4063, 4076, 4089, 4102, 4115, 4128, 4141, 4154, 4167, 4180, 4193, 4206, 4219, 4232, 4245, 4258, 4271, 4284, 4297, 4310, 4323, 4336, 4349, 4362, 4375, 4388, 4401, 4414, 4427, 4440, 4453, 4466, 4479, 4492, 4505, 4518, 4531, 4544, 4557, 4570, 4583, 4596, 4609, 4622, 4635, 4648, 4661, 4674, 4687, 4700, 4713, 4726, 4739, 4752, 4765, 4778, 4791, 4804, 4817, 4830, 4843, 4856, 4869, 4882, 4895, 4908, 4921, 4934, 4947, 4960, 4973, 4986, 4999, 5012, 5025, 5038, 5051, 5064, 5077, 5090, 5103, 5116, 5129, 5142, 5155, 5168, 5181, 5194, 5207, 5220, 5233, 5246, 5259, 5272, 5285, 5298, 5311, 5324, 5337, 5350, 5363, 5376, 5389, 5402, 5415, 5428, 5441, 5454, 5467, 5480, 5493, 5506, 5519, 5532, 5545, 5558, 5571, 5584, 5597, 5610, 5623, 5636, 5649, 5662, 5675, 5688, 5701, 5714, 5727, 5740, 5753, 5766, 5779, 5792, 5805, 5818, 5831, 5844, 5857, 5870, 5883, 5896, 5909, 5922, 5935, 5948, 5961, 5974, 5987, 6000, 6013, 6026, 6039, 6052, 6065, 6078, 6091, 6104, 6117, 6130, 6143, 6156, 6169, 6182, 6195, 6208, 6221, 6234, 6247, 6260, 6273, 6286, 6299, 6312, 6325, 6338, 6351, 6364, 6377, 6390, 6403, 6416, 6429, 6442, 6455, 6468, 6481, 6494, 6507, 6520, 6533, 6546, 6559, 6572, 6585, 6598, 6611, 6624, 6637, 6650, 6663, 6676, 6689, 6702, 6715, 6728, 6741, 6754, 6767, 6780, 6793, 6806, 6819, 6832, 6845, 6858, 6871, 6884, 6897, 6910, 6923, 6936, 6949, 6962, 6975, 6988, 7001, 7014, 7027, 7040, 7053, 7066, 7079, 7092, 7105, 7118, 7131, 7144, 7157, 7170, 7183, 7196, 7209, 7222, 7235, 7248, 7261, 7274, 7287, 7300, 7313, 7326, 7339, 7352, 7365, 7378, 7391, 7404, 7417, 7430, 7443, 7456, 7469, 7482, 7495, 7508, 7521, 7534, 7547, 7560, 7573, 7586, 7599, 7612, 7625, 7638, 7651, 7664, 7677, 7690, 7703, 7716, 7729, 7742, 7755, 7768, 7781, 7794, 7807, 7820, 7833, 7846, 7859, 7872, 7885, 7898, 7911, 7924, 7937, 7950, 7963, 7976, 7989, 8002, 8015, 8028, 8041, 8054, 8067, 8080, 8093, 8106, 8119, 8132, 8145, 8158, 8171, 8184, 8197, 8210, 8223, 8236, 8249, 8262, 8275, 8288, 8301, 8314, 8327, 8340, 8353, 8366, 8379, 8392, 8405, 8418, 8431, 8444, 8457, 8470, 8483, 8496, 8509, 8522, 8535, 8548, 8561, 8574, 8587, 8600, 8613, 8626, 8639, 8652, 8665, 8678, 8691, 8704, 8717, 8730, 8743, 8756, 8769, 8782, 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12682, 12695, 12708, 12721, 12734, 12747, 12760, 12773, 12786, 12799, 12812, 12825, 12838, 12851, 12864, 12877, 12890, 12903, 12916, 12929, 12942, 12955, 12968, 12981, 12994, 13007, 13020, 13033, 13046, 13059, 13072, 13085, 13098, 13111, 13124, 13137, 13150, 13163, 13176, 13189, 13202, 13215, 13228, 13241, 13254, 13267, 13280, 13293, 13306, 13319, 13332, 13345, 13358, 13371, 13384, 13397, 13410, 13423, 13436, 13449, 13462, 13475, 13488, 13501, 13514, 13527, 13540, 13553, 13566, 13579, 13592, 13605, 13618, 13631, 13644, 13657, 13670, 13683, 13696, 13709, 13722, 13735, 13748, 13761, 13774, 13787, 13800, 13813, 13826, 13839, 13852, 13865, 13878, 13891, 13904, 13917, 13930, 13943, 13956, 13969, 13982, 13995, 14008, 14021, 14034, 14047, 14060, 14073, 14086, 14099, 14112, 14125, 14138, 14151, 14164, 14177, 14190, 14203, 14216, 14229, 14242, 14255, 14268, 14281, 14294, 14307, 14320, 14333, 14346, 14359, 14372, 14385, 14398, 14411, 14424, 14437, 14450, 14463, 14476, 14489, 14502, 14515, 14528, 14541, 14554, 14567, 14580, 14593, 14606, 14619, 14632, 14645, 14658, 14671, 14684, 14697, 14710, 14723, 14736, 14749, 14762, 14775, 14788, 14801, 14814, 14827, 14840, 14853, 14866, 14879, 14892, 14905, 14918, 14931, 14944, 14957, 14970, 14983, 14996, 15009, 15022, 15035, 15048, 15061, 15074, 15087, 15100, 15113, 15126, 15139, 15152, 15165, 15178, 15191, 15204, 15217, 15230, 15243, 15256, 15269, 15282, 15295, 15308, 15321, 15334, 15347, 15360, 15373, 15386, 15399, 15412, 15425, 15438, 15451, 15464, 15477, 15490, 15503, 15516, 15529, 15542, 15555, 15568, 15581, 15594, 15607, 15620, 15633, 15646, 15659, 15672, 15685, 15698, 15711, 15724, 15737, 15750, 15763, 15776, 15789, 15802, 15815, 15828, 15841, 15854, 15867, 15880, 15893, 15906, 15919, 15932, 15945, 15958, 15971, 15984, 15997, 16010, 16023, 16036, 16049, 16062, 16075, 16088, 16101, 16114, 16127, 16140, 16153, 16166, 16179, 16192, 16205, 16218, 16231, 16244, 16257, 16270, 16283, 16296, 16309, 16322, 16335, 16348, 16361, 16374, 16387, 16400, 16413, 16426, 16439, 16452, 16465, 16478, 16491, 16504, 16517, 16530, 16543, 16556, 16569, 16582, 16595, 16608, 16621, 16634, 16647, 16660, 16673, 16686, 16699, 16712, 16725, 16738, 16751, 16764, 16777, 16790, 16803, 16816, 16829, 16842, 16855, 16868, 16881, 16894, 16907, 16920, 16933, 16946, 16959, 16972, 16985, 16998, 17011, 17024, 17037, 17050, 17063, 17076, 17089, 17102, 17115, 17128, 17141, 17154, 17167, 17180, 17193, 17206, 17219, 17232, 17245, 17258, 17271, 17284, 17297, 17310, 17323, 17336, 17349, 17362, 17375, 17388, 17401, 17414, 17427, 17440, 17453, 17466, 17479, 17492, 17505, 17518, 17531, 17544, 17557, 17570, 17583, 17596, 17609, 17622, 17635, 17648, 17661, 17674, 17687, 17700, 17713, 17726, 17739, 17752, 17765, 17778, 17791, 17804, 17817, 17830, 17843, 17856, 17869, 17882, 17895, 17908, 17921, 17934, 17947, 17960, 17973, 17986, 17999, 18012, 18025, 18038, 18051, 18064, 18077, 18090, 18103, 18116, 18129, 18142, 18155, 18168, 18181, 18194, 18207, 18220, 18233, 18246, 18259, 18272, 18285, 18298, 18311, 18324, 18337, 18350, 18363, 18376, 18389, 18402, 18415, 18428, 18441, 18454, 18467, 18480, 18493, 18506, 18519, 18532, 18545, 18558, 18571, 18584, 18597, 18610, 18623, 18636, 18649, 18662, 18675, 18688, 18701, 18714, 18727, 18740, 18753, 18766, 18779, 18792, 18805, 18818, 18831, 18844, 18857, 18870, 18883, 18896, 18909, 18922, 18935, 18948, 18961, 18974, 18987, 19000, 19013, 19026, 19039, 19052, 19065, 19078, 19091, 19104, 19117, 19130, 19143, 19156, 19169, 19182, 19195, 19208, 19221, 19234, 19247, 19260, 19273, 19286, 19299, 19312, 19325, 19338, 19351, 19364, 19377, 19390, 19403, 19416, 19429, 19442, 19455, 19468, 19481, 19494, 19507, 19520, 19533, 19546, 19559, 19572, 19585, 19598, 19611, 19624, 19637, 19650, 19663, 19676, 19689, 19702, 19715, 19728, 19741, 19754, 19767, 19780, 19793, 19806, 19819, 19832, 19845, 19858, 19871, 19884, 19897, 19910, 19923, 19936, 19949, 19962, 19975, 19988, 20001, 20014, 20027, 20040, 20053, 20066, 20079, 20092, 20105, 20118, 20131, 20144, 20157, 20170, 20183, 20196, 20209, 20222, 20235, 20248, 20261, 20274, 20287, 20300, 20313, 20326, 20339, 20352, 20365, 20378, 20391, 20404, 20417, 20430, 20443, 20456, 20469, 20482, 20495, 20508, 20521, 20534, 20547, 20560, 20573, 20586, 20599, 20612, 20625, 20638, 20651, 20664, 20677, 20690, 20703, 20716, 20729, 20742, 20755, 20768, 20781, 20794, 20807, 20820, 20833, 20846, 20859, 20872, 20885, 20898, 20911, 20924, 20937, 20950, 20963, 20976, 20989, 21002, 21015, 21028, 21041, 21054, 21067, 21080, 21093, 21106, 21119, 21132, 21145, 21158, 21171, 21184, 21197, 21210, 21223, 21236, 21249, 21262, 21275, 21288, 21301, 21314, 21327, 21340, 21353, 21366, 21379, 21392, 21405, 21418, 21431, 21444, 21457, 21470, 21483, 21496, 21509, 21522, 21535, 21548, 21561, 21574, 21587, 21600, 21613, 21626, 21639, 21652, 21665, 21678, 21691, 21704, 21717, 21730, 21743, 21756, 21769, 21782, 21795, 21808, 21821, 21834, 21847, 21860, 21873, 21886, 21899, 21912, 21925, 21938, 21951, 21964, 21977, 21990, 22003, 22016, 22029, 22042, 22055, 22068, 22081, 22094, 22107, 22120, 22133, 22146, 22159, 22172, 22185, 22198, 22211, 22224, 22237, 22250, 22263, 22276, 22289, 22302, 22315, 22328, 22341, 22354, 22367, 22380, 22393, 22406, 22419, 22432, 22445, 22458, 22471, 22484, 22497, 22510, 22523, 22536,

The opposition angle, which runs thru the center of the square, from east to west, equally dividing it, is one of the very strong angles because it equals one-half. Any stock moving up or down and reaching these prices will meet with any resistance and make tops or bottoms. These numbers are 6, 7, 18, 19, 30, 31, 42, 43, 54, 55, 66, 67, 78, 79, 90, 91, 102, 103, 114, 115, 126, 127, 138, 139.

Remember, when anything has moved three sections over from the beginning, it reaches the square of its own place, which is the first strong resistance. When it has moved six sections over, it reaches the opposition, or what equals the half-way point of its own place and meets still stronger resistance. Moving over nine places or sections from its own place, it reaches the $3/4$ point, another square. The 8th and 9th sections are the strongest and hardest points to pass because this is the "death" zone. The next and still stronger is the 12th section or column which ends at 144. Anything getting into this section meets the strongest resistance but once it moves out of this square and gets as much as 3 points into Square #2, that is, making 147, will indicate much higher. But after reaching this, it should not drop back to 141 or 3 points into Square #1.

When a stock gets into the Second Square of "12", it has faster moves, and when the time or number of months from any bottom or top moves into the Second Square, it is an indication of faster moves, both up and down.

Apply the same rule to the 3rd, 4th, 5th and 6th Squares. In the 3rd and 4th squares of the Master "12", you will find that most of the big bull and bear campaigns culminate, when measured by months, which determines the division, according to time. All of the other rules given you to apply to space movements, angles and time, can be used with the Master "12" tables.

SQUARE OF NINE

You have already had the MASTER SQUARE OF TWELVE explained, which represents days, weeks, months and years, and the measurements of TIME in the square of twelve or the square of the circle; also used to measure price movements and resistance levels.

The SQUARE OF NINE is very important because nine digits are used in measuring everything. We cannot go beyond 9 without starting to repeat and using the 0. If we divide 360° by 9, we get 40, which measures 40° , 40 months, 40 days, 40 weeks or 40 months, and shows why bottoms and tops often come out on these angles measured by one-ninth of the circle. This is why the children of Israel were 40 years in the wilderness.

If we divide our 20-year period, or 240 months, by 9, we get $26\frac{2}{3}$ months, making an important angle of $26\frac{2}{3}^\circ$, months, days or weeks. Nine times 9 equals 81, which completes the First Square of Nine. Note the angles and how they run from the main center. The Second Square of Nine is completed at 162. Note how this is in opposition to the main center. The Third square of Nine is completed at 243, which would equal 243 months or 3 months over our 20-year

period and accounts for the time which often elapses before the change in the Cycle, sometimes running over 3 months or more. The Fourth Square of Nine ends at 324. Note the angles of 45° cross at 325, indicating a change in cycles here. To complete the 360° requires Four Squares of Nine and 36 over. Note that 361 equals a Square of 19 times 19, thus proving the great value of the Square of Nine in working out the important angles and proving up discrepancies.

Beginning with "1" at the center, note how 7, 21, 43, 73, 111, 157, 211, 273 and 343 all fall on a 45° angle. Going the other way, note that 5, 13, 31, 57, 91, 133, 183, 241, and 307 fall on an angle of 45° . Remember there are always four ways you can travel from a center following an angle of 45° , or an angle of 180° or an angle of 90° , which all equal about the same when measured on a flat surface. Note that 8, 23, 46, 77, 116, 163, 218, 281 and 352 are all on an angle from the main center; also note that 4, 15, 34, 61, 96, 139, 190, 249, and 316 are on an angle from the main center, all of these being great resistance points and measuring out important time factors and angles.

Study the SQUARE OF NINE very carefully in connection with the MASTER TWELVE and 360° CIRCLE CHART.

SIX SQUARES OF NINE

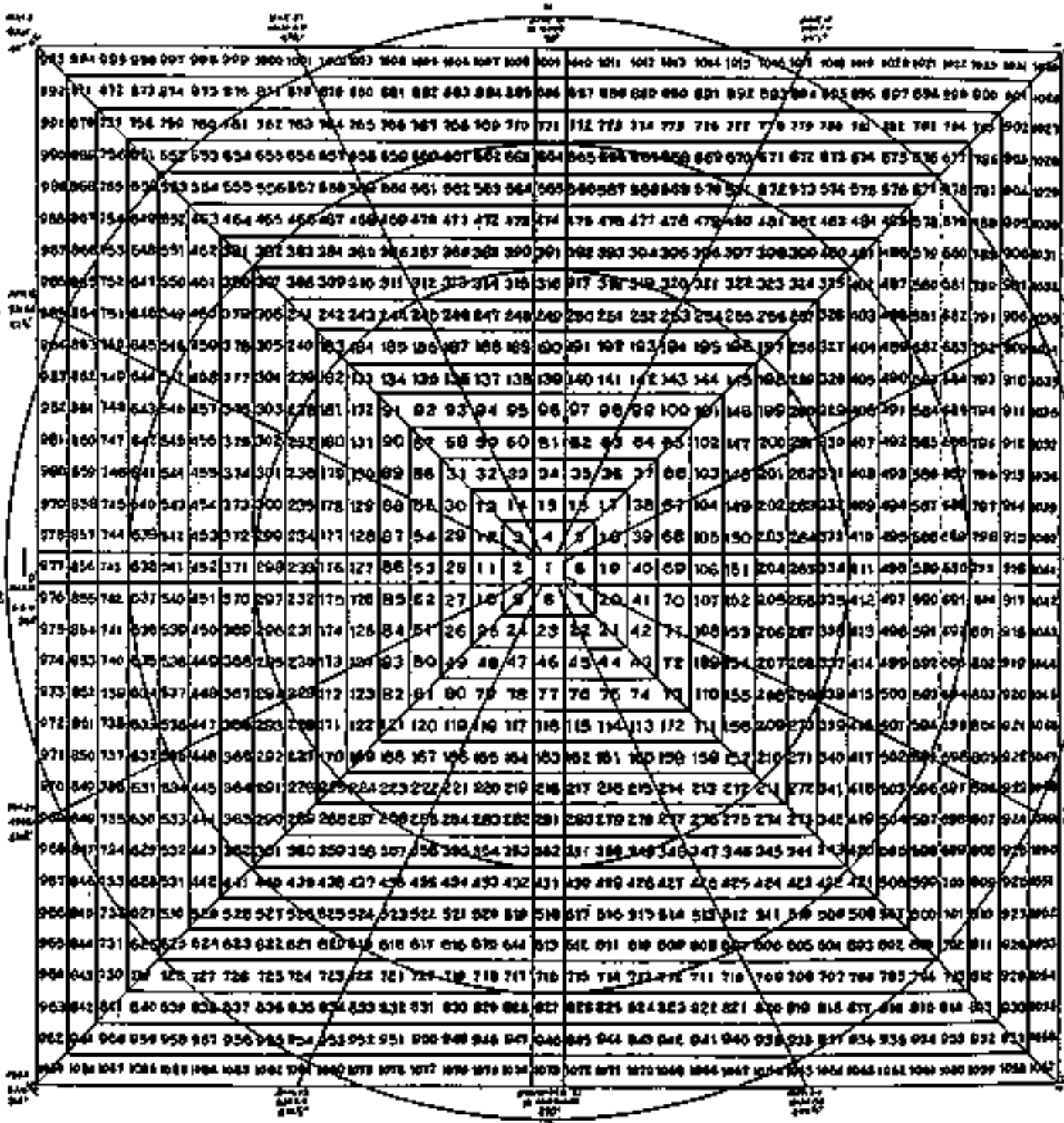
You will receive six Permanent Charts, each containing 81 numbers. The First Square of Nine runs from 1 to 81. Everything must have a bottom, top, and four sides to be a square or cube. The first Square running up to 81 is the bottom, base, floor or beginning point. Squares 2, 3, 4, and 5 are the four sides, which are equal and contain 81 numbers. The Sixth Square of Nine is the top and means that it is times times as referred to in the Bible, or a thing reproducing itself by being multiplied by itself. Nine times nine equals 81 and six times 81 equals 486. We can also use 9 times 81, which would equal 729.

The number 5 is the most important number of the digits because it is the balance or main center. There are four numbers on each side of it. Note how it is shown as the balancing or center number in the Square of Nine.

We square the Circle by beginning at 1 in the center and going around until we reach 360. Note that the Square of Nine comes out at 361. The reason for this is: It is 19 times 19, and the 1 to begin with and 1 over 360 represent the beginning and ending points. 361 is a transition point and begins the next circle. Should we leave the first space blank or make it "0", then we would come out at 360. Everything in mathematics must prove. You can begin at the center and work out, or begin at the outer rim and work in to the center. Begin at the left and work right to the center or to the outer rim or square.

Note the Square of Nine or the Square of the Circle, where we begin with 1 and run up the side of the column to 19, then continue to go across until we have made 19 columns, again the square of 19

2



10:30pm

by 19. Note how this proves up the circle. One-half of the circle is 180° . Note that in the grand-center, where all angles from the four corners and from the East, West, North and South reach gravity center, number 181 appears, showing that at this point we are crossing the Equator or Gravity center and are starting on the other half of the circle.

We have astronomical and mathematical proof of the whys and wherefores and the cause of the workings of geometrical angles. When you have made progress, proved yourself worthy, I will give you the Master Number and also the Master Word.

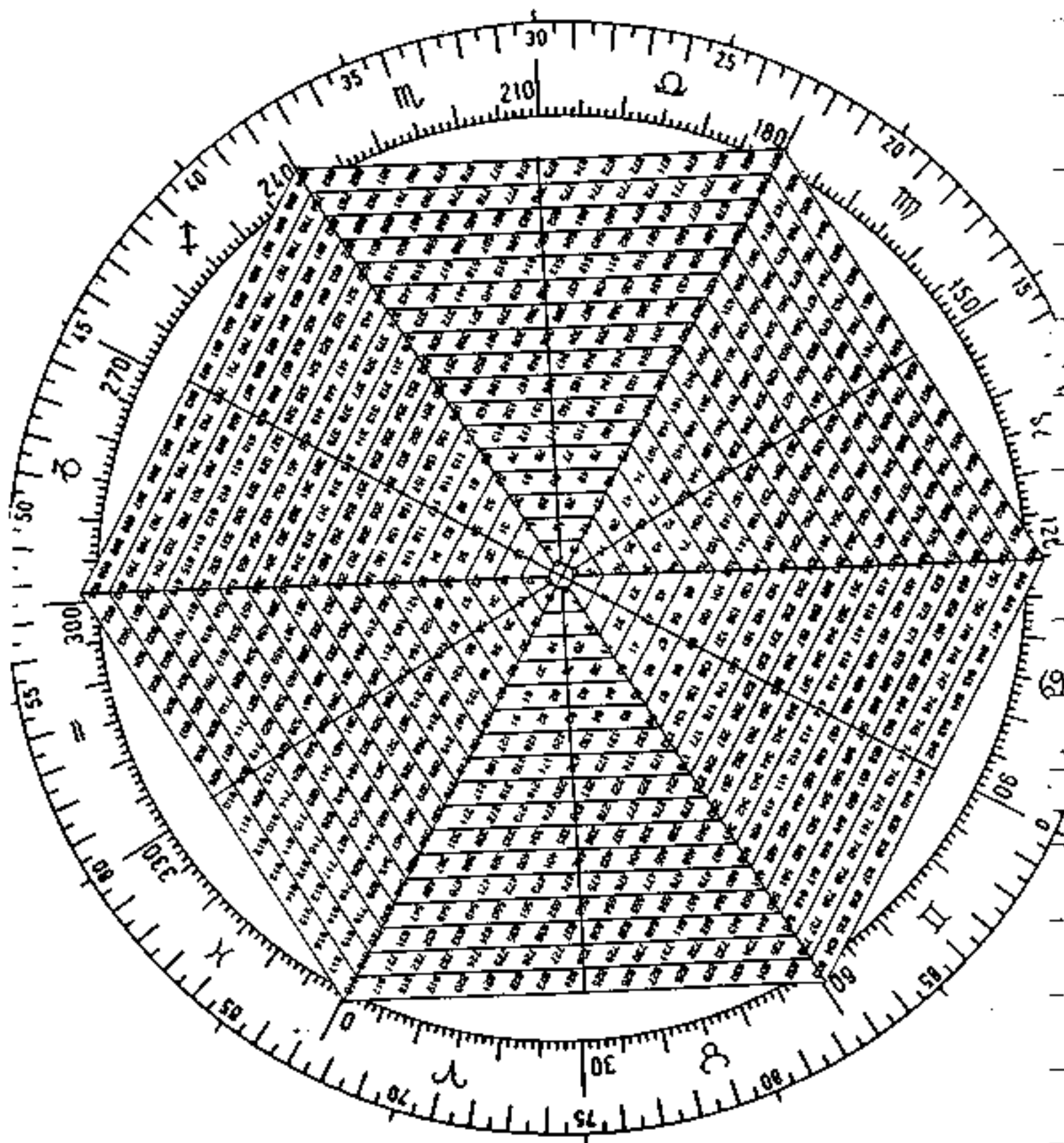
THE HEXAGON CHART

Since everything moves in a circle and nothing moves in straight lines, this chart is to show you how the angles influence stocks at very low levels and very high levels and why stocks move faster the higher they get, because they have moved out to where the distance between the angles of 45° are so far apart that there is nothing to stop them and their moves are naturally rapid up and down.

We begin with a circle of "1" in the center and while this only contains 1, yet the circle is 360° just the same. We then place a circle of circles around this circle and six circles complete the second circle, making a gain of 6 over the first one, ending the second circle at 7, making 7 on this angle a very important month, year, and week as well as day, the seventh day being sacred and a day of rest. The third circle is completed at 19. The fourth circle around is completed at 37, a gain of 18 over the previous circle. The fifth circle is completed at 61, a gain of 24 over the previous circle. The sixth circle is completed at 91, a gain of 30 over the previous circle, and the seventh circle at 127, a gain of 36 over the last circle. Note that from the first the gain is 6 each time we go around. In other words, when we have traveled six times around we have gained 36. Note that this completes the first Hexagon and as this equals 127 months, shows why some campaigns will run 10 years and seven months, or until they reach a square of the Hexagon, or the important last angle of 45° .

The eighth circle around is completed at 169, a gain of 42 over the first. This is a very important angle and an important time factor for more reasons than one. It is 14 years and one month, or double our Cycle of 7 years. Important tops and bottoms culminate at this angle as you will see by going over your charts.

The ninth circle is completed at 217, a gain of 48 over the previous circle. The tenth circle is completed at 271, a gain of 54. Note that 271 is the 9th circle from the first, or is the third 90° angle or 270° , three-fourths of a circle, a very strong point. All this is confirmed by the Master Twelve Chart, by the four seasons and by the Square of Nine Chart, and also confirmed by the Hexagon Chart, showing that mathematical proof is always exact no matter how many ways or from what directions you figure it.



The eleventh circle is completed at 331, a gain of 60 over the last circle. The twelfth circle is completed at 397, which completes the Hexagon, making a gain in 11 circles of 66 from the beginning. 66 months, or 5 years and six months, marks the culmination of major campaigns in stocks. Note how often they culminate on the 60th month, then have a reaction, and make a second top or bottom in the 66th month. Note the number 66 on the Master Twelve Chart. Note it on the Square of Nine and see that 66 occurs on an angle of 180° on the Hexagon Chart, all of which confirms the strong angle at this point.

We have an angle of 66° , one of $67\frac{1}{2}$, and one of 68, confirming this point to be doubly strong for tops and bottoms or space movements up or down.

Note the number 360 on the Hexagon Chart. It completes a circle of 360° . From our beginning point this occurs at an angle of 150° on the Hexagon Chart going around, but measuring from the center, it would equal an angle of 90° or 180° , making this a strong point, hard and difficult to pass, and the ending of one campaign and the beginning of another.

Again with the Center of the Hexagon Chart at "1" notice that 7, 9, 37, 61, 91, 127, 169, 217, 271, 331, and 397 are all on this direct angle and are important points in time measurement. Beginning with "1" and following the other angle, note that 2, 9, 22, 41, 66, 97, 134, 177, 226, 281, and 342 are all on the same angle of 90° , or an angle of 60° and 240° as measured by the Hexagon Chart.

Go over this Chart and the important angles each way and you will see why resistance is met either on days, weeks, months, or years, and why stocks stop and make tops and bottoms at these strong important points according to time and price.

When any stock has passed out above 120° or especially above 127° or 127 points and gone out of the square of the first Hexagon, its fluctuations will be more rapid and it will move faster up and down. Notice near the center that in traveling from 6 to 7 it strikes the angle of 180° or 90° , but when the stock gets out to 162, it can travel up to 169 before striking another strong angle. That is why fast moves occur up and down as stocks get higher and as they move from a center in time.

Remember that everything seeks the center of gravity and important tops and bottoms are formed according to centers and measurements of time from a center, base or beginning point, either top or bottom. The angles formed going straight up and across, may form just the same going across as the stock travels over for days, weeks, months or years. Thus, a stock going up to 22 $\frac{1}{2}$ would strike an angle of $22\frac{1}{2}^\circ$. If it moves over 22 $\frac{1}{2}$ days, 22 $\frac{1}{2}$ weeks, or 22 $\frac{1}{2}$ months, it would also strike an angle of $22\frac{1}{2}^\circ$, and the higher it is when those angles are struck and the angle it hits going up, the greater the resistance to be met. Reverse the rule going down.

Market movements are made just the same as any other thing which

is constructed. It is just the same as constructing a building. First the foundation has to be laid and then the four sides have to be completed and last, but not least of all, the top has to be put on. The cube or hexagon proves exactly the law which works because of time and space in the market. When a building is put up it is built according to a square or hexagon. It has four walls or four sides, a bottom and a top; therefore, it is a cube.

In working out the 20-year Cycle in the stock market, the first 60° , or 5 years, from the beginning forms the bottom of the cube. The second 60° , running to 120, completes the first angle or the first side and runs out the 10-year Cycle. The third 60° , or the second side, ends 15 years or 180° . It is very important because we have the building half completed and must meet the strongest resistance at this point. The fourth 60° , or the end of 20 years or 240 months, completes the third side. We are now two-thirds around the building, a very strong point which culminates and completes our 20-year Cycle. The fifth 60° , or 300th point, days, weeks, or months, completes 25 years, a repetition of the first 5 years, but it completes the fourth side of our building and is a very important angle. The 6th 60° or 360° , completing the circle and ending 30 years as measured by our Time Factor, which runs 1° per month on an angle of 45° , completes the top. This is a complete cube and we begin over again.

Study this in connection with the Hexagon Chart. It will help you.

W. D. Gamm

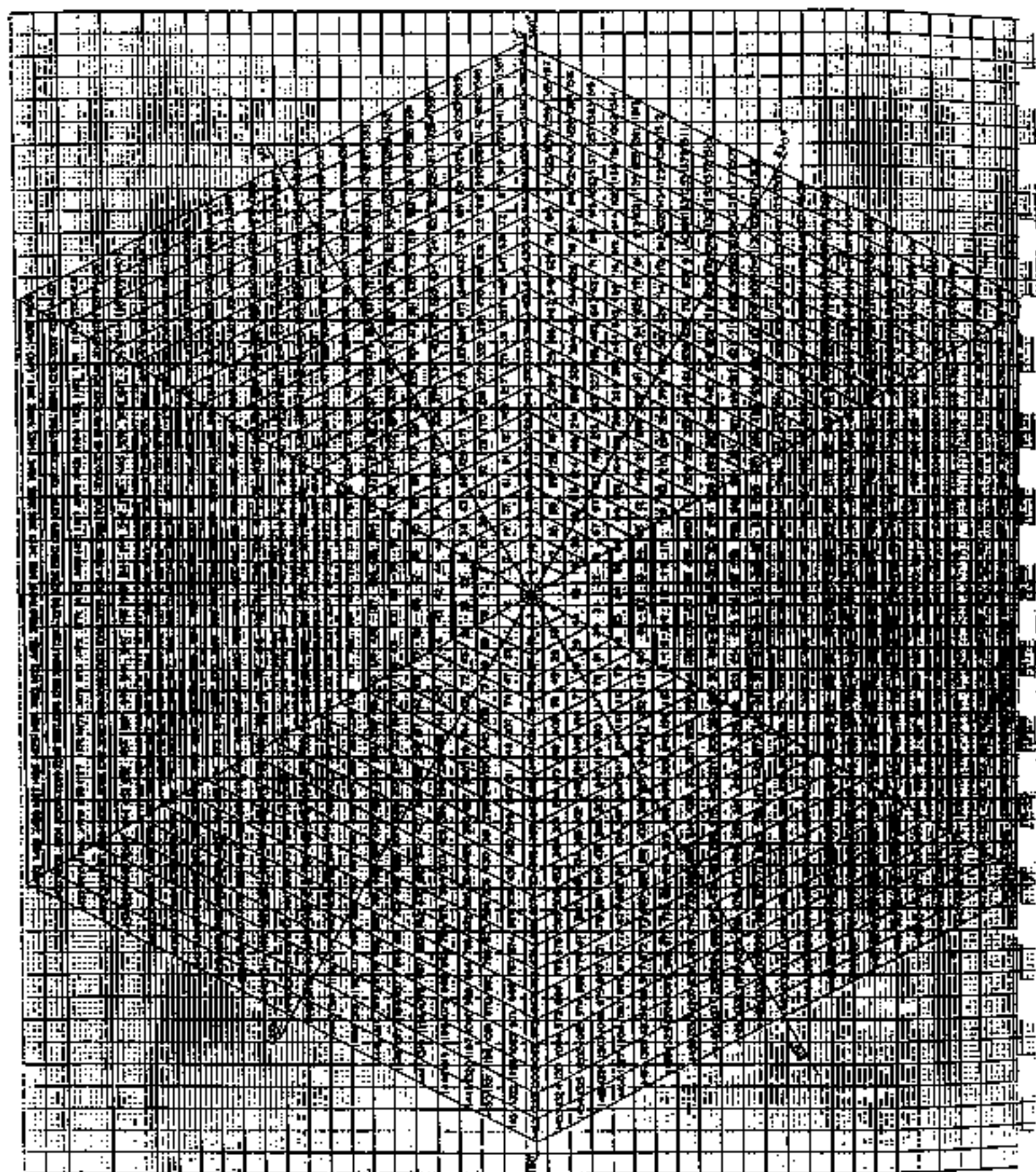
Double numbered squares

The large units are price
The smaller units are time

700 to 6460	\$1008.00	5.2160	Gain \$8.208.00
1500	\$2.160.00	5855	Gain \$6.293.60
Every 70 Points to 1 month			
70 Points = \$100.80			
5855	\$8.446.60		
6400	\$9.216.00	70	\$1008.00

* 5460 →
78 time units →
(2) 70pts/month

5460	5390	5320	5250	5180	5110	5040	4970	4900	4830
78	77	76	75	74	73	72	71	70	69
5530	5460	5390	5320	5250	5180	5110	5040	4970	4900
79	78	77	76	75	74	73	72	71	70
5600	5530	5460	5390	5320	5250	5180	5110	5040	4970
80	79	78	77	76	75	74	73	72	71
5670	5600	5530	5460	5390	5320	5250	5180	5110	5040
81	80	79	78	77	76	75	74	73	72
5740	5670	5600	5530	5460	5390	5320	5250	5180	5110
82	81	80	79	78	77	76	75	74	73
5810	5740	5670	5600	5530	5460	5390	5320	5250	5180
83	82	81	80	79	78	77	76	75	74
5880	5810	5740	5670	5600	5530	5460	5390	5320	5250
84	83	82	81	80	79	78	77	76	75
5950	5880	5810	5740	5670	5600	5530	5460	5390	5320
85	84	83	82	81	80	79	78	77	76
6020	5950	5880	5810	5740	5670	5600	5530	5460	5390
86	85	84	83	82	81	80	79	78	77
6090	6020	5950	5880	5810	5740	5670	5600	5530	5460
87	86	85	84	83	82	81	80	79	78



W. D. GANN

91 WALL STREET
NEW YORK

SCIENTIFIC ANALYSIS
OF ANALYTICAL REPORTS
TICKS AND COMMODITIES
AND OF "TRENDS OF THE STOCK MARKET"
BY "THE TRENCH TRENDS THE AM"

MEMBER
AMERICAN ECONOMIC ASS'n
ROYAL ECONOMIC SOCIETY
CIVIL ASSOCIATES
"CANTON NEW YORK"

MASTER CHARTS (continued)

THE MASTER CHART OF 360°.

This chart begins at "0" and runs around the circle to "360". We first divide by two and get 180°; then divide 180 by 2 and we get 90°; then divide 90 by 2 and we get 45°; then divide 45 by 2 and we get 22½°; then 22½ by 2 and we get 11¼°; then divide 11¼ by 2 and we get 5-5/8°, - all of which form the important angles around the circle.

I have only shown all the important angles from 3-3/4 to 360°. All the angles drawn in red because they are the important square angles,

After dividing the circle by 2, the next important number to divide by is 3. Dividing by 3, we get 120°, 240° and 360°, making the important triangle points. We then divide 120 by 2 and get the angles of 60°, 150°, 210°, 300° and 330°. We divide 60 by 2 and get the angles of 30° and their proportionate parts around. Then divide 30 by 2 and get the angles of 15° and their proportionate angles around the circle. Then divide 15 by 2 and we get the angles at 7½° around the circle.

24 hours in a day divided into the circle equals 15° for each hour. There are 48 divisions of the circle of 7½°, which are important in measuring day, weekly and monthly time periods.

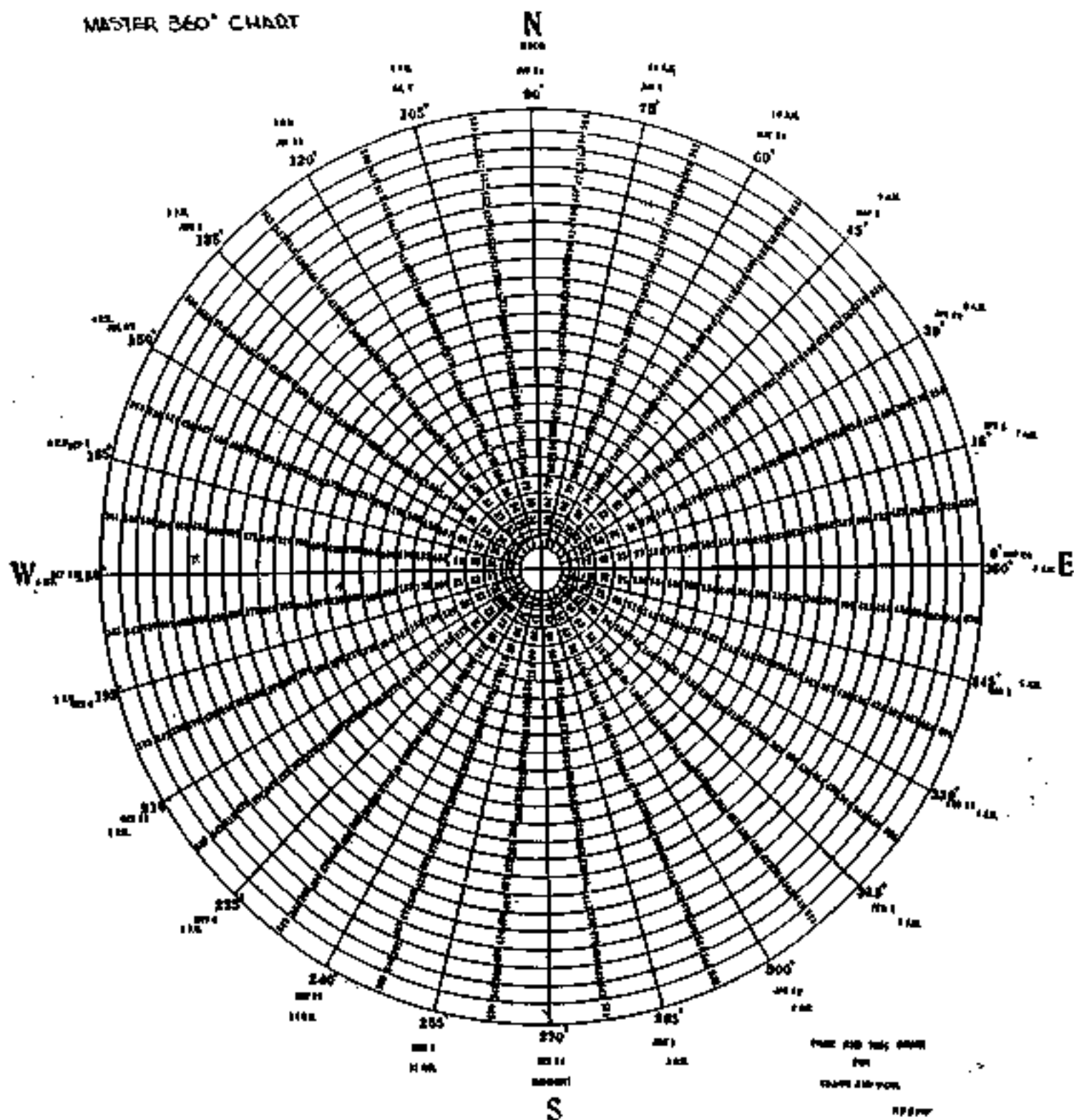
Study the Major Chart of 360° carefully and you will see why cycles repeat. When anything has traveled up 180°, it starts to go down to the opposite point and each angle up to 180° is the opposite on the other side of the circle. That is why tops and bottoms come out at each one of these degrees. For instance, top coming out at 90° or 90 months at one time will occur 90° or 90 months apart and there will be similar tops causing rapid fluctuations and fast moves up and down because this angle is so acute that stocks can not remain long before starting in the opposite direction.

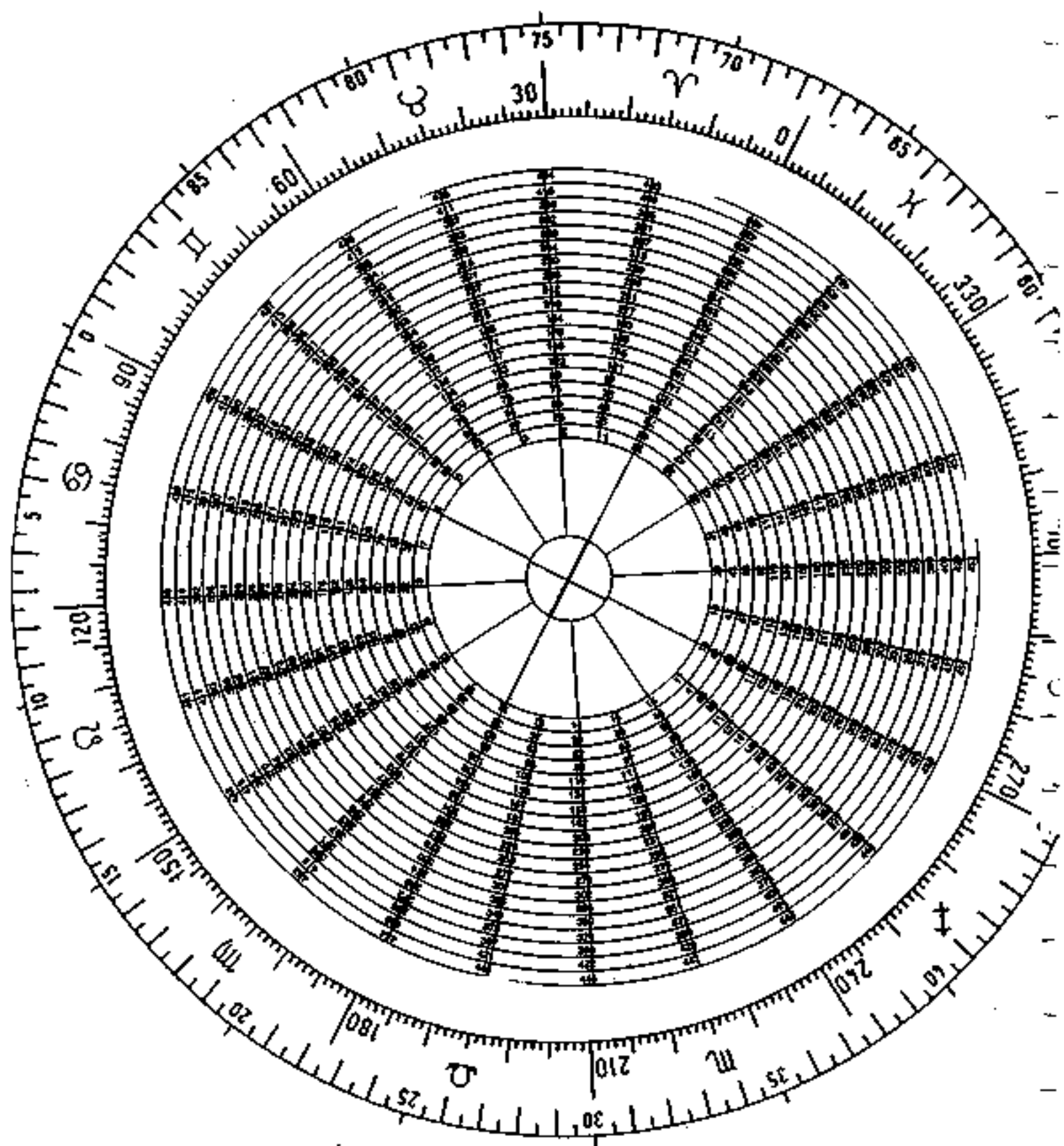
Note the culminations that occur every 14 and 15 years, 180 months, or when an 180° angle or an angle straight up and down is reached. How rapid the moves are when they near this angle and how fast they move up and down and how quickly reverse. It is the same with the 45°, the 135°, the 225° and the 315° angle. Note how this proves the important campaigns when tops and bottoms work out in regularity an equal number of months apart. Note the 22½° and then follow the next important angle, the 30° angle, then the 45°, then note the 60 and 57½° are close together but both very strong angles. Note also 112½° and 120°, both strong angles and close together, indicating important tops and bottoms. Also note 150° and 157½°, strong angles close together, indicating important tops and bottoms, and so go all the way around.

When the circle is divided by 2 and 3 and redivided, we get the following important angles and measurements of time, space and volume around the circle of 360°:-

Mr. Timothy Sullivan & Associates, Inc.

MASTER 360° CHART





5-5/8, 7-1/2, 11-1/4, 15, 16-7/8, 22-1/2, 27-7/8, 30, 33-3/4, 37-1/2, 39-3/8, 45, 50-5/8, 56-1/4, 60, 61-7/8, 67-1/2, 73-1/8, 75, 78-3/4, 82-1/2, 84-3/8, 90, 95-5/8, 101-1/4, 105, 106-7/8, 112-1/2, 118-1/8, 120, 123-3/4, 129-3/8, 135, 140-6/8, 146-1/4, 150, 152-7/8, 157-1/2, 163-1/8, 168-3/4, 174-3/8, 180, 185-5/8, 191-1/4, 196-7/8, 202-1/2, 208-1/8, 210, 213-3/4, 219-3/8, 225, 230-5/8, 236-1/4, 240, 241-7/8, 247-1/2, 253-1/8, 258-3/4, 264-3/8, 270, 275-5/8, 281-1/4, 286-7/8, 292-1/2, 298-1/8, 300, 303-3/4, 309-3/8, 315, 320-5/8, 326-1/4, 330, 331-7/8, 337-1/2, 343-1/8, 348-5/4, 354-3/8, 360, which completes the circle.

These points are all made by a division of angles and are measurements of one-half, one-third, one-fourth, one-eighth, one-sixteenth, one-thirty-second and one-sixty-fourth.

Compare these points with your Master Twelve Chart, your Square of Nine Chart, your Hexagon Chart and your Major Chart of 360°. You will see how they all confirm the important angles and time factors.

The number "7" being so important in determining the culmination in weeks, days, months and years, we must divide the circle by 7 to get the important points, or the one-seventh points in the circle, which are vital and important angles.

The first one-seventh of 360 equals 51-3/7, the second equals 102-6/7, the third equals 154-2/7, the fourth equals 205-5/7, the fifth equals 257-1/7, the sixth equals 308-4/7 and the seventh completes the circle, equalling 360 degrees, days, weeks, months or years. If you divide each of these points by 2, you will also get other important and valuable angles which will confirm and correspond to the other angles in the other charts.

One-seventh of a year or one-seventh of a circle shows why so many fast market movements culminate in the 49th day or the 52nd day and why the 7th week is so very important in culminations and also the 7th month as well as the 7th year.

1½ times 51-3/7 equals 77-1/8 and shows why the angles are so very strong around that point and why the 77th day, 77th week and 77th month are so important for culminations.

MASTER 360° CIRCLE CHART SQUARED

The Master 360 Circle Chart when squared is 90 x 90, and contains 8,100 cells, zones or spaces. Therefore, the square of 360 will contain 32,400 spaces. This shows you why a stock fluctuates up and down so many times over the same territory, because it is working out the number of cells or vibrations of each space in a square. For example:-

1/8 of 90 equals 1,012-1/2
 1/4 of 90 equals 2,025
 1/2 of 90 equals 4,050
 3/4 of 90 equals 6,075
 1/3 of 90 equals 2,700
 2/3 of 90 equals 5,400

Master 360° Circle Chart

		1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5
12	24	360	720	1080	1440	1800	2160	2520	2880	3240	3600	3960	4320	4680	5040	5400	5760	6120
	23	345	705	1065	1425	1785	2145	2505	2865	3225	3585	3945	4305	4665	5025	5385	5745	6105
11	22	330	660	990	1320	1650	1980	2310	2640	2970	3300	3630	3960	4290	4620	4950	5280	5610
	21	315	630	945	1260	1575	1890	2205	2520	2835	3150	3465	3780	4095	4410	4725	5040	5355
10	20	300	600	900	1200	1500	1800	2100	2400	2700	3000	3300	3600	3900	4200	4500	4800	5100
	19	285	570	855	1140	1425	1710	2000	2285	2570	2855	3140	3425	3710	4000	4285	4570	4855
9	18	270	540	810	1080	1350	1620	1890	2160	2430	2700	2970	3240	3510	3780	4050	4320	4590
	17	255	510	765	1020	1275	1530	1785	2040	2295	2550	2805	3060	3315	3570	3825	4080	4335
8	16	240	480	720	960	1200	1440	1680	1920	2160	2400	2640	2880	3120	3360	3600	3840	4080
	15	225	450	675	900	1125	1350	1575	1800	2025	2250	2475	2700	2925	3150	3375	3600	3825
7	14	210	420	630	840	1050	1260	1470	1680	1890	2100	2310	2520	2730	2940	3150	3360	3570
	13	195	390	585	780	975	1170	1365	1560	1755	1950	2145	2340	2535	2730	2925	3120	3315
6	12	180	360	540	720	900	1080	1260	1440	1620	1800	1980	2160	2340	2520	2700	2880	3060
	11	165	330	495	660	825	990	1155	1320	1485	1650	1815	1980	2145	2310	2475	2640	2805
5	10	150	300	450	600	750	900	1050	1200	1350	1500	1650	1800	1950	2100	2250	2400	2550
	9	135	270	405	540	675	810	945	1080	1215	1350	1485	1620	1755	1890	2025	2160	2295
4	8	120	240	360	480	600	720	840	960	1080	1200	1320	1440	1560	1680	1800	1920	2040
	7	105	210	315	420	525	630	735	840	945	1050	1155	1260	1365	1470	1575	1680	1785
3	6	90	180	270	360	450	540	630	720	810	900	990	1080	1170	1260	1350	1440	1530
	5	75	150	225	300	375	450	525	600	675	750	825	900	975	1050	1125	1200	1275
2	4	60	120	180	240	300	360	420	480	540	600	660	720	780	840	900	960	1020
	3	45	90	135	180	225	270	315	360	405	450	495	540	585	630	675	720	765
1	2	30	60	90	120	150	180	210	240	270	300	330	360	390	420	450	480	510
	1	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225	240	255
		30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°	30°	60°	90°	120°	150°

The square of 360, or 360 times 360, equals 32,400.

$1/4$ of 360 equals 8,100
 $1/3$ of 360 equals 10,860
 $1/2$ of 360 equals 16,200
 $2/3$ of 360 equals 21,600
 $3/4$ of 360 equals 24,300
 $7/8$ of 360 equals 28,350.

These points are very important to use for volume of sales as well as time and price measurements.

Suppose you want to know the number of days required to fill or work out the square of 90. There are 365 days in a year. 20 years will give you 7,300 days, and in counting, the leap years will run a little over this. Therefore, about 22 years, 2 months and 10 days are required to work out each vibration in the square of 90.

THE SPIRAL CHART

The Spiral Chart represents the correct position, time and space of anything that begins at zero and begins to move round and round. It shows just exactly how the numbers increase as the spiral moves round and round, and why stocks move faster as they grow older, or swing so much more rapidly as the price reaches higher levels. At center, beginning point or zero, it requires 45° to represent one point. When the stock traveled seven times around from the center, it then required seven points to strike a 45° angle. When it has traveled around the spiral twelve times, it will then require a space of ten points before striking a 45° angle. It would also mean that the stock could move in one direction ten months without striking anything to cause any very great reaction. On this chart, we have only shown the 45, 60, 90, 120, 135, 180, 225, 240, 270, 300, 315 and 360 degree angles. This shows the division of the circle by 2, 4 and 8, and also shows the one-third point and the two-thirds point; being the vital and most important angles, we place them so you can see how space or time makes rapid fluctuations.

NEW YORK STOCK EXCHANGE PERMANENT CHART.

This Chart is a square of 20, or 20 up and 20 over, making a total of 400, which can be used to measure days, weeks, months or years, and to determine when tops and bottoms will be made against strong angles, as indicated on this Permanent Chart.

For example:- The New York Stock Exchange was incorporated May 17th, 1792. Therefore, we begin at "0" on May 17, 1792. 1793 ends on 1, when the Stock Exchange was 1 year old. 1812 will come out on 20;

1832 on 40	1892 on 100
1852 on 60	1912 on 120
1872 on 80	1932 on 140

Note that 139, or 1931, strikes the 45° angle, running from 20 down, and that this is in the 7th zone, or the 7th space over, which indicates that 1931 is the ending of a bear campaign, and the starting of a bull market. But we must watch out for a break around May and June, 1931, when this angle is

NYSE

PERMANENT CHART

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	400
19	39	59	79	99	119	139	159	179	199	219	239	259	279	299	319	339	359	379	399
18	38	58	78	98	118	138	158	178	198	218	238	258	278	298	318	338	358	378	398
17	37	57	77	97	117	137	157	177	197	217	237	257	277	297	317	337	357	377	397
16	36	56	76	96	116	136	156	176	196	216	236	256	276	296	316	336	356	376	396
15	35	55	75	95	115	135	155	175	195	215	235	255	275	295	315	335	355	375	395
14	34	54	74	94	114	134	154	174	194	214	234	254	274	294	314	334	354	374	394
13	33	53	73	93	113	133	153	173	193	213	233	253	273	293	313	333	353	373	393
12	32	52	72	92	112	132	152	172	192	212	232	252	272	292	312	332	352	372	392
11	31	51	71	91	111	131	151	171	191	211	231	251	271	291	311	331	351	371	391
10	30	50	70	90	110	130	150	170	190	210	230	250	270	290	310	330	350	370	390
9	29	49	69	89	109	129	149	169	189	209	229	249	269	289	309	329	349	369	389
8	28	48	68	88	108	128	148	168	188	208	228	248	268	288	308	328	348	368	388
7	27	47	67	87	107	127	147	167	187	207	227	247	267	287	307	327	347	367	387
6	26	46	66	86	106	126	146	166	186	206	226	246	266	286	306	326	346	366	386
5	25	45	65	85	105	125	145	165	185	205	225	245	265	285	305	325	345	365	385
4	24	44	64	84	104	124	144	164	184	204	224	244	264	284	304	324	344	364	384
3	23	43	63	83	103	123	143	163	183	203	223	243	263	283	303	323	343	363	383
2	22	42	62	82	102	122	142	162	182	202	222	242	262	282	302	322	342	362	382
1	21	41	61	81	101	121	141	161	181	201	221	241	261	281	301	321	341	361	381

hit at the end of the 139th year.

You will notice that the numbers which divide the square into equal parts, run across 10, 30, 50, 70, 90, 110, etc., and that the year 1802 comes out on 10, the year 1822 on 30, the year 1842 on 50, the year 1862 on 70. Note that the year 1861, when the Civil War broke out, was on the number 69, which is a 45° angle. Then note that 1882 ended in May on the 90° angle, and at the $1/2$ point, 180° angle, running horizontally across.

Again in 1902, it was at 110; the $1/2$ point, and in 1903 and 1904, hit the 45° angle.

Note that the years 1920 and 1921 hit the 45° angle on No. 129 and 1922, the first year of the bull market, was at 130 at the $1/2$ point.

Note that 1929 was on the 137th number, or 137th month, and hit an angle of 45° , and that the year 1930 was at the $1/2$ point on the 4th square, a strong Resistance Point, which indicated a sharp, severe decline.

Again, 138 is at the $1/2$ point on the Master 12 Chart.

1933 will be on 141, in the 8th Zone, and at the center or $1/2$ point of the 2nd quarter of the Square of 20.

The years 1934 and 1935, ending in May, will be on 142 and 143, and 1935 will come out on the 45° angle at the grand center in the 8th Zone and at the $1/2$ point of the 2nd square, going to $1/2$ of the total square, which will indicate a decline and bottom for an advance to run up in 1936, with 1937 striking at 145, which is $1/4$ th of the column on the way up in the square.

If you will study the weeks, months, as well as the years, and apply them to these important points and angles, you will see how they have determined the important tops and bottoms in the past campaigns.

UNITED STATES STEEL NAME CHART

The name, United States Steel, contains 17 letters. Therefore, to make up a Permanent Square for United States Steel according to its name, will require 17×17 , or a square of 289. Therefore, 17, which is really $2\frac{1}{2}$ times the lowest price at which Steel ever sold is an important point.

The prices 34, 51, 68, 85, 102, 119, 136, 153, 170, 187, 204, 221, 238, 255, 272 are all important because they strike important vibrations on the name and angles of U. S. Steel.

The basic number or low point, for example 9, the lowest digit on U. S. Steel, and its vibrations according to its name, all cause slight variations at times from other stocks, because each stock works according to its own base, beginning point, numbers and name.

U. S. Steel works good to the natural angles and Master Chart, because its digit is 9 and comes out on exact 9 vibrations all the way through, finishing at 261, which was 28 nines from its beginning point, or lowest level.

A study of all these various tables will help you to understand Resistance Levels.

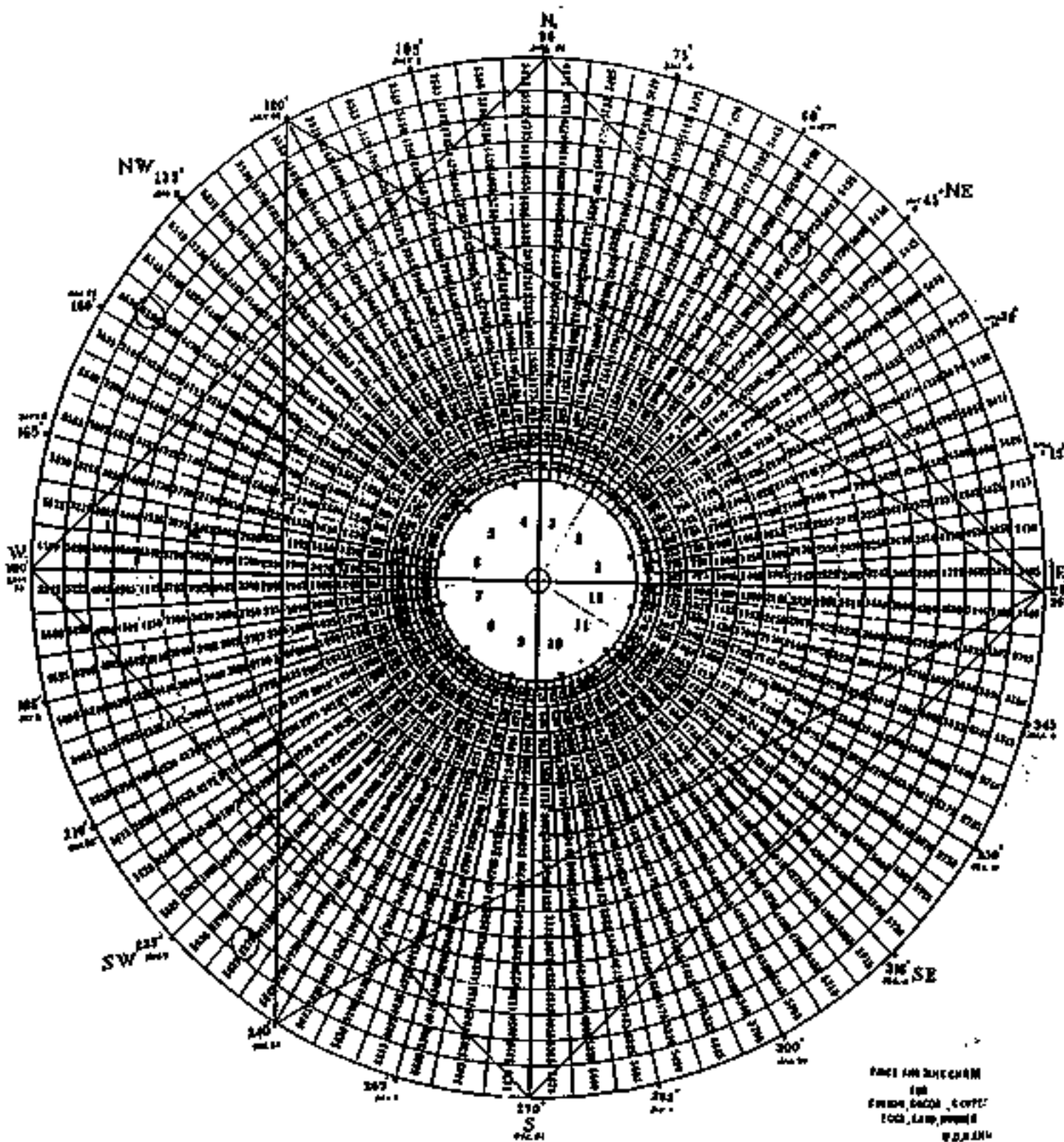
UNITED STATES PERMANENT MASTER CHART

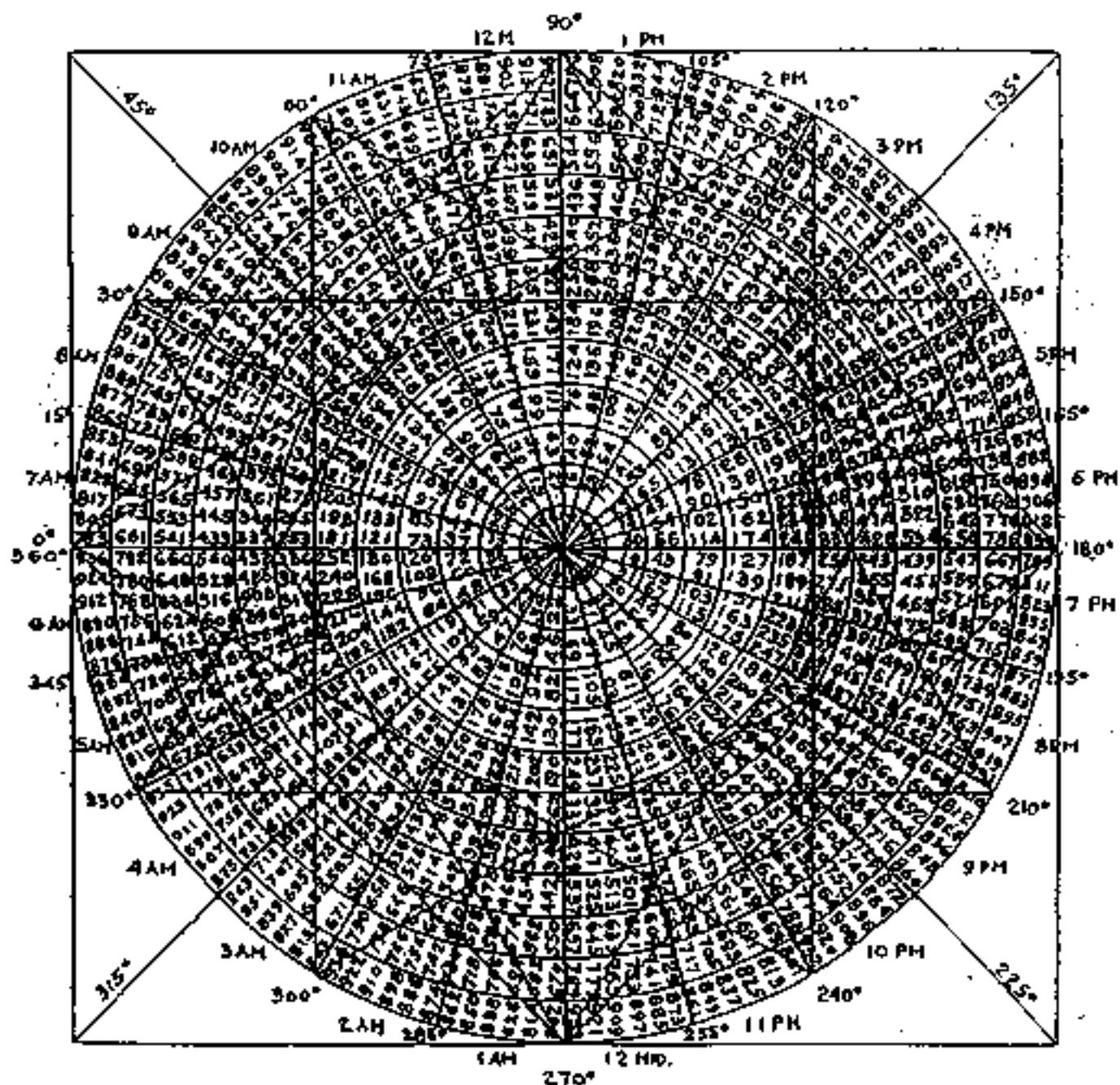
We use the square of 7 for the United States because the name of America contains 7 letters, and this square is 49, a very important and fatal number,

We begin the United States Chart on October 12th, 1492. Make up these Squares of 7 and put on the years. You will note how this indicates the panic years in the United States and the years of prosperity.

You can also make up a square of 21 x 21, which is the number of letters in the name of United States of America. These angles and important points will all come out about the same as when we use America, because three times 7 is 21. However, you will get some stronger angles and more important points if you make up the square of 21 x 21, which will run out to 441, with the $1/2$ point at 220- $1/2$.

The more you study these Permanent Charts, the more you will appreciate their great value, and will see that numbers do determine everything in the future, and that geometrical angles and mathematical points measure every Resistance level, time, price, space or volume.





774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000
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Master Price & Time Chart
 Cotton, Coffee, Cocoa, etc.,
 and Grains

MASTER CHART

MAY 50Y BEANS

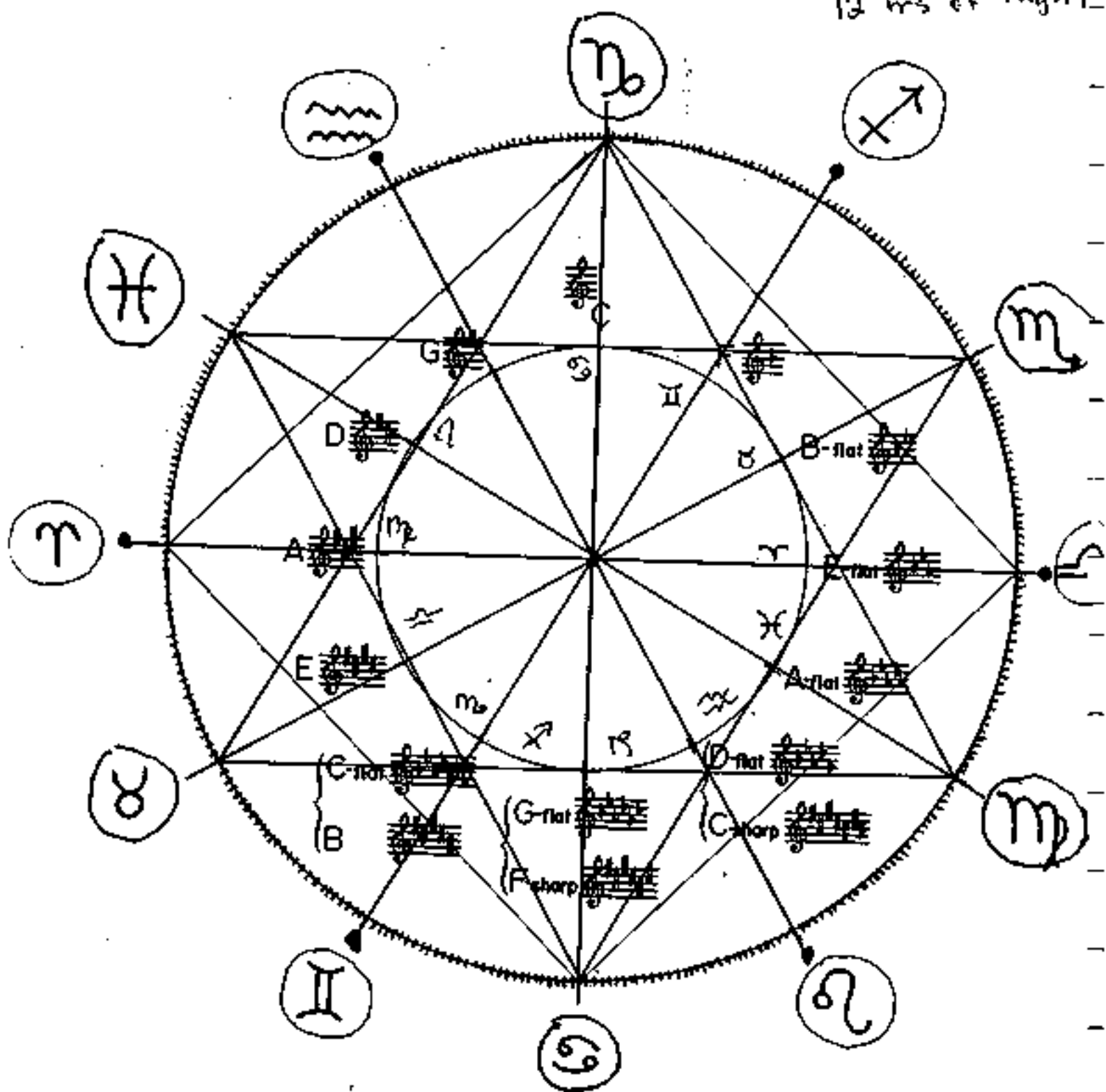
PRICE & TPIR

443	442	441	440	439	438	437	436	435	434	433	432	431	430	429	428	427	426	425	424
368	367	366	365	364	363	362	361	360	359	358	357	356	355	354	353	352	351	350	423
369	300	299	298	297	296	295	294	293	292	291	290	289	288	287	286	285	284	349	424
370	301	240	239	238	237	236	235	234	233	232	231	230	229	228	227	226	283	348	42
371	302	241	188	187	186	185	184	183	182	181	180	179	178	177	176	225	282	347	420
372	303	242	189	144	143	142	141	140	139	138	137	136	135	134	175	224	281	346	41
373	304	243	190	145	108	107	106	105	104	103	102	101	100	133	174	223	280	345	41
374	305	244	191	146	109	80	79	78	77	76	75	74	99	132	173	222	279	344	417
375	306	245	192	147	110	81	60	59	58	57	56	73	98	131	172	221	278	343	41
376	307	246	193	148	111	82	61	48	47	46	55	72	97	130	171	220	277	342	4
377	308	247	194	149	112	83	62	49	44	45	54	71	96	129	170	219	276	341	414
378	309	248	195	150	113	84	63	50	51	52	53	70	95	128	169	218	275	340	4
379	310	249	196	151	114	85	64	65	66	67	68	69	94	127	168	217	274	339	4
380	311	250	197	152	115	86	87	88	89	90	91	92	93	126	167	216	273	338	411
381	310	251	198	153	116	117	118	119	120	121	122	123	124	125	166	215	272	337	4
382	313	252	199	154	155	156	157	158	159	160	161	162	163	164	165	214	271	336	4
383	314	253	200	201	202	203	204	205	206	207	208	209	210	211	212	213	270	335	404
384	315	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	334	4
385	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	4
386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	40

The image shows a document page that is almost entirely illegible due to extreme noise and degradation. The layout appears to be a table or ledger with multiple columns and rows, but the text within these cells is completely unreadable. The page is covered in a dense pattern of black and white noise, which obscures any original content. There are no discernible figures, tables, or text blocks that can be accurately transcribed or described.

SQR of 9 Pythagorean OVERLAY

12 TRIBES of Isr.
12 Months in a Yr.
12 Apostles of Jee.
12 Signs of the zodiac
12 fundamental vibrat
or musical notes.
12 hrs of Day
12 hrs of night.



"Succession of 5ths"

C D E F G A B (Pythagorean "Diatonic Major" musical scale)

Coffin River Jan 30 Oct 2 1936

4300	500	530	550	570	590	610	630	650	670	690	710	730	750	770	790	810	830	850	870	890	910	930	950	970	990	1010	1030	1050	1070	1090	1110	1130	1150	1170	1190	1210	1230	1250	1270	1290	1310	1330	1350	1370	1390	1410	1430	1450	1470	1490	1510	1530	1550	1570	1590	1610	1630	1650	1670	1690	1710	1730	1750	1770	1790	1810	1830	1850	1870	1890	1910	1930	1950	1970	1990	2010	2030	2050	2070	2090	2110	2130	2150	2170	2190	2210	2230	2250	2270	2290	2310	2330	2350	2370	2390	2410	2430	2450	2470	2490	2510	2530	2550	2570	2590	2610	2630	2650	2670	2690	2710	2730	2750	2770	2790	2810	2830	2850	2870	2890	2910	2930	2950	2970	2990	3010	3030	3050	3070	3090	3110	3130	3150	3170	3190	3210	3230	3250	3270	3290	3310	3330	3350	3370	3390	3410	3430	3450	3470	3490	3510	3530	3550	3570	3590	3610	3630	3650	3670	3690	3710	3730	3750	3770	3790	3810	3830	3850	3870	3890	3910	3930	3950	3970	3990	4010	4030	4050	4070	4090	4110	4130	4150	4170	4190	4210	4230	4250	4270	4290	4310	4330	4350	4370	4390	4410	4430	4450	4470	4490	4510	4530	4550	4570	4590	4610	4630	4650	4670	4690	4710	4730	4750	4770	4790	4810	4830	4850	4870	4890	4910	4930	4950	4970	4990	5010	5030	5050	5070	5090	5110	5130	5150	5170	5190	5210	5230	5250	5270	5290	5310	5330	5350	5370	5390	5410	5430	5450	5470	5490	5510	5530	5550	5570	5590	5610	5630	5650	5670	5690	5710	5730	5750	5770	5790	5810	5830	5850	5870	5890	5910	5930	5950	5970	5990	6010	6030	6050	6070	6090	6110	6130	6150	6170	6190	6210	6230	6250	6270	6290	6310	6330	6350	6370	6390	6410	6430	6450	6470	6490	6510	6530	6550	6570	6590	6610	6630	6650	6670	6690	6710	6730	6750	6770	6790	6810	6830	6850	6870	6890	6910	6930	6950	6970	6990	7010	7030	7050	7070	7090	7110	7130	7150	7170	7190	7210	7230	7250	7270	7290	7310	7330	7350	7370	7390	7410	7430	7450	7470	7490	7510	7530	7550	7570	7590	7610	7630	7650	7670	7690	7710	7730	7750	7770	7790	7810	7830	7850	7870	7890	7910	7930	7950	7970	7990	8010	8030	8050	8070	8090	8110	8130	8150	8170	8190	8210	8230	8250	8270	8290	8310	8330	8350	8370	8390	8410	8430	8450	8470	8490	8510	8530	8550	8570	8590	8610	8630	8650	8670	8690	8710	8730	8750	8770	8790	8810	8830	8850	8870	8890	8910	8930	8950	8970	8990	9010	9030	9050	9070	9090	9110	9130	9150	9170	9190	9210	9230	9250	9270	9290	9310	9330	9350	9370	9390	9410	9430	9450	9470	9490	9510	9530	9550	9570	9590	9610	9630	9650	9670	9690	9710	9730	9750	9770	9790	9810	9830	9850	9870	9890	9910	9930	9950	9970	9990	10010	10030	10050	10070	10090	10110	10130	10150	10170	10190	10210	10230	10250	10270	10290	10310	10330	10350	10370	10390	10410	10430	10450	10470	10490	10510	10530	10550	10570	10590	10610	10630	10650	10670	10690	10710	10730	10750	10770	10790	10810	10830	10850	10870	10890	10910	10930	10950	10970	10990	11010	11030	11050	11070	11090	11110	11130	11150	11170	11190	11210	11230	11250	11270	11290	11310	11330	11350	11370	11390	11410	11430	11450	11470	11490	11510	11530	11550	11570	11590	11610	11630	11650	11670	11690	11710	11730	11750	11770	11790	11810	11830	11850	11870	11890	11910	11930	11950	11970	11990	12010	12030	12050	12070	12090	12110	12130	12150	12170	12190	12210	12230	12250	12270	12290	12310	12330	12350	12370	12390	12410	12430	12450	12470	12490	12510	12530	12550	12570	12590	12610	12630	12650	12670	12690	12710	12730	12750	12770	12790	12810	12830	12850	12870	12890	12910	12930	12950	12970	12990	13010	13030	13050	13070	13090	13110	13130	13150	13170	13190	13210	13230	13250	13270	13290	13310	13330	13350	13370	13390	13410	13430	13450	13470	13490	13510	13530	13550	13570	13590	13610	13630	13650	13670	13690	13710	13730	13750	13770	13790	13810	13830	13850	13870	13890	13910	13930	13950	13970	13990	14010	14030	14050	14070	14090	14110	14130	14150	14170	14190	14210	14230	14250	14270	14290	14310	14330	14350	14370	14390	14410	14430	14450	14470	14490	14510	14530	14550	14570	14590	14610	14630	14650	14670	14690	14710	14730	14750	14770	14790	14810	14830	14850	14870	14890	14910	14930	14950	14970	14990	15010	15030	15050	15070	15090	15110	15130	15150	15170	15190	15210	15230	15250	15270	15290	15310	15330	15350	15370	15390	15410	15430	15450	15470	15490	15510	15530	15550	15570	15590	15610	15630	15650	15670	15690	15710	15730	15750	15770	15790	15810	15830	15850	15870	15890	15910	15930	15950	15970	15990	16010	16030	16050	16070	16090	16110	16130	16150	16170	16190	16210	16230	16250	16270	16290	16310	16330	16350	16370	16390	16410	16430	16450	16470	16490	16510	16530	16550	16570	16590	16610	16630	16650	16670	16690	16710	16730	16750	16770	16790	16810	16830	16850	16870	16890	16910	16930	16950	16970	16990	17010	17030	17050	17070	17090	17110	17130	17150	17170	17190	17210	17230	17250	17270	17290	17310	17330	17350	17370	17390	17410	17430	17450	17470	17490	17510	17530	17550	17570	17590	17610	17630	17650	17670	17690	17710	17730	17750	17770	17790	17810	17830	17850	17870	17890	17910	17930	17950	17970	17990	18010	18030	18050	18070	18090	18110	18130	18150	18170	18190	18210	18230	18250	18270	18290	18310	18330	18350	18370	18390	18410	18430	18450	18470	18490	18510	18530	18550	18570	18590	18610	18630	18650	18670	18690	18710	18730	18750	18770	18790	18810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Native Eggs
Cotton Eggs, etc.

March Coffee Rio
October 2, 1906 Low 300

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54.5(2) ϕ or 540

Eggs October Low May 28 1932
 High 585 Sept 10 1931
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1. *Staphylococcus aureus*

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EVEN SQUARES FOR COTTON

See Page 170 - Cont.

210	269	165	165	750	179	168	751	154	100	164	155	152	153	203
6600	4910	4810	4860	4830	4800	4770	4740	4710	4680	4650	4590	4560	4530	6090
221	165	117	116	105	719	113	112	110	104	101	100	100	100	202
6630	4950	3570	3480	3450	3420	3390	3360	3330	3300	3270	3240	3210	3180	6060
222	166	118	118	107	720	114	113	111	105	102	101	101	101	201
6660	4980	3540	3450	3420	3390	3360	3330	3300	3270	3240	3210	3180	3150	6030
223	167	119	119	108	721	115	114	112	106	103	102	102	102	200
6690	5010	3570	3480	3450	3420	3390	3360	3330	3300	3270	3240	3210	3180	6000
224	168	120	120	109	722	116	115	113	107	104	103	103	103	199
6720	5040	3600	3510	3480	3450	3420	3390	3360	3330	3300	3270	3240	3210	5970
225	169	121	121	110	723	117	116	114	108	105	104	104	104	198
6750	5070	3630	3540	3510	3480	3450	3420	3390	3360	3330	3300	3270	3240	5940
226	170	122	122	111	724	118	117	115	109	106	105	105	105	197
6780	5100	3660	3570	3540	3510	3480	3450	3420	3390	3360	3330	3300	3270	5910
227	171	123	123	112	725	119	118	116	110	107	106	106	106	196
6810	5130	3690	3600	3570	3540	3510	3480	3450	3420	3390	3360	3330	3300	5880
228	172	124	124	113	726	120	119	117	111	108	107	107	107	195
6840	5160	3720	3630	3600	3570	3540	3510	3480	3450	3420	3390	3360	3330	5850
229	173	125	125	114	727	121	120	118	112	109	108	108	108	194
6870	5190	3750	3660	3630	3600	3570	3540	3510	3480	3450	3420	3390	3360	5820
230	174	126	126	115	728	122	121	119	113	110	109	109	109	193
6900	5220	3780	3690	3660	3630	3600	3570	3540	3510	3480	3450	3420	3390	5790
231	175	127	127	116	729	123	122	120	114	111	110	110	110	192
6930	5250	3810	3720	3690	3660	3630	3600	3570	3540	3510	3480	3450	3420	5760
232	176	128	128	117	730	124	123	121	115	112	111	111	111	191
6960	5280	3840	3750	3720	3690	3660	3630	3600	3570	3540	3510	3480	3450	5730
233	177	129	129	118	731	125	124	122	116	113	112	112	112	190
6990	5310	3870	3780	3750	3720	3690	3660	3630	3600	3570	3540	3510	3480	5700
234	178	130	130	119	732	126	125	123	117	114	113	113	113	189
7020	7050	7080	7110	7140	7170	7200	7230	7260	7290	7320	7350	7380	7410	5670

Mr. J. Edgar Hoover, Director, Federal Bureau of Investigation

	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
12	360	720	1080	1440	1800	2160	2520	2880	3240	3600	3960	4320
11	345	690	1035	1380	1725	2070	2415	2760	3105	3450	3795	4140
10	330	660	990	1320	1650	1980	2310	2640	2970	3300	3630	3960
9	315	630	945	1260	1575	1905	2235	2565	2895	3225	3555	3885
8	300	600	900	1200	1500	1800	2100	2400	2700	3000	3300	3600
7	285	570	855	1140	1425	1710	2000	2280	2565	2850	3135	3420
6	270	540	810	1080	1350	1620	1890	2160	2430	2700	2970	3240
5	255	510	765	1020	1275	1530	1785	2040	2295	2550	2805	3060
4	240	480	720	960	1200	1440	1680	1920	2160	2400	2640	2880
3	225	450	675	900	1125	1350	1575	1800	2025	2250	2475	2700
2	210	420	630	840	1050	1260	1470	1680	1890	2100	2310	2520
1	195	390	585	780	945	1110	1275	1440	1605	1770	1935	2100

	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
12	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
11	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
10	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
9	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
8	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
7	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
6	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
5	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
4	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
3	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
2	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3
1	1/2	1/3	1/4	1/5	2/5	1/2	2/3	3/4	3/5	2/3	1/2	1/3

APPENDIX G

MARKET VOLUME

Market Volume

In the back of this work, you will find a two-page course that I believe was written by Gann Titled **MARKET TRENDS**. This course deals with market trends and the correlation between transactional volume. Gann says, "The trend of stock prices is influenced by supply and demand. The object of following a chart is to determine whether stocks are in a period of accumulation, or distribution; whether stocks are being marked up or down, or whether they are being left alone". It is suggested that you read this small course at this time and also Gann's course on Form Reading and also my notes from Ted Warren's Book: **HOW TO MAKE THE STOCK MARKET MAKE MONEY FOR YOU**, which is also included with this book.

Now that you have had time to review the above referenced material, I will discuss my opinion of volume and it's significance in forming an opinion on the markets. Because there is always a seller for every buyer, and a buyer for every seller on any given day with the exception of Limit moves and Panics, volume simply records the total number of units that resulted in a transaction between the buyer and seller. The volume data in this format has limited uses but there are several books that have been written by various authors discussing how to interpret market movements from this raw form of the data. My personal view is that volume needs to be converted so that it more accurately represents buying pressure and selling pressure or supply vs. demand. Because we already know that you must have a buyer for every seller and vice versa before any volume can be recorded, we need to determine how this number of units relates to price fluctuations. If we take the fundamental view that the only reason prices advance is simply because there was a greater demand to purchase the stock or commodity than there was a willingness to sell the item at the current existing price. Prices decline because there is a greater willingness to sell the stock or commodity verses the demand for the item at the current price. Selling panics occur when there are no buyers for the item at the current price or even substantially lower prices (zero demand driven). Buying panics occur when there are no sellers at the current price and is obviously a supply driven scenario.

Many analysts have argued that the supply of any stock or commodity is definable simply because you can not instantly create more coffee, corn, oil or shares of a particular stock. These analysts believe that the available supply of the item is what creates the demand that will eventually show up as volume or a transaction between buyer and seller. However, this point of view is not entirely accurate because short selling creates fictional or phantom units of the particular item and therefore distorts or enlarges the available supply of the item. This means that to make volume understandable, we must look at it as it relates to the desire to buy or sell any item at a given moment and we must also quantify this desire.

What I like to do to determine this desire is to look at the percent change up or down on a given day and multiply this fraction by the volume on that day. For example: lets say that Cisco Systems, Inc. (CSCO) is up \$1.00 today on 55 million shares from 77% the day before. The % change is 1 divided by 77% or 0.0128. This fraction multiplied by the volume of 55 million shares is 707,395 and represents the lack of selling desire or excess demand. If the price had declined \$1.00 on this volume it would represent the lack of buying desire or a greater supply. So based on the advancing scenario, we would assume that of the 55 million shares traded this day, that desire to buy exceeded the willingness to sell by 707,395 shares. This number will now let us separate the Supply & Demand forces as it relates to whole volume. $55,000,000$ divided by 2 is $27,500,000$ and this represents perfect balance of desire to sell and willingness to buy. $27,500,000 - 707,395 = 26,792,605$ (supply) and $27,500,000 + 707,395 = 28,207,395$ (demand). You can plot these numbers on graph paper or have a spread sheet program like Microsoft Excel maintain the chart and data. This will give you a much better visual picture of the psychology in the market as it pertains to Gann's rules on volume. Another valuable calculation is to take the supply volume numbers and divide them by the total number of shares that the company actually has issued (for a commodity, use the Government estimates of supply). This ratio will show the percentage of turnover in the stock or how actively the stock is changing hands. When a stock is in "strong" hands, this number is low because the big block share holders have little if any willingness to sell. When distribution is taking place (Form reading + Cycle Timing), this number will show a pretty dramatic increase compared to it's relative history during the up trend or "Mark-up" phase. The market is churning as the inventory is moving from strong to weak hands. This information should help you with volume analysis.

MARKET TRENDS

PRICE AND VOLUME TRADING POINTERS:

THE VOLUME RULE

The trend of stock prices is influenced by supply and demand. The object of following a chart is to determine whether stocks are in a period of accumulation, or distribution; whether they are being marked up or down, or whether they are being left alone. Stocks have habits and characteristics which are distinct as those of human beings or animals. It is worth while becoming acquainted with those habits in order to anticipate stock movements.

1. When prices advance rapidly and maintain their advance, it is perfectly obvious that great strength is shown.

2. When prices decline slowly and with small volume of transactions, that shows there is no great quantity of stocks for sale.

3. Big volumes of transactions on advances are an indication of big demand for stocks, and heavy volume on declines shows PRESSURE, or stocks for sale. Sometimes prices and volume both point the same direction, and sometimes they are apparently opposed.

4. Covering short sales often causes a sharp ADVANCE in prices, but with DECREASING VOLUME.

5. At the culmination of a big slump, prices may advance very slowly but on big VOLUME of transactions or trading. This action generally indicates distribution.

6. When prices persistently refuse to go down on evident bear news, the market is either oversold or stocks have drifted into strong hands. This condition usually appears after a considerable decline.

7. When bull news fails to bull the market it shows an overbought condition; that the buying power is pretty well used up. This action appears at the top of a big rise. Watch increasing interest rates as danger signals. News nearly always has some effect in increasing transactions on advances or declines.

8. The action at the bottom generally occurs in the following manner. Prices decline rapidly on a big volume of trading for a period of weeks with relatively wide fluctuations in the leading speculative stocks from day to day. Then from near bottom there is a sharp rally for several days caused by short covering. After the market grows dull and drifts slowly downward for some weeks with transactions much smaller than on the previous decline, and with a narrower range of daily fluctuations. A SOLID bottom is always made on small volume. One of the major chart formations usually appears if there is a worth while bull move ahead.

9. The action at the top generally occurs in the following manner: At the top, large volume IS SIMPLY A MEANS OF ATTRACTING BUYERS, and disguising the hand of the operator. Usually there is abnormally large volume within a small radius. After a season of wild activity on the bull side, people pyramid on their profits, and overload at the top. As there is no one left to buy, the inevitable break follows, after which dullness sets in and operators play short while the public sees leading stocks drop 100 points and hopes for the upward turn that never comes. With corporation profits at the expense of the lambs, great buildings are constructed.

10. In a bull move, VOLUME MUST INCREASE ON ADVANCES, and the rest of the market offer some support, or at least not oppose. Get in as near the beginning of a move as possible. However, a safer method is to wait for the FIRST REACTION and note the stocks action there.

11. In a bull move reactions must be accompanied by decreasing volume; that is, the reactions must have less volume than the advances. In a BEAR MOVE vice versa is true, while rallies should have least volume.

12. Each advance should be of longer duration, and reach a new high on larger volume than the preceding advance. In other words a new high on smaller volume indicates at least temporary weakness.

These details can be checked most successfully by keeping graph charts of specific stocks and also noting daily Dow Jones trend and volume reports.

... Confidential ...

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FORM READING and RULES FOR DETERMINING TREND OF STOCKS

REQUIREMENTS FOR SUCCESS IN SPECULATION OR INVESTMENT

KNOWLEDGE

One of the vital and most important factors for making a success in speculation or anything else is KNOWLEDGE. The well-posted man or the man who knows his business is the man who succeeds. Therefore, put it down as a rule that hard work in acquiring knowledge will surely bring success in speculation or business.

PATIENCE

After you have acquired knowledge you will need to learn PATIENCE, if you have not already learned the value of patience. You must learn to wait for a definite indication of a change in trend before buying or selling. You must not guess or gamble on hope or fear. You must have ability to act and to act quickly at the right time, acting after you have acquired knowledge and know it is the time to act.

COURAGE

You must have the COURAGE to act. Courage and boldness will come after you have acquired knowledge because you will have confidence in proven rules and confidence in your ability. Therefore, the acquiring of knowledge will give you the courage for action at the time when the real opportunity comes.

STUDY

A man who will not work hard and STUDY and pay in advance for success will never get it. If you will put in the time, study, and go over the records of the Dow-Jones Industrial averages from 1892 to date, you will be convinced that the rules work and that you can make money by following the main trend of the market.

FORM READING

Sighty-five per cent of what any of us learn is from what we see. It has been well said, "One picture is worth a thousand words." That is why FORM READING or the reading of various formations at different periods of time is so valuable. The future is but a repetition of the past. The same formation at tops or bottoms or intermediate points at different times indicates the trend of the market. Therefore, when you see the same picture or formation in the market the second and third time, you know what it means and can determine the trend.

You do not have to accept my word that the rules I give you will work in the future as they have in the past but you owe it to yourself to prove by past records that these rules work; then you will have the faith to follow them and make money.

CAPITAL REQUIRED

The first point to consider in operating any method on the stock market is the amount of capital required, with which you can trade and never lose your capital and over a period of 5, 10 or 15 years be able to make profits, because a method that will make profits and never lose your capital is the kind of a method that every man should follow to make a success.

As a general rule, I have always considered it advisable to use at least \$3,000 capital for every 100 shares of stock traded in and to limit stop loss orders to not more than 3 points on every 100 shares. In this way you will be able to make 10 trades on your capital and the market would have to beat you 10 consecutive times to wipe out your capital, which it will not do. Whatever amount of capital you use to trade with, follow this rule: Divide your capital into 10 equal parts and never risk more than 10% of your capital on any one trade. Should you lose for 3 consecutive times, then reduce your trading unit and only risk 10% of your remaining capital. If you follow this rule, your success is sure.

On stocks selling around \$15 to \$30 per share, you can start with a capital of \$1,500. The first trade should be made at a time when you can place a stop loss order not more than 3 points away and you should try to start when your risk will only be 1 point. In other words, with a capital of \$1,500 you must figure that you would be able to make at least 7 to 10 trades and the market would have to beat you 7 to 10 consecutive times to wipe out your trading capital. With this Method it is impossible for that to happen, provided you follow the rules and trade on definite indications.

This Method will make the most money trading in high-priced stocks, and for trading in stocks selling above \$100 per share, you should use a capital of \$4,000 and adhere strictly to all the rules.

If you want to start trading in small units of stock, use a capital of \$300 for each 10 shares and never risk more than 3 points on the initial trade. Try to make the first trade, if possible, where your stop loss order will not be more than 1 or 2 points. Never risk more than one-tenth of your capital on any one trade.

KIND OF CHARTS TO USE

A busy man or specialist should keep a weekly high and low chart on the Dow Jones 30 Industrial averages and the 15 Public Utility averages and should keep

weekly high and low charts on 5 to 10 of the leading active stocks in the different groups. He could also keep up monthly high and low charts on a few stocks selling below 20 or below 10 and watch these different low-priced stocks when they indicate a change in trend. By crossing old tops and showing activity, it would indicate a good time to buy them.

MAJOR AND MINOR TRENDS

You will always make the most money by following the main trend of the market, altho to say that you must never trade against the trend means that you will miss a lot of intermediate moves which will make big profits, but your rule must be: When you are trading against the trend, wait until one of your rules gives you a definite indication of a buying or selling point at bottom or top, where you can place close stop loss orders.

There are always two trends--a major trend and a minor trend. The minor trend is a reversal of the main trend, which lasts for a short period of time. When the main trend is down, it is much safer to sell stocks short on rallies at a point where the rules indicate that they are top than it is to buy on a reaction. In a Bull Campaign or advancing market, it is much safer to wait for minor reactions and buy when the rules indicate that it is time to buy than it is to sell short on rallies. You will always make the most money by waiting for a definite indication of the trend before buying and selling.

TREND LINE INDICATIONS

GREEN TREND LINE

This Green Trend Line indicates uptrend, either the minor or the main trend.

We use the GREEN TREND LINE for an advancing market or when a stock or the averages are making higher tops and higher bottoms each week. The first week that a higher bottom and a higher top is made than a previous week, the Green Trend Line is moved up to the top of that week. Then, as long as the stock or the averages make higher bottoms and higher tops, the Green Trend Line continues to be moved up to the high point of each week.

RED TREND LINE

The first week that a stock or the averages make a lower bottom than the previous week, the Trend Line changes to red and is moved down to the low of that week and continues to be moved down as long as the stock or the averages make lower bottoms. This Red Trend Line means that the minor or the main trend has turned down and that you should follow the RED TREND LINE until it reverses. The first week a higher bottom and a higher top are made, the Trend Line changes again to green.

BUY WHEN THE RED TREND LINE CHANGES TO GREEN

SELL WHEN THE GREEN TREND LINE CHANGES TO RED

For study purposes we have used the Dow-Jones 30 Railroad Stock averages ... weekly high and low ... from 1898 to July, 1914, because the rails were leaders during that period and moved faster, making a wider range, than the Industrial Stocks, therefore, were better trend indicators and more profitable to trade in.

You should follow the trend lines and apply the rules given here for buying points:

THREE BUYING POINTS

1 - BUY AT OLD BOTTOMS OR OLD TOPS

When a stock declines to an old bottom or to an old top, it is always a buying point with a stop loss order. In fact, you should never buy unless you can figure where to place a stop loss order 1 to 3 points away and in high-priced stocks never more than 5 points away.

Remember, it is safe to buy when a stock reacts to old tops the first, second or third time, but when it declines to the same level the fourth time, it is dangerous to buy as it nearly always goes lower.

Buy when a stock declines 1 to 3 points under old tops or old bottoms. However, a stock is always strongest if it holds just around the old tops or old bottoms and does not break 1 to 3 points under. Holding slightly higher than these old levels is a still stronger indication.

Stocks selling above \$100 a share-- After they cross old tops they can react 5 points under the old tops but not more, and if the market is really strong, they should not go as much as 5 points under, except in very rare cases when the market is in a wide trading range and very active.

2 - SAFER BUYING POINT

Buy when a stock crosses former tops or crosses a series of tops of previous weeks, showing that the minor or the main trend has turned up as indicated by the Green Trend Line.

3 - SAFEST BUYING POINT

Buy on a secondary reaction after a stock has crossed previous weekly tops and the advance exceeds the greatest rally on the way down from the top.

Buy when the first rally from the extreme bottom exceeds in time the greatest rally in the preceding Bear Campaign.

Buy when the period of time exceeds the last rally before extreme lows were reached. If the last rally was 3 or 4 weeks, when the advance from the bottom is more than 3 or 4 weeks, consider the trend has turned up and stocks are a safer buy on a secondary reaction. Examples later will prove this rule.

THREE SELLING POINTS

When we refer to selling points we mean to either sell out long stocks or sell short.

1 - SELL AT OLD TOPS OR OLD BOTTOMS

An important point to sell out longs and go short is at old tops or when a stock rallies to old bottoms the first, second or third time. As a rule, if

is risky to sell the fourth time that a stock advances to the same level because it nearly always goes higher. When you sell short, place stop loss orders 1, 2, or 3 points above old tops or old bottoms.

When prices are at high levels above \$100, the averages can go 5 points above old tops or 5 points above old bottoms without changing the main trend. But this seldom happens, as a study of previous tops and bottoms will prove. As a rule, when the market is weak and the main trend down, the rally stops right under old bottoms and should not go more than 2 points above them. If it goes 3 points above, it is an indication that the market is strong and likely to go higher. If it declines under these old levels, this indicates that the market is very weak.

2 - SAFER SELLING POINT

Sell when a stock breaks the low of a previous week or a series of bottoms of previous weeks as indicated by the Trend Line.

3 - SAFEST SELLING POINT

Sell on a secondary rally after the stock has broken the previous bottoms of several weeks or has broken the bottom of the last reaction, turning trend down. This secondary rally nearly always comes after the first sharp decline in the first section of a Bear Campaign.

Sell after the first decline exceeds the greatest reaction in the preceding Bull Campaign or the last reaction before final top.

Sell when the period of time of the first decline exceeds the last reaction before final top of the Bull Campaign. Example: If a stock has advanced for several months or for one year or more and the greatest reaction has been four weeks ... which is an average reaction in a Bull Market ... then after top is reached and the first decline runs more than 4 weeks, it is an indication of a change in the minor trend or the main trend. The stock will be a safer short sale on any rally because you will be trading with the trend after it has been definitely defined.

FORMATIONS AT BOTTOMS AND TOPS

By studying stock formations of the past you will be able to determine what is going to happen when similar formations occur in the future, just as you know that there is going to be a rainstorm when you see a heavy dark cloud form.

After accumulation or distribution at bottom or top has been completed, there is a BREAKAWAY POINT. When you buy or sell stocks at this point, you make money very quickly.

Study the volume of sales, the space and price movements and the last and most important time period. Similar action of the market occurs around the same month years apart. When we come to the Lesson on "Volume" it will give you more rules and information.

Study the different types of bottom formations-- Sharp, double, triple, flat and ascending bottoms.

SINGLE "V" OR SHARP BOTTOM

This formation can be a sharp, fast decline followed by a fast advance, or even a slow decline followed by a quick rally from the bottom with no secondary reactions until it advances to higher levels.

Example: July 26, 1910, the Dow-Jones 20 Railroad averages had a sharp decline to 105½; then rallied sharply to 114½ ... a 9-point advance with no reactions ... and did not later go back to make a double bottom.

"U" BOTTOM OR FLAT BOTTOM

This "U" bottom is a formation where a stock remains for 3 to 10 weeks or more in a narrow trading range, making about the same top and bottom levels several times; then when it crosses the intermediate tops, it has formed a "U" or flat bottom and is at the breakaway point ... a safe place to buy.

Example: March 12 to April 30, 1898 ... The 20 Railroad averages held between 56 and 60, hitting the bottom level 4 times. This was a flat bottom and when they crossed 60, the averages indicated higher prices.

"W" BOTTOM OR DOUBLE BOTTOM

When a stock declines and makes bottom; then rallies for 2 to 3 weeks or more; declines and makes a bottom around the same level the second time; then advances and crosses the previous top, it has formed a "W" or double bottom. It is safe to buy when it crosses the top or middle of the "W" ... which is the BREAKAWAY POINT.

Example: December 23, 1899, low 72½. June 23, 1900, low 73.

"W V" BOTTOM OR TRIPLE BOTTOM

This is a third higher bottom after a double bottom or three bottoms near the same level. It is safe to buy when a stock has formed a "W" and a "V" on the side and crosses the second top of the "W".

Example: Dec. 23, 1899 - 1st bottom at 72½
 Jun. 23, 1900 - 2nd bottom at 73
 Sep. 28, 1900 - 3rd bottom at 73½
 This formed a "W V" bottom and after the averages advanced above 76, the BREAKAWAY POINT ... a runaway advance followed, the market advancing 44 points in 28 weeks.

"W W" BOTTOM OR 4-BOTTOM FORMATION

This formation shows first, second, third and fourth bottoms. The safest point to buy is at the BREAKAWAY POINT or when a stock crosses the middle point of the second "W".

Example: Aug. 8, 1903, low 90½ rallied to 98½ on Aug. 22, 1903
 Oct. 17, 1903, low 99-3/8 rallied to 99-5/8 on Jan. 23, 1904
 Mar. 19, 1904, low 91½ rallied to 97½ on Apr. 16, 1904
 May 21, 1904, low 93½

When the averages crossed $97\frac{1}{2}$, the middle of the second "W" they went on thru the top of $99-5/8$, top of the first "W". This was the BREAKAWAY POINT and the market advanced 27 points without reacting to 99 again.

TOP FORMATIONS

You should study the various types of tops-- Sharp, flat, double, triple, and descending tops.

SINGLE "A" OR SHARP TOP

After a prolonged advance or at the end of a Bull Campaign, the averages or individual stocks often make a single sharp top ... advancing 17 to 28 weeks or more with only small reactions lasting sometimes 10 days to two weeks ... then follows a sharp, quick decline. It is safe to sell on a subsequent or secondary rally and safer to sell when it breaks the last leg of the "A" or when it breaks the bottom of the first sharp decline.

Example: The 20 Railroad averages advanced from $117\frac{1}{2}$ to $129\frac{1}{2}$ in 16 weeks from May 24 to September 13, 1903; then had a sharp decline to October 18, and only rallied to $123\frac{1}{2}$. When they broke $118\frac{1}{2}$, it was an indication of weakness and the Bear Market continued.

"□" TOP OR FLAT TOP

When the market makes several tops near the same level and the bottoms on reactions are near the same level ... holding in a narrow trading range ... it forms a "□" top or flat top. It is safe to sell short when it breaks under the series of weekly bottoms.

Example: May 13 to July 29, 1911, the Railroad averages held in a range of $2\frac{1}{2}$ points for 8 weeks, forming a flat top, and the market was a safe short sale when the averages broke 121, under the low point of this narrow range.

"M" TOP OR DOUBLE TOP

When a stock or the averages reach top after a substantial advance; then react 3 to 7 weeks or more and rally again to around the same top, an "M" or double top is formed. Then, when it declines and breaks under the low of the last reaction or under the middle of the "M", it is safe to sell short.

Example: 1906, September 15 high $137\frac{1}{2}$; November 17 low $131\frac{1}{2}$; December 15 high $137\frac{1}{2}$; then followed a quick decline, breaking the low of $131\frac{1}{2}$, and a Bear Market followed.

"M A" TOP OR TRIPLE TOP

This formation occurs when a stock or the averages make 3 tops near the same level or the 2nd and 3rd tops are slightly lower. When these formations are made at tops after a long advance, they are signals for a major decline. The more time between tops, the stronger the indications for a big decline. It is safe to sell when a stock breaks the last bottom or breaks the end of the "M" and safer when it breaks the bottom of the "A" ... which is the BREAKAWAY POINT.

Example: 1st top - Jan. 19, 1906, high 138-3/8 ... end of Bull Market;
 May 5, 1906, sharp decline to 130 1/2;
 2nd top - Sep. 22, 1906, high 137 1/2, slightly lower than Jan. 1906 top;
 Nov. 17, 1906, low of 8-point reaction;
 3rd top - Dec. 15, 1906, high 137 1/2, near same level ... a signal for
 a big decline. Panic followed in 1907 and the averages de-
 clined 57 points in 11 months.

"M" OR 4-TOP FORMATION

This formation occurs when a stock or the averages make 4 tops at the same lev-
 el or slightly lower. It is safest to sell short when a stock breaks the second
 point of the second "M" or the low of the last reaction.

Example: 1st top - July 22, 1911, high 124;
 2nd top - Aug. 17, 1912, high 124;
 Sep. 14, 1912, reacted to 120 1/2;
 3rd top - Oct. 5, 1912, high 124-3/8;
 Nov. 2, 1912, reacted to 119-5/8;
 4th top - Nov. 9, 1912, last rally to 122 1/2;
 When the averages broke under 119-5/8, low of
 the last reaction, this was the BREAKAWAY
 POINT and a big decline followed.

SECTIONS OF MARKET CAMPAIGNS

A Bull or Bear Campaign in stocks or the averages runs out in 3 to 4 sections--

BULL MARKET

1st Section - Advance after final bottom; then a secondary reaction,

2nd Section - Advance to higher levels, above the highs of previous
 weeks and of the first advance; then a reaction.

3rd Section - Advance to new high for the move. In many cases this
 means the end of the campaign, but you must watch for
 a definite indication before deciding that the 3rd run
 up means a change in the main trend.

4th Section - Often four sections are run out and this 4th move or
 run-up is the most important to watch for the end of a
 Bull Campaign and a change in trend.

Minor Bull Campaigns of short duration, running one year or less, often run out
 in two sections, especially if the first section is from a sharp bottom. There-
 fore, always watch the action of the market after the second advance to see if it
 is forming a top and gives indications of a change in trend.

BEAR MARKET

A Bear Campaign runs opposite to a Bull Campaign--

1st Section - There is a sharp, severe decline which changes the main

trend; then a secondary rally on which stocks are safer short sales. That marks the end of the 1st Section.

2nd Section - Then there is a second decline to lower prices, followed by a moderate rally.

3rd Section - A third decline or move to still lower prices, which may be the end of the campaign.

4th Section - There is often a 4th move, when you must watch closely for bottom. In determining whether it is final bottom, you use all of the other rules ... watching old tops and old bottoms for definite indication that the main trend is ready to change.

Minor Bear Campaigns of short duration, running one year or less, often run out in two sections, especially if the 1st Section is from a sharp top. Therefore, always watch the action of the market after the 2nd decline to see if it is forming a bottom and gives indication of a change in trend.

In extreme cases, like 1929 and the Bear Campaign which followed from 1929 to 1932, there are as many as 7 sections up or down, but this is abnormal and unusual and only occurs many years apart.

Go back over all of the campaigns we have worked out and you will see how these sections or moves run out.

HOW TO DETERMINE CHANGE IN THE MAIN TREND

SPACE MOVEMENT

RULE FOR DETERMINING CHANGE IN TREND ACCORDING TO SPACE MOVEMENTS--

When a decline in points exceeds the greatest decline of a previous reaction by one or more points, it is an indication of a change in trend.

When the market has run out 3 or more sections in a Bull Campaign, go back over the record and find out what the greatest reaction has been in any section, whether 10, 15, 20, 30 points or more. Suppose the averages have been advancing for a long time and the greatest reaction in the Bull Market has been 10 points and the market has reached the 3rd or 4th Section of the campaign. Then the first time the averages or the price of a stock breaks back more than 10 points, or more than the greatest reaction, it is an indication that the main trend has changed or will change soon. This does not mean that a rally cannot take place after this definite signal of reversal has been given, as usually after the first signal of a change in trend is given, there is a secondary rally in a Bull Market and time has to be allowed at top for distribution to take place. Therefore, just because you get a definite indication that the main trend has changed, do not jump to the conclusion that you can sell short right at that time and that there will be no rally. Always sell on rallies, if possible, altho there are times that you can sell at new low levels or when bottoms are broken. Judge this by applying all of your rules.

TIME PERIOD -- Another way to tell when the main trend is changing.

RULE-- When a campaign has only 3 or 4 sections and the TIME period on a reaction exceeds the greatest time of a previous reaction, consider that the main trend has changed.

Go back over the record and find the greatest time period from any minor top to the duration of a reaction in previous sections of the Bull Market. If you find that the greatest reaction has been about 4 weeks, then the first time the market declines consecutively for 5 weeks or more, it is an indication that the main trend has changed and that stocks will be short sales on a secondary rally.

Apply the same rules in a Bear Market. When the SPACE movement or number of points that a stock or the averages have rallied during a Bear Campaign is exceeded, the trend is changing. When the TIME period of the greatest reverse movement is exceeded, then the main trend is changing and a Bull Campaign is starting.

A reversal in SPACE movement after the second run or 2nd Section in a Bull or Bear Market, would not mean as important a change in trend as if it came after the 3rd or 4th Section had run out, either up or down.

Examples of all these rules will be referred to in the working out of each Bull and Bear Campaign from 1893 to 1933 so that you will know just how to apply the rules in future market movements.

MANY WEEKS IN NARROW RANGE

When a stock or the averages hold for 3 to 6 weeks or 10 to 15 weeks in a narrow range, then cross tops or break bottoms of previous weeks, the trend has changed and you should go with it. The longer the period of time in a narrow range, the greater the advance or the decline when they break out of the range either up or down.

Example: 1897, January 16 to March 20, the Railroad averages held in 2-point range; then broke the low of 52 and went lower. 1897, April 10 to May 23, held in a 2-point range for 6 weeks; then crossed the top at 61 and the Bull Market started.

ACCUMULATION OR DISTRIBUTION ON SIDE

After a market has advanced to the 3rd or 4th Section, then has a sharp decline and rallies, it will often remain for a long period of time in a range while DISTRIBUTION is taking place ON THE SIDE. The top of the range may be several points below the extreme high. In FORM READING it is very important to note the range from high to low in this zone of distribution. The stock is a short sale at the top of this range of distribution on the side and a safer sale when it breaks the low point of this range ... which is the BREAKAWAY POINT.

At the end of a Bear Campaign, after the first sharp advance there is a secondary reaction; then a long period of ACCUMULATION ON THE SIDE, with several moves up to the top of the range and back to the bottom of the range of accumulation. The market is a buy at the bottom of this range and a safer buy when the tops of this range are crossed as that is the BREAKAWAY POINT and a signal for a fast advance.

Examples of these sideways accumulation and sideways distribution will be shown in the different campaigns worked out.

LAST STAGE OF BULL OR BEAR MARKET

In fact, advancing markets in the last stage of the campaign reactions get smaller as stocks work to higher levels, until the final section or run has ended. Then comes a sharp, quick reaction and a reversal in trend.

In the last stage of a Bear Market, after all old bottoms and resistance levels have been broken, rallies get less or smaller as prices work lower. Therefore, people who buy have no chance to sell on rallies until the final bottom has been reached and the first rally takes place.

This is why it never pays to buck the trend in the last stage of a Bull Market or the last stage of a Bear Market.

RANGE OF BOTTOMS

Never consider that a major or a minor trend has reversed or changed until the bottoms of previous weeks have been broken or the tops of previous weeks have been crossed. The number of points that a stock or the averages should decline below a bottom to indicate a change in trend to lower levels, varies according to the price at which the averages or the stock is selling. We consider a range within 1 to 3 points a double or triple bottom or a double or triple top. In a strong market a stock will break only 1 point under a bottom and then rally and, in extreme cases, not more than 2 points. As a rule when bottoms are broken by 3 full points it is an indication for lower prices before any rally of importance.

RANGE OF TOPS

The same at the top. The range for double tops is about 3 points. These tops can be in a range of 1 to 3 points and still be considered double and triple tops. Advancing 1 to 2 points above an old top does not always indicate that the main trend has changed and that stocks are going up immediately but advancing 3 points above old tops is nearly always a definite indication that higher prices will follow before much reaction. At the end of Bull or Bear Markets some false moves are often made and quick reversals follow.

HOW FAR SHOULD STOCKS DECLINE BELOW OLD TOPS AFTER THEY ADVANCE ABOVE THEM?

In order to still show uptrend, after a stock or the averages have advanced above old tops, then reacted, when in strong position they will stop right around the old tops or sometimes go 1 to 2 points below the old tops but seldom more than 3. Regardless of how high a stock is selling, a decline of more than 5 points below the old top would indicate that the trend had reversed and at that immediate time the stock would not go higher but would go lower for awhile. It can decline 5 points under old tops and still be in a Bull Market, all depending on what section the market is in. A signal in the last section is most important.

HOW FAR STOCKS CAN GO ABOVE OLD BOTTOMS IN BEAR MARKETS

Reverse this rule in a Bear Market. When stocks advance to old bottoms, they are short sales because bottoms become tops and tops become bottoms. They should not go more than 1 to 3 points above the old bottoms and on an average should not go more than 3 points. Therefore, even when stocks are at high levels if they advance more than 5 points above an old bottom, it is an indication that they are going higher and are not going to work lower immediately with the main trend.

SEVERAL FAST MOVES UP OR DOWN OVER A RANGE OF 5 TO 7 OR 10 TO 12 POINTS

Whether the market is very active or in a slow trading range, all indications are more accurate and more valuable when the market is quite active.

After a stock or the averages have been advancing for some time and have run out 3 or 4 section, if there are several moves of 10 to 12 points up or down in a range, making several bottoms and several tops in this range, it indicates either accumulation or distribution. When the bottoms in a range of this kind are broken it is an indication of lower prices and when the tops in a range of this kind are crossed, it is an indication of higher prices. Note the range on sideways accumulation and the range on sideways distribution.

An advancing market may have several reactions of 10 to 12 points; then have a reaction of 20 to 24 points; then after an advance if it declines from any top more than 20 points, it will usually run 30 to 40 points. Go over the stocks or averages when they have been selling at very high levels and prove to yourself the value of this rule.

WHEN THE MARKET IS IN STRONGEST POSITION OR IN WEAKEST POSITION

A stock or the averages are in the strongest position after there has been a prolonged decline and the market starts making higher bottoms, especially after a sharp, fast decline, when rallies have been small. After the 2nd or 3rd higher bottom has been made and then the top of a previous rally has been crossed, the stock is in the strongest position. Rising bottoms always indicate strength and an advance usually starts from the 3rd or 4th higher bottom, that is, the big advance which runs for a long time with only small reactions. You make money quickest when you get in on a move of this kind.

Reverse this rule in a Bear Market. A market is weakest when it is making lower tops. The 3rd or 4th lower top is the safest place to sell. After the 3rd or 4th lower top, when it breaks the last low or previous bottom, it is in the weakest position and indicates that the main trend is down and declines will be faster.

HOW TO DETERMINE CHANGE IN MINOR TREND

MINOR ADVANCE

When a market is advancing and makes a top around the same level for 2 or more weeks, especially when the range is very narrow near top levels, then prices break under the bottom of 2 weeks or more, the minor trend has turned down and you would follow it until there is another definite indication of a change in trend.

MINOR REACTION

After the averages or a stock has been declining for several weeks or several months and prices make bottom 2 weeks or more around the same level and hold in a narrow trading range for 2 weeks or more and then cross the tops of 2 or 3 weeks on the upside, the minor trend has changed, at least temporarily, and you should go with it.

DULL MARKET

A dull market, in a narrow trading range at any point, indicates that it is getting ready for some kind of a change and you should follow it whichever way it breaks out, up or down, after these narrow, dull periods.

DURATION OF MINOR MOVES

RULE FOR TIME OF MINOR REACTION IN BULL MARKETS--

In an advancing market or Bull Campaign when indications are given for a minor reaction, prices will react 3 to 4 weeks but as a rule in the 4th week they will rally and close higher. In some cases there will only be a sharp, quick reaction of 2 weeks and then the main trend will be resumed, but after a decline of 3 to 4 weeks should the market have a minor rally; then break back under the bottom of the 3rd or 4th week, it will be an indication of a greater change in the trend and probably a change in the main trend.

In extreme cases, after a secondary rally in a Bull Market, prices will decline as much as 6 to 7 weeks, seldom more, before the main trend is resumed.

RULE FOR TIME OF MINOR RALLIES IN BEAR MARKETS--

These rules are reversed in a Bear Market. Rallies in a Bear Market last 2 to 4 weeks. Should a rally hold into the 5th week, it is likely to run into the 6th or 7th week; then you can watch for an important change in trend.

After a rally of 3 to 4 weeks and a secondary reaction in a Bear Market, should the market advance and cross the level made at the end of the 3 or 4 weeks' rally, then the trend is changing, at least temporarily, and higher prices ... even in a Bear Market ... are indicated.

HOW TO DETERMINE FIRST CHANGE IN TREND BY OPENING AND CLOSING PRICES

The Weekly high and low Chart with the opening and closing prices on it is one of the best charts to use for determining trend of stocks or the averages. That is why we have used the Weekly Chart on the Dow-Jones 30 Industrial averages and given all the indications according to it.

The closing price at the end of the week or at the end of any day is most important because regardless of how high or how low stocks have been during the week or during the day, the closing price shows exactly what they have lost or gained at the end of the period.

After a stock or the averages have been advancing for a considerable period of time and have reached an old top or a section of the campaign which indicates that a change in trend could take place, it is very important in very active markets to watch the opening price on Monday morning. Should the averages or a stock not sell more than one point above the opening price on Monday morning and then decline and close at or near the low levels of the week on Saturday, it is the first indication of a change in trend on the Weekly Chart. But don't fail to use your other rules and wait for the proper declines in TIME or SPACE or until previous weekly bottoms have been broken before deciding that there is a definite change in trend.

Reverse this rule at the bottom of a decline or after there has been a sharp, fast decline in a very active market. Watch the opening price on Monday morning. Should a stock decline rapidly to around the middle or latter part of the week, then after making bottom, reverse and rally quickly and on Saturday close at or near the opening on Monday morning or above this opening. It is an especially strong indication that the trend is reversing and that prices are going higher temporarily. But use all of your other rules ... the crossing of previous weekly tops ... the reversal of the SPACE movement in points ... also reversal in TIME .. before deciding that a definite change in trend is indicated.

It is also very important at the end of the week if prices close under old bottoms or under bottoms of previous weeks or close above old tops or above tops of previous weeks as it is an indication of weakness or strength.

WHEN STOCKS ADVANCE INTO NEW HIGH GROUND OR DECLINE INTO NEW LOW GROUND

When stocks advance to higher levels than they have been before in their previous history or decline to lower levels than they have ever sold before in their history, they are in new high or new low ground and you must have a rule to follow when stocks reach new high or new low record prices.

First, apply all of your other rules and do not buy or sell until there is a definite indication of a change in the main trend.

When a stock or the averages advance to a new high level that they have never sold at before in their history, it is an indication of an advance of 7, 10, 15, 20 to 24 points or more, especially if it is in the 1st or 2nd Section of the Bull Campaign. Should new highs be made in the 3rd or 4th Section, then the advance into new high ground may not be very many points before final high is reached and the trend changes.

You should watch the action of the market when it has advanced 7 points into new high territory, 10 points, again around 15 points, and on an extreme advance watch around 20 to 24 points, where there is likely to be resistance and top made. These are average moves and it depends upon the activity and the price of a stock whether it will stop on any of these points. By following the trend indications and rules you will be able to determine when the first move into new high territory has run out and the trend has changed.

Reverse the above rules in a Bear Campaign.

When the averages or stocks selling between 50 and 125 advance into new high ground, these average movements of 7, 10, 15, 20 and 24 points, work out quite accurately. After stocks advance above 125, they move faster and go higher than they do below 125. Above 200, the movement increases and the range is much wider. The same when stocks or the averages get above 300 ... they have a still greater range and faster moves, all of which depends upon how long the campaign has been running and how many moves or sections the campaign has run out and how high the stocks are from the last bottom or how far down from the last top.

WHEN TO USE DAILY HIGH AND LOW CHART

When markets are very active and fluctuating over a wide range, especially in the last stages of a Bull Market or the last fast decline in a Bear Market, you should keep up a Daily high and low Chart on the averages or individual stocks and put on the Trend Lines, applying the same rules as used for the Weekly Chart, because the Daily Chart will give you the first change in the minor trend, which may later be confirmed by the Weekly Chart into a change of the main trend. Full instructions will follow under "Resistance Levels."

Again I remind you ... Don't try to get ahead of the market. Don't guess if it is making a change in trend and be wrong. Wait until it shows a definite change in trend. You will be right when you form your judgment after definite indication is given according these rules.

July 6, 1939

DOW-JONES 20 RAILROAD AVERAGES

BOTTOM AND TOP FORMATIONS

1896 - 1914

1896	Aug. 8	Sharp bottom or single "V" - Low 42.
1897	Jan. 16	to 23 and March 20 - Flat double top "M".
	Apr.	Sharp bottom and 4 weeks narrow on side in 1-point range.
	Sep.	Sharp or "A" top at 1/2 point from 42 to 92.
	Nov.	Sharp or "V" bottom.
1898	Feb.	Flat top ... 2 weeks in 1-point range.
	Mar. 5	to April 23 - Flat bottom ... triple bottom near same level.
	Aug.	Sharp top.
	Oct. 1	to Nov. 5 - Flat bottom ... 6 weeks in 1-point range.
1899	Apr. 1	to 29 - Flat top ... 5 weeks in 2-point range.
	Jun. 3	Sharp bottom.
	Jul. 29	to Sep. 9 - Flat top ... 7 weeks in 2-point range; 1st lower top after April.
	Nov.	Sharp top ... 4 weeks on side in 2-point range ... 2nd lower top.
	Dec. 23	Sharp bottom.
1900	Apr. 7	Sharp top ... 4th lower top.
	Jun. 23	Sharp bottom ... 1st higher bottom and double bottom against Dec. 1899 low.
	Aug. 18	Flat intermediate or middle top.
	Sep. 29	Sharp bottom ... a triple bottom and 3rd higher bottom ... strong and good buying point.
1901	May 4	Sharp top.
	May 9	Sharp bottom ... panic.
	Jun. 15	Sharp top ... 2 narrow weeks on side ... a double top.
	Aug. 10	Sep. 14 and Oct. 12 - Sharp bottoms ... triple bottoms with 2nd and 3rd bottoms higher than the 1st ... strong.
1902	Sep. 13	A sharp top and sharp decline followed.
	Nov. 15	and Dec. 13 - Two sharp bottoms ... double bottoms.
1903	Jan. 10	Sharp top with 5 weeks on side in 2-point range.
	Jul. 11	Sharp top ... 2 weeks rally followed.
	Aug. 22	Sharp top ... 3 weeks on side in 2-point range.
	Sep. 19	to Nov. 14 - Triple bottoms ... 3rd bottom higher.
1904	Jan. 23	Sharp or "A" top.
	Feb. 27	to Mar. 19 - Flat bottom ... 3 weeks in 2-point range.
	Dec. 3	Sharp top ... 3 weeks' reaction.
1905	Mar. 18	and Apr. 16 - A double flat top.
	May 27	Sharp bottom.
	Sep.	to Nov. - Flat top but only declined 4 points.
1906	Jan. 19	Final sharp top ... sharp decline followed.
	May 6	Sharp bottom ... sharp rally followed.
	Jun. 16	Sharp top ... sharp decline followed.
	Jul. 7	Sharp bottom ... sharp rally followed.
	Sep. 18	to Oct. 13 - Flat top ... 7 weeks in 2-point range, just under January top.
	Nov. 17	to Dec. 22 - Flat top ... 4 weeks in 3-point range. A panicky decline followed this 3rd lower top.

BOTTOM AND TOP FORMATIONS 1896 - 1914 (Continued)

1907	Feb. 14	to 25 - A sharp bottom,
	Apr. 6	to May 11 - Sharp double tops.
	Jun. 9	Flat bottom ... 4 weeks in 2-point range.
	Jul. 6	to 27 - Flat top ... 4 weeks in 3-point range.
	Nov. 23	Sharp bottom and sharp rally followed.
1908	Jan. 18	Sharp bottom and sharp decline followed.
	Feb. 15	to Feb. 7 - Flat narrow bottom ... 3 weeks in 1½-point range.
	May 23	Sharp top.
	Jun. 27	Sharp bottom.
	Sep. 12	Sharp top.
	Sep. 22	Sharp bottom ... quick, sharp rally followed.
1909	Jan. 2	Sharp top.
	Feb. 27	Sharp bottom.
	Aug. 14	Sharp top followed by sharp reaction.
	Sep. 11	Sharp bottom.
	Sep. 18	to Oct. 9 - Flat top ... 4 weeks in 2-point range ... 1st lower top.
1910	Jul. 26	Sharp bottom ... end of Bear Market ... quick rally followed.
	Oct. 28	Sharp top.
	Dec. 10	Sharp bottom.
1911	Feb. 4	to 18 - Flat top ... 3 weeks in 2-point range.
	Feb. 11	and Apr. 23 - Double bottoms.
	Jun. 10	to Jul. 29 - Flat narrow top ... 7 weeks in 2-point range.
	Sep. 30	Flat bottom ... 4 weeks in 2-point range.
1912	Aug. 17	to Oct. 3 - Double tops ... narrow range ... 3rd top from July 23, 1911. A prolonged decline followed.
1913	Jun. 14	Sharp bottom.
	Nov. 15	to Dec. 20 - Flat double bottoms.
	Sep. 13	to 27 and
1914	Jan. 24	to Feb. 7 - Double flat tops ... narrow range.
	Apr. 25	A sharp bottom ... slow rally followed.
	May 23	to July 11 - Flat top ... 9 weeks in 2-point range ... a panicky decline followed.

BREAKAWAY POINTS IN BULL & BEAR MARKETS
DOW-JONES 30 INDUSTRIAL AVERAGES
STUDY THESE POINTS ON THE CHART

YEAR				WEEKS	POINTS
1897	June	5	to Sept. 4	12	16
1897	Oct.	2	Mch. 26, 1898	10	14
1898	May	7	Aug. 28	12	15
1898	Dec.	3	Apr. 29, 1899	9	20
1899	Dec.	9	Dec. 23	3	15
1900	Oct.	20	Apr. 20, 1901	10	18
1901	July	13	Oct. 14	15	14
1903	Mch.	28	Oct. 17	29	25
1904	July	9	Apr. 18, 1905	40	33
1905	June	24	Jan. 19, 1906	30	28
1906	Feb.	17	May 5	8	14
1906	July	28	Oct. 13	11	6
1907	Feb.	9	Mch. 25	7	16
1907	July	6	Nov. 15	19	25
1908	Feb.	29	Aug. 14	23	28
1909	Apr.	3	Oct. 2	25	16
1910	Jan.	3	Feb. 12	6	13
1910	Apr.	30	July 26	13	16
1910	Oct.	1	Oct. 29	4	6
1911	Aug.	5	Sept. 30	8	12
1911	Nov.	11	Sept. 7, 1912	46	15
1912	Dec.	7	Mch. 22, 1913	15	12
1913	Apr.	26	June 14	7	9
1913	July	19	Sept. 13	8	8
1913	Oct.	4	Dec. 20	11	6
1913	Dec.	27	Mch. 24, 1914	12	5
1914	July	25	Dec. 24	21	27
1915	Mch.	20	May 1	6	13
1915	July	24	Dec. 30	32	29
1916	Jan.	22	Apr. 29	14	10
1916	Aug.	12	Nov. 14	14	13
1916	Dec.	9	Feb. 3, 1917	8	12
1917	June	16	Dec. 22	26	30
1918	Jan.	5	Oct. 19	41	14
1919	Feb.	22	July 19	21	30
1919	Aug.	2	Aug. 23	3	10
1919	Aug.	30	Nov. 8	10	18
1919	Nov.	15	Feb. 26, 1920	17	23
1920	Mch.	6	Apr. 10	6	10
1920	Apr.	24	May 29	5	14
1920	July	24	Dec. 25	23	23
1921	May	21	Aug. 24	15	13
1921	Sept.	10	Oct. 28, 1922	59	13
1923	Mch.	24	Aug. 4	19	16
1923	Nov.	10	Feb. 9, 1924	13	10
1924	Mch.	23	May 24	10	9
1924	June	14	Aug. 23	16	14
1924	Nov.	6	Mch. 14, 1925	16	16

BREAKAWAY POINTS IN BULL & BEAR MARKETS
DOW-JONES 30 INDUSTRIAL AVERAGES (Cont.)
STUDY THESE POINTS ON THE CHART

YEAR			WEEKS		POINTS
1925	May	9	Feb. 27, 1926	41	36
1926	Mar.	1	Apr. 17	9	19
1926	June	5	Aug. 21	11	16
1926	Sept.	12	Oct. 16	5	12
1927	Apr.	16	Sept. 24	22	31
1927	Sept.	30	Oct. 15	2	14
1928	Mar.	17	May 12	8	14
1928	Aug.	18	Oct. 8	15	68
1928	Oct.	8	Oct. 22	2	24
1928	Oct.	22	Feb. 9, 1929	16	40
1929	Mar.	22	Mar. 30	1	4
1929	Apr.	30	May 11	2	8
1929	May	18	June 8	3	9
1929	July	8	Sept. 3	8	56
1929	Sept.	7	Nov. 13	9	160
1929	Nov.	16	Dec. 14	4	46
1930	Jan.	25	Apr. 17	13	36
1930	Apr.	19	July 5	11	64
1930	Sept.	20	Nov. 15	8	62
1930	Nov.	29	Dec. 20	3	27
1931	Feb.	7	Feb. 24	3	20
1931	Mar.	7	Jun. 8	12	64
1931	Jun.	9	Jul. 27	2	27
1931	Jul.	2	Aug. 8	6	20
1931	Aug.	31	Oct. 3	6	45
1931	Oct.	10	Nov. 9	4	24
1931	Nov.	16	Jan. 9, 1932	8	42
1932	Mar.	19	July 8	17	58
1932	July	16	Sept. 8	8	31
1932	Sept.	7	Oct. 30	7	16
1933	Feb.	4	Feb. 27	3	8
1933	Apr.	8	Jul. 17	17	44
1933	July	18	Jul. 21	1	19
1933	Sept.	23	Oct. 21	4	15
1933	Nov.	18	Feb. 5, 1934	12	15
1934	May	8	Jul. 26	12	17
1934	Nov.	24	Feb. 23, 1935	14	8
1935	Apr.	13	Apr. 6, 1936	52	54
1936	Apr.	10	Apr. 30	4	15
1936	May	30	Nov. 21	26	32
1936	Nov.	28	Dec. 26	4	4
1937	Jan.	16	Mar. 8	8	8
1937	Mar.	27	Jun. 17	14	22
1937	Jul.	3	Aug. 14	6	20
1937	Aug.	21	Oct. 19	9	65
1937	Oct.	19	Oct. 29	1	18
1937	Nov.	23	Jan. 15, 1938	7	15

BREAKAWAY POINTS IN BULL & BEAR MARKETS
 DOW-JONES 30 INDUSTRIAL AVERAGES (Cont.)
 STUDY THESE POINTS ON THE CHART

YEAR				WEEKS	POINTS
1938	Mar. 10	Mar. 31		3	28
1938	Jun. 25	Jul. 25		4	24
1938	Sept. 17	Sept. 26		1	10
1938	Sept. 28	Nov. 10		7	25
1939	Jan. 14	Jan. 28		2	14
1939	Mar. 18	Apr. 11		3	25
1939	Apr. 12	Jun. 9		8	15
1939	Jun. 10	Jun. 30		3	7
1939	Jun. 30	Jul. 25		3	12

TIME PERIODS FOR DIFFERENT FORMATIONS

DOW JONES 20 RAILROAD AVERAGES

The RUN OR ADVANCES

AFTER BREAKAWAY

IN A BULL MARKET

EXAMPLES:

Nov.	1898	to	April	1899.	in 21 weeks advanced 38 points after the breakaway.
Mar.	1902	to	Sept.	1902.	in 21 weeks advanced 15 points.
July 9,	1904	to	Mar.	1906.	in 36 weeks advanced 29 points.
June 24,	1905	to	Jan. 19,	1908.	in 27 weeks advanced 19 points.
Mar. 7,	1906	to	Sept. 12,	1908.	in 26 weeks advanced 20 points.
Mar. 20,	1909	to	Aug. 14,	1909.	in 21 weeks advanced 16½ points.

BREAKAWAY POINTS

IN A BEAR MARKET

EXAMPLES:

Feb. 21,	1903	to	Oct. 13,	1903.	in 33 weeks declined 30 points.
Dec. 22,	1906	to	Mar. 25,	1907.	in 15 weeks declined 26 points.
Aug. 10,	1907	to	Nov. 21,	1907.	in 15 weeks declined 24 points.
Jan. 15,	1910	to	July 26,	1910.	in 27 weeks declined 22 points.
Aug. 5,	1911	to	Sept. 30,	1911.	in 8 weeks declined 12 points.
Dec. 7,	1912	to	June 14,	1913.	in 27 weeks declined 17 points.
July 18,	1914	to	Dec. 23,	1914.	in 23 weeks declined 11 points.

HOW TO MAKE THE STOCK MARKET MAKE MONEY FOR YOU

BY
TED WARREN 1963

THE FOUR BASIC PHASES / CYCLES OF STOCK PRICES.

- (1) **ACCUMULATION** - Quiet low level period, stock is being accumulated from weak shareholders by the manipulators. Volume is comparatively low "wearing out" the weak shareholders resolve to hold the stock.
- (2) **MARKUP** - The stock is now in strong hands (manipulators & insiders) and the stock begins to rise.
- (3) **DISTRIBUTION** - Stock prices have neared or reached their peak, volume is high, the public is insanely excited ("Tulipmania") into buying at a pace that far exceeds the rate at which they sell. The stock is transferred from strong to weak hands.
- (4) **MARKDOWN** - The stock has been successfully distributed to the weak hands of the public and the price of the stock begins to inevitably fall.



"A stock is never a bargain unless it is in a range where it is changing from weak to strong hands; changing to those who KNOW what they are doing, and WHY."

"There can never be a sound base, unless a large percentage of the public has for one reason or another sold out."

"When a stock appears unable to rise above an apparent 'resistance level', this is almost positive proof that it eventually will, because this action induces public selling."

"To be a successful investor, you must learn to recognize what kind of action is discouraging to the public, and then form a contrary opinion."

"Buy a stock after a low price range or base of several years has formed, especially, after it has turned 'extremely quiet' (low volume) for several months."

"From a sound base, you should expect at least a 200% profit." Additional technical strength would justify higher profit expectations.

"The sounder the base, the better the chance of a very large rise." You can't build skyscrapers on weak foundations.

"MORAL : an investor should NEVER sell at a loss !"

" There is no better proof that a stock will go up, than when it acts as if it can't. "

LONG TERM HEAD & SHOULDERS BOTTOM FORMATIONS : " The depressing influence of a head and shoulders bottom, which is so necessary during the accumulation phase, lies in the fact that the public is frightened (*' the shakeout '*) during the first shoulder and the head. During the second shoulder, they sell out their positions in fear that the price will drop down again. They have lost hope in the stocks ability to go up. "

" You would not be justified in expecting much over a 100% rise from a head and shoulders bottom formation. Of course, if it is in conjunction with another base formation, then you may raise your sights in anticipating a much larger rise. "

GENERAL MARKET BREAKS : " There comes a time when *panic selling* hits the stock market. When the public panics, for a period of a month or so, it is a rare stock that does not drop in its value. Some will merely sag, while others will break wide open. For others, this break may be the beginning of a long, drawn out accumulation base period. NEVER consider buying these weak stocks after a sharp break no matter how low they may appear. Rather than buying stocks at what "publicly" appear to be low prices after a general break, check the charts closely for those few stocks that resisted the general market break. They will attract very little attention by their action (or lack of it), but they show one of the most positive indications of internal technical strength. Heavy selling fails to develop in these particular stocks because the public majority of Milquetoasts and Nervous Nellies have long ago sold out. *The minor selling that does appear is well taken by the Smart Money.* "

" An investorator must realize that a fair percentage of the public must be sold out before a stock has a chance for a profitable rise. "

" Long Term trend lines cannot be ignored ! The extremely steep downtrend lines, when they are broken by an upswing are the trend lines of least value (i.e., the least reliable). "

" The slow gradual downtrend line of at least two years duration, that forms after a stock has had a very large drop, and ends with extreme quietness, gives a near perfect buy signal when broken. "

UPTRENDS : " If you bought a stock on a sound base, you should seldom consider selling as long as the rise is small or merely slow, no matter how high it has moved. Slow uptrends when broken, should be ignored. Usually, they are only interruptions (or shakeouts) during the markup. The rise is too slow (and quiet) to be considered a distribution. "

SELLING YOUR STOCK POSITION : " There are two prerequisites to be met before you should consider selling your stock position : a fast rate of climb and the percentage of rise from its base. Distribution can seldom take place unless the public are insanely excited into buying at a pace far above that at which they are selling. This excitement can be generated by the mere rising of stock prices (Tulipmania). With proper assistance by the manipulators, when (in their opinion) the time is ripe, a rise will be accelerated and publicized to a degree that will give confidence to even the most timid Mr. Milquetoast. It is then that distribution takes place. There are a wide variety of distribution tops, both in periods of time and range of price. During these tops, there are heavy buying waves by the public (high volume). It is quite possible for the insiders to sell in six months what it took them sixteen years to accumulate. "

Another prerequisite is : " Has the stock risen to heights you were justified in believing it should go, according to the length of time and type of action during its previous base ? In other words, the sounder the foundation, the higher the rise to be expected. From a base of four years, you should expect a minimum rise of over 200% from its average base price. A longer base would indicate more. If, as the

early slow markup proceeded, it spent years at one or two price levels consolidating, this would be a positive buildup of more internal strength and you can expect another 100% - 300% rise based on the original base price. *

* You must sell when an uptrend line of over 45 degrees has been positively broken, providing that the percentage rise has been at or above the minimum reasonably expected according to its base formation. *

* NEVER sell a stock short when trading volume is extremely light (quiet) *

Buying stocks on margin is a smart way to leverage your purchasing power.

VOLUME : * Volume is always comparatively light (quiet) when a stock is under accumulation, the "*wearing out* " process of inducing the weak shareholders to sell out. The public selling is light, but their buying is even less. *

* Volume must be heavy during distribution. *

* Stocks can show technical strength by the lightness of its volume. The stock is in strong hands and there is little of it for sale. *

STOCK SPLITS : * I have a different viewpoint as to why most stocks are split. It is my opinion, that the real reason is not to have the shares more widely held ; it is to make the shares easier to be widely "*distributed*" at the top. It is easier to "distribute" more shares at a lower price than it is to "distribute" less shares of the stock at a higher price. The public is dollar conscious as to the stocks price but not dollar wise. *

The Futures Markets

When an event seems least likely to occur from a traders viewpoint, that's when it is most likely to happen.

To be successful in the markets (Stock & Futures) you must learn to interpret how the public's opinions are being manipulated and then you must form the opposite opinion.

" When an option appears to have lost any ability to rally, you should pay close attention to it. It is from these lengthy low areas that many of the unrealistic large moves start. "

" The profit potential in commodity trading is terrific, but the odds against becoming a successful speculator are almost insurmountable. The price actions of futures is so highly manipulative that at times a trader hardly knows whether he is coming or going. Taking a loss becomes a way of life and a trader literally gets to a point where he will run at the sight of his own shadow. Shakeouts and false starts invariably make him wrong at the right time (and right at the wrong time). What he thinks is rationalizing, becomes just plain emotionalism. "

" When the public plays the tables of Las Vegas, they know what they are up against. There remains only a dim hope that lady luck could smile upon them. But when people try to hit the jackpot in the wheat futures, they seriously think they can do it. In Reno and Las Vegas only the odds are against the public. On the Chicago Board of Trade and the other exchanges, the cards are stacked against them as well. "

" Fear is your worst enemy in the futures markets and the manipulators most powerful weapon. Fear can come under many different disguises and the masters of manipulation are experts at instilling fears. I now realize that fear of missing a move was primarily my worst weakness, causing me to get into the market much too often. "

" Only enter the futures markets after a proper formation such as an accumulation triangle, especially a flat topped triangle, or a long consolidation period of many weeks having a positive resistance level established to give the false impression of the commodity being unable to go up. If you are lucky enough to have a bottom (Head & Shoulders or W-Formation), or a good formation show up once a year among all these commodities, and play it heavily by buying on a scale-up during the early part of the move, you will become a millionaire in less time than I did. Playing for the small and intermediate moves is what kills you. "

Chart Patterns

Flat Top Triangle " Bear Trap " - This type of triangle formation is what really turns a Bull into a Bear to their sorrow. To the average trader, it seems obvious that it can not possibly go up (false resistance). Ironically, if too many Bears crowd into an option, they are the very ones who give it the strength to go far higher when they panic. Purchasing the option to cover their short sale. From a traders viewpoint, its obvious that the option can not go up. They do not understand that when it is obvious to them, it is obvious to the majority. And except for temporary periods, the majority are always wrong, because the insiders are on the opposing side, as in a contest. They know what they are doing and have the power to do it.

Accumulation Triangle - This type of formation brings a feeling of uncertainty as to the direction the market will take as the two oposing trend lines (Bulls & Bears) converge. This uneasiness can be seen as the formation matures and the commodity becomes less volatile indicating that it is now under accumulation. This price action insulls fear among speculators driving them out of the option which is exactly what the manipulators intended to do. The commodity is now ready for a big move since all the dead weight has been dumped overboard.

W - Formation " Double Bottom " - The technical strength of this formation lies in the fact that it creates a public fear that prices will continue lower. They have lost hope in it's ability to rise and therefore that's exactly what happens.

Head & Shoulders Bottom - This pattern frightens the speculators during the first shoulder and head. During the second shoulder, they sell out their positions in fear that the price will drop down again. They have lost hope in it's ability to rise.

Bull Flag Formation - " As momentum is gathered (during the markup) there may suddenly be a one day rise followed by a sharp setback almost as quickly, giving a weak appearance as if it has gone too high. Don't let this fool you, it is meant to look this way. You and others are considered to be excess baggage; this was meant to dump you overboard and to discourage buying." This type of formation is what causes people to sell out their positions too early as they become fearful of a big drop. This is how the manipulators get rid of all the free loaders only to attract them back into the option when they drive the price up hard for distribution.

Consolidation patterns - This quiet formation gives the false impression of the commodity being unable to rise. This false resistance level and quiet action is meant to "wear out" the speculators resolve to hold a position in the option. It is from these lengthy low consolidation areas that most of the really large moves start.

W.D. GANN. He is reputed to have made \$50 million in his 52-year career of trading in the futures and stock markets. He started trading with \$300 and made over \$25,000 in profits in his first year of active trading. He then took \$973 and made over \$30,000 in the Cotton market in 60 days time. Over a period of years, on each \$1,000 he started with, he made profits of \$26,500. A friend said, "He has made a great deal of money. I once saw him take \$130, and in less than a 1 month, run it up to over \$12,000. He has taken half a million dollars out of the markets in the past few years. He can compound money faster than any man I ever met". His commodity course and stock market courses sold for \$5,000 each in the 1930's.

"As a young man, I spent 9 months working day and night in the Astor Library of New York and in the British Museum of London, going over the records of stock transactions as far back as 1820. Charts are records of past market movements. The future is but a repetition of the past; there is nothing new. As the Bible says, 'The thing that hath been, it is that which shall be; and that which is done is that which shall be done, and there is no new things under the sun'. Knowledge of the markets is more important than money and with proper knowledge and rules a trader can always make a success in the markets.

What has happened in past markets will happen again in the future. Human nature does not change and that is one reason why history repeats, and markets act very much the same under certain conditions year after year and in the various cycle of time. The future is nothing but a repeat of the past, and charts are the only guide. Charts are of great value. Future market movements can be forecast by a study of past history and past movements. It is important to know what has happened in the past and for how many months, for example, important moves have run before there was a reaction.

I want to impress upon you strongly that if you expect to make a success in the markets you must put in plenty of time studying past price movements, because the more time you put in, the more knowledge you gain, and the more profits you will take out later. The time spent in study of price, time and past market movements, will give you a rich reward.

You should keep a YEARLY, MONTHLY and WEEKLY chart. The MONTHLY chart is the most important chart of all in determining the main trend. The next most important chart is the WEEKLY chart. You should keep a chart of MONTHLY prices for as far back as you can get them and then a WEEKLY chart anywhere from 6 to 12 months back. Watch for indications of activity on the MONTHLY and WEEKLY charts. By crossing old tops and showing activity, it is a good time to buy. Go with the trend. Wait until a market shows a definite trend.

If you study MONTHLY and WEEKLY charts you will note the different formations before any big advance takes place. Study the bottom formations at extreme low levels and note how long prices usually remain in the bottom range before the next advance begins. The big profits are made in the runs between accumulation and distribution. Therefore, you make money by waiting until a market PLAINLY DECLARES IT TREND than by getting in before it starts. When a market breaks out of its accumulation zone, it is a safe buy and you should make quick profits, because accumulation has already taken place.

The market advances once it breaks out of the trading range, with a ceiling, top or resistance level. When a commodity remains in a narrow trading range or sideways move, making about the same top and bottom levels several times, leave it alone until it shows some definite move out of this range. After accumulation, with considerable time consumed, the largest profits are made in the shortest time period by buying after the breakaway point. When a commodity holds for many weeks in a narrow range, then crosses tops of previous weeks, the trend has changed and you should go with it. The longer the period of time in a narrow range, the greater the advance. This sideways movement is nearly always accumulation.

When a commodity advances and crosses the highest price of its history, it usually indicates higher prices and is safe to buy. When tops that were made years previous are crossed, it usually means that the commodity is going much higher. Therefore, it is very important to watch old price levels. The longer the time that elapses between the breaking into new territory the greater the move you can expect, because the accumulated energy over a long period will produce a larger movement than if it only accumulated during a short time period. Market movements of importance, i.e., the long swings, require weeks and months to get ready, or for accumulation to be completed. There is always plenty of time to buy after the big move gets under way.

After a long period of time when prices advance above old highs or old tops, they are in a stronger position and indicate higher prices. This is the reason why you must have a chart a long ways back in order to see what position a commodity is in and at what stage it is. You should have a MONTHLY chart going back 15 or 20 years so you will know where the important tops and bottoms are. Remember the old Chinese proverb, "One picture is worth 10,000 words". You should study the picture of a commodity before you make a trade.

Acquiring knowledge will surely bring success in investment or speculation. After you have acquired knowledge you will need to learn patience. The value of patience is that you must learn to wait for a DEFINITE INDICATION OF A CHANGE IN TREND before buying or selling. You must not guess or gamble on hope or fear. You must have the ability to act and to act quickly at the right time, acting after you have acquired knowledge and know it is time to act. You must have the courage to act. Courage and boldness will come after you have acquired knowledge because you will have confidence in proven rules and confidence in your ability. Therefore, the acquiring of knowledge will give you the courage for action at the time when the real opportunity comes.

The main thing to do is to always GO WITH THE TREND and never buck the trend, regardless of how much capital you have. By trading with the main trend, you make greater profits. NEVER GUESS--LET THE MARKET TELL ITS OWN STORY! Buy or sell only on rules and definite indications. Stocks move faster at higher levels than they do at low levels. Therefore, as a general rule, you can make the most money by buying calls on stocks selling between \$75 and \$150 per share. Remember stocks are never too high to buy as long as the trend is up. Buy stocks in strong position. Always go with the trend. But do not overlook the fact that you must use a stop loss for your protection. You must trade in such a way that you will not be disturbed mentally by a loss, if it comes. You can always find new opportunities, so long as you have capital to operate with.

TIME is a most important factor in forecasting market movements and by studying the past records of markets you will be able to prove for yourself that history does repeat and that by knowing the past you can tell the future. The ancient hunters had a rule that when they were searching to locate an animal in this end, they always followed his tracks BACKWARD, figuring that it was the shortest route to his lair. The quickest way for you to learn how

to determine FUTURE market movements is to study the PAST.

It takes time to accumulate a commodity after a prolonged decline. By studying the record of past movements, you can tell about the time required at the bottom before a big change in trend should take place. When you learn the past history of a commodity, learn the way it is running, what cycle it is in, and then make a trade on definite knowledge, your chances for success are 100% greater. The most money is made when fast moves and extreme fluctuations occur at the end of major TIME cycles. Everything moves in cycles as a result of the natural law of action and reaction. By a study of the past, you will discover what cycles repeat in the future. In order to be accurate in forecasting the future, you must know the major cycle. When prices reach bottom, as a rule, a series of rapid fluctuations take place; then the market gradually narrows down and lays the foundation for another important move up.

When a stock advances to a new high level or crosses a former old top by 3 points, then if it is going higher, it should not react back 3 points below the former old high. When a stock advances into new historical high territory, it usually indicates higher prices and is safe to buy. If it is going higher the stock seldom reacts 3 points under the old high level.

Bert Castles, a large operator, made money and kept it. How did Castles do it? When he bought he always placed a stop loss order on stocks not more than 5 points away. This limited his loss if he was wrong. When he was right, he let his profits run until he had a definite reason to cash in. Successful investors have definite plans and rules, and follow them. Successful traders buy at the right time. They knew when to take profits. They did not overtrade--they knew the unexpected could happen, and they sold when everybody else was buying and bought when everyone else was selling. These are the reasons that they made a success and did not go broke. They used stops in case they were wrong and took a small loss. And they had the patience to wait for opportunities and to buy and sell at the right time. They followed rules.

For a trader to succeed, he must do the opposite of what he finds the general public doing. When extremes occur and everybody is radically bearish and can see no hope for prices ever advancing, that is the time to go against human nature and follow the charts for the main swing. For instance, the longer a market advances, especially when the main trend is up for weeks or months, the stronger it gets and the smaller the reactions in the last stage because in these last stages, hope gives way to euphoria. When a market is advancing, hope and optimism increase to the limit and traders buy, not using sound judgment, but buying on hope. This is the way the public trades. To make a success you must follow rules and trade opposite to the general public. You're in bad company, when the public is with you on a large scale.

It always happens that the higher stock prices go, the more people want to buy. This causes a final grand rush and a rapid advance in the last stages of a bull market. History repeats on Wall Street and what has happened in the past will happen again in the future. Stock market campaigns move in 3 to 4 sections or waves. Never consider that the market has reached final top when it makes the first section in a move up, because if it is a real bull market it will run at least 3 sections and possibly 4 before a final high is reached. In the grains, the big market moves, like volcanic eruptions, are over very quickly because they are the result of weeks, months or years of accumulation. When resistance is overcome and the accumulated forces are let loose, they move very rapidly until the energy is exhausted and the time is up. Guard against the human element...don't become impatient...don't be guided by your emotions. The big money is made in the last stage of a bull market when prices are feverishly active. But remember that sharp reactions follow from tops where there is a final grand rush after a fast move up. In wheat bull markets, when it comes to the finish or final grand rush and reaches the boiling point, there is a very fast move up. My rule is, wait to sell your wheat until it boils. Most grains finish a bull market campaign with a fast advance, lasting 6 or 7 weeks, sometimes as much as 10 weeks, moving up very fast. Remember one thing: prices are never so high but what they can go higher as long as the trend is up. My rule is to go with the trend and only change when the primary trend changes.

Commodities are much easier to forecast than stocks, because commodities always move with supply and demand--this is not always true with stocks. Commodities offer greater opportunity.

tunities for profit than stocks. You can forecast tops and bottoms of commodities with greater certainty than stocks. Commodities move faster and are more interesting than stocks. Notable speculators like Livermore, Armour, Patten and Crawford, have discovered after long experience that they can make money with greater certainty in commodities than stocks. Your percentage of gain is much greater in commodities than in stocks. Let me cite two examples I know of personally. Eugene Scales and Jordan of New Orleans started with \$300 and \$500, respectively, years apart, and both ran it up into a fortune of millions in the Cotton market. There are many similar examples.

The best kind of markets to trade in are those that are active and, according to past movements, follow the rules and a definite trend. I have always made a success trading in active, fast-moving markets. The big money is always made in trading markets that fluctuate over a wide range. Wide swings give great opportunities for making money. Soybeans should remain "king" for a long time to come and offer some unusual money-making opportunities. The best way to trade and the most money is made in long-pull trades, that is, following a definite trend as long as the trend is up. My experience has proved that it is more profitable to trade in commodities than stocks, and you can make larger profits on the same amount of capital with smaller risk. When you trade in commodities, you are trading in the necessities of life--commodity prices obey the law of supply and demand.

The greatest opportunities for profits occur at the end of the great time cycles when advances are very rapid and you can make large profits in a short period of time. It is important to note whether a commodity is making lower tops each year. If a commodity has made a lower top each year for 5 years, then makes a higher top than the previous year, it is a sign of reversal and indicates a long up cycle. Monthly moves can also be determined and forecasted by this same rule. You will always make the greatest profits by following the main trend and playing the long swing. You can never make money jumping in and out of a market. The big money is always made by following the main trend. It is much better to make 3 or 4 trades each year and make large profits most of the time, than it is to try to make dozens of trades and be wrong half the time, and make small profits on your winners, and end up with a net loss.

Pyramiding. Be careful about pyramiding at the wrong time. Wait until the stock or commodity is very active and has crossed resistance levels before buying more. It is always safer to pyramid after a market moves out of an accumulation zone. Select stocks with small number of shares outstanding to pyramid. Select commodities that show strong uptrends to pyramid.

The proper way to trade is to go with the trend, and pyramid while you are making profits. In pyramiding, your first risk should be your greatest. Suppose on your first trade, you risk \$500. If the trade moves \$500 in your favor, you can then buy a second lot and place a stop loss order \$500 away. And if it is caught, you will still be only \$500 loser, because you will be even on your first trade. Once a market gets away from the accumulation period and the trend is well-defined up, reactions are usually small. Wait until prices cross a previous top before buying a second lot.

There are two methods of pyramiding. One is to buy more just as soon as the market breaks into new territory or makes a new high. In a fast-running market, you can continue to buy every 10 or 20 points up when the market is moving in your favor. My method is to determine the reaction levels and how many points a market has reacted from temporary top levels. Find out whether these reactions are running 10 or 20 points; then buy your 1st, 2nd, 3rd or 4th lot for pyramiding on reactions from the tops, waiting 10 or 20 points according to the past reactions. My time rule, which will help you in pyramiding, is to determine the time of the first important reaction. There are many examples of a market reacting only 2 to 3 weeks from the time it started up and was good to buy every time it reacted 2 to 3 weeks from any top, until it made the final top in a big bull market. Determining the time of the reactions and measuring them this way will greatly increase your profits and enable you to follow the main trend of the market, sometimes for a year or more. Never start to pyramid after a long advance. The chances are against you. Begin pyramiding when the trend first turns up. Many a trader has begun at the bottom of a bull market to trade conservatively and accumulated a large amount of profits. Finally he begins to pyramid too heavily and too fast

near the top, with the result that when the trend turns he gets caught overloaded and loses all the profits he has made. Sad experience has taught me that it is better to be safe than sorry. In pyramiding, let "safety first" be your motto.

A pyramid should always be followed up with a stop loss order, no matter what method you use, because your profits must be protected. The more profits you have, however, the more room you can give the market to fluctuate, or have its reverse moves or reactions, that is, you can place your stop loss further away from the market so that a natural reaction will not disturb your pyramid. For example, suppose you have followed a market up and have a 100 pt profit on your original purchase. If the market has had a previous reaction of, say, 20 points, it could again react 20 points without changing the main trend, therefore your stop loss could be 25 points under the market, because if it was caught, you would not be losing part of your capital, but only a part of your paper profits--while in the early stages of your pyramid your stop loss would have to be closer in order to protect your original capital. These pyramiding rules, like other rules, work best on active, high-priced stocks or actively-traded commodities and should only be applied in these types of markets.

It is important to know how and when to pyramid. When a stock is at low levels, between 20 and 50, you can buy every 5 points up, that is, when you have 5 points' profit on the first trade. Buy an additional amount and place a stop so that you will not have any loss on the two trades if the market should reverse and catch the stop. When a stock is selling between 80 and 200, you should only buy 10 points apart. After you make the first commitment then wait until you have 10 points' profit, then buy a second lot. After you have bought a 4th or 5th lot in a pyramid, then you should decrease the trading unit. For example, if you have been buying 200 shares each 5 or 10 points up and have accumulated 1,000 shares, then it is safer to reduce your trading to 100 shares every 5 or 10 points up. The big money is made by always following the main trend of a stock. Never buy a second lot unless the first trade has moved in your favor.

Protecting Pyramided Profits. Suppose a stock breaks out on the upside, moves up fast and shows uptrend. You bought call options and have profits in them the first month. The trend still shows up and when it crosses previous tops or resistance levels, then you buy more calls the next month. You can now protect the profits that you have by buying put options under the market. Many traders do this when a move of 50 or 60 points takes place in 2 or 3 months. They continue to buy calls every month and then often buy puts to protect their profits instead of placing stop loss orders, thus making a large amount of profit on a very small risk. Suppose you have bought a call on a stock at 100 and it advances to 110; then you buy another call at a higher level, and then buy a put to protect your profit. You can continue to buy calls and buy puts, pyramiding all the way up, and not selling until you think the time has come for a change in the main trend. In May, 1937 Douglas Aircraft declined 30 points from the January high, affording excellent opportunities for pyramiding on the way down and staying short after you once had a profit in the put and a chance to protect it with a call. The percentage of profits to the risk was enormous.

Trading Rules. When you learn the rules and follow them, you eliminate trading on hope, fear and guesswork. Do not follow "inside information"--if you do, you will fail. Your success depends on knowing the right kind of rules and following them. In order to make a success in trading the markets, the trader must have definite rules and follow them. Then speculation can be made a profitable profession. Trading in commodities is not a gambling business, as some people think, but a practical, safe business when conducted on business principles. Like me, I know that anyone can make a success, if they follow rules. These rules have cost me 50 years of experience and hundreds of thousands of dollars to learn. You can make profit by strictly adhering to these rules. Make up your mind, if you are NOT going to follow rules don't trade. I have made a success and you, too, can make plenty of profits out of the commodity markets, if you study hard enough. The markets can be beaten and there is big money operating in commodities if you follow the rules. Here are the rules you must observe:

1. NEVER RISK MORE THAN 10% OF YOUR CAPITAL ON ANY ONE TRADE. The first and most important thing to know when you start to trade is the amount of capital required that will enable you to trade and never lose your capital. If you always have capital to start trading again,

can recoup any losses and make money. But if you risk all of your capital on one or two trades and lose it, then the chances are against you. "Safety first" is the rule to apply. Whatever amount of capital you use to trade with, divide it up into 10 equal parts and never risk more than 10% on any one trade. Should you lose for 3 consecutive times, then reduce your trading unit and only risk 10% of your remaining capital. If you follow this rule, your success is sure. In this way you will be able to make 10 consecutive trades on your capital and the markets would have to beat you 10 straight times to wipe out your capital, which they will not do.

2. NEVER OVERTRADE. The greatest and most important thing a trader must learn is that the unexpected can and often does happen.

3. ALWAYS BUY AND SELL "AT THE MARKET". Always use "market orders" to buy or sell.

4. FOLLOW UP YOUR TRADES WITH A STOP LOSS ORDER TO PROTECT YOUR PROFITS.

5. ACCUMULATE A SURPLUS AFTER EACH WINNING STREAK.

6. HAVE PATIENCE. Never get into a trade because you are tired of waiting. Never get out of a trade just because you have lost patience. Patience is one of the very important qualifications for success. In sum, you must have the patience to wait for the right opportunities; then you must have the patience to get in right; then you must have the patience to hold on.

7. NEVER TAKE SMALL PROFITS. You must hold on unless your stop is hit.

8. NEVER BUY JUST BECAUSE THE PRICE IS LOW.

9. ACT AT THE RIGHT TIME. Have the nerve to "pull the trigger"--to buy and sell at the right time. Knowledge gives you nerve, makes you bold and enables you to act at the proper time. When you fail to buy or sell at the right time, you become afraid--and fear is a detrimental trading influence.

10. TIME TO STAY OUT OF THE MARKETS. Important: you cannot trade continuously. There are times you should stay out of the markets. There are many times it will pay you to keep on the sidelines and wait for a real opportunity, which is sure to come if you wait.

Remember that all the rules work best when markets are near extreme high or extreme low levels. Changes in trend which occur between extreme high and low do not mean as much as a reversal when prices are near extreme high and extreme low points. Have the patience to wait for the choice opportunities to buy and sell at the right time. Don't delay or become impatient. Trust me when I say these rules are vital to your success.

In conclusion. Always remember that nerve is the equal of capital. In getting my initial trading experience, I have been broke many times, i.e., I have lost all of my money, but there never has been a time yet when I lost my nerve. I always believe in facing the future with nerve and hope. I always had the nerve to pyramid and press a market when my forecasting science showed that I was on the right side. One successful trader, who followed scientific forecasting methods, never makes more than 5 or 6 trades in a year. He only goes after the main swings and he follows it up and pyramids when the advance materializes as he expected. Great and unusual opportunities, when you can start at the bottom of a move, pyramid and make a fortune, occur every couple of years. If you are able to catch half of these major moves on conservative trades, your profits will be enormous. Do not try to catch the minor moves. Make speculation a business, not a gamble. Be patient. Be conservative. If big risks are required, do not make the trade. Wait for a genuine opportunity where you can buy and pyramid with a minimum of risk."

THE TICKER INVESTMENT DIGEST

Investment: The placing of capital in a more or less permanent way, mainly for the income to be derived therefrom.

Speculation: Operations wherein intelligent foresight is employed for the purpose of deriving a profit from price changes.

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No. 2

WILLIAM D. GANN

An Operator Whose Science and Ability Place Him in the Front Rank—
His Remarkable Predictions and Trading Record.

SOMETIME ago the attention of this magazine was attracted by certain long pull stock market predictions which were being made by William D. Gann. In a large number of cases Mr. Gann gave us in advance the exact points at which certain stocks and commodities would sell, together with prices close to the then prevailing figures which would not be touched.

For instance, when New York Central was 131 he predicted that it would sell at 145 before 129.

So repeatedly did his figures prove to be accurate, and so different did his work appear from that of any expert whose methods we had examined, that we set about to investigate Mr. Gann and his way of figuring out these predictions, as well as the particular use which he was making of them in the market.

The results of this investigation are remarkable in many ways.

It appears to be a fact that Mr. Gann has developed an entirely new idea as to the principles governing stock market movements. He bases his operations upon certain natural laws, which, though existing since the world began, have only in recent years been subjected to the will of man, and added to the list of so-called modern discoveries.

We have asked Mr. Gann for an outline of his work and have secured some remarkable evidence as to the results obtained therefrom. We submit this in full recognition of the fact that

in Wall Street a man with a new idea—an idea which violates the traditions and encourages a scientific view of the proposition—is not usually welcomed by the majority, for the reason that he stimulates thought and research. These activities said majority abhors.

Mr. Gann's description of his experience and methods is given herewith. It should be read with a recognition of the established fact that Mr. Gann's predictions have proved correct in a large majority of instances.

"For the past ten years I have devoted my entire time and attention to the speculative markets. Like many others, I lost thousands of dollars and experienced the usual ups and downs incidental to the novice who enters the market without preparatory knowledge of the subject.

"I soon began to realize that all successful men, whether lawyers, doctors, or scientists, devoted years of time to the study and investigation of their particular pursuit or profession before attempting to make any money out of it.

"Being in the brokerage business myself and handling large accounts, I had opportunities seldom afforded the ordinary man for studying the cause of success and failure in the speculations of others. I found that over ninety per cent of the traders who go into the market without knowledge or study usually lose in the end.

"I soon began to note the periodical

he has a firm grasp of the basic principles which govern stock market movements, and I do not believe any other man on earth can duplicate the idea or his method at the present time.

"Early this year he figured that the top of the advance would fall on a certain day in August and calculated the prices at which the Dow-Jones averages would then stand. The market culminated on the exact day and within four-tenths of one per cent. of the figures predicted."

"You and Mr. Gann must have cleaned up considerable money on all these operations," was suggested.

"Yes, we have made a great deal of money. He has taken half a million dollars out of the market in the past few years. I once saw him take \$130, and in less than one month run it up to cover \$12,000. He can compound money faster than any man I ever met."

"One of the most astonishing calculations made by Mr. Gann was during last summer (1909) when he predicted that September wheat would sell at \$1.20. This meant that it must touch that figure before the end of the month of September. At twelve o'clock, Chicago time, on September 30th (the last day) the option was selling below \$1.08, and it looked as though his prediction would not be fulfilled. Mr. Gann said 'If it does not touch \$1.20 by the close of the market it will prove that there is something wrong with my whole method of calculation. I do not care what the price is now, it must go there.' It is common history that September wheat surprised the whole country by selling at \$1.20 and no higher in the very last hour of the trading, closing at that figure."

So much for what Mr. Gann has said and done as evidenced by himself and others. Now as to what demonstrations have taken place before our representative:

During the month of October, 1909, in twenty-five market days, Mr. Gann made, in the presence of our representative, two hundred and eighty-six transactions in various stocks, on both the long and short side of the market. Two hundred and sixty-four of these transactions resulted in profits; twenty-two in losses.

The capital with which he operated was doubled ten times, so that at the end of the month he had one thousand per cent. of his original margin.

In our presence Mr. Gann sold Steel common short at 94 $\frac{1}{8}$, saying that it would not go to 95. It did not.

On a drive which occurred during the week ending October 29th, Mr. Gann bought Steel common at 86 $\frac{1}{4}$, saying that it would not go to 86. The lowest it sold was 86 $\frac{1}{8}$.

We have seen him give in one day sixteen successive orders in the same stock, eight of which turned out to be either the top or the bottom eighth of that particular swing. The above we can positively verify.

Such performances as these, coupled with the foregoing, are probably unparalleled in the history of the Street.

James R. Keene has said, "The man who is right six times out of ten will make his fortune." Here is a trader, who, without any attempt to make a showing (for he did not know the results were to be published), establishes a record of over ninety-two per cent profitable trades.

Mr. Gann has refused to disclose his method at any price, but to those scientifically inclined he has unquestionably added to the stock of Wall Street knowledge and pointed out infinite possibilities.

We have requested Mr. Gann to figure out for the readers of *The Ticker* a few of the most striking indications which appear in his calculations. In presenting these we wish it understood that no man, in or out of Wall Street, is infallible.

Mr. Gann's figures at present indicate that the trend of the stock market should, barring the usual rallies, be toward lower prices until March or April, 1910.

He calculates that May wheat, which is now selling at \$1.02, should not sell below 99c. and should sell at \$1.45 next spring.

On cotton, which is now at about the 15c. level, he estimates that, after a good reaction from these prices, the commodity should reach 18c. in the spring of 1910. He looks for a corner in the March or May option.

Whether these figures prove correct or not, will in no sense detract from the record which Mr. Gann has already established.

Mr. Gann was born in Lufkin, Texas, and is thirty-one years of age. He is a gifted mathematician, has an extraordinary memory for figures, and is an expert Tape Reader. Take away his science and he would beat the

orbit, just as the advancing year ever brings the rose of spring, so do the properties of the elements periodically recur as the weight of the atoms rises."

"From my extensive investigations, studies and applied tests, I find that not only do the various stocks vibrate, but that the driving forces controlling the stocks are also in a state of vibration. These vibratory forces can only be known by the movements they generate on the stocks and their values in the market. Since all great swings or movements of the market are cyclic they act in accordance with the periodic law.

"Science has laid down the principle that 'the properties of an element are a periodic function of its atomic weight.' A famous scientist has stated that 'we are brought to the conviction that diversity in phenomenal nature in its different kingdoms, is most intimately associated with numerical relationship. The numbers are not intermixed, chaotically and accidentally, but are subject to regular periodicity. The changes and developments are also seen to be in many cases undulatory.'

"Thus, I affirm, every class of phenomena, whether in nature or in the stock market, must be subject to the universal law of causation and harmony. Every effect must have an adequate cause.

"If we wish to avert failure in speculation we must deal with causes. Everything in existence is based on exact proportion and perfect relationship. There is no chance in nature, because mathematical principles of the highest order lie at the foundation of all things. Faraday said: 'There is nothing in the Universe but mathematical points of force.'

"Vibration is fundamental; nothing is exempt from this law; it is universal, therefore applicable to every class of phenomena on the globe.

"Through the Law of Vibration every stock in the market moves in its own distinctive sphere of activities, as to intensity, volume and direction; all the essential qualities of its evolution are characterized in its own rate of vibration. Stocks, like atoms, are really centers of energies, therefore they are controlled mathematically. Stocks create their own field of action and power; power to attract and repel, which principle explains why certain stocks at times lead the market and

'turn dead' at other times. Thus to speculate scientifically it is absolutely necessary to follow natural law.

"After years of patient study I have proven to my entire satisfaction as well as demonstrated to others that vibration explains every possible phase and condition of the market."

In order to substantiate Mr. Gann's claims as to what he has been able to do under this method, we called upon Mr. William E. Gilley, an Inspector of Imports, 16 Beaver street, New York. Mr. Gilley is well-known in the down-town district. He himself has studied stock market movements for twenty-five years, during which time he has examined every piece of market literature that has been issued and procurable in Wall Street. It was he who encouraged Mr. Gann to study out the scientific and mathematical possibilities of the subject. When asked what had been the most impressive of Mr. Gann's work and predictions, he replied as follows:

"It is very difficult for me to remember all the predictions and operations of Mr. Gann which may be classed as phenomenal, but the following are a few: In 1908 when Union Pacific was 168 $\frac{1}{2}$ he told me that it would not touch 169 before it had a good break. We sold it short all the way down to 152 $\frac{3}{4}$, covering on the weak spots and putting it out again on the rallies, securing twenty-three points profit out of an eighteen-point move.

"He came to me when United States Steel was selling around 50 and said 'This Steel will run up to 58 but it will not sell at 59. From there it should break 16 $\frac{3}{4}$ points. We sold it short around 58 $\frac{3}{4}$ with a stop at 59. The highest it went was 58 $\frac{3}{4}$. From there it declined to 41 $\frac{1}{4}$ —17 $\frac{1}{2}$ points.

"At another time wheat was selling at about 89c. He predicted that the May option would sell at \$1.35. We bought it and made large profits on the way up. It actually touched \$1.35 $\frac{1}{2}$.

"When Union Pacific was 172, he said it would go to 184 $\frac{3}{4}$ but not an eighth higher until it had had a good break. It went to 184 $\frac{3}{4}$ and came back from there eight or nine times. We sold it short repeatedly with a stop at 185 and were never caught. It eventually came back to 172 $\frac{1}{2}$.

"Mr. Gann's calculations are based on natural law. I have followed his work closely for years. I know that

recurrence of the rise and fall in stocks and commodities. This led me to conclude that natural law was the basis of market movements. I then decided to devote ten years of my life to the study of natural law as applicable to the speculative markets and to devote my best energies toward making speculation a profitable profession. After exhaustive researches and investigations of the known sciences, I discovered that the Law of Vibration enabled me to accurately determine the exact points to which stocks or commodities should rise and fall within a given time. The working out of this law determines the cause and predicts the effect long before the Street is aware of either. Most speculators can testify to the fact that it is looking at the effect and ignoring the cause that has produced their losses.

"It is impossible here to give an adequate idea of the Law of Vibration as I apply it to the markets, however, the layman may be able to grasp some of the principles when I state that the Law of Vibration is the fundamental law upon which wireless telegraphy, wireless telephones and phonographs are based. Without the existence of this law the above inventions would have been impossible.

"In order to test out the efficiency of my idea I have not only put in years of labor in the regular way, but I spent nine months working night and day in the Astor Library of New York and in the British Museum of London, going over the records of stock transactions as far back as 1820. I have incidentally examined the manipulations of Jay Gould, Daniel Drew, Commodore Vanderbilt and all the other important Wall Street manipulators from that time to the present day. I have examined every quotation of Union Pacific prior to and from the time of E. H. Harriman's securing control, and can say that of all the manipulations in the history of Wall Street, Mr. Harriman's was the most masterly. The figures show that, whether unconsciously or not, Mr. Harriman worked strictly in accordance with natural law.

"In going over the history of markets and the great mass of related statistics, it soon becomes apparent that certain laws govern the changes and variations in the value of stocks and there exists a periodic or cyclic law, which is at the back of all these movements. Observation has shown that

there are regular periods of intense activity on the Exchange followed by periods of inactivity. Mr. Henry Hall, in his recent book devoted much space to 'Cycles of Prosperity and Depression' which he found recurring at regular intervals of time. The law which I have applied will not only give these long cycles or swings, but the daily and even hourly movements of stocks. By knowing the exact vibration of each individual stock I am able to determine at what point each will receive support and at what point the greatest resistance is to be met.

"Those in close touch with the markets have noticed the phenomena of ebb and flow, or rise and fall in the value of stocks. At certain times a stock will become intensely active, large transactions being made in it; at other times this same stock will become practically stationary or inactive with a very small volume of sales. I have found that the Law of Vibration governs and controls these conditions. I have also found that certain phases of this law govern the rise in a stock and an entirely different rule operates on the decline.

"While Union Pacific and other railroad stocks which made their high prices in August were declining, United States Steel common was steadily advancing. The Law of Vibration was at work, sending a particular stock on the upward trend, whilst others were trending downward.

"I have found that in the stock itself exists its harmonic or inharmonic relationship to the driving power or force behind it. The secret of all its activity is therefore apparent. By my method I can determine the vibration of each stock and by also taking certain time values into consideration I can in the majority of cases tell exactly what the stock will do under given conditions.

"The power to determine the trend of the market is due to my knowledge of the characteristics of each individual stock and a certain grouping of different stocks under their proper rates of vibration. Stocks are like electrons, atoms, and molecules, which hold persistently to their own individuality in response to the fundamental Law of Vibration. Science teaches 'that an original impulse of any kind finally resolves itself into periodic or rhythmical motion,' also 'just as the pendulum returns again in its swing, just as the moon returns in its

market on his intuitive tape reading alone.

Endowed as he is with such qualities, we have no hesitation in predicting that within a comparatively few years Wm. D. Gann will receive full recognition as one of Wall Street's leading operators.

R. D. W.

Note—Since the above forecast was made, Cotton has suffered the expected decline, the extreme break having been 120 points. The lowest on May wheat thus far has been \$1.01½. It is now selling at \$1.06¼.
